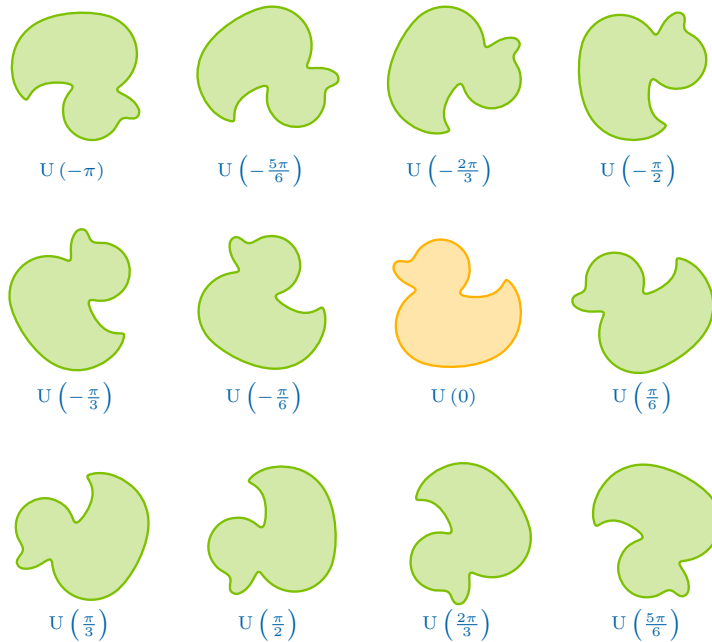




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Faculty of Mathematics

Linear Algebra



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MMXXVI

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Foreword

These lecture notes have been written for the linear algebra course which is taught on the second semester of the first of the bachelor of Mathematics for Data Science at the University of Vienna. Xiaoye and Olga hold the copyrights to this document and as such, its content, either partially or as a whole, cannot be redistributed or used without our explicit consent.

Acknowledgements

We would like to thank Prof. Hermann Schichl for helping in making the connection between the content of the previous linear algebra course and the current one. The content which is in the current notes is very classical, and several excellent books and lecture notes have inspired us in the writing. For the main text, the book by [Allaire and Kaber \(2002\)](#) on numerical linear algebra has been a source of inspiration. Many exercises are inspired from the lecture notes by [Amores Lázaro \(2018\)](#) from Universidad Complutense de Madrid.

This version of these lecture notes is in the process of being written, and as a result there will be many smaller and larger errors. We welcome any feedback, from anyone, that may help improve the text. In that light we already gratefully acknowledge the suggestions from Allard van Belois, Markus Ohrenhofer, Diana Artemova, Amadeus Vasyľ Pfund, Maximilian Bayerle, Jasper Jonkhans, Leonie Pribil.

Revisions

These notes will be updated as the course progresses. A record of revisions follows for convenience. We welcome any feedback that may help improve the text.

Date	Changes
27th February 2026	Initial version containing Chapter 1 and most of Chapter 2.
15th March 2026	Correct typos and refine Chapter 2.
14th April 2025	Add first part of Chapter 3
23rd April 2026	Add and streamline exercises in Chapter 2, add end of Chapter 3
8th May 2026	Add Chapter 4 on determinants
14th May 2026	Add Hadamard's inequality and exercises on determinants
19th May 2026	Add first part of Chapter 5 with a few exercises
26th May 2026	Add Section 5.3 and further exercises on eigenvalues
17th June 2026	Add triangularization and spectral theorems
29th June 2026	Add SVD

Further References

- Plenty of textbooks cover the content of this course: [Axler \(2024\)](#), [Strang \(2021\)](#).
- Many concepts that we discuss in the course are explained in an intuitive and visual way in these series of videos by [3Blue1Brown](#):

<https://www.3blue1brown.com/topics/linear-algebra>

- An interesting [lecture](#) by Prof. Alex Townsend connected to SVD, with nice intuitions on the importance of low rank matrices.

Chapter 1

Vector Spaces

1.1 The structure of a vector space

We wish to work on sets where it is possible to speak about the “sum” of two elements.

Definition 1.1: Binary operation $+$ on a set E

A binary operation on a set E is a mapping

$$\begin{aligned} + : E \times E &\rightarrow E \\ (x, y) &\mapsto x + y \end{aligned}$$

which produces an element E out of two elements from E .

The symbol “+” is just notation. Its meaning has to be adapted depending on the context as we see in the following examples.

Example 1.2

- Addition “+” on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$,
 - multiplication “ $\cdot$ ” on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$,
 - subtraction “-” on $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ but not on $\mathbb{N}$,
 - division “/” on $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^*$ but not on $\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C} \dots$
-

When $(E, +)$ has enough structure, it becomes what we call a group.

Definition 1.3: (Commutative) Group

We say that $(E, +)$ is a group when E is a set, and $+$ is a binary operation on E satisfying:

1. Associativity: $(x + y) + z = x + (y + z), \quad \forall x, y, z \in E$
2. Identity element (aka. neutral element): There exists $0_E \in E$ such that $x + 0_E = x$
3. Inverse element: for every $x \in E$, there exists $\tilde{x} \in E$ such that $x + \tilde{x} = 0_E$. We usually denote it by $-x$.

If, in addition, the elements commute, then we speak about a commutative group (aka. Abelian group):

4. Commutativity: $x + y = y + x, \quad \forall x, y \in E$
-

The following are examples and counter-examples of (commutative) groups (in ex. 1.1 we ask you to prove these statements).

Example 1.4

- The set $GL(n, \mathbb{R})$ of invertible matrices of size $n \times n$ with matrix multiplication as binary operation is a group, which is not commutative.
- Commutative groups:
 - $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$,
 - $(\mathbb{Q}^*, \cdot)$, $(\mathbb{R}^*, \cdot)$, $(\mathbb{R}_+^*, \cdot)$, $(\mathbb{C}^*, \cdot)$,
- Counter-examples: $(\mathbb{R}_+, +)$ and $(\mathbb{Z}^*, \cdot)$ are not groups (since they lack the existence of inverse elements)

On the group $(E, +)$, we wish to speak not only about binary operations within E , but also about the product $\lambda \cdot x$ between any element $x \in E$ and another element λ from a field $\mathbb{K}$. This is very useful to introduce a notion of scaling. For example, when $\mathbb{K} = \mathbb{R}$ we will be able to speak about the ‘triple’ or ‘half’ of $x \in E$, or to multiply an element of E by $\sqrt{2}$, or π . This motivates the following definition.

Definition 1.5: Scalar multiplication

A scalar multiplication on E over the field $\mathbb{K}$ is a map

$$\begin{aligned} \cdot : \mathbb{K} \times E &\rightarrow E \\ (\lambda, x) &\mapsto \lambda \cdot x \end{aligned}$$

which produces an element in E out of a scalar λ and an input element $x \in E$.

To be absolutely clear in certain definitions, we may sometimes overload the notation of multiplication, and write $\cdot_{\mathbb{K}}$ to emphasize on the field with which one works. For example, we use this notation in Section 1.2. The elements from $\mathbb{K}$ are often called ‘scalars’, and the elements of E are called ‘vectors’. It is important that binary operation $+$ is compatible with scalar multiplication $\cdot$ in the following sense.

Definition 1.6: Compatibility between $\cdot$ and $+$

We say that $\cdot$ is compatible with $+$ if and only if for all $x, y \in E$ and all $\lambda, \mu \in \mathbb{K}$:

1. Identity element of the scalar multiplication: $1_{\mathbb{K}} \cdot x = x$ ($1_{\mathbb{K}}$ being the multiplicative identity of $\mathbb{K}$)
2. Compatibility of scalar multiplication with field multiplication: $(\lambda\mu) \cdot x = \lambda \cdot (\mu \cdot x)$.
3. Distributivity of scalar multiplication w.r.t. vector addition: $\lambda \cdot (x + y) = \lambda \cdot x + \lambda \cdot y$
4. Distributivity of scalar multiplication w.r.t. field addition: $(\lambda + \mu) \cdot x = \lambda \cdot x + \mu \cdot x$.

We are now ready to define ‘an environment’ where the set E plays ‘in harmony’ with binary operations and scalar multiplication: vector spaces.

Definition 1.7: Vector space

Given a field $\mathbb{K}$, a set E , a binary operations $+$: $E \times E \rightarrow E$ and a scalar multiplication $\cdot$: $\mathbb{K} \times E \rightarrow E$. We call the triplet $(E, +, \cdot)$ a $\mathbb{K}$ -vector space if and only if $(E, +)$ is a commutative group, and $\cdot$ is compatible with the binary operation $+$. That is, for all $x, y, z \in E$ and all $\lambda, \mu \in \mathbb{K}$:

1. $(x + y) + z = x + (y + z)$
2. $\exists 0_E \in E$ s.t. $x + 0_E = x$
3. $\forall x \in E, \exists \tilde{x} \in E$ s.t. $x + \tilde{x} = 0_E$ (denoted by $-x$)
4. $x + y = y + x$
5. $1_{\mathbb{K}} \cdot x = x$
6. $(\lambda\mu) \cdot x = \lambda \cdot (\mu \cdot x)$
7. $\lambda \cdot (x + y) = \lambda \cdot x + \lambda \cdot y$
8. $(\lambda + \mu) \cdot x = \lambda \cdot x + \mu \cdot y$

Remark 1.8. • The eight properties that a vector space should satisfy are often referred to as *Axioms of vector spaces*.

- We may observe that a $\mathbb{K}$ -vector space $(E, +, \cdot)$ is never empty: since $(E, +)$ is assumed to be a group, then there always exists a neutral element 0_E which belongs to E .
- It is normal to find that the structure of a vector space is very abstract. However, this is done on purpose so that we are able to deal with plenty of different situations with the same machinery (see Section 1.2 for examples).
- In a vector space, we can multiply vectors $x \in E$ by scalars $\lambda \in \mathbb{K}$. However, the construction of vector spaces does not come with a notion of multiplication between vectors. We are going to come back to this notion later on.

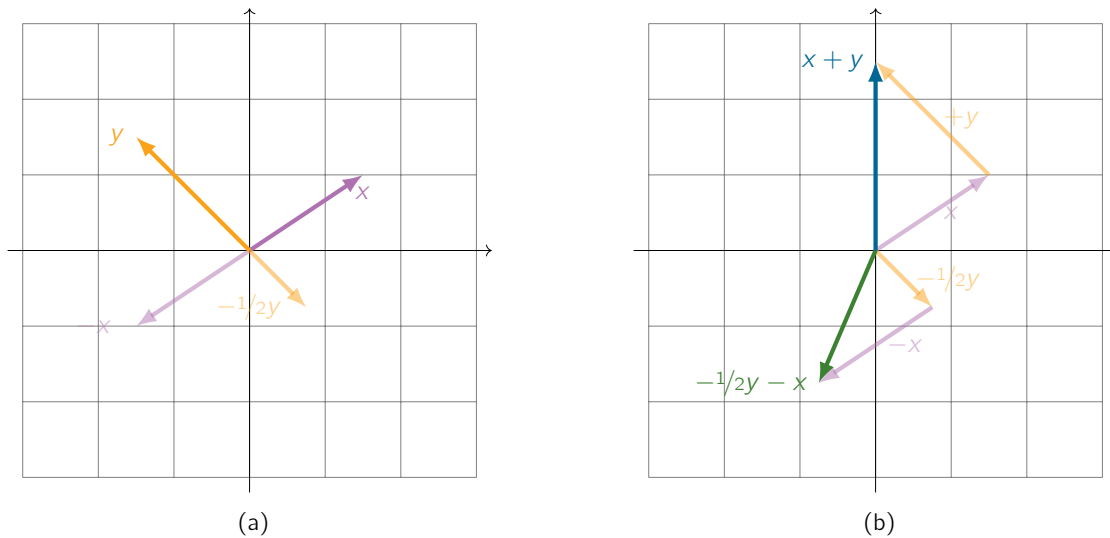


Figure 1.1: Graphical representation of scalar multiplication and addition of vectors in $\mathbb{R}^2$

1.2 Fundamental examples of vector spaces

We next list some of the most common vector spaces. We only provide the constructions for the summation and scalar multiplication, and leave the proofs as Section 1.11.2.

1.2.1 Vector spaces over $E = \mathbb{K}^n$ where $\mathbb{K}$ is a field

Given a field $\mathbb{K}$, we consider the set $E = \mathbb{K}^n := \overbrace{\mathbb{K} \times \dots \times \mathbb{K}}^{n \text{ times}}$, equipped with the vector addition $+_E$ and scalar multiplication $\cdot_E$ as defined below:

$$\begin{aligned}(x_1, \dots, x_n) +_E (y_1, \dots, y_n) &:= (x_1 +_{\mathbb{K}} y_1, \dots, x_n +_{\mathbb{K}} y_n), \\ \lambda \cdot_E (x_1, \dots, x_n) &:= (\lambda \cdot_{\mathbb{K}} x_1, \dots, \lambda \cdot_{\mathbb{K}} x_n).\end{aligned}$$

Then $(E, +_E, \cdot_E)$ is a $\mathbb{K}$ -vector space.

It holds that:

- $(\mathbb{R}^n, +_{\mathbb{R}^n}, \cdot_{\mathbb{R}^n})$ with standard vector addition, and coordinatewise scalar multiplication,

$$\begin{aligned}(x_1, \dots, x_n) +_{\mathbb{R}^n} (y_1, \dots, y_n) &:= (x_1 +_{\mathbb{R}} y_1, \dots, x_n +_{\mathbb{R}} y_n) \\ \lambda \cdot_{\mathbb{R}^n} (x_1, \dots, x_n) &:= (\lambda \cdot_{\mathbb{R}} x_1, \dots, \lambda \cdot_{\mathbb{R}} x_n)\end{aligned}$$

is an $\mathbb{R}$ -vector space.

- $(\mathbb{C}^n, +, \cdot)$ with the operations of complex addition and multiplication is a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Let $n \in \mathbb{N}$. The elements of $\mathbb{K}^n$ are the n -tuples of $\mathbb{K}$. If x is an element of $\mathbb{K}^n$, we will usually write

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (x_1, \dots, x_n)^T \quad (\text{a vertical column}),$$

where the x_i are elements of $\mathbb{K}$ for $i = 1, \dots, n$. If $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and $y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ are two vectors in $\mathbb{K}^n$, and if $\lambda \in \mathbb{K}$, we define the binary operation and scalar multiplication as

$$x +_{\mathbb{K}^n} y = \begin{pmatrix} x_1 +_{\mathbb{K}} y_1 \\ \vdots \\ x_n +_{\mathbb{K}} y_n \end{pmatrix}, \quad \text{and} \quad \lambda \cdot_{\mathbb{K}^n} x = \begin{pmatrix} \lambda \cdot_{\mathbb{K}} x_1 \\ \vdots \\ \lambda \cdot_{\mathbb{K}} x_n \end{pmatrix},$$

which we usually write in short without specifying the spaces for the operations, and without the multiplication symbol

$$x + y = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad \text{and} \quad \lambda x = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix},$$

We can then show that $(\mathbb{K}^n, +_{\mathbb{K}^n}, \cdot_{\mathbb{K}^n})$ is a $\mathbb{K}$ -vector space with neutral element $0_{\mathbb{K}^n} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ (we will prove this claim in Section 1.11.2).

Example 1.9

- $(\mathbb{R}^n, +_{\mathbb{R}^n}, \cdot_{\mathbb{R}^n})$ with standard vector summation, and coordinatewise scalar multiplication,

$$\begin{aligned}(x_1, \dots, x_n) +_{\mathbb{R}^n} (y_1, \dots, y_n) &:= (x_1 +_{\mathbb{R}} y_1, \dots, x_n +_{\mathbb{R}} y_n) \\ \lambda \cdot_{\mathbb{R}^n} (x_1, \dots, x_n) &:= (\lambda \cdot_{\mathbb{R}} x_1, \dots, \lambda \cdot_{\mathbb{R}} x_n)\end{aligned}$$

is an $\mathbb{R}$ -vector space.

- $(\mathbb{C}^n, +, \cdot)$ with the same operations of summation and multiplication as above is a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

1.2.2 Vector space over the set of matrices $\mathbb{K}^{p,n}$ with $p, n \in \mathbb{N}$

Let $A = (a_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}}$ and $B = (b_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}}$ be two matrices with coefficients in $\mathbb{K} = \mathbb{C}, \mathbb{R}, \mathbb{Q}$, and having p rows and n columns. Let $\lambda \in \mathbb{K}$, and define the binary and scalar operations as

$$A + B = (a_{ij} +_{\mathbb{K}} b_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}}, \quad \text{and} \quad \lambda \cdot A = (\lambda \cdot_{\mathbb{K}} a_{ij})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}}.$$

We can prove that $(\mathbb{K}^{p,n}, +, \cdot)$ is a $\mathbb{K}$ -vector space where the neutral element is the zero matrix $0_{\mathbb{K}^{p,n}} = (0_{\mathbb{K}})_{\substack{1 \leq i \leq p \\ 1 \leq j \leq n}}$.

We take the chance of reminding certain notions from the previous linear algebra course:

- It is important to understand Gaussian elimination, and its connection with LU decomposition.
- It is also important to know the basic operations surrounding matrix transposition and conjugate-transposition. For every $A, B \in \mathbb{K}^{p,n}$, $C \in \mathbb{K}^{n,k}$ and $\lambda \in \mathbb{K}$:

- $A^T \in \mathbb{K}^{n,p}$, and $A^H = \overline{A}^T \in \mathbb{K}^{n,p}$
- $AC \in \mathbb{K}^{p,k}$
- $(A + B)^T = A^T + B^T$
- $(\lambda A)^T = \lambda A^T$
- $(A^T)^T = A$
- $(AC)^T = C^T A^T$

- The trace of a square matrix: For every $A, B \in \mathbb{K}^{n,n}$ and $\lambda \in \mathbb{K}$:

- $\text{Tr}(A) = \sum_{i=1}^n a_{i,i}$
- $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$
- $\text{Tr}(\lambda A) = \lambda \text{Tr}(A)$
- $\text{Tr}(AB) = \text{Tr}(BA)$

1.2.3 Polynomial spaces $\mathbb{K}[X]$

Let $\mathbb{K}$ be a field (for us, it will be $\mathbb{R}, \mathbb{C}$). A polynomial over $\mathbb{K}$ with coefficients $a_0, \dots, a_n \in \mathbb{K}$ and with $a_n \neq 0_{\mathbb{K}}$ is the finite sum

$$p(X) = \sum_{k=0}^n a_k X^k, \quad a_k \in \mathbb{K},$$

where X is an indeterminate. This means that X is a symbol, not a number. It needs to be further specified depending on the context (we explain the most common situations at the end of the section).

Definition 1.10: $\mathbb{K}[X]$ and degree of a polynomial

The set of polynomials over $\mathbb{K}$ is defined as

$$\mathbb{K}[X] := \left\{ \sum_{k=0}^{\infty} a_k X^k : a_k \in \mathbb{K} \text{ and } \exists n \in \mathbb{N}_0 \text{ s.t. } a_k = 0_{\mathbb{K}} \text{ for } k > n \right\}.$$

The degree of a polynomial $p \in \mathbb{K}[X]$ is

$$\deg(p) := \max\{n \in \mathbb{N}_0 : a_n \neq 0_{\mathbb{K}}\}.$$

To obtain a vector space over $\mathbb{K}[X]$, we define summation and scalar multiplication as follows. Let $p(X) = \sum_{k=0}^{\infty} a_k X^k$ and $q(X) = \sum_{k=0}^{\infty} b_k X^k$ be two elements of $\mathbb{K}[X]$, and let $\lambda \in \mathbb{K}$. Then we define

$$s(X) := p(X) + q(X) = \sum_{k=0}^{\infty} (a_k + b_k) X^k = \sum_{k=0}^{\max(\deg(p), \deg(q))} (a_k + b_k) X^k$$

It holds that $s(X) \in \mathbb{K}[X]$ and $\deg(s) = \max(\deg(p), \deg(q))$. We also define scalar multiplication as

$$\lambda \cdot_{\mathbb{K}[X]} p(x) = \sum_{k=0}^{\infty} (\lambda \cdot_{\mathbb{K}} a_k) X^k.$$

We can then prove that $(\mathbb{K}[X], +, \cdot)$ is a $\mathbb{K}$ vector space.

Let us now discuss the usual choices for X , and how this affects what we then understand by $p(X)$:

- The most common situation is to take X as a subset of $\mathbb{R}$ or $\mathbb{C}$ and $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. We then view a polynomial $p(X) \in \mathbb{K}[X]$ as a mapping where we can evaluate values at every $x \in X$,

$$p(X) : X \rightarrow \mathbb{K}$$

$$x \mapsto p(X)(x) := \sum_{k=0}^{\infty} a_k x^k.$$

We usually write $p(x)$ instead of $p(X)(x)$.

- Another relevant situation occurs when we take $X = \mathbb{K}^{n \times n}$. Similarly than in the previous case, we then view $p(X) \in \mathbb{K}[X]$ as the mapping

$$p(X) : \mathbb{K}^{n \times n} \rightarrow \mathbb{K}^{n \times n}$$

$$M \mapsto p(X)(M) := \sum_{k=0}^{\infty} a_k M^k, \quad (1.1)$$

and most often use the shorthand notation $p(M)$.

It is important to note that by convention we define the zero-th power of a matrix M to be $M^0 := \text{Id}$ (similar to how $x^0 = 1$ in the case of real numbers), therefore for polynomials that has a constant term λ_0 , it simply means $\lambda_0 M^0 = \lambda \text{Id}$.

- Since square matrices are the expression of endomorphisms on a given basis (see Lemma 1.35), one can make a further abstraction and define polynomials over endomorphisms. Let $(E, +, \cdot)$ be a vector space over the field $\mathbb{K}$, and define the set of endomorphisms over E as

$$\mathcal{L}(E) := \{T : E \rightarrow E : T \text{ is linear}\}.$$

Then we can choose $X = \mathcal{L}(E)$ and view $p(X)$ as

$$p(X) : \mathcal{L}(E) \rightarrow \mathcal{L}(E) \quad (1.2)$$

$$T \mapsto p(X)(T) := \sum_{k=0}^{\infty} a_k T^k.$$

1.2.4 Spaces of functions $\mathcal{F}(A, \mathbb{K})$ where A is a nonempty set and $\mathbb{K}$ is a field

Let A be a nonempty set and $\mathbb{K}$ a field. We define the set of functions from A to $\mathbb{K}$ as

$$\mathcal{F}(A, \mathbb{K}) := \{f : A \rightarrow \mathbb{K}\}.$$

For any two elements f and $g \in \mathcal{F}(A, \mathbb{K})$, we define summation and scalar multiplication by $\lambda \in \mathbb{K}$ as

$$\begin{aligned} f +_{\mathcal{F}(A, \mathbb{K})} g &: A \rightarrow \mathbb{K} \\ x &\mapsto f(x) +_{\mathbb{K}} g(x), \end{aligned}$$

and

$$\begin{aligned}\lambda \cdot_{\mathcal{F}(A, \mathbb{K})} f &: A \rightarrow \mathbb{K} \\ x &\mapsto \lambda \cdot_{\mathbb{K}} f(x).\end{aligned}$$

We can then prove that $(\mathcal{F}(A, \mathbb{K}), +, \cdot)$ is a $\mathbb{K}$ -vector space where the neutral element in $\mathcal{F}(A, \mathbb{K})$ is the zero function

$$\begin{aligned}0_{\mathcal{F}(A, \mathbb{K})} &: A \rightarrow \mathbb{K} \\ x &\mapsto 0_{\mathbb{K}}.\end{aligned}$$

Example 1.11

- We will often encounter the space $\mathcal{F}(\mathbb{R}, \mathbb{R})$ of functions from $\mathbb{R}$ to $\mathbb{R}$,
 - the space $\mathcal{F}(\mathbb{N}, \mathbb{R})$ of functions from $\mathbb{N}$ to $\mathbb{R}$, which is called the space of real sequences $\{x_n\}_{n=1}^{\infty}$, $x_n \in \mathbb{R}$, $\forall n \in \mathbb{N}$.
-

1.2.5 Spaces of functions $\mathcal{F}(A, E)$ where A is a nonempty set and $(E, +_E, \cdot_E)$ is a $\mathbb{K}$ -vector space

Let A be a nonempty set and $(E, +_E, \cdot_E)$ be a $\mathbb{K}$ -vector space. For any two applications f and g from A to E , we can define

$$\begin{aligned}f +_{\mathcal{F}(A, E)} g &: A \rightarrow E \\ x &\mapsto f(x) +_E g(x),\end{aligned}$$

and

$$\begin{aligned}\lambda \cdot_{\mathcal{F}(A, E)} f &: A \rightarrow E \\ x &\mapsto \lambda \cdot_E f(x).\end{aligned}$$

These operations define a binary operation, and scalar multiplication of elements of $\mathcal{F}(A, E)$ with scalars from $\mathbb{K}$. We can then prove that $(\mathcal{F}(A, E), +_E, \cdot_E)$ is a $\mathbb{K}$ -vector space.

Example 1.12

The space $\mathcal{F}(A, \mathbb{R}^3)$ of functions from a set A to $\mathbb{R}^3$ is an $\mathbb{R}$ -vector space.

1.2.6 Cartesian product of $\mathbb{K}$ -vector spaces

Let $n \in \mathbb{N}$ and for $1 \leq i \leq n$, let $(E_i, +_i, \cdot_i)$ be $\mathbb{K}$ -vector spaces. On the cartesian product $E := \prod_{i=1}^n E_i := E_1 \times \cdots \times E_n$:

- For all $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n) \in E$, we can define the binary operation as

$$(x_1, \dots, x_n) +_E (y_1, \dots, y_n) := (x_1 +_{E_1} y_1, \dots, x_n +_{E_n} y_n)$$

- For every $\lambda \in \mathbb{K}$, we can also define scalar multiplication as

$$\lambda \cdot_E (x_1, \dots, x_n) = (\lambda \cdot_{E_1} x_1, \dots, \lambda \cdot_{E_n} x_n)$$

One can then prove that $(\prod_{i=1}^n E_i, +, \cdot)$ is a $\mathbb{K}$ -vector space (... and that is the topic of Section 1.11.2).

1.3 Linear combination, generating family, free family, basis

Definition 1.13: Linear Combination

Let E be a $\mathbb{K}$ -vector space. Given vectors $x_1, \dots, x_n \in E$, a *linear combination* of these vectors is any vector of the form

$$v = \lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n, \quad \text{with } \lambda_1, \dots, \lambda_n \in \mathbb{K}, \text{ and } n \in \mathbb{N}.$$

Definition 1.14: Generating Set (Spanning Set)

Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be a finite subset of vectors of E with $n \in \mathbb{N}$. We say that $\mathcal{X}$ a *generating set* (or *spanning set*) of E if any element of E can be expressed as a linear combination of the elements of $\mathcal{X}$:

$$\forall x \in E, \exists \lambda_1, \dots, \lambda_n \in \mathbb{K} \text{ s.t. } x = \sum_{i=1}^n \lambda_i x_i.$$

Equivalently,

$$E = \text{span}\{x_i\}_{i=1}^n.$$

Definition 1.15: Linearly Independent Set (Free Family)

Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be a finite subset of vectors of E with $n \in \mathbb{N}$. We say that $\mathcal{X}$ is a *linearly independent set* (or a *free family*) if the only linear combination that gives the zero vector is the trivial one:

$$\lambda_1 x_1 + \dots + \lambda_n x_n = 0 \text{ for some } \lambda_1, \dots, \lambda_n \in \mathbb{K} \Rightarrow \lambda_1 = \dots = \lambda_n = 0.$$

Definition 1.16: Basis

A set of vectors $\mathcal{B} = \{x_1, \dots, x_n\} \subset E$ is a *basis* of E if it is both generating and linearly independent.

Proposition 1.17: Coordinate of a vector on a given basis

Let $\mathcal{B} = \{x_1, \dots, x_n\} \subset E$ be a basis of E . Then for any vector $x \in E$ there exist unique scalars $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that

$$x = \sum_{i=1}^n \lambda_i x_i.$$

The tuple of scalars

$$[x]^{\mathcal{B}} := (\lambda_1, \dots, \lambda_n)^{\top} \in \mathbb{K}^n$$

is called the *coordinate vector* of x with respect to the basis $\mathcal{B}$, and the linear mapping

$$\begin{aligned} \text{Vec}_{\mathcal{B}} : E &\rightarrow \mathbb{K}^n \\ x &\mapsto [x]^{\mathcal{B}} \end{aligned}$$

is a isomorphism.

Proof. Let $x \in E$. Since $\mathcal{B}$ is a generating set, there exist $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that $x = \sum_{i=1}^n \lambda_i x_i$. To show that the λ_i are unique, let's assume that there exists other scalars $\tilde{\lambda}_1, \dots, \tilde{\lambda}_n \in \mathbb{K}$ such that $x = \sum_{i=1}^n \tilde{\lambda}_i x_i$. Taking the difference between the two linear combinations, we get $\sum_{i=1}^n (\tilde{\lambda}_i - \lambda_i) x_i = 0$. Since $\mathcal{B}$ is a linearly independent set, we get $\tilde{\lambda}_i - \lambda_i = 0 \iff \tilde{\lambda}_i = \lambda_i$. In this way, we have proven that for every $x \in E$, there exists a unique vector $\text{Vec}(x) = [x]^{\mathcal{B}} = (\lambda_1, \dots, \lambda_n)^{\top}$, which shows that the mapping Vec is a bijection. Since the mapping is also linear, we conclude that Vec is an isomorphism. $\square$

A couple of key points related to the coordinate representation of vectors are:

- The coordinates depend on the chosen basis.
- The coefficients on a given basis are unique as we have just proven.
- The coordinates provide a concrete representation of abstract vectors as elements of $\mathbb{K}^n$.

Proposition 1.18

Let $x, y \in E$. Then

$$[\alpha x + \beta y]^{\mathcal{E}} = \alpha [x]^{\mathcal{E}} + \beta [y]^{\mathcal{E}}, \quad \forall \alpha, \beta \in \mathbb{K}.$$

Proof. We simply use the definition of the coordinate representation and the fact that $\mathbb{R}^n$ is a vector space. $\square$

1.4 Subspaces, direct sum

In the previous linear algebra course, the notions of subspaces, direct sum, etc were introduced. We are writing the essential facts here.

Definition 1.19: Subspace

Let $(E, +, \cdot)$ be a $\mathbb{K}$ -vector space, and let $F \subset E$ be a subset of E . We say that F is a subspace of E if and only if F is nonempty and stable by linear combinations:

1. $F \neq \emptyset$
 2. $\forall x, y \in F, x + y \in F$
 3. $\forall x \in F, \lambda \in \mathbb{K}, \lambda x \in F$.
-

The above conditions are equivalent to the following characterization, which is a bit more concise, and it is the one mostly used in practice. We suggest that you prove that both conditions are equivalent as an exercise.

Proposition 1.20

$F \subset E$ is a subspace of E if and only if:

1. $0 \in F$
 2. $\forall x, y \in F, \forall \lambda, \mu \in \mathbb{K}, \lambda x + \mu y \in F$.
-

Example 1.21

Here are some examples and counter-examples of subspaces (you should make sure that you know how to prove these statements):

- A line going through the zero element of a vector space E : for every $x \in E$, $F_x = \{\lambda x : \lambda \in \mathbb{K}\}$.
 - The set $\mathcal{C}(\mathbb{R}, \mathbb{R})$ of continuous functions is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.
 - The subset $\mathbb{K}_n[X]$ of polynomials of degree at most n is a subspace of $\mathbb{K}[X]$.
 - The set $\text{GL}_n(\mathbb{K})$ of square invertible matrices of size n is not a subspace of $\mathbb{K}^{n \times n}$.
-

Proposition 1.22

Let F and G be two subspaces of a vector space E . Then:

- $F \cap G$ is a subspace of E
- $F \cup G$ is in general not a subspace of E
- $F + G := \{x + y : x \in F, y \in G\}$ is a subspace of E , and we have the Grassmann formula

$$\dim(F + G) = \dim(F) + \dim(G) - \dim(F \cap G).$$

To convince oneself that $F \cup G$ is in general not a subspace, one can try to draw an example if $\mathbb{R}^2$, and then prove it in the general case (see ex. 1.35).

Definition 1.23: Direct sum

Let E be a vector space. The sum of k subspaces $F = F_1 + \cdots + F_k$ is said to be direct, and is written $F = F_1 \oplus \cdots \oplus F_k$ when for every $x \in F$ there exists a unique decomposition

$$x = \sum_{i=1}^k x_i, \quad \text{with } x_i \in F_i \text{ for } 1 \leq i \leq k.$$

The sum is said to be complementary when it is direct and it spans the whole space: $E = F_1 \oplus \cdots \oplus F_k$.

Theorem 1.24: Characterization of direct sum

Let $F = \sum_{i=1}^k F_i$ be a sum of subspaces of E . The following statements are equivalent:

1. For every $1 \leq i \leq k$,

$$\left(\sum_{\substack{j=1 \\ j \neq i}}^k F_j \right) \cap F_i = \{0\}.$$

2. If $0 = \sum_{i=1}^k x_i$ with $x_i \in F_i$ for $1 \leq i \leq k$, then $x_1 = \cdots = x_k = 0$.
 3. The subspaces are in direct sum: for every $x \in F$, there exists a unique set $(x_1, \dots, x_k) \in F_1 \times \cdots \times F_k$ such that $x = \sum_{i=1}^k x_i$.
-

Proof.

□

1.5 Linear Maps

Linear Algebra carries the adjective linear because it mostly focuses on vector spaces (stable by *linear* combinations), and also because *linear* maps are a very important object of study. Here is what we mean by a linear map.

Definition 1.25: Linear Map

Let E and F be two $\mathbb{K}$ -vector spaces. The mapping $f : E \rightarrow F$ is linear if and only if

$$f(\lambda x + \mu y) = \lambda f(x) + \mu f(y), \quad \forall x, y \in E, \lambda, \mu \in \mathbb{K}.$$

We denote by $\mathcal{L}(E, F)$ the set of all linear mappings from E to F . When $F = E$, the mapping is called an endomorphism, and the set of endomorphisms is denoted as $\mathcal{L}(E) := \mathcal{L}(E, E)$.

Proposition 1.26

Let E and F be two $\mathbb{K}$ -vector spaces, and $f \in \mathcal{L}(E, F)$. Then:

- $\mathcal{L}(E, F)$ is a subspace of $\mathcal{F}(E, F)$ (with addition and scalar multiplication as described Section 1.2.5), and this implies that $\mathcal{L}(E, F)$ is a vector space.
- The kernel and image sets

$$\ker(f) := \{x \in E : f(x) = 0\}$$

$$\operatorname{im}(f) := \{y \in F : \exists x \in E \text{ s.t. } y = f(x)\}$$

are subspaces of E .

Example 1.27

The mapping from $\mathbb{R}^3$ to $\mathbb{R}^2$ which, to every $(x_1, x_2, x_3) \in \mathbb{R}^3$ gives $(x_1, x_2 + x_3)$ is linear. The mapping giving $(x_1 x_2, x_2 + x_3)$ is nonlinear.

1.6 Dual Space

For every $\mathbb{K}$ -vector space E we can attach a notion of dual space, which plays a fundamental role in many mathematical developments.

Definition 1.28: Linear Form, Dual Space

Let E be a $\mathbb{K}$ -vector space. The dual space of E is the set $E^* := \mathcal{L}(E, \mathbb{K})$. Every $f \in E^*$ is called a linear functional.

The underlying intuition is that elements of the dual space are “machines” that take a vector and return a number.

Example 1.29

- Let $E = \mathbb{R}^3$, and consider $f, g, h : E \rightarrow \mathbb{R}$ defined as $f(x_1, x_2, x_3) = x_1 + x_2$ and $g(x_1, x_2, x_3) = x_1 + |x_2|$ and $h(x_1, x_2, x_3) = x_1 + 1$. Then $f \in E^*$ but $g, h \notin E^*$.

- If $E = \mathcal{F}(\mathbb{R}, \mathbb{R})$ and $a \in \mathbb{R}$, then the mapping

$$\begin{aligned} \text{Eval}_a : E &\rightarrow \mathbb{R} \\ f &\mapsto f(a) \end{aligned}$$

is a linear functional over E , namely $\text{Eval}_a \in E^*$.

- If $E = \mathcal{C}(\mathbb{R}, \mathbb{R})$, then the mapping

$$\begin{aligned} \text{Int}_{[0,1]} : E &\rightarrow \mathbb{R} \\ f &\mapsto \int_0^1 f(x) dx \end{aligned}$$

is a linear functional over E , namely $\text{Int}_{[0,1]} \in E^*$.

- If $E = \mathbb{R}[X]$, then the mapping

$$\begin{aligned} \text{Diff} : \mathbb{R}[X] &\rightarrow \mathbb{R} \\ P &\mapsto P(1) - P(0) \end{aligned}$$

is a linear functional over E .

Theorem 1.30: Existence of Dual Basis

Let E be a $\mathbb{K}$ -vector space of finite dimension $n \in \mathbb{N}$, and let $\mathcal{E} = \{e_1, \dots, e_n\}$ be a basis of E . For every $1 \leq i \leq n$, we define

$$\begin{aligned} e_i^* : E &\rightarrow \mathbb{K} \\ x &\mapsto [x]_i^{\mathcal{E}}, \end{aligned}$$

which is the mapping that returns the i -th coordinate component of the elements of E when expressed in the basis $\mathcal{E}$. Then $e_i^* \in E^*$ and

$$\mathcal{B}^* := \{e_1^*, \dots, e_n^*\}$$

is a basis of E^* , which is called the *dual basis* of $\mathcal{E}$. Furthermore,

$$e_i^*(e_j) = \delta_{ij}, \quad \forall 1 \leq i, j \leq n.$$

Proof. The mapping e_i^* is linear by Proposition 1.18, and the fact that component extraction is also a linear application. We next prove that $\mathcal{B}^*$ is a basis of E^* by showing that it is both a linearly independent and generating set.

- Linear Independence: Let $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that $\lambda_1 e_1^* + \dots + \lambda_n e_n^* = 0$. Then, $(\lambda_1 e_1^* + \dots + \lambda_n e_n^*)(e_i) = \lambda_i = 0$ for all $1 \leq i \leq n$, which shows that $\mathcal{B}^*$ is a linearly independent set.
- Generating set: Let $\varphi \in E^*$. For every $x = \sum_{i=1}^n [x]_i^{\mathcal{E}} e_i \in E$, it holds that

$$\varphi(x) = \varphi(x_1 e_1 + \dots + x_n e_n) = [x]_1^{\mathcal{E}} \varphi(e_1) + \dots + [x]_n^{\mathcal{E}} \varphi(e_n) = \varphi(e_1) e_1^*(x) + \dots + \varphi(e_n) e_n^*(x),$$

where we have used that $[x]_i^{\mathcal{E}} = e_i^*(x)$ by definition of the e_i^* . Since the equality holds for every $x \in E$, we obtain the equality of linear functionals

$$\varphi = \varphi(e_1) e_1^* + \dots + \varphi(e_n) e_n^*.$$

It expresses φ as linear combination of the elements $e_i^* \in E^*$, thus showing that $\varphi \in \text{span}\{\mathcal{B}^*\}_{i=1}^n$. Since this holds for every $\varphi \in E^*$, we have shown that $E^* = \text{span}\{\mathcal{B}^*\}$.

□

From Theorem 1.30 we immediately deduce the following result.

Corollary 1.31

If E is finite-dimensional, then $\dim(E) = \dim(E^*)$.

We next define the concept of annihilator of a subspace.

Definition 1.32

Let E be a $\mathbb{K}$ -vector space of dimension n , and let W be a subspace of E . The annihilator W° of W is the set of linear functionals that vanish on W .

$$W^\circ := \{\varphi \in E^* : \varphi(x) = 0, \forall x \in W\}.$$

The annihilator of a subspace $W \subset E$ is a subspace of E^* .

Example 1.33

Let

$$E = \mathbb{R}^3, \quad W = \text{span}\{(1, 0, 0), (0, 1, 0)\}.$$

Thus W is the xy -plane:

$$W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}.$$

A linear functional $f \in E^*$ has the form

$$f(x, y, z) = ax + by + cz, \quad a, b, c \in \mathbb{R}.$$

For f to belong to the annihilator W° , we require

$$f(w) = 0 \quad \text{for all } w \in W.$$

Since every $w \in W$ has the form $(x, y, 0)$, this gives

$$f(x, y, 0) = ax + by = 0 \quad \text{for all } x, y \in \mathbb{R}.$$

Hence

$$a = 0, \quad b = 0,$$

and therefore the annihilator is

$$W^\circ = \{f \in E^* \mid f(x, y, z) = cz, \quad c \in \mathbb{R}\}.$$

Equivalently,

$$W^\circ = \text{span}\{(0, 0, 1)\},$$

meaning that “only the last coordinate survives” to the annihilator.

Theorem 1.34

Let E be a $\mathbb{K}$ -vector space of dimension n , and let W be a subspace of E . Then

$$\dim(E) = \dim(W) + \dim(W^\circ).$$

Proof. Suppose that $1 \leq \dim(W) = k < n$, and let $(e_1, \dots, e_k)$ be a basis of W . We complete this basis with $n - k$ vectors $(e_{k+1}, \dots, e_n)$ to obtain a basis of E which we call $\mathcal{B}$. We introduce the dual basis $\mathcal{B}^* = \{e_1^*, \dots, e_n^*\}$, and observe that $W^\circ = \text{span}\{e_{k+1}^*, \dots, e_n^*\}$. Indeed, we clearly have that $e_{k+1}^*, \dots, e_n^* \in W^\circ$. Conversely, any element $\mu \in W^\circ$ cannot have components in $e_1^*, \dots, e_k^*$ since $\mu(e_1) = \dots = \mu(e_k) = 0$. We therefore have $\dim(W^\circ) = n - k$. $\square$

1.7 Matrix Representation of Linear Mappings

Linear mappings acting on finite-dimensional spaces can be handled with matrices. This section is to explain how this works. The following lemma shows that the action a linear map $f \in \mathcal{L}(E, F)$ can be expressed in terms of a matrix $\mathbb{K}^{p,n}$ when E and F are vector spaces of dimension p and n respectively. This is somehow the extension of the idea that in finite dimensional spaces we can represent vectors with coordinates from $\mathbb{K}^n$ (see Proposition 1.17).

Lemma 1.35: Matrix representation of a linear map

Let:

- E and F be two finite-dimensional $\mathbb{K}$ -vector spaces (of dimension p and n respectively),
- $\mathcal{E} = \{e_1, \dots, e_p\}$ and $\mathcal{F} = \{u_1, \dots, u_n\}$ be two bases of E and F respectively.

The mapping

$$\begin{aligned} \text{Mat} : \mathcal{L}(E, F) &\rightarrow \mathbb{K}^{p \times n} \\ f &\mapsto \text{Mat}(f) = [f]_{\mathcal{E}}^{\mathcal{F}} := ([f(e_1)]^{\mathcal{F}} \mid \dots \mid [f(e_p)]^{\mathcal{F}}) \end{aligned} \quad (1.3)$$

is an isomorphism. For every $f \in \mathcal{L}(E, F)$, $[f]_{\mathcal{E}}^{\mathcal{F}}$ is called the matrix representation of f in the bases $\mathcal{E}$ and $\mathcal{F}$.

Proof. We next prove that Mat is both surjective and injective.

Injectivity: Let $f, g \in \mathcal{L}(E, F)$ such that $\text{Mat}(f) = \text{Mat}(g)$. Then $[f(e_i)]^{\mathcal{F}} = [g(e_i)]^{\mathcal{F}}$ for all $i = 1, \dots, p$. Since the coordinate representation is unique, it follows that this equality is also true at the level of the vectors, $f(e_i) = g(e_i)$. Let now $x \in E$ such that $x = \sum_{i=1}^p c_i e_i$. Then $f(x) = \sum_{i=1}^p c_i f(e_i) = \sum_{i=1}^p c_i g(e_i) = g(x)$. Since this holds for all $x \in E$, it follows that $f = g$, and we have proven injectivity.

Surjectivity: Let $A \in \mathbb{K}^{p \times n}$. We are going to show that there exists $f_A \in \mathcal{L}(E, F)$ such that $\text{Mat}(f_A) = A$. For this, we define a mapping $f_A : E \rightarrow F$ such that

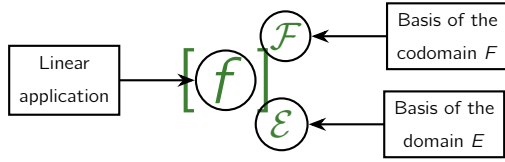
$$[f_A(x)]^{\mathcal{F}} := A[x]^{\mathcal{E}}, \quad \forall x \in E. \quad (1.4)$$

The mapping f_A is linear since, by construction, $[f_A(x)]^{\mathcal{F}} = 0_{\mathbb{R}^n}$ and therefore $f_A(0) = 0$. Also, for all $x, y \in E$ and $\mu, \lambda \in \mathbb{K}$, $[f_A(\mu x + \lambda y)]^{\mathcal{F}} = A[\mu x + \lambda y]^{\mathcal{E}} = \lambda A[x]^{\mathcal{E}} + \mu A[y]^{\mathcal{E}} = \lambda [f_A(x)]^{\mathcal{F}} + \mu [f_A(y)]^{\mathcal{F}}$. This shows surjectivity. $\square$

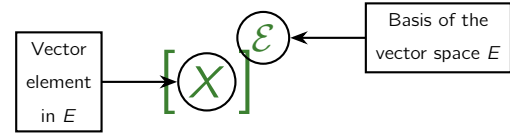
The matrix representation of a linear map, and its proof are very important, especially in connection to how to work with these maps in practice. Here are a few important observations in connection to this:

- The lemma shows that it is equivalent to work with linear maps on abstract spaces, and with matrices as long as we have prescribed bases in the input and output spaces.
- The proof of surjectivity of Mat shows that as soon as bases are prescribed, the action of linear maps boils down to matrix-vectors products, see (1.4). That part of the proof also shows that for every $A \in \mathbb{K}^{p \times n}$, the associated linear map is given by f_A as defined in Eq. (1.4).

- The matrix $[f]_{\mathcal{E}}^{\mathcal{F}}$ is such that each column j carries $[f(e_j)]^{\mathcal{F}}$, which is the image by f of the j -th input basis element from $\mathcal{E}$ when we express it with coordinates of the basis of $\mathcal{F}$.
 $[f(x)]^{\mathcal{F}} = [f]_{\mathcal{E}}^{\mathcal{F}} [x]^{\mathcal{E}}$
- We emphasize that, in this lecture, the input basis is always written at the bottom right, and the output basis at the top right.



(a) Notation for the matrix representation of an application $f : E \rightarrow F$ with respect to the bases $\mathcal{E}$ and $\mathcal{F}$.



(b) Notation for the matrix representation of an element $x \in E$ w.r.t. basis $\mathcal{E}$.

Figure 1.2: Overview of the matrix representation notations.

Example 1.36

Many interesting (and very visual) linear maps are obtained from appropriate choices on the matrices A . Here are some examples for $E = F = \mathbb{R}^2$ and using the canonical basis:

- If $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, then $f_A(x, y) = (-x, y)$ is the symmetry with respect to the x -axis. We define the concept of symmetry and projection rigorously in Section 1.8.
- Let $\theta \in \mathbb{R}$, and $A = U(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$, then f_A is the rotation of an angle θ around the origin. This is exactly the transformation that we applied to the duck in the front page!
- Can you find other matrices that lead to very interpretable linear maps?

We next discuss how certain operations on linear maps translate at the level of their matrix representation.

Proposition 1.37: evaluation = matrix vector product

Let $f \in \mathcal{L}(E, F)$ and let $\mathcal{E}$ and $\mathcal{F}$ be two bases of E and F respectively. Then

$$[f(x)]^{\mathcal{F}} = [f]_{\mathcal{E}}^{\mathcal{F}} [x]^{\mathcal{E}}, \quad \forall x \in E.$$

Proof. Let $x \in E$, and denote $[x]^{\mathcal{E}} = (c_1, \dots, c_n)^T$ its coordinate representation in the basis $\mathcal{E}$. Then $f(x) = \sum_{j=1}^n c_j f(e_j)$, and by Proposition 1.18, $[f(x)]^{\mathcal{F}} = \sum_{j=1}^n c_j [f(e_j)]^{\mathcal{F}} = [f]_{\mathcal{E}}^{\mathcal{F}} [x]^{\mathcal{E}}$. To obtain the last equality, we have used the definition of $[f]_{\mathcal{E}}^{\mathcal{F}}$ from Eq. (1.3), and we have observed that the linear combination can be expressed as a matrix-vector product. $\square$

Proposition 1.38: matrix of linear combination = linear combination of matrices

Let:

- E, F be two finite $\mathbb{K}$ -dimensional spaces
- $f, g \in \mathcal{L}(E, F)$
- $\alpha, \beta \in \mathbb{K}$
- $\mathcal{E}, \mathcal{F}$ bases of E and F respectively.

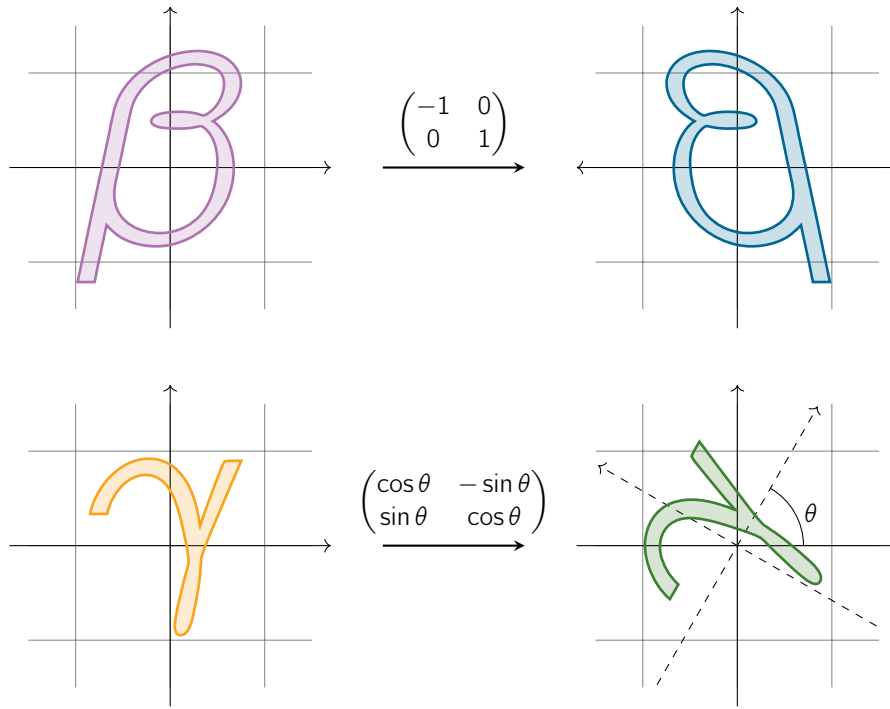


Figure 1.3: Illustration of the actions of symmetry and rotation matrices on the real plane.

It holds that

$$[\alpha f + \beta g]_{\mathcal{E}}^{\mathcal{F}} = \alpha[f]_{\mathcal{E}}^{\mathcal{F}} + \beta[g]_{\mathcal{E}}^{\mathcal{F}}.$$

Proof. It suffices to use the definition of $\text{Mat}(\alpha f + \beta g) = [\alpha f + \beta g]_{\mathcal{E}}^{\mathcal{F}}$ in terms of matrices, and use linearity of the space of matrices. $\square$

Proposition 1.39: matrix of composition = matrix product

Let:

- E, F, G be three finite-dimensional $\mathbb{K}$ -vector spaces
- $f \in \mathcal{L}(E, F)$ and $g \in \mathcal{L}(F, G)$
- $\mathcal{E}, \mathcal{F}, \mathcal{G}$ bases of E, F, G

It holds that

$$[g \circ f]_{\mathcal{E}}^{\mathcal{G}} = [g]_{\mathcal{F}}^{\mathcal{G}}[f]_{\mathcal{E}}^{\mathcal{F}}.$$

Proof. Let $x \in E$. On the one hand, by Proposition 1.37, we have that

$$[(g \circ f)(x)]^{\mathcal{G}} = [g \circ f]_{\mathcal{E}}^{\mathcal{G}}[x]^{\mathcal{E}}.$$

On the other hand, using the nested structure of the composition, and again using Proposition 1.37, we also have that

$$[(g \circ f)(x)]^{\mathcal{G}} = [g(f(x))]^{\mathcal{G}} = [g]_{\mathcal{F}}^{\mathcal{G}}[f(x)]^{\mathcal{F}} = [g]_{\mathcal{F}}^{\mathcal{G}}[f]_{\mathcal{E}}^{\mathcal{F}}[x]^{\mathcal{E}}.$$

Therefore,

$$[g \circ f]_{\mathcal{E}}^{\mathcal{G}}[x]_{\mathcal{E}}^{\mathcal{F}} = [g]_{\mathcal{F}}^{\mathcal{G}}[f]_{\mathcal{E}}^{\mathcal{F}}[x]_{\mathcal{E}}^{\mathcal{F}}, \quad \forall x \in E,$$

which leads to the desired equality. $\square$

The following figures illustrate the usefulness of our notation in combination with Proposition 1.37 and Proposition 1.39:

$$[f(x)]_{\mathcal{F}}^{\mathcal{F}} = [f]_{\mathcal{F}}^{\mathcal{F}}[x]_{\mathcal{E}}^{\mathcal{F}} \quad [g \circ f]_{\mathcal{E}}^{\mathcal{G}} = [g]_{\mathcal{F}}^{\mathcal{G}}[f]_{\mathcal{E}}^{\mathcal{F}}$$

Figure 1.4: The notation that we use emphasizes the underlying relations between vector spaces, linear applications and their matrix representations given by propositions 1.37 and Proposition 1.39.

Proposition 1.40: Matrix representation of inverse applications

Let $\mathcal{E}$ and $\mathcal{F}$ be bases of vector spaces E and F , respectively, and let $f : E \rightarrow F$ be an invertible linear application. Then the inverse application $f^{-1} : F \rightarrow E$ has the following matrix representation:

$$[f^{-1}]_{\mathcal{F}}^{\mathcal{E}} = \left([f]_{\mathcal{E}}^{\mathcal{F}}\right)^{-1}.$$

Proof. Left as an exercise. $\square$

The case of the identity map has a very clear and relevant meaning. When $f = \text{Id} : E \rightarrow E$, and we have input and output bases $\mathcal{E} = \{e_1, \dots, e_n\}$ and $\mathcal{F} = \{f_1, \dots, f_n\}$ respectively, then applying Eq. (1.3) to the present case yields

$$[\text{Id}]_{\mathcal{E}}^{\mathcal{F}} := ([e_1]_{\mathcal{F}}^{\mathcal{F}} \mid \dots \mid [e_n]_{\mathcal{F}}^{\mathcal{F}}). \quad (1.5)$$

This means, in words, the columns of $[\text{Id}]_{\mathcal{E}}^{\mathcal{F}}$ are the vectors of the input basis $\mathcal{E}$ written in the coordinates of the basis $\mathcal{F}$. This matrix tells us how the coordinates of a vector transform when we switch from one basis to another. Indeed, applying Proposition 1.37, it follows that

$$[x]_{\mathcal{F}}^{\mathcal{F}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}}[x]_{\mathcal{E}}^{\mathcal{F}}, \quad x \in E. \quad (1.6)$$

This justifies why $[\text{Id}]_{\mathcal{E}}^{\mathcal{F}}$ is called the matrix of change of basis. Since $[\text{Id}]_{\mathcal{E}}^{\mathcal{F}}$ is invertible (make sure you know how to prove it!), applying Proposition 1.40 we obtain that $[\text{Id}]_{\mathcal{F}}^{\mathcal{E}} = [\text{Id}^{-1}]_{\mathcal{F}}^{\mathcal{E}} = \left([\text{Id}]_{\mathcal{E}}^{\mathcal{F}}\right)^{-1}$. Left-multiplying Eq. (1.6) by this matrix, we deduce

$$[x]_{\mathcal{E}}^{\mathcal{E}} = [\text{Id}]_{\mathcal{F}}^{\mathcal{E}}[x]_{\mathcal{F}}^{\mathcal{F}}, \quad x \in E.$$

So we have effectively swapped the roles of $\mathcal{E}$ and $\mathcal{F}$, and verified that formula Eq. (1.6) stays coherent. We next record the notion of change of basis matrix in the following definition.

Definition 1.41: Matrix of change of basis

Let E be an n -dimensional vector space, and let an input and an output basis $\mathcal{E} = \{e_1, \dots, e_n\}$ and $\mathcal{F} = \{f_1, \dots, f_n\}$. The matrix of change of basis from $\mathcal{E}$ to $\mathcal{F}$ is the invertible matrix $[\text{Id}]_{\mathcal{E}}^{\mathcal{F}}$. Its columns are the vectors of $\mathcal{E}$ written in the basis $\mathcal{F}$ (see Eq. (1.5)). The matrix converts coordinate vectors expressed the basis $\mathcal{E}$ to coordinate vectors in $\mathcal{F}$ (see Eq. (1.6)).

The previous definition allows us to change very efficiently from one basis to another one the expression of endomorphisms as we record in the next proposition.

Proposition 1.42

Let E be an n -dimensional vector space, let $\mathcal{E} = \{e_1, \dots, e_n\}$ and $\mathcal{F} = \{f_1, \dots, f_n\}$ be two bases, and let $T \in \mathcal{L}(E)$. Then

$$[T]_{\mathcal{F}}^{\mathcal{F}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [T]_{\mathcal{E}}^{\mathcal{E}} [\text{Id}]_{\mathcal{F}}^{\mathcal{E}}.$$

Proof. Let f_j be the j -th basis element of $\mathcal{F}$. By Proposition 1.37, it holds that

$$[T(f_j)]_{\mathcal{E}}^{\mathcal{E}} = [T]_{\mathcal{E}}^{\mathcal{E}} [f_j]_{\mathcal{E}}^{\mathcal{E}}. \quad (1.7)$$

In addition, by Eq. (1.6), we have that $[T(f_j)]_{\mathcal{E}}^{\mathcal{E}} = [\text{Id}]_{\mathcal{F}}^{\mathcal{E}} [T(f_j)]_{\mathcal{F}}^{\mathcal{F}}$. Inserting this formula into Eq. (1.7) and using the invertibility of $[\text{Id}]_{\mathcal{F}}^{\mathcal{E}}$, we deduce $[T(f_j)]_{\mathcal{F}}^{\mathcal{F}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [T]_{\mathcal{E}}^{\mathcal{E}} [f_j]_{\mathcal{E}}^{\mathcal{E}}$. We now assemble $[T]_{\mathcal{F}}^{\mathcal{F}}$ by gathering all the vectors as matrix columns, and obtain

$$[T]_{\mathcal{F}}^{\mathcal{F}} = ([T(f_1)]_{\mathcal{F}}^{\mathcal{F}} \mid \dots \mid [T(f_n)]_{\mathcal{F}}^{\mathcal{F}}) = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [T]_{\mathcal{E}}^{\mathcal{E}} ([f_1]_{\mathcal{E}}^{\mathcal{E}} \mid \dots \mid [f_n]_{\mathcal{E}}^{\mathcal{E}}) = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [T]_{\mathcal{E}}^{\mathcal{E}} [\text{Id}]_{\mathcal{F}}^{\mathcal{E}}.$$

□

1.8 Projection and Symmetries

We now turn to projections, a specific type of linear maps which arises in many contexts.

Definition 1.43: Projection mappings

Let E be a vector space, a linear map $P : E \rightarrow E$ is called a projection mapping if and only if $P^2 = P$.

For an interesting way of understanding why projections are defined in this seemingly unintuitive way, we refer to the "steamroller analogy" in [this math stackexchange blog](#).

Remark 1.44. An equivalent¹ definition of projection mappings is that $P|_{\text{im}(P)} = \text{Id}$, i.e. the mapping P restricted to its image $\text{im}(P)$ is the identity application.

Example 1.45: Projection mappings

Let $\mathbb{R}^2$, the linear mapping $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with matrix representation

$$[P]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

is a projection operator since

$$[P^2]_{\mathcal{E}}^{\mathcal{E}} = ([P]_{\mathcal{E}}^{\mathcal{E}})^2 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} = [P]_{\mathcal{E}}^{\mathcal{E}}.$$

We can also verify that it is a projection mapping using Remark 1.44, since $\text{im } P = \text{span}\{v\}$ with $[v]_{\mathcal{E}}^{\mathcal{E}} = (1, 0)^{\top}$, and

$$[P]_{\mathcal{E}}^{\mathcal{E}} \begin{pmatrix} \lambda \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda \\ 0 \end{pmatrix} \Rightarrow P|_{\text{im } P} = \text{Id}.$$

Let us next define two special types of projections.

¹Do not take our word for this, try and prove it!

Definition 1.46: Projection onto a subspace

Let E be a vector space with subspaces $F, G \subset E$ such that $F \oplus G = E$. The projection on F associated to the direct sum $F \oplus G$ is the linear application

$$P_{F \oplus G} : E \rightarrow F \subset E$$

$$x = x_F + x_G \mapsto x_F,$$

where, for every $x \in E$, we use the unique decomposition $x = x_F + x_G$ with respect to the direct sum.

Definition 1.47: Symmetry w.r.t. a subspace

Let E be a vector space with subspaces $F, G \subset E$ such that $F \oplus G = E$. The symmetry with respect to F associated to the direct sum $F \oplus G$ is the linear application

$$S_{F \oplus G} : E \rightarrow E$$

$$x = x_F + x_G \mapsto x_F - x_G$$

Remark 1.48. (i) It can be easily verified that $P_{F \oplus G}$ as defined in Definition 1.46 satisfies the definition of projection mapping $P^2 = P$.

(ii) For direct sum of the form $F_1 \oplus \dots \oplus F_k = E$, if we wish to define the projection and symmetry onto/with respect to the subspace F_1 associated with this direct product, it suffices to write $G = F_2 \oplus \dots \oplus F_k$ and define the operators P and S as above with the direct sum $F_1 \oplus G = E$.

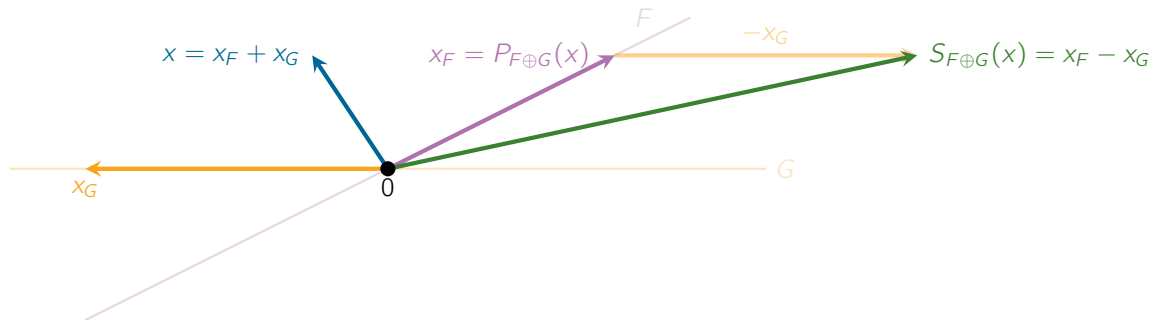


Figure 1.5: Illustration of the linear applications defined in Definitions 1.46 and 1.47.

At a glance, the projection P may seem to be only dependent on the choice of the subspace F , but in reality, it depends on the pair of subspaces F and G that make up the direct sum (see fig. 1.6). We will see in the next section that there is a unique projection that solely depends on the choice of the subspace it is "projecting" onto.

We have the following proposition

Proposition 1.49

Let $P_{F \oplus G}$, $S_{F \oplus G}$ be projection and symmetric mappings onto/with respect to F associated to the direct sum $E = F \oplus G$, then:

$$\ker(P_{F \oplus G}) = G, \quad \text{im}(P_{F \oplus G}) = F, \quad \ker(S_{F \oplus G}) = \{0\}, \quad \text{and} \quad \text{im}(S_{F \oplus G}) = E.$$

Proof. We shall prove the first two set equalities by double inclusion.

(i) ($\ker P = G$)

For $x = x_G \in G$, from the definition of the projection mapping we have that $P(x) = P(0 + x_G) = 0$, therefore $G \subset \ker P$. Conversely, if $x \in \ker P$, we first decompose $x = x_F + x_G$, and from the fact that $P(x) = x_F = 0$ we know that $x = 0 + x_G = x_G \in G$, therefore $\ker P \subset G$.

(ii) ($\text{im } P = F$)

If $x \in \text{im } P$, then there exists $y \in E$ such that $P(y) = x$, writing the direct sum decomposition $y = y_F + y_G$ we get that $x = P(y) = y_F \in F$, therefore $\text{im } P \subset F$. Conversely, for $x \in F$, since P is a projection, we get that $P(x) = \text{Id}(x) = x$ (why?), which implies that $x \in \text{im } P$, therefore $F \subset \text{im } P$.

(iii) ($\ker S = 0$ and $\text{im } S = E$)

Observe that $S^2 = \text{Id}$, therefore $S^{-1} = S$ and S is an invertible linear application, consequently $\ker S = \{0\}$ and $\text{im } P = E$.

□



Figure 1.6: Here we consider two direct sums of $\mathbb{R}^2$. The first is $\mathbb{R}^2 = F \oplus G$, the second $\mathbb{R}^2 = F \oplus H$. We see that even though both projections are onto the subspace F , different choices of direct sum lead to different projection images.

1.9 Metric Spaces

When working with elements from a nonempty set E , it is often crucial that we can make sense of what it means for two elements to be close or far away. We can make this precise by the concept of a distance. A distance is nothing more than a function which takes a pair of points of E as an input, and produces a real number as output satisfying the following properties.

Definition 1.50

Let E be a nonempty set. A function $d : E \times E \rightarrow \mathbb{R}$ is called a distance on E if and only if:

1. positivity:

$$d(x, y) \geq 0, \quad \forall x, y \in E,$$

2. non-degeneracy: for all $x, y \in E$

$$d(x, y) = 0 \quad \Rightarrow \quad x = y,$$

3. symmetry:

$$d(x, y) = d(y, x), \quad \forall x, y \in E,$$

4. triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z), \quad \forall x, y, z \in E$$

5. reflexivity:

$$d(x, x) = 0, \quad \forall x \in E$$

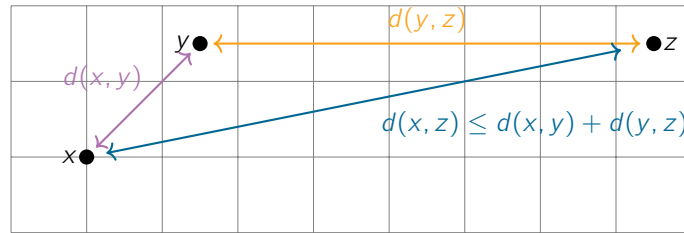


Figure 1.7: Illustration of the triangle inequality for the usual distance on the real plane.

Usually, conditions (ii) and (v) are merged to the conditions that for all $(x, y) \in E^2$, $d(x, y) = 0$ if and only if $x = y$.

Definition 1.51

A metric space (E, d) is a pair of a set E and a distance function $d : E \times E \rightarrow \mathbb{R}$.

Example 1.52

Let $E = \{ \text{red, yellow, blue} \}$ be the set of primary colors. The following table describes a function $d : E \times E \rightarrow \mathbb{R}$ described by the table below (for instance $d(\text{yellow, blue}) = 3$).

	red	yellow	blue
red	0	1	2
yellow	1	0	3
blue	2	3	0

We can show that d is a distance function on E and that (E, d) is a metric space.

1.10 Normed Vector Spaces

1.10.1 Definition

Great examples of metric spaces are normed vector spaces. These are $\mathbb{K}$ -vector spaces where the notion of distance is given through a norm.

Definition 1.53: Norm and normed vector space

Let $(E, +, \cdot)$ be a $\mathbb{K}$ -vector space. A norm is a mapping $\|\cdot\|_E : E \rightarrow \mathbb{R}$ satisfying:

1. positivity: For all $x \in E$, it holds that $\|x\|_E \geq 0$,
2. non-degeneracy: for all $x \in E$, if $\|x\|_E = 0$, then $x = 0_E$,
3. homogeneity: for all $\lambda \in \mathbb{R}$ and all $x \in E$, $\|\lambda x\|_E = |\lambda| \|x\|_E$,
4. triangle inequality: for all $(x, y) \in E$, it holds that $\|x + y\|_E \leq \|x\|_E + \|y\|_E$.

$(E, \|\cdot\|_E)$ is called a normed vector space.

The following proposition tells us that a normed vector space $(E, \|\cdot\|_E)$ is a metric space because we can define a distance from the norm.

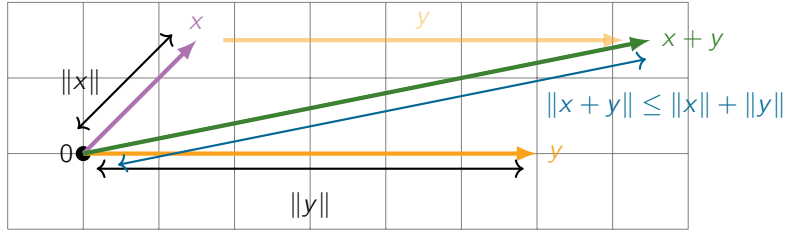


Figure 1.8: Illustration of the triangle inequality for the usual euclidean norm on the real plane.

Proposition 1.54

Let $(E, \|\cdot\|_E)$ be a normed vector space. Define the function

$$\begin{aligned} d_{\|\cdot\|_E} : E \times E &\rightarrow \mathbb{R} \\ (x, y) &\mapsto d_{\|\cdot\|_E}(x, y) = \|x - y\|_E. \end{aligned}$$

Then $d_{\|\cdot\|_E}$ is a distance on E , and $(E, d_{\|\cdot\|_E})$ is a metric space.

Proof. We need to show that $d_{\|\cdot\|_E}$ satisfies all properties of a distance on E . This is left as an exercise. $\square$

An interesting observation is that

$$\|x\|_E = \|x - 0_E\|_E = d_{\|\cdot\|_E}(x, 0_E), \quad \forall x \in E.$$

In other words, the norm of a vector $x \in E$ is given by its distance to the neutral element 0_E . This is why we usually interpret $\|x\|_E$ as the length of the vector. The identity is also one of the many manifestations revealing the extent to which the neutral element plays a major role in linear vector spaces: everything is measured with respect to the origin.

As the following set of examples show, one can define plenty of different norms on the same vector space (there are different ways to measure the size of vectors and distances).

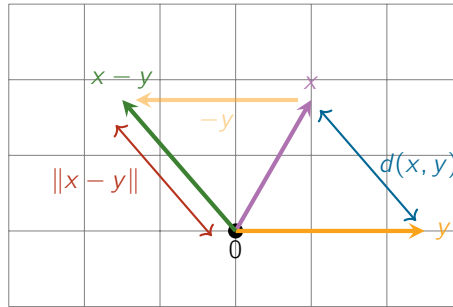


Figure 1.9: Illustration of the notion of distance induced by a norm.

Lemma 1.55

The following vector spaces can be endowed with norms and be seen as normed vector spaces:

1. We can endow $\mathbb{R}^n$ and $\mathbb{C}^n$ with different norms:

$$\begin{aligned} \|(x_1, \dots, x_n)\|_p &:= \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad 1 \leq p < +\infty \\ \|(x_1, \dots, x_n)\|_\infty &:= \max\{|x_1|, \dots, |x_n|\}. \end{aligned} \tag{1.8}$$

2. We can endow the space of matrices $\mathbb{K}^{q,n}$ with the following norms,

$$\|A\|_p := \left(\sum_{i=1}^q \sum_{j=1}^n |a_{i,j}|^p \right)^{1/p}, \quad \forall A \in \mathbb{K}^{q,n}, \quad p \geq 1, \quad (1.9)$$

$$\|A\|_\infty := \max_{i,j} |a_{i,j}|$$

3. The $\mathbb{R}$ -vector space of continuous functions $\mathcal{C}([0, 1]; \mathbb{R})$ over $[0, 1]$ and with values in $\mathbb{R}$ can be endowed with the following norms:

$$\|f\|_p := \left(\int_0^1 |f(x)|^p dx \right)^{1/p}, \quad 1 \leq p < +\infty$$

$$\|f\|_\infty := \max_{x \in [0,1]} |f(x)|$$

The 2-norm is usually called the Frobenius, Schur or Hilbert-Schmidt norm.

Showing that the above norms are indeed satisfying all necessary properties can become difficult for general values of $1 \leq p < \infty$. The proof is easier for the cases $p \in \{1, 2, \infty\}$.

An important notion is the one of equivalence between norms. It expresses the idea that two norms may be different, but that the notion of size between them is preserved up to a certain scaling. Consequently, if one of the norms indicates that a vector is very small, then the vector will continue to be small in the other norm.

Definition 1.56: Norm Equivalence

Let E be a $\mathbb{K}$ -vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be two norms on E . We say that $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are equivalent if there exists two constants $0 < \underline{c} \leq \bar{c}$ such that

$$\underline{c}\|x\|_\alpha \leq \|x\|_\beta \leq \bar{c}\|x\|_\alpha, \quad \forall x \in E.$$

We would like to insist on the order of the quantifiers in the above definitions: since the statement is that there exists two constants $0 < \underline{c} \leq \bar{c}$ such that for all $x \in E$ the inequality is satisfied, this means that $\underline{c}$ and $\bar{c}$ are independent of x .

There is a fundamental theorem about norm equivalences which is the following. Its proof is relatively technical and lengthy, and requires notions which we have not introduced. We will thus skip its proof.

Theorem 1.57: All norms are equivalent in finite dimension

Let E be a $\mathbb{K}$ -vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then:

- If E is finite dimensional, then all norms are equivalent,
- If E is infinite dimensional, then there always exists norms that are not equivalent.

1.10.2 The space of bounded linear maps, operator norm, continuous dual

In Lemma 1.55 we have given norms for very usual vector spaces, but we did not address the case of the space of linear mappings $\mathcal{L}(E, F)$. For this case, when $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ are two normed vector spaces, we can also define a notion of norm and of normed vector space for $\mathcal{L}(E, F)$, but the subtlety is that this only holds for a subset of $\mathcal{L}(E, F)$ which we next define.

Definition 1.58: Bounded Linear Mappings

Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be two normed vector spaces. A linear mapping $T \in \mathcal{L}(E, F)$ is called *bounded* if there exists a constant $C > 0$ such that

$$\|T(x)\|_F \leq C \|x\|_E, \quad \forall x \in E. \quad (1.10)$$

The space of bounded linear mappings is

$$\mathcal{B}(E, F) := \{T \in \mathcal{L}(E, F) : T \text{ is bounded}\}. \quad (1.11)$$

Lemma 1.59

It holds that

$$\mathcal{B}(E, F) \subseteq \mathcal{L}(E, F),$$

and:

- with infinite-dimensional spaces the inclusion may be strict,
 - when E and F are finite dimensional, equality holds.
-

We skip the proof of the above lemma, but we next provide an example illustrating that we indeed sometimes have a strict inclusion when dealing with infinite-dimensional spaces.

Example 1.60: The derivative operator

Let

$$E = C^1([0, 1]), \quad F = C([0, 1]),$$

both equipped with the supremum norm

$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|.$$

We define the derivative operator as

$$T : E \rightarrow F, \quad T(f) = f'.$$

We have that $T \in \mathcal{L}(E, F)$. Indeed, for any $f, g \in C^1([0, 1])$ and $a, b \in \mathbb{K}$,

$$T(af + bg) = (af + bg)' = af' + bg' = aT(f) + bT(g).$$

Hence T is linear.

We next show that the operator is not bounded. For this, we consider the sequence of functions

$$f_n(x) = \frac{\sin(nx)}{n}, \quad \forall n \geq 1.$$

First observe that

$$\|f_n\|_\infty = \sup_{x \in [0, 1]} \left| \frac{\sin(nx)}{n} \right| \leq \frac{1}{n} \rightarrow 0.$$

However

$$T(f_n) = f'_n(x) = \cos(nx),$$

and therefore

$$\|T(f_n)\|_\infty = \sup_{x \in [0, 1]} |\cos(nx)| = 1.$$

Thus

$$\frac{\|T(f_n)\|_\infty}{\|f_n\|_\infty} \geq n \rightarrow \infty.$$

Hence there cannot exist a constant C such that

$$\|T(f)\|_\infty \leq C\|f\|_\infty \quad \text{for all } f \in C^1([0, 1]).$$

Therefore T is not bounded:

$$T \in \mathcal{L}(E, F) \quad \text{but} \quad T \notin \mathcal{B}(E, F).$$

In summary, here are the main distinctions between $\mathcal{L}(E, F)$ and $\mathcal{B}(E, F)$:

Space	Definition	Structure
$\mathcal{L}(E, F)$	all linear maps	algebraic
$\mathcal{B}(E, F)$	bounded linear maps	analytic/topological

The fact that some linear maps from $\mathcal{L}(E, F)$ can be unbounded is a behavior which is totally admissible, but it is rather pathological. This is the main motivation to introduce the subset $\mathcal{B}(E, F)$. It turns out that on this nicer set, we can define a notion of norm to give $\mathcal{B}(E, F)$ the structure of a normed vector space.

Theorem 1.61

The space $\mathcal{B}(E, F)$ endowed with the *operator norm*

$$\|T\|_{\text{op}} := \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \sup_{\|x\|_E = 1} \|T(x)\|_F = \sup_{x \in E - \{0_E\}} \frac{\|T(x)\|_F}{\|x\|_E}, \quad \forall T \in \mathcal{B}(E, F) \quad (1.12)$$

is a normed vector space, and it holds that

$$\mathcal{B}(E, F) = \{T \in \mathcal{L}(E, F) : \|T\|_{\text{op}} < +\infty\}. \quad (1.13)$$

In addition, for any $T \in \mathcal{B}(E, F)$, $\|T\|_{\text{op}}$ is the smallest constant $C > 0$ such that (1.10) holds.

Proof. We first prove the three equalities given in Eq. (1.12):

- We first show that

$$\sup_{\|x\|_E = 1} \|T(x)\|_F = \sup_{x \in E - \{0_E\}} \frac{\|T(x)\|_F}{\|x\|_E}, \quad \forall T \in \mathcal{B}(E, F).$$

For this, let $T \in \mathcal{B}(E, F)$ and $x \neq 0_E$. Thus the vector $y := \frac{x}{\|x\|_E}$ has norm 1, and by linearity

$$\|T(y)\|_F = \frac{\|T(x)\|_F}{\|x\|_E}.$$

Consequently,

$$\sup_{\|y\|_E = 1} \|T(y)\|_F = \sup_{x \neq 0_E \in E} \frac{\|T(x)\|_F}{\|x\|_E},$$

and this shows the first equality that we want to prove.

- We next show that

$$\sup_{\|x\|_E \leq 1} \|T(x)\|_F = \sup_{\|x\|_E = 1} \|T(x)\|_F, \quad \forall T \in \mathcal{B}(E, F).$$

Since $\{x \in E : \|x\|_E = 1\} \subseteq \{x \in E : \|x\|_E \leq 1\}$ we immediately obtain

$$\sup_{\|x\|_E = 1} \|T(x)\|_F \leq \sup_{\|x\|_E \leq 1} \|T(x)\|_F.$$

To show the reverse inequality, we let $x \in E$ be such that $\|x\|_E \leq 1$ and $x \neq 0$, and define $y = \frac{x}{\|x\|_E}$ so that $\|y\|_E = 1$. By linearity,

$$\|T(x)\|_F = \|x\|_E \|T(y)\| \leq \|T(y)\|.$$

Hence

$$\sup_{\|x\|_E \leq 1} \|T(x)\|_F \leq \sup_{\|y\|=1} \|T(y)\|.$$

Combining the two inequalities yields the result.

We now show that $\|\cdot\|_{\text{op}}$ is a norm in $\mathcal{B}(E, F)$. For this, let $T \in \mathcal{B}(E, F)$.

(i) Positivity and definiteness. By definition, $\|T\|_{\text{op}}$ is a supremum of nonnegative numbers, so $\|T\|_{\text{op}} \geq 0$. If $\|T\|_{\text{op}} = 0$, then $\|T(x)\|_F = 0$ for all $\|x\|_E \leq 1$, hence $T(x) = 0$ on the unit ball. For any $y \in E$ with $y \neq 0$, set $x := y/\|y\|_E$, so $\|x\|_E = 1$, and by linearity

$$T(y) = T(\|y\|_E x) = \|y\|_E T(x) = 0.$$

Also $T(0_E) = 0_F$, so $T = 0$. Conversely, if $T = 0$ then clearly $\|T\|_{\text{op}} = 0$.

(ii) Absolute homogeneity. For any scalar $\alpha \in \mathbb{K}$,

$$\|\alpha T\|_{\text{op}} = \sup_{\|x\|_E \leq 1} \|(\alpha T)(x)\|_F = \sup_{\|x\|_E \leq 1} |\alpha| \|T(x)\|_F = |\alpha| \sup_{\|x\|_E \leq 1} \|T(x)\|_F = |\alpha| \|T\|_{\text{op}}.$$

(iii) Triangle inequality. For all x with $\|x\|_E \leq 1$,

$$\|(T_1 + T_2)(x)\|_F = \|T_1(x) + T_2(x)\|_F \leq \|T_1(x)\|_F + \|T_2(x)\|_F.$$

Taking the supremum over $\{x : \|x\|_E \leq 1\}$ gives

$$\|T_1 + T_2\|_{\text{op}} = \sup_{\|x\|_E \leq 1} \|(T_1 + T_2)(x)\|_F \leq \sup_{\|x\|_E \leq 1} \|T_1(x)\|_F + \sup_{\|x\|_E \leq 1} \|T_2(x)\|_F = \|T_1\|_{\text{op}} + \|T_2\|_{\text{op}}.$$

The three norm axioms are satisfied, hence $\|\cdot\|_{\text{op}}$ is a norm on $\mathcal{B}(E, F)$.

We finish by showing that Eq. (1.13) holds. Let $T \in \mathcal{B}(E, F)$. By definition Eq. (1.11) of $\mathcal{B}(E, F)$, T is bounded, therefore there exists $C > 0$ such that $\frac{\|T(x)\|_F}{\|x\|_E} \leq C$ for all $x \neq 0 \in E$. Taking the supremum in $x \neq 0 \in E$ shows that $\|T\|_{\text{op}} \leq C < +\infty$, which in turn implies that $T \in \{T \in \mathcal{L}(E, F) : \|T\|_{\text{op}} < +\infty\}$, and therefore $\mathcal{B}(E, F) \subseteq \{T \in \mathcal{L}(E, F) : \|T\|_{\text{op}}\}$. To show the reverse inclusion, we let $T \in \mathcal{L}(E, F)$ be such that $\|T\|_{\text{op}} < +\infty$. Therefore, by the property of the supremizer, it holds that $\frac{\|T(x)\|_F}{\|x\|_E} < \infty$ for all $x \neq 0 \in E$. In addition $\|T(0)\|_F = 0$. This shows that T is bounded, thus a member of $\mathcal{B}(E, F)$, and concludes the proof. $\square$

Lemma 1.62: Boundedness and continuity.

For linear mappings between normed spaces $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ the following are equivalent:

1. T is bounded,
2. $\|T\|_{\text{op}} < +\infty$,
3. T is continuous, that is:
 - for all $x \in E$,
 - for all $\epsilon > 0$,
 - there exists $\delta > 0$ such that
 - for all $y \in E$ such that $\|x - y\|_E \leq \delta \Rightarrow \|T(x) - T(y)\|_F \leq \epsilon$.

Therefore

$$\begin{aligned}\mathcal{B}(E, F) &= \{T \in \mathcal{L}(E, F) : T \text{ is bounded} \} \\ &= \{T \in \mathcal{L}(E, F) : \|T\|_{\text{op}} < +\infty\} \\ &= \{T \in \mathcal{L}(E, F) : T \text{ is continuous} \}.\end{aligned}$$

The results that we have just stated for general spaces $\mathcal{B}(E, F)$ can be particularized to the case of the dual space by taking $F = (\mathbb{K}, |\cdot|_{\mathbb{K}})$. This yields two notions of dual spaces.

Definition 1.63: Algebraic vs Topological dual

For a normed $\mathbb{K}$ -vector space $(E, \|\cdot\|_E)$, there are two notions of dual spaces:

- The algebraic dual is

$$E^* := \mathcal{L}(E, \mathbb{K}) = \{\ell : E \rightarrow \mathbb{K} : \ell \text{ is linear} \}$$

as already defined in Definition 1.28.

- The topological or continuous dual is the space of bounded linear functionals,

$$E' := \mathcal{B}(E, \mathbb{K}) = \{\ell \in E^* : \|\ell\|_{\text{op}} < +\infty\} \subseteq E^*. \quad (1.14)$$

By Theorem 1.61, it is a normed vector space when endowed with the operator norm Eq. (1.12), which is often called the dual norm.

1.11 Exercises

We have split exercises into the ones that are a reminder of concepts from the previous course (Section 1.11.1), and the ones having new content from the current lecture (Section 1.11.2).

1.11.1 Exercises to remember the previous linear algebra course

Exercise 1.1 (Groups). Prove the statements from Example 1.4 and Section 1.2.

Exercise 1.2 (Vector spaces). Prove that the spaces from Section 1.2 are vector spaces.

Exercise 1.3. (Sigma notation) Let $a_1, \dots, a_n$ be elements of a field $\mathbb{K}$, prove the following formula:

$$\left(\sum_{i=1}^n a_i\right)^2 = \sum_{i=1}^n (a_i)^2 + 2 \sum_{1 \leq i < j \leq n} a_i a_j$$

Exercise 1.4. (Matrix multiplication) Let $A \in \mathbb{K}^{n \times n}$, with $A_{ij} = 1$, $\forall i, j$. What is the expression of $A^{(m)} = \overbrace{A \cdot \dots \cdot A}^{m \text{ times}}$?

Exercise 1.5. (Invertibility of product of matrices)

- Given invertible square matrices A and B , prove that AB is invertible and that $(AB)^{-1} = B^{-1}A^{-1}$.
- Prove that, if A and B are square matrices such that AB is an invertible matrix, then A and B are invertible matrices.

(**Indication:** M is invertible $\iff \ker(M) = \{0\}$).

Exercise 1.6. (Triangular and diagonal matrices)

- Prove that if A and B are upper triangular matrices, then AB and BA are also upper triangular matrices. Prove the same for lower triangular matrices. Does $AB = BA$ hold in this case?

- (ii) Show that, for any invertible lower/upper triangular matrix, the inverse is also lower/upper triangular.
- (iii) Prove that if A and B are diagonal matrices, then AB is also diagonal. Does $AB = BA$ in this case?

Exercise 1.7. (Transposition and invertibility)

- (i) Given matrices $A, B \in \mathbb{K}^{n \times m}$ and $\lambda, \mu \in \mathbb{K}$, prove that $(\lambda A + \mu B)^T = \lambda A^T + \mu B^T$.
- (ii) Given $A \in \mathbb{K}^{n \times k}$ and $B \in \mathbb{K}^{k \times m}$, prove that $(AB)^T = B^T A^T$.
- (iii) If A is a square invertible matrix, prove that $(A^{-1})^T = (A^T)^{-1}$, i.e. the transposition of the inverse is the inverse of the transposition. (**Indication:** use the previous point.)
- (iv) Given a symmetric matrix S (i.e. $S^T = S$), is the matrix $S^{(k)} := \overbrace{S \cdot \dots \cdot S}^{k \text{ times}}$ also symmetric? Similarly, is $A^{(k)}$ skew-symmetric when A is skew-symmetric (i.e. $A^T = -A$)?

Exercise 1.8. (Trace of a matrix)

Given square matrices A, B and $\lambda, \mu \in \mathbb{K}$, prove the following identities:

$$\text{Tr}(\lambda A + \mu B) = \lambda \text{Tr}(A) + \mu \text{Tr}(B), \quad \text{Tr}(A^T) = \text{Tr}(A), \quad \text{and} \quad \text{Tr}(AB) = \text{Tr}(BA).$$

Conclude from the first identity that the set of matrices $\{A \in \mathbb{K}^{n \times n} : \text{Tr} A = 0\} \subset \mathbb{K}^{n \times n}$ is a vector space.

Exercise 1.9. (Transposed and adjoint matrices)

Prove that the adjoint operation has properties analogous to the ones in problem 1.7 (now $\mathbb{K} = \mathbb{C}$). Furthermore prove that for $A \in \mathbb{K}^{n \times m}$

$$\text{Tr}(A^T A) = \sum_{i=1}^n \sum_{j=1}^m (A_{ij})^2, \quad \text{Tr}(A^\dagger A) = \sum_{i=1}^n \sum_{j=1}^m |A_{ij}|^2.$$

Thus $A = 0$ is equivalent to $\text{Tr}(A^T A) = 0$ (in the case of $\mathbb{K} = \mathbb{R}$) or $\text{Tr}(A^\dagger A) = 0$ (in the case of $\mathbb{K} = \mathbb{C}$).

Exercise 1.10. (Compatible and determined systems) Given $\mathbb{K} = \mathbb{R}$ and the following systems of equations

$$\begin{pmatrix} 1 & h \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ h \end{pmatrix}, \quad \begin{pmatrix} h & 0 & 1 \\ 0 & h & 0 \\ 1 & 0 & h \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} h \\ 0 \\ h \end{pmatrix},$$

where $h \in \mathbb{R}$. Determine for each of the systems the value of h for which the system is compatible and/or determined.

Exercise 1.11. (Invertibility) Calculate, depending on the value of h , the inverse of the following matrix A :

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & h & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

Exercise 1.12. (Invertibility)

Let $c = \cos(\theta)$ and $s = \sin(\theta)$. Determine the values of $\theta \in [0, 2\pi)$ for which the following matrix A is not invertible:

$$A = \begin{pmatrix} c & s \\ s & c \end{pmatrix}.$$

Furthermore, for values of θ such that A is invertible, calculate A^{-1} . (**Indication:** This exercise may seem intimidating at first, but one should maybe first start by ruling out specific cases of c and s . For example, what would be the possible values of s if $c = 0$? Furthermore, one may use the trigonometric identity $\cos^2(\theta) - \sin^2(\theta) = \cos(2\theta)$.)

Exercise 1.13. (Inverse matrix)

Calculate the inverse of the following upper triangular matrix

$$T = \begin{pmatrix} 1 & 1 & \dots & 1 & 1 \\ 0 & 1 & \dots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 1 \\ 0 & 0 & \dots & 0 & 1 \end{pmatrix} \in \mathbb{K}^{n \times n}.$$

Exercise 1.14. (Basis)

For the field $\mathbb{K} = \mathbb{C}$ we have the following matrix:

$$a = \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Study whether the vectors (a_1, a_2) form a basis for the vector space $\mathbb{C}^2$.

Exercise 1.15. (Independent set and generating set) Given a generating set $(a_1, \dots, a_p)$ and a independent set $(b_1, \dots, b_q)$ of a subspace $\mathbb{F}$ of $\mathbb{K}^n$. Prove that by reordering $(a_1, \dots, a_p)$, there exist $a_1, \dots, a_h$ (with $h \leq p$) such that $(b_1, \dots, b_q, a_1, \dots, a_h)$ forms a basis for $\mathbb{F}$.

Exercise 1.16. (Solution subspace)

Given the matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & h & 0 \end{pmatrix}$$

Consider the spaces $\mathbb{F} = \{x \in \mathbb{R}^3 \mid Ax = 0\}$ and $\mathbb{G}$ generated by the columns of A . Calculate the dimension of $\mathbb{F}$ and $\mathbb{G}$ and give a basis for each of these spaces.

Exercise 1.17. Repeat ex. 1.16 for the following matrix:

$$A = \begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Exercise 1.18. (Implicit form)

In this exercise we consider the matrix

$$A = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}.$$

We define the subspace $\mathbb{F} \subset \mathbb{R}^3$ as the space $\mathbb{F} = \text{span}\{a_1, a_2, a_3\}$. Here a_i refers to the i -column of the matrix A .

Find the implicit equation that defines $\mathbb{F}$, i.e. find $P, Q, S \in \mathbb{R}$ such that $\mathbb{F}$ can be defined as follows:

$$\mathbb{F} = \{(y_1, y_2, y_3)^\top \in \mathbb{R}^3 : Py_1 + Qy_2 + Sy_3 = 0\}$$

Exercise 1.19. Let $(\lambda_1, \dots, \lambda_m)$ and $(\mu_1, \dots, \mu_m)$ be two sequences with $\lambda_i, \mu_j \in \mathbb{K}$, with at least one nonzero λ_i and μ_j . We construct the matrix A with $A_{ij} = \lambda_i \mu_j$. What is the rank of A ?

Exercise 1.20. (Grassmann's formula) Prove that two distinct planes $\mathbb{F}, \mathbb{G}$ in a three dimensional vector space $\mathbb{E}$ necessarily intersects at a line.

Exercise 1.21. (Change of basis)

Let bases $\mathcal{U}$ and $\mathcal{V}$ of $\mathbb{R}^2$ be given by

$$\mathcal{U} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}, \quad \mathcal{V} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

Let $x, y \in \mathbb{R}^2$ be such that

$$[x]^{\mathcal{E}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad [y]^{\mathcal{U}} = \begin{pmatrix} 3 \\ -1 \end{pmatrix},$$

where $\mathcal{E}$ is the standard basis on $\mathbb{R}^2$ and $[z]^{\mathcal{X}}$ indicates the coordinates of z with respect to the basis $\mathcal{X}$.
Give the coordinates of x in $\mathcal{U}$ and of y in $\mathcal{V}$.

Exercise 1.22. (Change of basis)

Let $h, k \in \mathbb{R} \setminus \{0\}$, and bases $\mathcal{U}$ and $\mathcal{V}$ of $\mathbb{R}^3$ given by

$$\mathcal{U} = \left\{ \begin{pmatrix} 1 \\ h \\ k \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \text{and} \quad \mathcal{V} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} h \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} k \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Determine the set $\mathbb{F}$ of all the vectors in $\mathbb{R}^3$ that has the same coordinates in both $\mathcal{U}$ and $\mathcal{V}$. Is $\mathbb{F}$ a vector subspace of $\mathbb{R}^3$? (**Indication:** try and define a linear mapping such that its kernel is the set of vectors that we are looking for.)

1.11.2 Exercises with content from the current course

Exercise 1.23. (Norm Equivalences in $\mathbb{R}^n$)

Show that $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are equivalent norms in $\mathbb{R}^n$ (of course, without using Theorem 1.57!).

Exercise 1.24. (Reverse Triangle Inequality)

From the triangle inequality, derive the *reverse triangle inequality*, namely

$$|||x| - |y|| \leq \|x - y\|, \quad \forall (x, y) \in \mathbb{K}$$

where $\mathbb{K} = \mathbb{R}^n$ or $\mathbb{C}^n$ and $\|\cdot\|$ is any norm in $\mathbb{R}^n$.

Exercise 1.25. (Product of polynomials)

Let $P(X) = \sum_k \lambda_k X^k$ and $Q(X) = \sum_k \mu_k X^k$ be two polynomials of $\mathbb{K}[X]$, furthermore let $\deg P = p$ and $\deg Q = q$.

Prove that

$$P(X) \cdot Q(X) = \sum_{k=0}^p \sum_{m=0}^q \lambda_k \mu_m X^{k+m}.$$

Conclude from this that $\deg(P(X) \cdot Q(X)) = \deg(P) + \deg(Q)$. (**Note:** if we stand by the definition $\deg(0(X)) = -\infty$ and use the convention $-\infty + n = -\infty$, $\forall n \in \mathbb{N}$ and $-\infty + (-\infty) = -\infty$, then this identity also works for products that involves the zero polynomial $0(X)$.)

Exercise 1.26. (Compatibility of $P(X)(A)$ with sum and product of polynomials)

In Eq. (1.1) we have seen that for a polynomial $P(X) = \sum_{k=0}^{\infty} \lambda_k X^k \in \mathbb{K}[X]$, we can view it as a mapping

$$P(X) : \mathbb{K}^{n \times n} \rightarrow \mathbb{K}^{n \times n},$$

$$A \mapsto P(A) = P(X)(A) := \sum_{k=0}^{\infty} \lambda_k A^{(k)}.$$

Given $A \in \mathbb{K}^{n \times n}$, $P(X) = \sum_{k=0}^{\infty} \lambda_k X^k$ and $Q(X) = \sum_{k=0}^{\infty} \mu_k X^k$

(i) prove that

$$(P(X) + Q(X))(A) = P(X)(A) + Q(X)(A)$$

that is, the image of A via the mapping $P(X) + Q(X)$ is equal to the sum of the images $P(X)(A)$ and $Q(X)(A)$.

(ii) Prove analogously that

$$(P(X) \cdot Q(X))(A) = P(X)(A) \cdot Q(X)(A).$$

Obtain as a corollary that $P(A)$ and $Q(A)$ commute for any matrix $A \in \mathbb{K}^{n \times n}$.

Exercise 1.27. (Matrix polynomial and invertibility)

Given a polynomial $P(X) = \lambda_0 + \lambda_1 X + \dots + \lambda_p X^p$ of degree $\deg(P) = p \geq 1$ and a matrix A such that $P(A) = 0$. Prove that if $\lambda_0 \neq 0$, then A is invertible.

Exercise 1.28. (Matrix polynomial, nilpotent matrix and inverse matrix)

Recall that we call a matrix M **nilpotent** if there exists $k \in \mathbb{N}$ such that $M^k = O$. The smallest of these k is referred to as the index of M .

Given $n \in \mathbb{N}$ and non-negative integer $q \leq n$, we call a matrix $A \in \mathbb{K}^{n \times n}$ a " q -strictly triangular matrix", if $A_{ij} = 0, \forall i > j - q$. For example, below are a 1-strictly triangular matrix and 2-strictly triangular matrix, respectively:

$$\begin{pmatrix} 0 & a & d & f \\ 0 & 0 & b & e \\ 0 & 0 & 0 & c \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & x & z \\ 0 & 0 & 0 & y \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

(i) Prove that the product of a p -strictly triangular matrix and a q -strictly triangular matrix is a $(p + q)$ -strictly triangular matrix. Obtain as a corollary that a 1-strictly triangular matrix is nilpotent with index less or equal to n (**Indication:** an n -strictly triangular matrix is the zero matrix).

(ii) Prove by induction that the shift matrix:

$$N = \left(\begin{array}{c|c} 0 & \text{Id}_{n-1} \\ \hline 0 & 0 \end{array} \right) = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \in \mathbb{K}^{n \times n}.$$

satisfies:

$$N^{(k)} = \overbrace{N \cdot N \cdot \dots \cdot N}^{k \text{ times}} = \left(\begin{array}{c|c} 0 & \text{Id}_{n-k} \\ \hline 0 & 0 \end{array} \right) = \begin{pmatrix} \overbrace{0 \ 0 \ \dots \ 0}^{k \text{ columns}} & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & \dots & \dots & \dots & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \dots & \dots & \dots & \dots & 0 \end{pmatrix}.$$

Is N nilpotent? If so, what is the index of N ?

(iii) Calculate the inverse of the following matrix:

$$A = \begin{pmatrix} 1 & t & t^2 & t^3 & \dots & t^{n-1} \\ 0 & 1 & t & t^2 & \dots & t^{n-2} \\ 0 & 0 & 1 & t & \dots & t^{n-3} \\ 0 & 0 & 0 & 1 & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & t \\ 0 & 0 & 0 & 0 & \dots & 1 \end{pmatrix} \in \mathbb{K}^{n \times n}$$

(**Indication:** Instead of performing the row operations, use what we've derived in the previous points and the algebraic identity $(1 - X) \left(\sum_{k=0}^{n-1} X^k \right) = (1 - X^n)$. What happens when $X = tN$?).

Exercise 1.29. (Vector space)

On $\mathbb{R}^2$ we define the following operations

$$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2), \quad \lambda \odot (x_1, x_2) = (\lambda x_1, 0).$$

Which vector space axioms does $(\mathbb{R}^2, \oplus, \odot)$ satisfy? Is it a vector space?

Exercise 1.30. (Real-valued polynomials as a vector space)

We consider the vector space of polynomials $\mathbb{E} = \mathbb{R}[x]$, determine whether the subsets $\mathbb{F}_i \subset \mathbb{E}$ are subspaces.

- (i) $\mathbb{F}_1 = \{p(x) \in \mathbb{E} : p(1) = 0\}$.
- (ii) $\mathbb{F}_2 = \{p(x) \in \mathbb{E} : p(1) = 1\}$.
- (iii) $\mathbb{F}_3 = \{p(x) \in \mathbb{E} : p(1) = p(2) = 0\}$.
- (iv) $\mathbb{F}_4 = \{p(x) \in \mathbb{E} : p'(1) = 0\}$, where p' denotes the first derivative of p .
- (v) $\mathbb{F}_5 = \{p(x) \in \mathbb{E} : p'(1) = 1\}$.
- (vi) $\mathbb{F}_6 = \{p(x) \in \mathbb{E} : p(x) \text{ is divisible by } x^2 + x + 1\}$.
- (vii) $\mathbb{F}_7 = \{p(x) \in \mathbb{E} : \deg(p) \leq 3\}$.
- (viii) $\mathbb{F}_8 = \{p(x) \in \mathbb{E} : \deg(p) = 3\}$.

Exercise 1.31. (Sequences as a vector space)

Let $\mathbb{E} = \mathcal{F}(\mathbb{N}, \mathbb{R}) = \{(x_n)_{n=1}^\infty : x_n \in \mathbb{R}, \forall n \in \mathbb{N}\}$ be the set of all real sequences $(x_n)_{n=1}^\infty$ (as defined in Example 1.11).

- (i) Prove that $\mathbb{E}$, when equipped with the following operations:

$$(x_n)_{n=1}^\infty + (y_n)_{n=1}^\infty := (x_n + y_n)_{n=1}^\infty, \quad \lambda(x_n)_{n=1}^\infty := (\lambda x_n)_{n=1}^\infty,$$

is a vector space and that $\dim \mathbb{E} = \infty$ (**Indication:** for the last part, it suffices to give a countable infinite set of linearly independent real sequences).

- (ii) Prove that the set of all convergent sequences $\mathbb{F} = \mathcal{F}_c(\mathbb{N}, \mathbb{R}) := \{(x_n)_{n=1}^\infty \in \mathbb{E} : \lim_{n \rightarrow \infty} x_n \in \mathbb{R}\}$ is a subspace of $\mathbb{E}$. Does $\dim \mathbb{F} = \infty$?
- (iii) Prove that the application

$$\begin{aligned} \mathcal{L} : \mathbb{F} &\rightarrow \mathbb{R} \\ (x_n)_{n=1}^\infty &\mapsto \lim_{n \rightarrow \infty} x_n \end{aligned}$$

which maps convergent sequences to their limit, is a linear application from $\mathbb{E}$ to $\mathbb{R}$. Obtain as a corollary that the set $\mathbb{G} = \mathcal{F}_0(\mathbb{N}, \mathbb{R}) := \{(x_n)_{n=1}^\infty \in \mathbb{F} : \lim_{n \rightarrow \infty} x_n = 0\}$ of real sequences that converges to zero is a vector subspace of $\mathbb{F}$. Does $\dim \mathbb{G} = \infty$?

Exercise 1.32. (Fibonacci sequences as a vector space)

We call a real sequence $\{x_n\}_{n=0}^\infty \in \mathcal{F}(\mathbb{N}, \mathbb{R})$ a "Fibonacci sequence" if

$$x_0, x_1 \in \mathbb{R}, \quad \text{and} \quad x_n = x_{n-2} + x_{n-1}, \quad \forall n \geq 2.$$

- (i) Prove that the set $\text{Fib}(\mathbb{R})$ of all Fibonacci sequences is a vector space.
- (ii) Prove that the linear mapping

$$\begin{aligned} \Phi : \text{Fib}(\mathbb{R}) &\rightarrow \mathbb{R}^2 \\ \{x_n\}_{n=0}^\infty &\mapsto (x_0, x_1)^\top \end{aligned}$$

is an isomorphism. Conclude that $\text{Fib}(\mathbb{R})$ is a two dimensional real vector space and that

$$\{u_n\}_{n=0}^\infty = (1, 0, 1, 1, 2, 3, 5, 8, \dots), \quad \{v_n\}_{n=0}^\infty = (0, 1, 1, 2, 3, 5, 8, 13, \dots),$$

form a basis on $\text{Fib}(\mathbb{R})$.

- (iii) Show, using the fact that $\phi = 1+\sqrt{5}/2$ and $\psi = 1-\sqrt{5}/2$ are roots of the polynomial $\lambda^2 - \lambda - 1$, that the sequences $\{x_n\}_{n=0}^{\infty}$, $\{y_n\}_{n=0}^{\infty}$ where

$$x_n = \phi^n \quad \text{and} \quad y_n = \psi^n, \quad \forall n = 0, 1, 2, \dots$$

are Fibonacci sequences.

- (iv) Do the sequences $\{x_n\}_{n=0}^{\infty}$ and $\{y_n\}_{n=0}^{\infty}$ defined in the previous part form a basis on $\text{Fib}(\mathbb{R})$? If so, express the *standard Fibonacci sequence* $\{v_n\}_{n=0}^{\infty} = (0, 1, 1, 2, 3, 5, 8, 13, \dots)$ as a linear combination of $\{x_n\}_{n=0}^{\infty}$ and $\{y_n\}_{n=0}^{\infty}$. Conclude that each term of the standard Fibonacci sequence can be written in closed form as

$$v_n = \frac{1}{\sqrt{5}} [\phi^n - \psi^n], \quad \forall n \in \{0, 1, 2, 3, \dots\}.$$

(**Indication:** The isomorphism defined in (ii) may help you solve this part.)

Exercise 1.33. (Bases and regular matrices)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be a basis of a vector space $\mathbb{E}$. Let $A \in \mathbb{K}^{n \times n}$ and vectors $v_j = \sum_{i=1}^n A_{ij} u_i$. Prove that $\mathcal{V} = \{v_1, \dots, v_n\}$ is also a basis if and only if A is invertible.

Exercise 1.34. (Rank-nullity theorem)

From ex. 1.8 we know that the set $\mathbb{F} = \{A \in \mathbb{K}^{n \times n} : \text{Tr } A = 0\}$ is a subspace of the vector space $\mathbb{K}^{n \times n}$. What is the dimension of $\mathbb{F}$?

Exercise 1.35. (Union of subspaces are not always a vector space)

Let $\mathbb{F}, \mathbb{G}$ be subspaces of the vector space $\mathbb{E}$. Prove that their union $\mathbb{F} \cup \mathbb{G}$ is a subspace if and only if $\mathbb{F} \subset \mathbb{G}$ or $\mathbb{G} \subset \mathbb{F}$.

Exercise 1.36. (Direct sum in space of square matrices)

Let $\mathbb{E} = \mathbb{K}^{n \times n}$ with $\mathbb{K}$ a field with characteristic different than 2. Consider the subspaces of symmetric and skew-symmetric matrices $\mathbb{S} = \{S \in \mathbb{E} : S^T = S\}$, $\mathbb{A} = \{A \in \mathbb{E} : A^T = -A\}$. Prove that $\mathbb{E} = \mathbb{S} + \mathbb{A}$. Does it also hold that $\mathbb{E} = \mathbb{S} \oplus \mathbb{A}$?

Exercise 1.37. (Direct sum of triangular matrices)

Let $\mathbb{E} = \mathbb{K}^{n \times n}$, we define $\mathbb{F}_1, \mathbb{F}_2 \subset \mathbb{E}$ as the subspaces of upper and lower triangular matrices, respectively. Determine whether $\mathbb{E} = \mathbb{F}_1 \oplus \mathbb{F}_2$. Does the result vary if we replace $\mathbb{F}_1$ by $\mathbb{F}'_1$ the subspace of 1-strictly triangular matrices (as defined in ex. 1.28)?

Exercise 1.38. (Distances are not always induced by norms)

From Proposition 1.54 we have seen that for any normed vector space $(\mathbb{E}, \|\cdot\|_{\mathbb{E}})$, we can define a distance on $\mathbb{E}$ $d_{\|\cdot\|_{\mathbb{E}}}(x, y) = \|x - y\|_{\mathbb{E}}$ such that $(\mathbb{E}, d_{\|\cdot\|_{\mathbb{E}}})$ is a metric space. We shall see with this exercise that not every distance can be expressed in terms of a norm $\|x - y\|$.

- (i) Let X be an arbitrary non-empty set, we define the *discrete distance* $\rho : X \times X \rightarrow \mathbb{R}$, given by

$$\rho(x, y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}.$$

Prove that (X, ρ) is a metric space.

- (ii) We now set $X = \mathbb{R}$ with the distance $\rho : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as defined above, prove by contradiction that ρ cannot be written as $\rho(x, y) = \|x - y\|$, for any $\|\cdot\| : \mathbb{R} \rightarrow \mathbb{R}$ defined on $\mathbb{R}$. (**Indication:** Suppose that $\rho(x, y) = \|x - y\|$ for some $\|\cdot\|$, then $\|1\| = \rho(0, 1) = 1$, and $\|2\| = |2| \cdot \|1\| = 2$. What does this contradict?)

- (iii) Similarly to the previous point, prove this generalized result:

Given $\mathbb{E} \neq \{0\}$ a $\mathbb{R}$ -vector space and $d(\cdot, \cdot) : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ a *bounded metric* (i.e. there exists $K > 0$ such that $d(x, y) \leq K$, $\forall x, y \in \mathbb{E}$). Then $d(\cdot, \cdot)$ cannot be written as $d(x, y) = \|x - y\|$, for any $\|\cdot\| : \mathbb{E} \rightarrow \mathbb{R}$.

- (iv) Prove that for any norm $\|\cdot\| : \mathbb{E} \rightarrow \mathbb{R}$ defined on a $\mathbb{R}$ -vector space $\mathbb{E} \neq \{0\}$, the function $d(\cdot, \cdot) : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ given by

$$d(x, y) := \frac{\|x - y\|}{1 + \|x - y\|}$$

defines a distance on $\mathbb{E}$ but is not induced by a norm. (**Indication:** to see that $d(x, y)$ as defined above satisfies the triangular inequality, it might be useful to first prove that the function $\phi(t) = t/(1+t)$ is monotone increasing on the positive real numbers and subadditive, i.e. $\phi(a+b) \leq \phi(a) + \phi(b)$, for all $a, b \geq 0$)

Exercise 1.39. (Vector space of polynomials and its standard basis)

Let $\mathbb{E} = \mathbb{K}_n[X]$ be the vector space of $\mathbb{K}$ -polynomials ($\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$) p of degree $\deg p \leq n$.

- (i) Prove that the set $\mathcal{E} := \{1, X, X^2, \dots, X^n\}$ is a generating family on $\mathbb{E}$.
- (ii) Prove that $\mathcal{E}$ is a linearly independent set, and therefore it is a basis on $\mathbb{E}$. From now on, we will refer to $\mathcal{E}$ as the standard basis of such vector spaces. (**Indication:** Recall the Fundamental Theorem of Algebra, specifically, a $\mathbb{K}$ -polynomial of degree n can have at most n distinct roots in $\mathbb{K}$.)

Exercise 1.40. (Matrix representation of linear applications and vector elements)

Let $\mathbb{E} := \mathbb{R}^n$, and $\mathcal{E} = \{e_i\}_{i=1}^n \subset \mathbb{E}$ be the standard basis of $\mathbb{E}$.

- (i) Let $\mathbb{F} := \mathbb{R}_{n-1}[x]$ be the vector space of real polynomials p of degree $\deg p \leq n-1$, construct a representation of $\mathbb{F}$ in $\mathbb{E}$. (**Indication:** assume that we have proven that $\{1, x, x^2, x^3, \dots, x^{n-1}\} \subset \mathbb{F}$ forms a basis).
- (ii) Let $D : \mathbb{F} \rightarrow \mathbb{F}$ be the *derivative mapping*: $D(p) = p'$ (where p' stands for the polynomial derivative of p). Prove that D is an endomorphism on $\mathbb{F}$ (i.e. $D \in \mathcal{L}(\mathbb{F})$) and give the corresponding matrix representation $[D]_{\mathcal{E}}^{\mathcal{E}}$. What is $\ker D$?
- (iii) We now define the following application:

$$\begin{aligned} L : \mathbb{F} &\rightarrow \mathbb{F} \\ p &\mapsto p' + \alpha p \end{aligned}$$

where $\alpha \in \mathbb{R} \setminus \{0\}$. Prove that $L \in \mathcal{L}(\mathbb{E})$, give $[L]_{\mathcal{E}}^{\mathcal{E}}$.

- (iv) Let $q(x) = 5x^3 + 4x^2 + 5$, find $p(x) \in \mathbb{R}_3$ such that $p'(x) + 6p(x) = q(x)$.
- (v) Can we conclude from (iii) the following: given $\alpha \in \mathbb{R} \setminus \{0\}$, for each $q \in \mathbb{R}_{n-1}[x]$ there exists a unique $p \in \mathbb{R}_{n-1}[x]$ such that $p' + \alpha p = q$?

Exercise 1.41. (Matrix representation of trigonometric functions)

Let $\mathbb{E}$ be real functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy the equation $f''(x) = -f(x)$, $\forall x \in \mathbb{R}$ (i.e. functions whose second derivative is equal to its opposite over the entire real line). Suppose that we've proven that the functions $\mathcal{E} := \{\cos x, \sin x\}$ form a basis on $\mathbb{E}$.

- (i) Let $D : \mathbb{E} \rightarrow \mathbb{E}$ be the *derivative mapping* $D(f(x)) = f'(x)$. Give the matrix representation of D with respect to $\mathcal{E}$.
- (ii) For fixed $y \in \mathbb{R}$, does $f_y(x) := \cos(x+y) \in \mathbb{E}$? If so, express f_y as a linear combination of $\cos x$ and $\sin x$. (**Indication:** It is possible that you have learnt this identity in highschool. If not, write $\cos(x+y) = f_y(x) = c_1 \cos x + c_2 \sin x$, substitute specific values of x to f_y and f'_y to find out the values of c_i).
- (iii) Do the same for $g_y(x) := \sin(x+y)$.
- (iv) For $y \in \mathbb{R}$, we define the *phase shift operator* T_y , where for $f \in \mathbb{E}$

$$T_y(f(x)) = f(x+y).$$

Use the above to prove that $\text{im } T_y \subset \mathbb{E}$ and give the matrix representation $[T_y]_{\mathcal{E}}^{\mathcal{E}}$. Conclude that if f is solution to $f'' = -f$, then f is necessarily 2π -periodic (i.e. $f(x) = f(x+2\pi)$, $\forall x \in \mathbb{R}$).

Exercise 1.42. (Dual space and dual basis in $\mathbb{R}^n$)

In $\mathbb{E} = \mathbb{R}^n$ we are given the basis $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$, where $v_j = \sum_{k=1}^j k e_k$. Express the dual basis $\mathcal{V}^*$ in terms of the vectors e_j^* in the dual space $\mathbb{E}^*$.

Exercise 1.43. (Dual space and dual basis in $\mathbb{R}_n[X]$)

Let $\mathbb{E} := \mathbb{R}_1[X]$ be the space of real polynomials $p(X)$ with $\deg p \leq 1$ (Suppose that we have already proven that the standard basis on $\mathbb{E}$, i.e. $\{1, X\}$, is indeed a basis). Consider functions $f, g, h : \mathbb{E} \rightarrow \mathbb{K}$ defined by

$$f(p(X)) = p(0), \quad g(p(X)) = p(1), \quad \text{and} \quad h(p(X)) = \int_0^1 p(X) dX, \quad p(X) \in \mathbb{R}_1[X].$$

- (i) Prove that $f, g, h \in \mathbb{E}^*$, and that f and g form a basis on $\mathbb{E}^*$.
- (ii) Express h as a linear combination of f and g , is the linear combination unique?
- (iii) Give the dual basis $\{e_1^*, e_2^*\}$ of the standard basis $\{1, X\}$ (**Indication:** try and obtain these vectors as a linear combination of the basis vectors $\{f, g\}$).

Exercise 1.44. (Dual space)

The following is a similar but perhaps more laborious exercise to the previous one (ex. 1.43).

Let $\mathbb{E} = \mathbb{C}_2[X]$ be the space of **complex** polynomials $p(X)$ with $\deg p \leq 2$ (suppose that $\{1, X, X^2\}$ is proven to be a basis on $\mathbb{E}$), and let $f, g \in \mathbb{E}^*$ be as defined below:

$$f(p(X)) = p(1), \quad \text{and} \quad g(p(X)) = p(i),$$

where $i \in \mathbb{C}$ is the imaginary unit element.

- (i) Are f and g linearly independent?
- (ii) If the previous point is affirmative, give a third element $h \in \mathbb{E}^*$ such that $\{f, g, h\}$ forms a basis on $\mathbb{E}^*$.
- (iii) Does the system of equations

$$\begin{cases} f(p(X)) = i, & \text{and} \\ g(p(X)) = 1 \end{cases}$$

have any solution for $p \in \mathbb{E}$? if there is, give all the polynomials that are solution of the system. Is the solution unique?

Exercise 1.45. (Dual space and annihilator)

Let $\mathbb{E} = \mathbb{R}_n[X]$ with $n \geq 1$, and let $f \in \mathbb{E}^*$ be defined by $f(p(X)) = p(1)$.

- (i) Let $\mathbb{F} = \{p(X) \in \mathbb{E} : p(1) = 0\}$. Find a basis of $\mathbb{F}$. What is $\dim \mathbb{F}$?
- (ii) Verify that $f \in \mathbb{F}^\circ$, is f a basis for $\mathbb{F}^\circ$?

Chapter 2

Inner-Product Spaces

2.1 Inner Product

We have seen that in normed vector spaces we have information about how the norm behaves with respect to the operations of addition and scalar multiplication (see points 3 and 4 in Definition 1.53). For the addition, we only have an inequality, though. In this chapter, we will see that, for certain classes of norms, we can give more information about the behavior of this sum. These special norms are obtained through an inner product (aka. scalar product), which is the main idea of this chapter. The spaces having these special norms are called inner-product spaces. In addition to this type of information, the notion of an inner product will give rise to plenty of additional concepts like the notion of angle, and orthogonality.

Let us start by defining the notion of inner product, and then see how it connects to norms.

Definition 2.1: Inner product (aka. scalar product)

Let E be an $\mathbb{K}$ -vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. An inner product on E is a map $\langle \cdot, \cdot \rangle_E : E \times E \rightarrow \mathbb{K}$ satisfying for all $x, y, z \in E$ and $\lambda \in \mathbb{K}$:

1. linearity with respect to the first variable:

$$\langle \lambda x + \mu y, z \rangle_E = \lambda \langle x, z \rangle_E + \mu \langle y, z \rangle_E, \quad \forall x, y, z \in E \text{ and } \lambda, \mu \in \mathbb{K},$$

2. conjugate symmetry:

$$\langle x, y \rangle_E = \overline{\langle y, x \rangle_E}, \quad \forall x, y \in E$$

3. positive-definiteness:

$$\langle x, x \rangle_E \geq 0, \quad \forall x \in E \quad \text{and} \quad \langle x, x \rangle_E = 0 \iff x = 0_E.$$

From an inner product, we can always build a norm.

Lemma 2.2: Norm induced by an inner product

Let E be an $\mathbb{K}$ -vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and $\langle \cdot, \cdot \rangle_E$ be an inner product over E . The map

$$\begin{aligned} \|\cdot\|_E : E &\rightarrow \mathbb{R} \\ x &\mapsto \|x\|_E := \langle x, x \rangle_E^{1/2}, \end{aligned} \tag{2.1}$$

is a norm on E which is called the norm induced by the inner product.

The proof of this statement requires the following inequality, for which it is hard to overstate how fundamental it is in mathematics.

Lemma 2.3: Cauchy-Schwarz inequality

Let E be an $\mathbb{K}$ -vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and let $\langle \cdot, \cdot \rangle_E$ be defined as in Eq. (2.1). Then

$$|\langle x, y \rangle_E| \leq \|x\|_E \|y\|_E,$$

and equality holds if and only if x and y are linearly dependent.

Proof. We first start with the real case $\mathbb{K} = \mathbb{R}$. Let us define the map $p : \lambda \in \mathbb{R} \mapsto \|x + \lambda y\|^2$, which is nonnegative for every $\lambda \in \mathbb{R}$. Then

$$p(\lambda) = \|x + \lambda y\|^2 = \langle x + \lambda y, x + \lambda y \rangle = \|x\|^2 + 2\lambda \langle x, y \rangle + \lambda^2 \|y\|^2.$$

Thus p is a polynomial of degree two in λ . Since $p(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$, then its discriminant must be nonpositive, which translates into,

$$4|\langle x, y \rangle|^2 \leq 4\|x\|^2 \|y\|^2,$$

and yields the desired inequality. In addition, in case of an equality, this means that $|\langle x, y \rangle| = \|x\| \|y\|$. The discriminant is equal to zero, and p has a unique root μ . From the fact that $p(\mu) = 0$, it follows that $\|x + \mu y\| = 0$ which is equivalent to $x + \mu y$ (linear dependence).

We now consider the case $\mathbb{K} = \mathbb{C}$, and let $x, y \in E$. Since $\langle x, y \rangle \in \mathbb{C}$, it admits a polar decomposition of the form $\langle x, y \rangle = \rho e^{i\theta}$ with $\rho = |\langle x, y \rangle| \geq 0$ and $\theta \in \mathbb{R}$. We now let $x' = e^{i\theta} x$, and define for all $\lambda \in \mathbb{R}$

$$\begin{aligned} p(\lambda) &:= \|x' + \lambda y\|^2 \\ &= \langle x' + \lambda y, x' + \lambda y \rangle \end{aligned} \tag{2.2}$$

$$\begin{aligned} &= \|x'\|^2 + \|\lambda y\|^2 + \langle x', \lambda y \rangle + \langle \lambda y, x' \rangle \\ &= \|x'\|^2 + \|\lambda y\|^2 + 2\Re(\langle x', \lambda y \rangle) \\ &= \|x\|^2 + \lambda^2 \|y\|^2 + 2\rho\lambda. \end{aligned} \tag{2.3}$$

From Eq. (2.3), it follows that p is a real-valued polynomial in λ . In addition, from Eq. (2.2) we have that $p(\lambda) \geq 0$ due to the nonnegativity of the inner product (see point 3 in Definition 2.1). Therefore, the discriminant of p is nonpositive, and we conclude following the same lines as for the real case. $\square$

We are now well equipped to prove Lemma 2.2.

Proof of Lemma 2.2. We have to prove that the mapping defined in Eq. (2.1) satisfies all the properties of a norm. The only nontrivial point is the triangle inequality. For this, we let $x, y \in E$ and write

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \\ &= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2 \\ &= \|x\|^2 + 2\Re(\langle x, y \rangle) + \|y\|^2 \\ &\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \end{aligned}$$

By the Cauchy-Schwarz inequality,

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

$\square$

We are now ready to rigorously define the notion of inner-product spaces, and subclasses of it.

Definition 2.4: Inner-Product, Hilbert and Euclidean Spaces

- An inner-product space is a normed space $(E, \|\cdot\|_E)$ where the norm is induced by an inner product $\langle \cdot, \cdot \rangle_E$. We usually denote it as $(E, \langle \cdot, \cdot \rangle_E)$.
- A Hilbert space is an inner-product space which is also complete^a (i.e. any Cauchy sequence of

elements of E converges in this norm to a limiting element which is also in E).

- A Euclidean space is a finite-dimensional, real inner-product space.

^aYou will learn about completeness in your analysis, topology and functional analysis courses.

From Proposition 1.54, Lemma 2.2 and Definition 2.4, we can devise the following diagram:

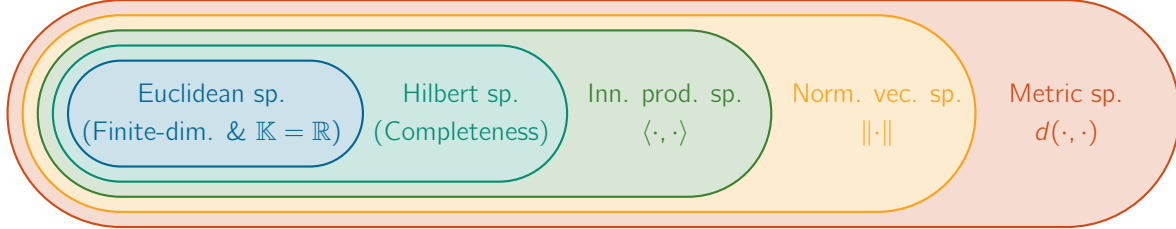


Figure 2.1: Useful diagram to remind the reader the essential properties and relations of the spaces we have mentioned.

2.2 Examples

Canonical/Euclidean inner product in $\mathbb{R}^n$: In $\mathbb{R}^n$, the most classical inner product is defined as

$$\langle (x_1, \dots, x_n)^\top, (y_1, \dots, y_n)^\top \rangle_{\mathbb{R}^n} := \sum_{i=1}^n x_i y_i, \quad \forall (x, y) \in \mathbb{R}^n \times \mathbb{R}^n.$$

We sometimes denote it with the subscript 2 because it induces the norm $\|\cdot\|_2$ that we introduced in Eq. (1.8). $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{\mathbb{R}^n})$ is a Euclidean space, and also a Hilbert space.

With this inner product, the length of a vector and orthogonality are the classical ones. For example, in $\mathbb{R}^3$ the length of a vector $x \in \mathbb{R}^3$ is the distance to the origin, and this is given by the norm $\|x\|_2 = (x_1^2 + x_2^2 + x_3^2)^{1/2}$.

Canonical inner product in $\mathbb{C}^n$: In $\mathbb{C}^n$, the canonical inner product is defined as

$$\langle (z_1, \dots, z_n)^\top, (\zeta_1, \dots, \zeta_n)^\top \rangle_{\mathbb{C}^n} := \sum_{i=1}^n z_i \bar{\zeta}_i, \quad \forall (z, \zeta) \in \mathbb{C}^n \times \mathbb{C}^n.$$

When z and $\zeta \in \mathbb{R}^n$, we have that $\langle z, \zeta \rangle_{\mathbb{C}^n} = \langle z, \zeta \rangle_{\mathbb{R}^n}$. Also, $(\mathbb{C}^n, \langle \cdot, \cdot \rangle_{\mathbb{C}^n})$ is a Hilbert space.

Frobenius inner product in $\mathbb{K}^{m \times n}$: It is defined as

$$\langle A, B \rangle_F := \sum_{i=1}^n \sum_{j=1}^m a_{i,j} \bar{b}_{i,j} = \text{Tr}(B^H A), \quad (2.4)$$

and it induces the Frobenius norm (see Eq. (1.9) with $p = 2$, also see ex. 1.9). This yields a structure of a Hilbert space (if $\mathbb{K} = \mathbb{C}$) and a Euclidean space (if $\mathbb{K} = \mathbb{R}$).

Scalar product in function spaces: As an example, let us consider $E = \mathcal{C}([a, b]; \mathbb{K}^n)$ with $a < b \in \mathbb{R}$. In this space, we can define an inner product as

$$\langle f, g \rangle_{\mathcal{C}([a, b]; \mathbb{K}^n)} := \int_a^b \langle f(x), g(x) \rangle_{\mathbb{K}^n} dx, \quad \forall f, g \in \mathcal{C}([a, b]; \mathbb{K}^n).$$

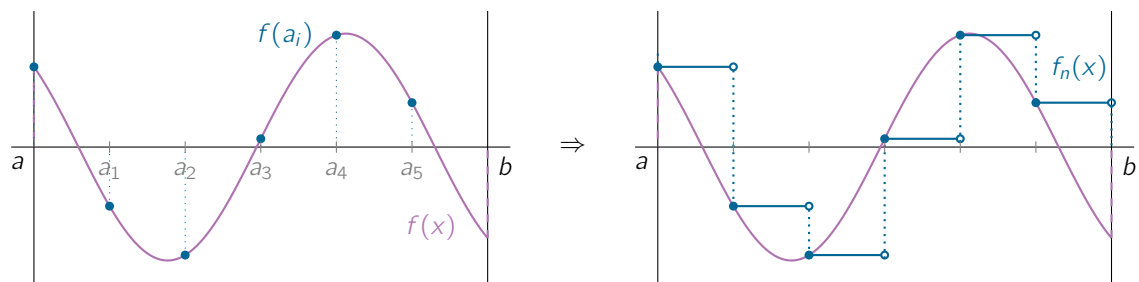
For example, when $n = 1$ and $\mathbb{K} = \mathbb{C}$, this reads as

$$\langle f, g \rangle_{\mathcal{C}([a, b]; \mathbb{C})} := \int_a^b f(x) \overline{g(x)} dx, \quad \forall f, g \in \mathcal{C}([a, b]; \mathbb{C}). \quad (2.5)$$

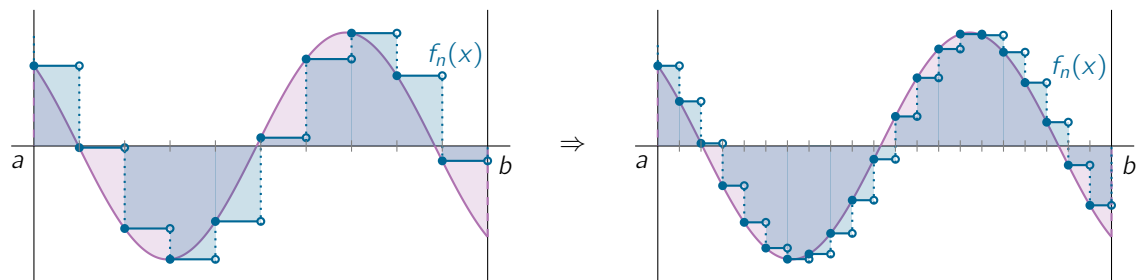
Such an inner product gives a Hilbert space structure to $\mathcal{C}([a, b]; \mathbb{K}^n)$.

Remark 2.5. It is interesting to think about how to compute norms from functions, e.g., the one from Eq. (2.5). In full generality, we will not be able to compute the integral neither analytically nor exactly (our success heavily depends on the input functions f, g). One possibility to deal with this is to approximate the continuous functions with piecewise functions as follows. For a given $f \in \mathcal{C}([a, b]; \mathbb{C})$, we set a number of subintervals $n \geq 1$ and define its piecewise approximation as

$$f_n(x) := \sum_{i=0}^{n-1} f(a_i) \mathbb{1}_{[a_i, a_{i+1})}(x), \quad \text{with } a_k = a + \frac{k}{n}(b-a) \text{ for } k \in \{0, \dots, n-1\}.$$



(a) Piece-wise approximation $f_n(x)$ of a continuous function $f(x)$ over a closed interval $[a, b]$ with $n = 6$ intervals.



(b) We can see that as n becomes larger, the integral of the piece-wise approximation f_n (the shaded area) approximates to the integral of f .

Figure 2.2: Illustration of the piece-wise approximation

This means that, on a subinterval $[a_i, a_{i+1}) \subset [0, 1]$, we use the value of f at the beginning of the subinterval, $f(a_i)$. We can then exactly evaluate the inner product between f_n and g_n since

$$\langle f, g \rangle_{\mathcal{C}([a, b]; \mathbb{C})} = \int_a^b f_n(x) \overline{g_n(x)} dx = \sum_{i=0}^{n-1} \int_{a_i}^{a_{i+1}} f(x_i) \overline{g(x_i)} \mathbb{1}_{[a_i, a_{i+1})}(x) dx = \frac{1}{n} \sum_{i=0}^{n-1} f(x_i) \overline{g(x_i)}.$$

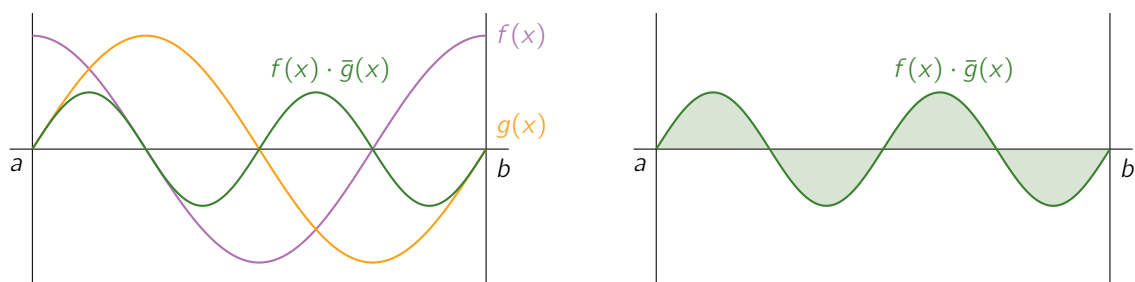


Figure 2.3: Illustration of two functions $\cos(2\pi x)$, $\sin(2\pi x) \in \mathcal{C}([0, 1], \mathbb{C})$, and the inner product $\langle f, g \rangle_{\mathcal{C}([0, 1], \mathbb{C})} = \int_0^1 f(x) \overline{g(x)} dx = 0$

2.3 Which norms arise from an inner product? The Jordan-von Neumann theorem

In many applications, inner-products play a very central and useful role, and motivate the question: which norms arise from an inner product, and which ones do not? The answer is given by the Jordan-von Neumann theorem, whose proof is based on two fundamental identities: the parallelogram identity and the polarization identity. The parallelogram identity shows a necessary condition for a norm to come from an inner product.

Proposition 2.6: Parallelogram Identity

Let $(E, \langle \cdot, \cdot \rangle_E)$ be a $\mathbb{K}$ -inner-product vector space, and let $\|\cdot\|$ be its induced norm. Then

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad \forall x, y \in E.$$

Proof. It suffices to develop $\|x + y\|^2 + \|x - y\|^2 = \langle x + y, x + y \rangle + \langle x - y, x - y \rangle = \dots$ □

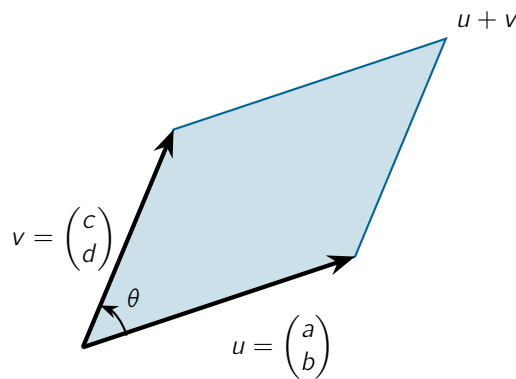


Figure 2.4: Illustration of the parallelogram identity $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ on $\mathbb{R}^2$

The parallelogram identity expresses a purely geometric fact: in an inner-product space, the sum of the squares of the diagonals of a parallelogram equals the sum of the squares of its sides. We can see this clearly in $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$ as Fig. 4.1 illustrates. We see that we have

$$\|AC\|^2 + \|BC\|^2 = \|AB\|^2 + \|BC\|^2 + \|CD\|^2 + \|DA\|^2 = 2(\|AB\|^2 + \|BC\|^2).$$

We can thus raise the converse question: If a norm satisfies the parallelogram identity, does it necessarily come from an inner product? The answer relies on an explicit construction of the inner product from the norm, which is known as the polarization identity.

Lemma 2.7: Polarization identity

Let $(E, \|\cdot\|)$ be a $\mathbb{K}$ -vector space that satisfies the parallelogram identity. For a real vector space, define

$$\langle x, y \rangle := \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2) \tag{2.6}$$

and for a complex vector space define

$$\langle x, y \rangle := \frac{1}{4} \sum_{k=0}^3 i^k \|x + i^k y\|^2. \tag{2.7}$$

Then:

- (i) $\langle \cdot, \cdot \rangle$ is an inner product,

(ii) $\langle \cdot, \cdot \rangle$ is the only inner-product that induces the norm $\|\cdot\|_E$.

Proof. We give the proof for the real case:

(i) It is easy to see that $\langle \cdot, \cdot \rangle$ is symmetric and positive, we proceed by checking the additivity and homogeneity.

Additivity: We first give a few identities that can be directly derived from the parallelogram identity (simply applying eq. (2.6) on the pairs $\{u, v\}$, $\{w, t\}$,... etc.), let $x, y, z \in E$:

$$\begin{cases} u = x + z \\ v = y + z \end{cases} \Rightarrow \|x + z\|^2 + \|y + z\|^2 = \frac{1}{2} (\|x + y + 2z\|^2 + \|x - y\|^2) \quad (2.8)$$

$$\begin{cases} w = x - z \\ t = y - z \end{cases} \Rightarrow \|x - z\|^2 + \|y - z\|^2 = \frac{1}{2} (\|x + y - 2z\|^2 + \|x - y\|^2) \quad (2.9)$$

$$\begin{cases} r = x + y + z \\ s = z \end{cases} \Rightarrow \|x + y + z\|^2 + \|z\|^2 = \frac{1}{2} (\|x + y + 2z\|^2 + \|x + y\|^2) \quad (2.10)$$

$$\begin{cases} p = x + y - z \\ q = -z \end{cases} \Rightarrow \|x + y - z\|^2 + \|z\|^2 = \frac{1}{2} (\|x + y - 2z\|^2 + \|x + y\|^2) \quad (2.11)$$

We then have

$$\begin{aligned} \langle x, z \rangle + \langle y, z \rangle &\stackrel{(2.6)}{=} \frac{1}{4} (\|x + z\|^2 + \|y + z\|^2 - \|x - z\|^2 - \|y - z\|^2) \\ &\stackrel{(2.8), (2.9)}{=} \frac{1}{4} \left[\frac{1}{2} (\|x + y + 2z\|^2 + \|x - y\|^2) - \frac{1}{2} (\|x + y - 2z\|^2 + \|x - y\|^2) \right] \\ &= \frac{1}{4} \left[\frac{1}{2} \|x + y + 2z\|^2 - \frac{1}{2} \|x + y - 2z\|^2 \right] \\ &\stackrel{(2.10), (2.11)}{=} \frac{1}{4} \left[\left(\|x + y + z\|^2 + \|z\|^2 - \frac{1}{2} \|x + z\|^2 \right) \right. \\ &\quad \left. - \left(\|x + y - z\|^2 + \|z\|^2 - \frac{1}{2} \|x + z\|^2 \right) \right] \\ &= \frac{1}{4} (\|x + y + z\|^2 - \|x + y - z\|^2) \\ &\stackrel{(2.6)}{=} \langle x + y, z \rangle \end{aligned}$$

which is the additivity we wish to prove.

Homogeneity: We wish to prove that

$$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle, \quad \forall \lambda \in \mathbb{R}.$$

Firstly, using the additivity of $\langle \cdot, \cdot \rangle$ that we have just proved, it is easy to see that for natural numbers¹:

$$\langle nx, y \rangle = n \langle x, y \rangle, \quad \forall n \in \mathbb{N}. \quad (2.12)$$

For negative natural numbers $-n$, we have that

$$\langle -nx, y \rangle + n \langle x, y \rangle \stackrel{(2.12)}{=} \langle -nx, y \rangle + \langle nx, y \rangle \stackrel{(\text{Add.})}{=} \langle -nx + nx, y \rangle = \langle 0, y \rangle = 0.$$

Or equivalently:

$$\langle -nx, y \rangle = -n \langle x, y \rangle, \quad \forall n \in \mathbb{N}. \quad (2.13)$$

Equations (2.12) and (2.13) can be summarised as

$$\langle zx, y \rangle = z \langle x, y \rangle, \quad \forall z \in \mathbb{Z}. \quad (2.14)$$

¹It is a simple proof by induction exercise, try and do it!

Now we turn to rational numbers. Let $q \in \mathbb{Q}$, then there exists $n, m \in \mathbb{Z}$ with $m \neq 0$ such that $q = n/m$. Therefore for $x, y \in E$ we get

$$m\langle qx, y \rangle \stackrel{(2.14)}{=} \langle m \cdot (qx), y \rangle \stackrel{(mq=n)}{=} \langle nx, y \rangle \stackrel{(2.14)}{=} n\langle x, y \rangle$$

Dividing both sides by m we get

$$\langle qx, y \rangle = q\langle x, y \rangle, \quad \forall q \in \mathbb{Q}. \quad (2.15)$$

Lastly, let $\lambda \in \mathbb{R}$, from analysis we know that there exists a sequence of rational numbers $\{q_n\}_{n=1}^{\infty} \subset \mathbb{Q}$ that converges to λ . For each $n \in \mathbb{N}$, thanks to (2.15) we have

$$\langle q_n x, y \rangle = q_n \langle x, y \rangle \quad (2.16)$$

On the right hand side we have

$$\lim_{n \rightarrow \infty} q_n \langle x, y \rangle = \lambda \langle x, y \rangle.$$

The left hand side is trickier: we first expand the expression:

$$\langle q_n x, y \rangle = \frac{1}{4} [\|q_n x + y\|^2 - \|q_n x - y\|^2] \quad (2.17)$$

Using the reverse triangle inequality we see that

$$\| \|q_n x + y\| - \|\lambda x + y\| \| \leq \| (q_n - \lambda) \cdot x \| = \underbrace{|q_n - \lambda|}_{\xrightarrow{n \rightarrow \infty} 0} \cdot \|x\| \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \lim_{n \rightarrow \infty} \|q_n x + y\| = \|\lambda x + y\|,$$

similarly $\lim_{n \rightarrow \infty} \|q_n x - y\| = \|\lambda x - y\|$, putting this back to (2.17) we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \langle q_n x, y \rangle &= \lim_{n \rightarrow \infty} \left[\frac{1}{4} \|q_n x + y\|^2 - \|q_n x - y\|^2 \right] = \frac{1}{4} \left[\lim_{n \rightarrow \infty} \|q_n x + y\|^2 - \lim_{n \rightarrow \infty} \|q_n x - y\|^2 \right] \\ &= \frac{1}{4} [\|\lambda x + y\|^2 - \|\lambda x - y\|^2] = \langle \lambda x, y \rangle \end{aligned}$$

Going back to (2.16) we have that

$$\langle \lambda x, y \rangle \xleftarrow{n \rightarrow \infty} \langle q_n x, y \rangle = q_n \langle x, y \rangle \xrightarrow{n \rightarrow \infty} \lambda \langle x, y \rangle \quad (\text{Uniqueness of limit}) \Rightarrow \langle \lambda x, y \rangle = \lambda \langle x, y \rangle.$$

Which is the homogeneity condition that we wanted to prove.

(ii) We first prove that the inner-product (2.6) induces the norm. This is true since we have

$$\langle x, x \rangle \stackrel{(2.6)}{=} \frac{1}{4} (\|x + x\|^2 + \|x - x\|^2) = \frac{1}{4} (\|2x\|^2 + \|0\|^2) = \frac{1}{4} (4\|x\|^2) = \|x\|^2.$$

We now show that (2.6) is the unique product that induces the norm. Let $\omega(\cdot, \cdot) : E \times E$ be an arbitrary inner product on E that satisfies

$$\|x\|^2 = \omega(x, x), \quad \forall x \in E. \quad (2.18)$$

For simplicity, we assume that $\mathbb{K} = \mathbb{R}$ so that ω is symmetric (for $\mathbb{K} = \mathbb{C}$ we need to minimally adapt the proof for ω being conjugate symmetric). Then for $x, y \in E$, from (2.18) we see that

$$\begin{aligned} \|x + y\|^2 &= \omega(x + y, x + y) = \omega(x, x) + \omega(y, y) + 2\omega(x, y) = \|x\|^2 + \|y\|^2 + 2\omega(x, y), \quad \text{and} \\ \|x - y\|^2 &= \omega(x - y, x - y) = \omega(x, x) + \omega(y, y) - 2\omega(x, y) = \|x\|^2 + \|y\|^2 - 2\omega(x, y), \end{aligned}$$

Combining both lines we see that

$$\|x + y\|^2 - \|x - y\|^2 = 4\omega(x, y) \iff \omega(x, y) = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2) \stackrel{(2.6)}{=} \langle x, y \rangle.$$

We have thus proven that any inner product on E that satisfies the property (2.18) is necessarily equal to $\langle \cdot, \cdot \rangle$. Or in other words, $\langle \cdot, \cdot \rangle$ is the unique inner product that has said property.

For the case where $\mathbb{K} = \mathbb{C}$, the lemma can be proved in analogous manner, although with more complex (no pun intended) calculations. $\square$

The polarization identity shows that, once the parallelogram law holds for a normed vector space, an inner product is guaranteed to exist, and is the unique inner product in the sense that it induces the norm. We can now state the main characterization of inner-product spaces.

Theorem 2.8: Jordan von-Neumann

Let $\|\cdot\|$ be a norm on a $\mathbb{K}$ -vector space E . Then we have the equivalence:

$$\|\cdot\| \text{ is induced by an inner product } \Leftrightarrow \|\cdot\| \text{ satisfies the parallelogram identity.}$$

Proof. The direction $\Leftarrow$ holds thanks to the parallelogram identity. The direction $\Rightarrow$ holds by the polarization identity, which gives us an explicit construction of the inner-product inducing the given norm. $\square$

2.4 Orthogonality

In this section $(E, \langle \cdot, \cdot \rangle_E)$ is a $\mathbb{K}$ -inner-product vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

2.4.1 Definition

Since we keep calling vectors to the elements of the E , one may wonder if can define a notion of alignment or angle between them. This is possible thanks to the inner product.

Definition 2.9: Angle between 2 vectors

The angle $\alpha \in [0, \pi]$ between two vectors $x, y \in E$ is defined through its cosine as

$$\cos(\alpha) := \frac{\langle x, y \rangle}{\|x\| \|y\|} \tag{2.19}$$

This definition makes sense since, by the Cauchy-Schwarz inequality, $\left| \frac{\langle x, y \rangle}{\|x\| \|y\|} \right| \leq 1$, therefore we can interpret this quantity as the cosine of an angle. We may note that we recover the classical definition of angle between vectors in $\mathbb{R}^n$.

Interesting things happen when the vectors ‘become perpendicular’ in the sense that $\alpha = \pi/2$. This happens when $\langle x, y \rangle = 0$ (assuming that $x, y \neq 0$). This gives rise to the general notion of orthogonality and orthonormality, which are key concepts in inner-product spaces.

Definition 2.10: Orthogonality and Orthonormality

Let $(E, \langle \cdot, \cdot \rangle_E)$ be a $\mathbb{K}$ -inner-product vector space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let W and T be two subspaces of E .

- We say that two vectors $x, y \in E$ are orthogonal if $\langle x, y \rangle = 0$, and we often denote this as $x \perp y$.
 - We say that x is orthogonal to W ($x \perp W$) if $\langle x, w \rangle = 0$ for all $w \in W$.
 - We say that W and T are orthogonal ($W \perp T$) if $\langle w, t \rangle = 0$ for all $w \in W$ and $t \in T$.
 - We say that a set of vectors $\{v_1, \dots, v_n\} \subset E$ is orthonormal if $\langle v_i, v_j \rangle = \delta_{ij}$.
-

2.4.2 Orthonormal bases

Let E be an n -dimensional Euclidean space and $X = \{e_1, \dots, e_n\}$ be a basis of E . For $x = \sum_{i=1}^n \alpha_i e_i$, and $y = \sum_{j=1}^n \beta_j e_j$, the scalar product reads

$$\langle x, y \rangle = \left\langle \sum_{i=1}^n \alpha_i e_i, \sum_{j=1}^n \beta_j e_j \right\rangle = \sum_{i,j=1}^n \alpha_i \beta_j \langle e_i, e_j \rangle.$$

This computation is in general quite cumbersome because it requires evaluating the n^2 inner products $\langle e_i, e_j \rangle$ for $1 \leq i, j \leq n$ (and, if we remember Eq. (2.5), computing each product can be expensive depending on the nature of E). So the numerical complexity is of order $\mathcal{O}(n^2)$. One could consider reducing the complexity by finding a basis with as many orthogonal vectors as possible. Independently of our choice, since $\langle e_i, e_i \rangle = \|e_i\|^2 > 0$, the complexity will be at least n . The basis for which all inner-products vanish except for the diagonal ones is called an orthogonal basis.

Definition 2.11

Let E be an n -dimensional Euclidean space and $X = \{e_1, \dots, e_n\}$ be a basis of E . We say that X is an orthogonal basis if

$$\langle e_i, e_j \rangle = 0, \quad \forall i \neq j.$$

The basis is said to be orthonormal if

$$\langle e_i, e_j \rangle = \delta_{ij}, \quad \forall 1 \leq i, j \leq n.$$

In an orthonormal basis, the computation of the scalar product and the norm becomes as simple as

$$\langle x, y \rangle = \sum_{i,j=1}^n \alpha_i \beta_j \langle e_i, e_j \rangle = \sum_{i,j=1}^n \alpha_i \beta_j \delta_{ij} = \sum_{i=1}^n \alpha_i \beta_i, \quad \|x\|^2 = \sum_{i=1}^n \alpha_i^2$$

It is important to realize that the value of the inner product does not depend on the basis. This is the topic of ??.

Proposition 2.12

Let E be a Euclidean space and $\mathcal{E}_n = \{e_1, \dots, e_n\}$ an orthonormal basis of E . Then

$$x = \sum_{i=1}^n \langle x, e_i \rangle e_i, \quad \text{and} \quad \|x\|^2 = \sum_{i=1}^n |\langle x, e_i \rangle|^2.$$

Proof. Since $\mathcal{E}_n$ is a basis, there exist $(c_1, \dots, c_n) \in \mathbb{R}^n$ such that $x = \sum_{i=1}^n c_i e_i$. Taking the inner product with e_j yields $c_j = \langle x, e_j \rangle$ for all $1 \leq j \leq n$. To prove the formula with the norm, we simply write $\|x\|^2 = \langle x, x \rangle$ and develop the sum using that $x = \sum_{i=1}^n \langle x, e_i \rangle e_i$. $\square$

2.4.3 Gram-Schmidt orthonormalization

The following theorem guarantees that one can always transform a basis of a euclidean space into an orthonormal basis. The procedure is constructive, and is very commonly used in practical applications.

Theorem 2.13: Gram-Schmidt orthonormalization

Any euclidean space E has an orthonormal basis.

Proof. Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be a basis of E . We define by induction the following vectors,

$$\begin{aligned} e_1 &:= \frac{x_1}{\|x_1\|} \\ e_i &:= \frac{x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\|}, \quad 2 \leq i \leq n. \end{aligned} \quad (2.20)$$

Our goal is to prove that $\mathcal{E}_n = \{e_1, \dots, e_n\}$ is an orthonormal basis of E . For this, we start by noticing that each $e_i \in \text{span}\{x_1, \dots, x_i\}$. This implies that for every $1 < i \leq n$, $\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\| \neq 0$ given that $\sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j \in \text{span}\{x_1, \dots, x_{i-1}\}$. This shows that definition (2.20) of the e_i is well-posed in the sense that the denominator does not vanish.

We are now going to show by induction that $\mathcal{E}_i$ is an orthonormal set for $1 \leq i \leq n$. We start with the case $i = 1$, for which it directly follows by the definition of e_1 that $\langle e_1, e_1 \rangle = \langle x_1, x_1 \rangle / \|x_1\|^2 = 1$. For the inductive step, let $1 < i \leq n$, and assume that $\mathcal{E}_{i-1}$ is an orthonormal set. By the definition of e_i , we have that

$$\langle e_i, e_i \rangle = \frac{\langle x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j, x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j \rangle}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\|^2} = 1.$$

For any $1 \leq k < i$,

$$\begin{aligned} \langle e_i, e_k \rangle &= \frac{\langle x_i, e_k \rangle - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle \langle e_j, e_k \rangle}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\|} \\ &= \frac{\langle x_i, e_k \rangle - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle \delta_{j,k}}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\|} \quad (\text{induction hypothesis: } \mathcal{E}_{i-1} \text{ is an orthonormal set}) \\ &= \frac{\langle x_i, e_k \rangle - \langle x_i, e_k \rangle}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle e_j\|} \\ &= 0. \end{aligned}$$

This allows us to conclude that $\mathcal{E}_n$ is an orthonormal set. To show that $\mathcal{E}_n$ is a basis of E , it suffices to show that it is a linearly independent family. Let $\lambda_1, \dots, \lambda_n \in \mathcal{K}$ be such that $\lambda_1 e_1 + \dots + \lambda_n e_n = 0$. Multiplying scalarly by e_i and using that $\mathcal{E}_n$ is an orthonormal set, we deduce that $\lambda_i = 0$ for $1 \leq i \leq n$. Thus $\mathcal{E}_n$ is linearly independent, and therefore a basis of E . $\square$

2.4.4 Orthogonal Projections

In inner-product spaces, there is a special type of subspace projection (see Definition 1.46) which has particular properties that are connected to the notion of orthogonality. To present them, we start by introducing the notion of orthogonal complement.

Definition 2.14: Orthogonal Complement

Let E be an inner-product space and $F \subset E$ a subspace of E . We denote

$$F^\perp := \{x \in E : \langle x, f \rangle = 0, \forall f \in F\} \quad (2.21)$$

the orthogonal complement of F .

Here are a few properties of orthogonal complements.

Proposition 2.15

Let F be a subspace of a euclidean space E . Then:

- (i) $E = \{0\}^\perp$ and $E^\perp = \{0\}$.

- (ii) $F^\perp$ is a nonempty subspace of E
- (iii) $E = F \oplus F^\perp$
- (iv) $(F^\perp)^\perp = F$

Proof. 1. To prove that $E^\perp = \{0\}$, we remark that if $x \in E^\perp$, then $\|x\| = 0$, thus $x = 0$.

2. $F^\perp$ is non empty since $0 \in F^\perp$. The fact that $F^\perp$ is a subspace can easily be verified and is left as an exercise.
3. Suppose that the dimension of F is p with $0 \leq p \leq n$. If $p = 0$ or $p = n$, the statement is proven thanks to point (i). Let us now deal with the case $1 \leq p < n$. To prove the first statement, we consider a basis of F , to which we add linearly independent vectors to obtain a basis of E . We next orthonormalize the basis by Gram-Schmidt orthonormalization. This gives us an orthonormal basis $\mathcal{E} = \{e_1, \dots, e_n\}$ where the first p vectors form a basis of F . By using Eq. (2.21), we can then show that

$$F^\perp = \text{span}\{e_{p+1}, \dots, e_n\}.$$

We thus conclude that $E = F \oplus F^\perp$ and $\dim(E) = \dim(F) + \dim(F^\perp)$.

4. We start by observing that $\dim((F^\perp)^\perp) = \dim(E) - \dim(F^\perp)$. Then, we easily verify that $F \subset (F^\perp)^\perp$, which leads to the conclusion that $(F^\perp)^\perp = F$. □

Projections onto subspaces $F \subset E$ involving the decomposition $E = F \oplus F^\perp$ are called orthogonal projections.

Definition 2.16: Orthogonal Projection

Let $E = F \oplus F^\perp$ for a subspace $F \subset E$. The mapping

$$\begin{aligned} P_F &= P_{F \oplus F^\perp} : E \rightarrow F \subset E \\ x &= x_F + x_{F^\perp} \mapsto x_F \end{aligned}$$

is called the orthogonal projector onto F , and x_F is called the orthogonal projection of $x \in E$ onto F .

Example 2.17

In $\mathbb{R}^3$, the orthogonal projection of $(3, 2, 4)$ onto the plane $\{x = 0\}$ is $(0, 2, 4)$, and its projection onto $\{x + y + z = 0\}$ is $(0, -1, 1)$.

The following result summarizes the main properties of orthogonal projectors.

Proposition 2.18

Let F be a subspace of a euclidean vector space E . Then:

1. P_F is the projection onto F in the sense that $\text{im } P_F = F$ and $P_F \circ P_F = P_F$.
2. Orthogonality: For all $x \in E$ it holds that

$$\begin{aligned} x - P_F(x) &\perp F \\ x &= P_F(x) + P_{F^\perp}(x) \end{aligned}$$

3. Contraction: For all $x \in E$, $\|P_F(x)\| \leq \|x\|$.

4. If $f \in \mathcal{L}(E)$ is a linear map such that $f = f \circ f$ and $\text{im}(f) \perp \ker(f)$, then f is the orthogonal projection onto $\text{im}(f)$.
5. Minimizing property: For all $x \in E$,

$$\|x - P_F(x)\| = \min_{y \in F} \|x - y\|. \quad (2.22)$$

Proof. 1. This property simply follows from the definition of the orthogonal projector P_F as the subspace projection $P_{F \oplus F^\perp}$.

2. Every $x \in E = F \oplus F^\perp$ can be uniquely decomposed into $x = x_F + x_{F^\perp}$, and by the definition of P_F and $P_{F^\perp}$, $x_F = P_F(x)$ and $x_{F^\perp} = P_{F^\perp}(x)$. Thus $x = P_F(x) + P_{F^\perp}(x)$. Therefore $x - P_F(x) \in F^\perp$, which is equivalent to $x - P_F(x) \perp F$.

3. For every $x \in E$,

$$\|x\|^2 = \langle x, x \rangle = \langle P_F(x) + P_{F^\perp}(x), P_F(x) + P_{F^\perp}(x) \rangle = \|P_F(x)\|^2 + \|P_{F^\perp}(x)\|^2 \geq \|P_F(x)\|^2.$$

4. We start by proving that $E = \text{im}(p) + \ker(p)$. Every $x \in E$ can be written as $x = p(x) + x - p(x)$. Since $p(x - p(x)) = 0$, it follows that $x - p(x) \in \ker(p)$. In addition, the decomposition of x into $p(x) \in \text{im}(f)$ and $x - p(x) \in \ker(f)$ is unique because if $x = a + b$ with $b \in \ker(p)$, then $p(x) = p(a) = a$. We conclude by the definition of orthogonal projection.

5. For any $f \in F$, it holds that

$$\begin{aligned} \|x - f\|^2 &= \|x - P_F(x) + P_F(x) - f\|^2 \\ &= \|x - P_F(x)\|^2 + \|P_F(x) - f\|^2 \\ &\geq \|x - P_F(x)\|^2. \end{aligned}$$

Therefore $\|x - P_F(x)\| = \min_{f \in F} \|x - f\|$.

□

Proposition 2.19

Let (E, ω) with $\dim E = n$ be a finite dimensional inner product space with inner product ω . Let $F \subset E$ be a subspace of E spanned by a basis $\mathcal{F} = \{f_1, \dots, f_p\}$. Then the coordinates of the orthogonal projection $P_F : E \rightarrow F$ in $\mathcal{F}$ are given by

$$[P_F(u)]^{\mathcal{F}} = [\omega]_{\mathcal{F}\mathcal{F}}^{-1} q, \quad \text{with } q = (\omega(u, f_i))_{i=1}^p. \quad (2.23)$$

Proof. Let $u \in E$. The orthogonal projection of u onto F can be written as

$$P_F(u) = \sum_{i=1}^p [P_F(u)]_i^{\mathcal{F}} f_i. \quad (2.24)$$

For shorthand notation, in this proof we write $\lambda = [P_F(u)]^{\mathcal{F}}$. Since $u - P_F(u) \perp F$, it follows that

$$\omega(u - P_F(u), f_i) = 0, \quad 1 \leq i \leq p. \quad (2.25)$$

Inserting Eq. (2.24) into Eq. (2.25), and rearranging, we are lead to the system of equations

$$[\omega]_{\mathcal{F}\mathcal{F}} \lambda = q.$$

Since the f_i are linearly independent, then $[\omega]_{\mathcal{F}\mathcal{F}}$ is invertible, and this leads to Eq. (2.23). □

2.4.5 Application to supervised learning and least-squares

Supervised learning is a paradigm in machine learning where the available data consists of a pairing of inputs with known, correct, outputs. What is unknown is exactly how the mapping between those inputs and outputs works. The goal is to infer, from the available data, the general structure of the mapping in the hope that this will generalize to unseen situations. Formally we can state the problem as follows.

Definition 2.20: The supervised learning problem

Given an unknown function $f : X \rightarrow Y$ between spaces X and Y , find a good approximation of f using only a dataset of N samples:

$$\mathcal{D} = \{ (x_i, y_i) \}_{i=1}^N \quad \text{with} \quad y_i = f(x_i) \quad \text{for all } i = 1 \dots N.$$

The space X is also called the *feature space* and its elements are referred to as feature vectors. The space Y is also called the *label space* and its elements are referred to as labels. We denote the dimensions of these spaces by $d = \dim X$ and $q = \dim Y$.

The general approach to solving this problem consists of three main steps.

1. Choose a model function $F : X \times W \rightarrow Y$ parametrized by a parameter space W (W stands for weights, since that is how parameters are often referred to). The goal is to find an appropriate value $w^* \in W$ for the parameters such that $F(x; w^*)$ is as close as possible to $f(x)$ for every $x \in X$.
2. We need to choose a measure of quality for the model's output, so we need to choose a *loss function* $\ell : Y \times Y \rightarrow \mathbb{R}$, where $\ell(y_1, y_2)$ indicates 'how different' y_1 and y_2 are. With this choice, $\ell(F(x; w^*), f(x))$ will indicate how close $F(x; w^*)$ and $f(x)$ are. Of course, since we only have the samples $\mathcal{D}$, we can only evaluate the loss for the x in the dataset $\mathcal{D}$.
3. Based on the dataset and the loss function ℓ , choose $w \in W$ so that $F(\cdot; w)$ is the 'best' possible approximation of the target function f . One of the most common strategies is to minimize the loss on the dataset, i.e. to search for

$$w^* = \arg \min_{w \in W} \sum_{i=1}^N \ell(F(x_i; w), y_i). \quad (2.26)$$

This high-level approach leaves some open questions, however:

- How to choose the model F ?
- How to choose the loss function ℓ ?
- How to find minima of (2.26)?

None of these questions have canonical answers. In practice they are approached by trial-and-error and heuristics. Nonetheless, these choices will have a large impact on the final result even though they are in no way supported by the available data or some form of first principle. The collective set of assumptions we make to be able to proceed with the problem is called *inductive bias* in the machine learning field.

At this stage, it is legitimate to ask: what is the connection between supervised learning, and what we have been discussing until now in the course? In fact, there are plenty of connections. We develop the main ones by focusing on very relevant and common settings concerning the spaces X , Y , and the choices for F , ℓ . An extremely common setting is the one in which $X = \mathbb{R}^p$ and $Y = \mathbb{R}^q$. To simplify the discussion, we will focus on the case $p = q = 1$. Then one common choice for the model function F is to work with polynomials of degree k , which can be expressed in the form (here $W = \mathbb{R}^{k+1}$)

$$F : \mathbb{R} \times \mathbb{R}^{k+1} \rightarrow \mathbb{R}$$

$$(x, w) \mapsto F(x, w) := w_0 + w_1 x + w_2 x^2 + \dots + w_k x^k.$$

For the loss function, we select any norm in $Y = \mathbb{R}$, so the most obvious one is the squared absolute value $\ell(y_1, y_2) := |y_1 - y_2|^2$. In this case, Eq. (2.26) reads

$$\begin{aligned} w^* &= \arg \min_{w \in \mathbb{R}^{k+1}} \sum_{i=1}^N |F(x_i, w) - y_i|^2 \\ &= \arg \min_{w \in \mathbb{R}^{k+1}} \sum_{i=1}^N |w_0 + w_1 x_i + w_2 x_i^2 + \cdots + w_k x_i^k - y_i|^2 \end{aligned} \quad (2.27)$$

Introducing the matrix and vector

$$A = \begin{pmatrix} 1 & x_1 & \cdots & x_1^k \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_N & \cdots & x_N^k \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix},$$

and remembering the formula for the Euclidean norm in $\mathbb{R}^N$, we can express Eq. (2.27) as

$$\min_{w \in \mathbb{R}^{k+1}} \|Aw - y\|_{\mathbb{R}^N}^2. \quad (2.28)$$

This expression looks very similar to Eq. (2.22) on the minimizing property of orthogonal projections. We will now reformulate Eq. (2.28) in the exact same language that we used for Eq. (2.22). For this, we set $E = (\mathbb{R}^N, \|\cdot\|_2)$ and denote by $\mathcal{E} = \{e_1, \dots, e_N\}$ the canonical basis of $\mathbb{R}^N$. We then choose the subspace $F = \text{span}\{f_1, \dots, f_k\}$ with

$$f_j = \sum_{i=1}^N a_{i,j} e_i, \quad \Leftrightarrow \quad [f_j]^\mathcal{E} = \begin{pmatrix} a_{1,j} \\ \vdots \\ a_{N,j} \end{pmatrix} = \begin{pmatrix} x_1^{j-1} \\ \vdots \\ x_N^{j-1} \end{pmatrix}, \quad \forall 1 \leq j \leq k.$$

Consequently, Eq. (2.28) is equivalent to

$$\min_{f \in F} \|f - y\|_{\mathbb{R}^N}^2. \quad (2.29)$$

At this point, we will not develop further this setting, but there are several points to comment on.

- **Terminology:** Problem (2.29) involving the square of a norm as the loss function is called a least squares problem.
- **Dimensions:** There is an interplay between the dimension N (number of available data points) and the dimension of F . Observe that $\dim(F) \leq k + 1$. When $\dim(F) < N$, there exists at least one solution to this minimization problem which is $P_F(y)$ as we have proven in Eq. (2.22). When $\dim(F) \geq N$, the solution to the problem is to choose $f = y$ since then the loss function becomes 0. One may thus think that a clever thing to do is to choose k sufficiently large so that $\dim(F) \geq N$. However, this is in general not a good choice when we have noise in the data, that is, when we are not given $y_i = f(x_i)$ exactly, but $y_i = f(x_i)\delta_i$ where δ_i is a random noise term. It may also sometimes happen that $N \gg 1$ is so large, that we cannot even consider setting $\dim(F) = N$ due to computational limitations.
- **Generalizing the example:** To go beyond, you could think how to generalize this example to higher dimensions p and q . We can see that it would suffice to define a model function $F : \mathbb{R}^p \times \mathbb{R}^k \rightarrow \mathbb{R}^q$ which works in multi-dimensions, and to appropriately choose a norm as a loss function. Here, you could think how to write our polynomial Ansatz in this case. Also, how would you write the same problem with feed-forward neural networks?

2.5 Enhanced structure of dual spaces in inner-product spaces

2.5.1 Riesz Theorem

A remarkable feature of Euclidean spaces is that there is a one-to-one correspondence between elements of E and linear functionals of E^* thanks to the inner-product.

Theorem 2.21: Riesz Theorem

Let $(E, \langle \cdot, \cdot \rangle_E)$ be a Hilbert space in $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. For every $\ell \in E'$, there exists exactly one vector u such that

$$\ell(x) = \langle x, u \rangle_E, \quad \forall x \in E,$$

and

$$\|\ell\|_{\text{op}} = \|u\|_E.$$

We say that u is the Riesz representer of ℓ in $(E, \langle \cdot, \cdot \rangle_E)$.

Importantly, for complex inner-product spaces, u is always located in the antilinear coordinate of the inner-product.

Proof. If $\ell = 0$, then $u = 0$ works and is unique, and $\|\ell\|_{\text{op}} = 0 = \|u\|_E$. Assume henceforth that $\ell \neq 0$.

Let $K := \ker \ell = \{x \in E : \ell(x) = 0\}$. Then K is a subspace of E and, since $\ell \neq 0$, we have $K \neq E$. In a finite-dimensional inner-product space,

$$E = K \oplus K^\perp.$$

Pick $v \in K^\perp$ with $v \neq 0$ (possible because $K^\perp \neq \{0\}$). Since $v \notin K$, we have $\ell(v) \neq 0$.

Existence: For our fixed $v \in K^\perp$, define

$$u := \frac{\overline{\ell(v)}}{\|v\|_E^2} v \quad (\text{overline denotes complex conjugation; in } \mathbb{R} \text{ it is the identity}).$$

We claim that $\ell(x) = \langle x, u \rangle_E$ for all $x \in E$. Indeed, for any $x \in E$ write uniquely $x = y + \alpha v$ with $y \in K$ and $\alpha \in \mathbb{K}$. Then $\ell(y) = 0$ and hence

$$\ell(x) = \ell(y) + \alpha \ell(v) = \alpha \ell(v).$$

On the other hand, since $y \perp v$,

$$\langle x, u \rangle_E = \langle y + \alpha v, \frac{\overline{\ell(v)}}{\|v\|_E^2} v \rangle_E = \alpha \frac{\overline{\ell(v)}}{\|v\|_E^2} \langle v, v \rangle_E = \alpha \overline{\ell(v)}.$$

Thus it remains to check that $\alpha \overline{\ell(v)} = \alpha \ell(v)$. With our convention for the inner-product (linear in the first argument and conjugate-linear in the second), the above computation yields

$$\langle \alpha v, u \rangle_E = \alpha \langle v, u \rangle_E = \alpha \langle v, \frac{\overline{\ell(v)}}{\|v\|_E^2} v \rangle_E = \alpha \ell(v),$$

so indeed $\ell(x) = \langle x, u \rangle_E$ for all $x \in E$.

Uniqueness: Suppose $u_1, u_2 \in E$ satisfy $\ell(x) = \langle x, u_1 \rangle_E = \langle x, u_2 \rangle_E$ for all x . Then $\langle x, u_1 - u_2 \rangle_E = 0$ for all $x \in E$. Taking $x = u_1 - u_2$ gives $\|u_1 - u_2\|_E^2 = 0$, hence $u_1 = u_2$.

Norm identity: For any $x \in E$, by Cauchy–Schwarz,

$$|\ell(x)| = |\langle x, u \rangle_E| \leq \|x\|_E \|u\|_E.$$

Taking the supremum over $x \neq 0$ yields $\|\ell\|_{\text{op}} \leq \|u\|_E$. Conversely, choose $x = u$. Then

$$\|\ell\|_{\text{op}} \geq \frac{|\ell(u)|}{\|u\|_E} = \frac{|\langle u, u \rangle_E|}{\|u\|_E} = \|u\|_E.$$

Therefore $\|\ell\|_{\text{op}} = \|u\|_E$, and concludes the proof. □

2.5.2 Adjoint Operators

Bounded linear maps in inner-product spaces have a “companion” map, called the adjoint operator.

Theorem 2.22: Adjoint Operator

Let $(E, \langle \cdot, \cdot \rangle_E)$ and $(F, \langle \cdot, \cdot \rangle_F)$ be inner-product spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let $T \in \mathcal{B}(E, F)$ be a bounded linear map. There exists a unique bounded linear map $T^* : F \rightarrow E$, called the *adjoint operator* of T , which is such that

$$\langle T(x), y \rangle_F = \langle x, T^*(y) \rangle_E \quad \text{for all } x \in E, y \in F. \quad (2.30)$$

Furthermore, it holds that $\|T\|_{\text{op}} = \|T^*\|_{\text{op}}$.

Proof. Fix $y \in F$, and define the linear functional $f_y(x) := \langle T(x), y \rangle_F$ for all $x \in E$. Since T is a bounded linear operator, by the Cauchy-Schwarz inequality we have

$$|f_y(x)| = |\langle T(x), y \rangle_F| \leq \|T(x)\|_F \|y\|_F \leq \underbrace{\|T\|_{\text{op}} \|y\|_F}_{=C < +\infty} \|x\|_F,$$

which shows that $f_y : E \rightarrow \mathbb{K}$ is bounded. Therefore, $f_y \in E'$ (the continuous dual of E , see Eq. (1.14)). By the Riesz representation theorem, there exists a unique $z \in E$ such that $f_y(x) = \langle x, z \rangle_E$. Therefore, for each $y \in F$, there exists a unique $z \in E$ such that $\langle x, z \rangle_E = \langle T(x), y \rangle_F$. Now, define $T^* : F \rightarrow E$ the mapping $T^*(y) = z$ for all $y \in F$. Thus,

$$\langle x, T^*(y) \rangle_E = \langle T(x), y \rangle_F, \quad \forall x \in E, y \in F.$$

The uniqueness of T^* follows again from the Riesz representation theorem. Suppose that there exists $T_1^*, T_2^* : F \rightarrow E$ with the adjoint property (2.30). This implies that $\langle (T_1^* - T_2^*)(x), y \rangle_F = 0$ for all x, y which, in turn, implies that $T_1^* - T_2^* = 0$. $\square$

The intuition behind the adjoint operator is the following. Fix $x \in E$ and $y \in F$ and suppose the vectors are not zero, then $\langle T(x), y \rangle_F$ tells us how much $T(x) \in F$ points in the direction $y \in F$. If we think in terms of the angle, the inner product is giving us an angle between these vectors, which is characterized by $\cos(\alpha_{T(x), y}) = \frac{\langle T(x), y \rangle_F}{\|T(x)\|_F \|y\|_F}$ (see Definition 2.9). This angle measures vectors in the output space. The adjoint operator allows us to retrieve a vector in the input space such that the angle is preserved. It answers the question: “what mapping should I apply to y so that I obtain a vector $T^*(y) \in E$ whose angle $\beta_{x, T^*(y)}$ with x is the same as what we had in the output space, thus having $\cos(\alpha_{T(x), y}) = \cos(\beta_{x, T^*(y)})$?”

In finite dimensions, the notion of adjoint operators is pivotal to understand what we are really doing when working with the transpose A^T (if $\mathbb{K} = \mathbb{R}$) or conjugate-transpose $A^H = \overline{A}^T$ (if $\mathbb{K} = \mathbb{C}$) of a matrix $A \in \mathbb{K}^{p \times n}$. The following result explains that the (conjugate-)transpose of a matrix is the matrix representation of the dual operator of a given linear map.

Lemma 2.23: Matrix representation

Let $(E, \langle \cdot, \cdot \rangle_E)$ and $(F, \langle \cdot, \cdot \rangle_F)$ be two inner-product spaces of finite dimension p and $n \in \mathbb{N}$ respectively. Let $\mathcal{E}$ and $\mathcal{F}$ be bases of E and F , and let $T \in \mathcal{L}(E, F)$ be a linear map. Then,

$$[T^*]_{\mathcal{F}}^{\mathcal{E}} = G_E^{-1}([T]_{\mathcal{E}}^{\mathcal{F}})^H G_F \in \mathbb{K}^{p \times n},$$

where

$$G_E := (\langle e_i, e_j \rangle_E)_{1 \leq i, j \leq p} \in \mathbb{K}^{p \times p}, \quad G_F := (\langle f_i, f_j \rangle_F)_{1 \leq i, j \leq n} \in \mathbb{K}^{n \times n},$$

are the so-called Gram-matrices of the vectors of $\mathcal{E}$ and $\mathcal{F}$ respectively. In particular, when $\mathcal{E}$ and $\mathcal{F}$ are orthonormal bases, then

$$[T^*]_{\mathcal{F}}^{\mathcal{E}} = ([T]_{\mathcal{E}}^{\mathcal{F}})^H \in \mathbb{K}^{p \times n},$$

Proof. We will prove the claim only for the cases when $\mathcal{E}$ and $\mathcal{F}$ are orthonormal bases. In this case, by

Proposition 2.12, we have that

$$\begin{aligned} T(e_i) &= \sum_{j=1}^n \langle T(e_i), f_j \rangle f_j, \quad 1 \leq i \leq p \\ T^*(f_j) &= \sum_{i=1}^p \langle T^*(f_j), e_i \rangle e_i, \quad 1 \leq j \leq n. \end{aligned}$$

Therefore

$$[T]_{\mathcal{F}}^{\mathcal{F}} = ([T(e_1)]^{\mathcal{F}} | \dots | [T(e_p)]^{\mathcal{F}}) = (\langle T(e_i), f_j \rangle_F)_{\substack{1 \leq i \leq p, \\ 1 \leq j \leq n}},$$

and

$$\begin{aligned} [T^*]_{\mathcal{F}}^{\mathcal{E}} &= ([T^*(f_1)]^{\mathcal{E}} | \dots | [T^*(f_n)]^{\mathcal{E}}) \\ &= (\langle T^*(f_i), e_j \rangle_E)_{\substack{1 \leq i \leq n \\ 1 \leq j \leq p}} \\ &= (\overline{\langle e_j, T^*(f_i) \rangle_E})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq p}} \quad (\text{by conjugate symmetry}) \\ &= (\overline{\langle T(e_j), f_i \rangle_F})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq p}} \quad (\text{by Eq. (2.30)}) \\ &= ([T]_{\mathcal{E}}^{\mathcal{F}})^H. \end{aligned}$$

□

2.5.3 Operators defined by relations with their adjoint

In this section $(E, \langle \cdot, \cdot \rangle_E)$ is an inner-product $\mathbb{K}$ vector space, and we focus on $\mathcal{B}(E, E)$, the space of bounded linear maps from E to E . Many important operators from this space are characterized by simple algebraic relations with their adjoint. They form a structured family with several classes which we next describe in the following table. Many properties are related to the eigenvalues of the operator, which is a concept which we will see later on. We nevertheless list the properties here in anticipation of what comes next. In the table, $T \in \mathcal{B}(E, E)$ and in the case where E is a finite dimensional space, we denote by $\mathbf{T} = [T]_{\mathcal{E}}^{\mathcal{E}}$ its matrix representation in an orthonormal basis $\mathcal{E}$ of E .

Class	Operator condition	Geometric meaning	Matrix interpretation (finite dim.)
Normal	$T \circ T^* = T^* \circ T$	adjoint commutes with operator orthogonal eigenbasis (finite dim.)	$TT^H = T^HT$ normal ($\mathbb{K} = \mathbb{C}$) $TT^T = T^TT$ normal ($\mathbb{K} = \mathbb{R}$)
Unitary	$T^* \circ T = T \circ T^* = \text{Id}_E$	preserves norm and inner product isometry of the space	$TT^H = T^HT = \text{Id}$ unitary ($\mathbb{K} = \mathbb{C}$) $TT^T = T^TT = \text{Id}$ orthogonal ($\mathbb{K} = \mathbb{R}$)
Self-adjoint	$T = T^*$	real eigenvalues orthogonal eigenvectors	$T = T^H$ Hermitian ($\mathbb{K} = \mathbb{C}$) $T = T^T$ symmetric ($\mathbb{K} = \mathbb{R}$)
Skew-adjoint	$T = -T^*$	purely imaginary spectrum infinitesimal rotations	$T = -T^H$ skew-Herm. ($\mathbb{K} = \mathbb{C}$) $T = -T^T$ skew-sym. ($\mathbb{K} = \mathbb{R}$)
Positive	$\langle Tx, x \rangle_E \geq 0$	non-negative quadratic forms defines an energy functional	positive Hermitian matrix eigenvalues ≥ 0 $x^T Tx \geq 0, \forall x \in \mathbb{R}^b$
Projection	$T^2 = T$	projects onto a subspace	$T^2 = T$ eigenvalues in $\{0, 1\}$
Ortho. Proj.	$T^2 = T; T = T^*$	projects onto a subspace	$T^2 = T$ and $T = T^H$ eigenvalues in $\{0, 1\}$

Unitary operators: At this point of the lecture, we are ready to comment on the properties of unitary operators. We will discuss the other operator classes later on in the course. We have the following theorem.

Theorem 2.24

Let $T \in \mathcal{B}(E, E)$. Then the following properties are equivalent:

1. $TT^* = T^*T = \text{Id}_E$,
2. T is bijective and $T^{-1} = T^*$.
3. T is surjective and it preserves inner products:

$$\langle T(x), T(y) \rangle_E = \langle x, y \rangle_E, \quad \forall x, y \in E.$$

4. T is surjective and it preserves the norm:

$$\|T(x)\|_E = \|x\|_E, \quad \forall x \in E.$$

5. $[T]_{\mathcal{E}}^{\mathcal{E}}$ is unitary ($\mathbb{K} = \mathbb{C}$) or orthogonal ($\mathbb{K} = \mathbb{R}$) if E is finite dimensional and $\mathcal{E} = \{e_i\}_{i=1}^n$ is an orthonormal basis.

If one of these properties hold, we say that T is a unitary operator ($\mathbb{K} = \mathbb{R}$) or orthogonal operator ($\mathbb{K} = \mathbb{R}$).

Proof. • $1 \Rightarrow 2$: Suppose $x \neq 0 \in \ker(T)$. Then $T^*T(x) = 0_E$ which implies that $x = 0_E$. This is a contradiction which implies that $\ker(T) = \{0_E\}$. Therefore T is injective.

We now show that T is surjective. Let $y \in E$, and choose $x = T^*(y)$. We then have $T(x) = (TT^*)(y) = y$ which shows that T is surjective.

It follows that T is bijective, thus invertible, and from the identity $TT^* = T^*T = \text{Id}_E$ we deduce that $T^{-1} = T^*$.

- $2 \Rightarrow 1$: Trivial.
- $1 \Rightarrow 3$: Let $x, y \in E$. By property (2.30) of the adjoint,

$$\langle T(x), T(y) \rangle_E = \langle x, (T^*T)(y) \rangle_E = \langle x, y \rangle_E.$$

In addition, we already proved earlier that having $TT^* = \text{Id}_E$ implies that T is surjective.

- $3 \Rightarrow 1$: It holds that $\langle x, (T^*T - \text{Id})(y) \rangle_E = 0$ for all $x \in E$. Therefore $(T^*T - \text{Id})(y) = 0$ for all $y \in E$, which implies that $T^*T = \text{Id}_E$. This also implies that T is injective as shown earlier. In addition, we are assuming that T is also surjective. Therefore T is bijective, thus invertible. From the fact that $T^*T = \text{Id}$ we deduce that $T^{-1} = T^*$ and therefore $TT^* = \text{Id}_E$, which concludes the proof.
- $3 \Rightarrow 4$: Trivial.
- $4 \Rightarrow 3$: Let $x, y \in E$ and suppose that E is a real vector space. By the polarization identity (2.6),

$$\langle T(x), T(y) \rangle = \frac{1}{4}(\|T(x) + T(y)\|_E^2 - \|T(x) - T(y)\|_E^2) = \frac{1}{4}(\|x + y\|_E^2 - \|x - y\|_E^2) = \langle x, y \rangle_E$$

- $1 \Leftrightarrow 5$: To prove this equivalence, we essentially need to use that $[T^*]_{\mathcal{E}}^{\mathcal{E}} = ([T]_{\mathcal{E}}^{\mathcal{E}})^H$.

□

Lemma 2.25

Let E be a finite dimensional, and $T \in \mathcal{B}(E, E)$ a unitary/orthogonal operator. Then $|\det([T]_{\mathcal{E}}^{\mathcal{E}})| = 1$ for any basis $\mathcal{E}$ of E .

Proof. We use the shorthand notation $T = [T]_{\mathcal{E}}^{\mathcal{E}}$. Taking determinants in the identity $T^H T = \text{Id}_n$, we obtain

$$\det(T^H T) = \det(\text{Id}_n) = 1.$$

By multiplicativity of the determinant,

$$\det(T^H T) = \det(T^H) \det(T).$$

Since $\det(T^H) = \det(T)$, it follows that

$$\det(T)^2 = 1.$$

Therefore,

$$|\det(T)| = 1.$$

□

2.6 Exercises

Exercise 2.1. (Parallelogram law & polarization identity)

Let $\mathbb{E} = \mathbb{R}^n$. Prove that $\|\cdot\|_1 : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ given by

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$$

is a norm on $\mathbb{R}^n$, but is not induced by an inner product for $n \geq 2$.

Exercise 2.2. (Lagrange's Method for obtaining orthonormal basis)

In $\mathbb{E} = \mathbb{R}^2$ we define the inner product

$$\omega(x, y) = (y_1 \ y_2) \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 y_1 + 2x_1 y_2 + 2x_2 y_1 + 5x_2 y_2$$

- (i) Let $\|\cdot\|$ be the norm associated to ω , express $\|x\|^2$ in terms of the coordinates x_i of $x = (x_1, x_2)^T$.
- (ii) By completing the square on the expression obtained in the previous point, verify that $\|x\|^2 = (x_1 + 2x_2)^2 + x_2^2$.
- (iii) Let $y = (y_1, y_2)$ where $y_1 = x_1 + 2x_2$ and $y_2 = x_2$. Write out the matrix equation $y = Qx$ that relates y and x .
- (iv) From the point of view of change of basis, the matrix Q transforms the coordinates $(x_1, x_2)^T$ from the standard basis $\mathcal{E} = \{e_1, e_2\}$ to the coordinates $(y_1, y_2)^T$ of a new basis $\mathcal{U} = \{u_1, u_2\}$. Obtain the coordinates of u_1 and u_2 with respect to the standard basis $\mathcal{E}$.
- (v) Verify that $\mathcal{U} = \{u_1, u_2\}$ is an orthonormal basis with respect to $\omega(\cdot, \cdot)$, that is, $\omega(u_i, u_j) = \delta_{ij}$.

Exercise 2.3. (Lagrange's Method) Following a similar procedure to ex. 2.2, obtain an orthonormal basis on $\mathbb{E} = \mathbb{R}^3$ with respect to the following inner product:

$$\omega(x, y) = y^T \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} x = 2x_1 y_1 + 2x_2 y_2 + x_3 y_3 + x_1 y_2 + x_2 y_1.$$

Exercise 2.4. Let $E = \mathbb{R}^3$, and consider the vectors $x, y, z \in \mathbb{R}^3$ expressed in coordinates from the canonical basis $\mathcal{E} = \{e_1, e_2, e_3\}$:

$$\begin{aligned} [x]^\mathcal{E} &= (1, 0, 1)^\top \\ [y]^\mathcal{E} &= (0, 3, 1)^\top \\ [z]^\mathcal{E} &= (-5, 1, 3)^\top \end{aligned}$$

Show that these vectors form a basis of $\mathbb{R}^3$, and compute the orthonormal basis obtained by their Gram-Schmidt orthonormalization.

Exercise 2.5. (Inner product and matrix representation)

Let $\mathbb{E} = \mathbb{R}^2$, and let $\omega(\cdot, \cdot) : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ be defined as

$$\omega\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = x_1 y_1 + 2x_2 y_2 + x_1 y_2 + x_2 y_1.$$

Prove that ω is an inner product on $\mathbb{R}^2$, and obtain the matrix S such that ω can be written as $\omega(x, y) = y^T S x$.

Exercise 2.6. (Trace of the matrix of an inner product)

On $\mathbb{E} = \mathbb{K}^n$ we have an inner product of the form $\omega(x, y) = y^H S x$. Show that $\text{Tr}(S) > 0$.

Exercise 2.7. (Inner product and positive definiteness)

In $\mathbb{R}^2$ we define $\omega(x, y) := x S y$, where

$$S = \begin{pmatrix} 1 & p \\ p & q \end{pmatrix}.$$

Prove that $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is an inner product if and only if $\Delta = q - p^2 > 0$. For which values of (p, q) is the standard basis an orthonormal basis w.r.t. ω ? Can ω be an inner product if $p < 0$? What about when $q \leq 0$?

Exercise 2.8. (Complex field as matrices)

Given $\mathbb{E} = \mathbb{R}^{2 \times 2}$ the space of real two-by-two matrices, we consider the subspace $\mathbb{F} = \text{span}(\text{Id}, J)$, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

We define the following bilinear form:

$$\omega(A, B) = \frac{1}{2} \text{Tr}(B^T A),$$

where $\text{Tr}(A)$ is the trace of a matrix A .

(i) Prove that ω is an inner product on $\mathbb{F}$, and that $\{\text{Id}, J\}$ forms an orthonormal basis w.r.t. ω .

(ii) We define the following linear function $\Phi : \mathbb{C} \rightarrow \mathbb{F}$ given by:

$$\Phi(z) = \Re(z) \cdot \text{Id} + \Im(z) \cdot J = \begin{pmatrix} \Re(z) & -\Im(z) \\ \Im(z) & \Re(z) \end{pmatrix} \in \mathbb{F}, \quad \forall z \in \mathbb{C},$$

where $\Re(z)$ and $\Im(z)$ are the real and imaginary part of z . Prove that Φ is an isomorphism between $\mathbb{C}$ and $\mathbb{F}$, and that for any complex numbers z_1, z_2 we have

$$\Phi(z_1 \cdot z_2) = \Phi(z_2) \cdot \Phi(z_1).$$

(iii) Let $\|\cdot\| : \mathbb{F} \rightarrow \mathbb{R}^+$ be the norm associated to ω , namely $\|\cdot\| := \sqrt{\omega(\cdot, \cdot)}$. Given a complex number $z \in \mathbb{C}$, what does $\|\Phi(z)\|$ represent? (**Indication:** One may first consider this: what does the $\Phi(z)^T$ represent?)

Exercise 2.9. (Properties of inner product)

(i) In $\mathbb{R}^2$ we define the following bilinear operation

$$\omega : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}, \quad \omega(x, y) = \epsilon(u, x)\epsilon(v, y) + \epsilon(u, y)\epsilon(v, x).$$

where $u, v \in \mathbb{R}^2 \setminus \{0\}$ and ϵ is the standard Euclidean inner product (dot product). Does ω constitute an inner product for any choice of u and v ? If it does, calculate an orthonormal basis w.r.t. ω .

(ii) Conclude from the previous point that $\omega(x, y) = y^T S x$ where $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ cannot be an inner product. What are the vectors $x \in \mathbb{E}$ such that $\omega(x, x) = 0$?

Exercise 2.10. (Inner product defined from an automorphism)

Let $A : \mathbb{E} \rightarrow \mathbb{E}$ be an automorphism defined on a vector space $\mathbb{E} := \mathbb{R}^n$, we know that $\omega : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ given by

$$\omega(x, y) := x^T A^T A y$$

is an inner product for $\mathbb{E}$. Prove that ω can be written alternatively as $\omega(x, y) = \epsilon(Ax, Ay)$, with ϵ being the standard inner product, and that the matrix representation of ω w.r.t. the standard basis $\mathcal{E} = \{e_i\}_{i=1}^n$ is $A^T A$.

Exercise 2.11. (Inner product defined from a set of dual elements.)

Let $\{f^1, \dots, f^n\} \in \mathbb{E}^*$ be an independent set of vectors in the dual space of $\mathbb{E} = \mathbb{R}^n$, we define the following inner product

$$\omega(x, y) = \sum_{h=1}^n f^h(x) f^h(y),$$

Supposed that, given a basis $\mathcal{U} = \{u_1, \dots, u_n\}$ of $\mathbb{E}$ and the corresponding dual basis $\mathcal{U}^* = \{u^1, \dots, u^n\}$ in $\mathbb{E}^*$, the dual elements f^i have the representation

$$[f^i]_{\mathcal{U}} = [a_1^i, \dots, a_n^i].$$

Prove that there exists the following alternative definition for ω :

$$\omega(x, y) = \left([x]^{\mathcal{U}}\right)^T A^T A [y]^{\mathcal{U}}, \quad \text{where} \quad A = \begin{pmatrix} a_1^1 & \dots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^n & \dots & a_n^n \end{pmatrix},$$

and consequently the matrix representation of ω w.r.t. the basis $\mathcal{U}$ is $[\omega]_{\mathcal{U}\mathcal{U}} = A^T A$.

Exercise 2.12. (Orthogonal projection and inner product)

Let $\mathbb{E} = \mathbb{R}^4$, and let $\langle \cdot, \cdot \rangle$ be the standard dot product on $\mathbb{E}$. Calculate the orthogonal projection of the vector $x = (1, 2, 1, 2)^T$ onto the subspace $\mathbb{F} = \text{span}\{(1, 2, 3, 4)^T, (4, 3, 2, 1)^T\}$ with respect to $\langle \cdot, \cdot \rangle$.

Exercise 2.13. (Matrix representation of an orthogonal projection)

Let $(\mathbb{E}, \omega)$ be an n -dimensional real inner product space with a subspace $\mathbb{F} \subset \mathbb{E}$ with $\dim \mathbb{F} = q \leq n \in \mathbb{N}$, where $\mathcal{E} = \{e_1, \dots, e_n\}$ denotes a basis of $\mathbb{E}$ and $\mathcal{F} = \{f_1, \dots, f_q\}$ a basis of $\mathbb{F}$. Furthermore, let $P_{\mathbb{F}} : \mathbb{E} \rightarrow \mathbb{F}$ be the orthogonal projection onto the subspace $\mathbb{F}$.

- (i) Let $\iota_{\mathbb{F}} : \mathbb{F} \hookrightarrow \mathbb{E}$ be the *inclusion mapping*, i.e. $\iota_{\mathbb{F}}(y) = y \in \mathbb{E}$, for all $y \in \mathbb{F}$. Prove that $[\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}$ is the matrix formed by the column vectors $[f_i]_{\mathcal{E}}^{\mathcal{E}}$.
- (ii) Since $\text{im}(P_{\mathbb{F}}) \subset \mathbb{F}$, express $P_{\mathbb{F}}(x)$ as a linear combination of the basis vectors $\mathcal{F}$. Using matrix representation, can we express $[P_{\mathbb{F}}(x)]_{\mathcal{E}}^{\mathcal{E}}$ in terms of the matrix $[\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}$ and a column vector of unknown coefficients $\lambda \in \mathbb{R}^q$?
- (iii) We know from the lectures that for all $x \in \mathbb{E}$, the vector $x - P_{\mathbb{F}}(x)$ is orthogonal to the subspace $\mathbb{F}$. Give the equation that expresses this idea.
- (iv) Using the two previous points, prove that $\left([\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right)^T [\omega]_{\mathcal{E}\mathcal{E}} [\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}} \lambda = \left([\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right)^T [\omega]_{\mathcal{E}\mathcal{E}} [x]_{\mathcal{E}}^{\mathcal{E}}$, for all $x \in \mathbb{E}$.
- (v) Is $\left([\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right)^T [\omega]_{\mathcal{E}\mathcal{E}} [\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}$ an invertible matrix? Why? (**Indication:** Give an argument in terms of ranks.)
- (vi) From the two previous points, prove that $[P_{\mathbb{F}}]_{\mathcal{E}}^{\mathcal{E}} = [\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}} \left[\left([\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right)^T [\omega]_{\mathcal{E}\mathcal{E}} [\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right]^{-1} \left([\iota_{\mathbb{F}}]_{\mathcal{F}}^{\mathcal{E}}\right)^T [\omega]_{\mathcal{E}\mathcal{E}}$.
- (vii) Using the expression above, verify that $P_{\mathbb{F}}$ is indeed a projection mapping. How would the formula for $[P_{\mathbb{F}}]_{\mathcal{E}}^{\mathcal{E}}$ change if $\mathcal{F}$ was a orthonormal basis for $\omega|_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{R}$?

Exercise 2.14. (Orthogonal Projection Formula)

Let $(\mathbb{E}, \langle \cdot, \cdot \rangle)$ be an inner product space with field $\mathbb{K}$, and let $\mathbb{F}$ be a finite-dimensional subspace of $\mathbb{E}$. Suppose that $\mathcal{U} = \{u_1, u_2, \dots, u_k\}$ is an *orthogonal* basis for $\mathbb{F}$ (i.e., $\langle u_i, u_j \rangle = 0$ for $i \neq j$, but the vectors are not necessarily normalized).

Let $v \in \mathbb{E}$ be an arbitrary vector. By definition, the orthogonal projection of v onto $\mathbb{F}$, denoted $w = P_{\mathbb{F}}(v)$, is the unique vector in $\mathbb{F}$ such that the error vector $(v - w)$ is orthogonal to every vector in the subspace $\mathbb{F}$.

- (i) Because $w \in \mathbb{F}$, it can be expressed as a linear combination of the basis vectors in $\mathcal{U}$. Write down this general expression for w using unknown scalar coefficients $\lambda_1, \dots, \lambda_k \in \mathbb{K}$.
- (ii) To guarantee that the vector $v - w = v - P_{\mathbb{F}}(v)$ is orthogonal to the entire subspace $\mathbb{F}$, it is sufficient to require that $v - w$ is orthogonal to every basis vector $u_i \in \mathcal{U}$ (why?). Write down the exact equation (using the inner product) that expresses the orthogonality between $v - w$ and a generic basis vector $u_i \in \mathcal{U}$.
- (iii) Substitute your expression for w from (i) into your orthogonality equation from (ii). Using the linearity of the inner product and the fact that $\mathcal{U}$ is an orthogonal basis, isolate and solve for the unknown coefficient λ_i .
- (iv) Substitute your explicit formula for λ_i back into your answer from (i) to conclude that:

$$P_{\mathbb{F}}(v) = \sum_{i=1}^k \frac{\langle v, u_i \rangle}{\|u_i\|^2} u_i$$

Exercise 2.15. (Inner products)

In this exercise we ask whether the following $\omega(\cdot, \cdot) : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ are inner products:

- (i) $\mathbb{E} = \mathbb{R}_n[X]$, with $\omega(P(X), Q(X)) = P(1)Q(1) + P(2)Q(2) + \dots + P(n)Q(n)$
- (ii) $\mathbb{E} = \mathbb{R}_n[X]$, with $\omega(P(X), Q(X)) = P(0)Q(0) + P(1)Q(1) + \dots + P(n)Q(n)$
- (iii) $\mathbb{E} = C[0, 1]$ is the space of continuous functions defined on $[0, 1]$ and $\omega(f, g) = \int_0^{1/2} f(t)g(t) dt$.

Exercise 2.16. (Inner product and orthogonality) In $\mathbb{E} = \mathbb{R}^2$ we define

$$\omega(x, y) = y^T \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} x.$$

- (i) Prove that ω is an inner product.
- (ii) Calculate the associated norm of $u = (1, 1)^T$.
- (iii) Construct an orthogonal basis that contains u and give an orthonormal basis on $\mathbb{E}$.

Exercise 2.17 (Rotations). The goal of this exercise is to show that rotations in 2d and 3d are unitary operators.

1. Let $E = (\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$ and let $\mathcal{E}$ be the canonical basis. Given $\theta \in \mathbb{R}$, we consider the mapping $R_\theta : E \rightarrow E$ whose matrix representation in $\mathcal{E}$ is

$$[R_\theta]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

This operation is the rotation of angle θ in the 2d plane. Show that R_θ is a unitary operator.

2. Let $E = (\mathbb{R}^3, \langle \cdot, \cdot \rangle_2)$.

- (a) Interpret the action of the mappings given by the following matrices

$$[R_{e_1, \alpha}]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{pmatrix}, \quad [R_{e_2, \beta}]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{pmatrix},$$

$$[R_{e_3, \gamma}]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

where $\alpha, \beta, \gamma \in \mathbb{R}$.

- (b) Let the mapping $R : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $R = R_{e_3, \gamma} \circ R_{e_2, \beta} \circ R_{e_1, \alpha}$. Explain the action of this mapping, and justify why one can interpret R as a rotation in 3d.
- (c) Give the matrix representation $[R]_{\mathcal{E}}^{\mathcal{E}}$.
- (d) Show that R is a unitary operator.

Exercise 2.18 (Inner product of continuous functions (Programming Exercise)). In this exercise, we want to illustrate that computing inner-products in infinite-dimensional spaces in practice is not an easy task. We illustrate what happens for $E = \mathcal{C}([0, 1], \mathbb{R})$.

- (a) Prove that $\mathcal{C}([0, 1], \mathbb{R})$ is a vector space over the field $\mathbb{K} = \mathbb{R}$.
- (b) Prove that the operator

$$\omega : \mathcal{C}([0, 1], \mathbb{R}) \times \mathcal{C}([0, 1], \mathbb{R}) \rightarrow \mathbb{R}$$

$$(f, g) \mapsto \omega(f, g) := \int_0^1 f(x)g(x)dx$$

is an inner product on $\mathcal{C}([0, 1])$.

In general, calculating analytically the inner product of two functions is not possible or difficult, so the inner product is approximated. One way to do so is by using Riemann sums. Consider a partition of $[0, 1]$ into n equal intervals with grid points

$$x_i^n = \frac{i}{n}, \quad i = 0, \dots, n$$

and step size $\Delta x = \frac{1}{n}$. We define the following Riemann sums:

- **Left Riemann sum:** $R_-(f) := \sum_{i=0}^{n-1} f(x_i^n) \Delta x$.
- **Right Riemann sum:** $R_+(f) := \sum_{i=0}^{n-1} f(x_{i+1}^n) \Delta x$.
- **Central Riemann sum:** $R_0(f) := \sum_{i=0}^{n-1} f(x_i^n + \frac{\Delta x}{2}) \Delta x$.

We next ask you to answer the following questions by programming a code that evaluates the above Riemann sums:

- (c) Let $n = 10$. Visualize the left, right and central Riemann sum approximation of the inner product of $f(x) = x^2$ and $g(x) = \sin(2\pi x)$. Plot the piecewise constant approximations corresponding to the left, right, and midpoint Riemann sums of f , g and fg , i.e. one plot for left Riemann sum, one for the right Riemann sum, etc. Compare the approximation of each method with the analytical solution.

We now consider the space $L^2([0, 1], \mathbb{R})$, containing all functions that are square integrable, i.e. $f \in L^2([0, 1], \mathbb{R}) \Leftrightarrow \int_0^1 |f(x)|^2 dx < \infty$.

- (d) Show that $\mathcal{C}([0, 1], \mathbb{R}) \subset L^2([0, 1], \mathbb{R})$.

$L^2([0, 1])$ also contains non-continuous functions, for example piecewise continuous functions.

- (e) Let $f(x) = \lfloor 3x \rfloor$ and $g(x) = x^2 \sin(2\pi x)$. Note the discontinuities of f at $x = \frac{1}{3}, \frac{2}{3}$. Visualize the three Riemann approximations and notice the difference.
- (f) Now increase n . Plot the error (i.e. difference of approximation with analytical solution) of the methods for both the functions of (c) and (e) against n for $n = \{10, 11, \dots, 30\}$. Discuss what you observe.
- (g) What would happen to the error if there are k discontinuities on $[0, 1]$?

Programming hints

When implementing the numerical approximations, it can be useful to structure your code into small reusable functions.

For example, you may define a Julia function that numerically approximates the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx$$

using a Riemann sum with n bins: `riemann(f, g, n, method)` with

- `f` and `g` are functions,
- `n` is the number of bins,
- `method` specifies the Riemann method (e.g. 'left', 'right', or 'midpoint').

Inside the function you can construct the grid points $x_i = i/n$, evaluate the product $f(x)g(x)$ at the appropriate sample points, multiply by $\Delta x = 1/n$ and sum the contributions.

Exercise 2.19 (Projections and Symmetries in $\mathbb{R}^2$ (Programming exercise)). Let $E = (\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$ with the canonical basis, and let the vector $u \in \mathbb{R}^2$ such that $[u]^\mathcal{E} = (1, 2)^\top$.

- (a) Implement a routine which computes the orthogonal projection $P_{F \oplus F^\perp}(x)$ of a given $x \in \mathbb{R}^2$, and where $F := \text{span}\{u\}$. Apply P to several vectors and visualize:
- the original vectors,

- their projections,
- the direction u .

(b) Verify numerically that:

$$P_{F \oplus F^\perp}^2(x) = P_{F \oplus F^\perp}(x) \quad \text{and} \quad P_{F \oplus F^\perp}^T(x) = P_{F \oplus F^\perp}(x), \quad \forall x \in \mathbb{R}^2.$$

(c) Construct the symmetry (reflection) matrix with respect to F defined as

$$S_{F \oplus F^\perp} = 2P_{F \oplus F^\perp} - I.$$

Apply it to several vectors $x \in \mathbb{R}^2$ and visualize the reflection in a similar way as in Fig. 1.3.

(d) We now turn to projections and symmetries that are not necessarily orthogonal. Let $G = \text{span}\{g\}$ be a subspace spanned by a vector $g \in \mathbb{R}^2$. Show that for the symmetry $S_{F \oplus G}$ as defined in Definition 1.47, it holds that

$$S_{F \oplus G} = 2P_{F \oplus G} - \text{Id}.$$

holds.

(e) Visualize the symmetry for three different projections in $\mathbb{R}^2$.

(f) Check numerically and visually that $\|x - P_{F \oplus F^\perp}(x)\|_2$ is minimized when $G = P^\perp$.

Chapter 3

Sesquilinear and Bilinear Forms

3.1 Definition and Motivations

In this chapter, the main object of study are sesquilinear forms, a concept which we next define.

Definition 3.1: Sesquilinear/Bilinear form

Let E be a $\mathbb{K}$ -vector space. We say that a mapping $b : E \times E \rightarrow \mathbb{K}$ is sesquilinear (resp. bilinear) if it is:

- Linear in the first variable,
- Conjugate linear (resp. linear) in the second variable.

In other words, for all $x, y \in E$ and all $\lambda, \mu \in \mathbb{K}$,

$$\begin{aligned} a(\lambda x + \mu y, z) &= \lambda a(x, z) + \mu a(y, z) \\ a(x, \lambda y + \mu z) &= \bar{\lambda} a(x, y) + \bar{\mu} a(x, z). \end{aligned}$$

Definition 3.2

Let E be a $\mathbb{K}$ -vector space. We denote by

$$\begin{aligned} \text{Bil}(E) &:= \{b : E \times E \rightarrow \mathbb{K} : b \text{ is bilinear} \} \\ \text{Sesq}(E) &:= \{b : E \times E \rightarrow \mathbb{K} : b \text{ is sesquilinear} \} \end{aligned}$$

the sets of bilinear and sesquilinear maps.

Lemma 3.3

$\text{Bil}(E)$ and $\text{Sesq}(E)$ are linear subspaces of $\mathcal{F}(E \times E, \mathbb{K})$.

Proof. Left as an exercise. □

Example 3.4

- Let $E = \mathbb{R}^n$ and let $\mathcal{E}$ be a basis of $\mathcal{E}$. We define $b : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ as $b(x, y) = \sum_{i=1}^n [x]_i^{\mathcal{E}} [y]_i^{\mathcal{E}}$ for all $x, y \in \mathbb{R}^n$. This is a bilinear map and equal to the Euclidean inner-product in $\mathbb{R}^n$.

- Similarly, for $E = \mathbb{C}^n$, $b : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ such that $b(x, y) = \sum_{i=1}^n [x]_i^{\varepsilon} \overline{[y]_i^{\varepsilon}}$. b is sesquilinear.
- Let $E = \mathcal{C}([0, 1], \mathbb{R})$ and define $b_1(f, g) := \int_0^1 f(t)g(t)dt$ and $b_2(f, g) = \left(\int_0^1 f(t)dt\right)\left(\int_0^1 g(t)dt\right)$ for all $f, g \in \mathcal{C}([0, 1], \mathbb{R})$. Both b_1 and b_2 are bilinear.
- Let E be a $\mathbb{K}$ -vector space, and choose two linear functionals $\varphi, \psi \in E^*$, the algebraic dual of E . Define $b : E \times E \rightarrow \mathbb{K}$ such that $b(x, y) = \varphi(x)\psi(y)$ for all $x, y \in E$. Then b is bilinear. This bilinear form is called the tensor product between φ and ψ , and we usually denote it as $b = \varphi \otimes \psi$. This construction will play a role later on in the chapter. At this stage, we may simply observe that all our previous examples can actually be written in this general form for specific choices of linear functionals.

There are several reasons why sesquilinear/bilinear forms are interesting:

1. **Generalization of linear forms:** Bilinear forms $a : E \times E \rightarrow \mathbb{K}$ are a natural generalization of linear maps from the dual space $f \in E^*$. In fact, one can also consider multilinear forms where the input space is $\underbrace{V \times V \times \cdots \times V}_{n \text{ times}}$ and the mapping is linear in each of the variables. In this course, we will focus only in the case with two input variables.
2. **Sesquilinear forms are a relaxation of the concept of inner products:** As you may recall, we said in Definition 2.1 that an inner-product is a mapping from $E \times E \rightarrow \mathbb{K}$ which is linear in the first variable, conjugate symmetric in the second variable, and positive definite. Observe that the first two properties are equivalent to being a sesquilinear form when $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$. When $\mathbb{K} = \mathbb{R}$, things simplify a bit, and the first two properties are equivalent to being a bilinear map. In sesquilinear/bilinear maps, we remove the positive-definiteness, and only ask for notions of (conjugate) linearity in each variable.
3. **Vector product:** As we saw in Chapter 1, vector spaces admit scaling, and addition. We have also seen in Chapter 2 that we can define geometric notions such as distances between 2 vectors (through metrics and norms), and their angle (through inner-products). However, we have not yet talked about how to multiply two vectors. Intuitively, this should be a fundamental binary operation that distributes over addition, and it should return an object living in a larger vector space. The latter condition can be understood via example: if we multiply two sides of a rectangle to find its area, we understand that the product lives in a different space because it has different units. This space is also larger in some sense than either of the constituent spaces (i.e., 2D instead of 1D). It turns out that the key concept to define multiplication between two vectors are precisely sesquilinear/bilinear forms.
4. **Optimization:** Another context where bilinear forms arise is optimization. We will justify the connection with a formal computation. Suppose that we wish to find the local minima of a function $f : E = \mathbb{R}^n \rightarrow \mathbb{R}$ which we assume to be twice continuously differentiable. A vector $x^* \in \mathbb{R}^n$ is a local minimum if and only if there exists a ball of radius $\rho > 0$ centered at x^* such that

$$f(x) \geq f(x^*), \quad \forall x \in \mathbb{R}^n \text{ s.t. } \|x - x^*\|_2 \leq \rho. \quad (3.1)$$

For vectors x that are close to x^* , a Taylor expansion of f around x^* gives us

$$\begin{aligned} f(x) &= f(x^*) + \langle \nabla f(x^*), x - x^* \rangle_2 + \langle x - x^*, H_f(x^*)(x - x^*) \rangle_2 + \mathcal{O}(\|x - x^*\|_2^3) \\ &= f(x^*) + \nabla f(x^*)^\top (x - x^*) + (x - x^*)^\top H_f(x^*)(x - x^*) + \mathcal{O}(\|x - x^*\|_2^3) \end{aligned}$$

where

$$\nabla f(x^*) = \left(\frac{\partial f}{\partial x_1}(x^*), \dots, \frac{\partial f}{\partial x_n}(x^*) \right)^\top \in \mathbb{R}^n, \quad \text{and} \quad H_f(x^*) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x^*) \right)_{1 \leq i, j \leq n} \in \mathbb{R}^{n \times n}$$

are respectively the gradient of f and the Hessian of f at x^* . From the theory of optimization, we know that a necessary condition for x^* to be a minimizer is that $\nabla f(x^*) = 0$ (this is the multidimensional generalization of $f'(x^*)=0$ in 1D). Therefore,

$$f(x) = f(x^*) + (x - x^*)^\top H_f(x^*)(x - x^*) + \mathcal{O}(\|x - x^*\|_2^3).$$

When x is very close to x^* , we can approximate $f(x)$ by $f(x^*) + (x - x^*)^\top H_f(x^*)(x - x^*)$ in a very accurate way. Replacing this expression in Eq. (3.1), we deduce that x^* is a local minimum if

$$(x - x^*)^\top H_f(x^*)(x - x^*) \geq 0, \quad \text{for all } x \text{ very close to } x^*.$$

By remarking that any vector $y \in \mathbb{R}^n$ can be written as $y = r \frac{x - x^*}{\|x - x^*\|_2}$ with $r \geq 0$, we observe that the above condition simplifies to

$$y^\top H_f(x^*)y \geq 0, \quad \forall y \in \mathbb{R}^n.$$

Now, here comes the connection with bilinear forms. Defining

$$b(z_1, z_2) := z_1^\top H_f(x^*)z_2, \quad \forall z_1, z_2 \in \mathbb{R}^n,$$

we remark that b is a symmetric bilinear form, and that x^* is a minimum if b is positive-definite, namely, if it defines an inner-product in $\mathbb{R}^n$. A similar development would give us that x^* is a local maximum if b is negative-definite. We will see in a future chapter that we can discover if b has such a property through the so-called eigenvalues of $H_f(x^*)$. If b was neither positive nor negative-definite, then x^* is not an optimum. In this case, one could further extract information about the behavior of f around x^* thanks to eigenvalue theory again.

3.2 Matrix representation in the finite dimensional case

In Lemma 1.35, we have seen that linear maps on finite dimensional spaces have a matrix representation which is tied to the choice of bases in the input and output spaces. In a similar spirit, we are now going to prove that bilinear forms also have a matrix representation.

Lemma 3.5: Matrix representation in finite dimensional spaces

Let E be n -dimensional $\mathbb{K}$ -vector space, $\mathcal{E} = \{e_1, \dots, e_p\}$ be a basis, and $b \in \text{Bil}(E)$ and $s \in \text{Sesq}(E)$. It holds that

$$b(x, y) = ([x]^\mathcal{E})^\top [b]_{\mathcal{E}\mathcal{E}} [y]^\mathcal{E}, \quad s(x, y) = ([x]^\mathcal{E})^\top [s]_{\mathcal{E}\mathcal{E}} \overline{[y]^\mathcal{E}}, \quad \forall x, y \in E. \quad (3.2)$$

where

$$[b]_{\mathcal{E}\mathcal{E}} := (b(e_i, e_j))_{1 \leq i, j \leq n}, \quad [s]_{\mathcal{E}\mathcal{E}} := (s(e_i, e_j))_{1 \leq i, j \leq n}$$

are called the matrix representation of b and s .

Proof. Let $s \in \text{Sesq}(E)$, and $x, y \in E$. It holds that

$$s(x, y) = \sum_{1 \leq i, j \leq n} ([x]_\mathcal{E})_i \overline{([y]_\mathcal{E})_j} s(e_i, e_j) = ([x]^\mathcal{E})^\top [s]_{\mathcal{E}\mathcal{E}} \overline{[y]^\mathcal{E}}.$$

□

Lemma 3.6: Uniqueness of matrix representation

Let E be n -dimensional $\mathbb{K}$ -vector space and $\mathcal{E} = \{e_1, \dots, e_p\}$ be a basis. The mappings

$$\begin{aligned} \text{BMat} : \text{Bil}(E) &\rightarrow \mathbb{K}^{n \times n} \\ b &\mapsto [b]_{\mathcal{E}\mathcal{E}} \end{aligned}$$

and

$$\begin{aligned} \text{SMat} : \text{Sesq}(E) &\rightarrow \mathbb{K}^{n \times n} \\ s &\mapsto [s]_{\mathcal{E}\mathcal{E}} \end{aligned}$$

are isomorphisms.

Proof. We next prove that $\mathbf{BMat}$ is linear, surjective and injective.

Injectivity: Let $b, \tilde{b} \in \text{Bil}(e_i, e_j)$ such that $[b]_{\mathcal{E}\mathcal{E}} = [\tilde{b}]_{\mathcal{E}\mathcal{E}}$. For any $x, y \in E$, we have that

$$b(x, y) = ([x]_{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} [y]_{\mathcal{E}} = ([x]_{\mathcal{E}})^{\top} [\tilde{b}]_{\mathcal{E}\mathcal{E}} [y]_{\mathcal{E}} = \tilde{b}(x, y).$$

Therefore $b = \tilde{b}$ and $\mathbf{BMat}$ is injective.

Surjectivity: Let $A \in \mathbb{K}^{n \times n}$, and define $b_A : E \times E \rightarrow \mathbb{K}$ such that $b(x, y) := ([x]_{\mathcal{E}})^{\top} A [y]_{\mathcal{E}}$. We can readily verify that $b \in \text{Bil}(E)$, and this shows that $\mathbf{BMat}$ is surjective.

The proof for the sesquilinear case follows the same lines, with due care for complex conjugation.

Linearity: Trivial. □

Lemma 3.7: Matrix representation under change of basis

Let E be n -dimensional $\mathbb{K}$ -vector space, and $b \in \text{Bil}(E)$ and $s \in \text{Sesq}(E)$. Then the matrix representation of b and s for two different bases $\mathcal{E}$ and $\mathcal{U}$ of E are connected as

$$[b]_{\mathcal{U}\mathcal{U}} = ([\text{Id}]_{\mathcal{U}}^{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} [\text{Id}]_{\mathcal{U}}^{\mathcal{E}}, \quad \text{and} \quad [s]_{\mathcal{U}\mathcal{U}} = ([\text{Id}]_{\mathcal{U}}^{\mathcal{E}})^{\top} [s]_{\mathcal{E}\mathcal{E}} \overline{[\text{Id}]_{\mathcal{U}}^{\mathcal{E}}}.$$

Proof. Let $x, y \in E$. By the matrix representation of $b(x, y)$ (see Eq. (3.2)),

$$\begin{aligned} b(x, y) &= ([x]_{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} [y]_{\mathcal{E}} \\ &= ([x]_{\mathcal{U}})^{\top} ([\text{Id}]_{\mathcal{U}}^{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} [\text{Id}]_{\mathcal{U}}^{\mathcal{E}} [y]_{\mathcal{U}} \quad (\text{since } [x]_{\mathcal{E}} = [\text{Id}]_{\mathcal{U}}^{\mathcal{E}} [x]_{\mathcal{U}}). \end{aligned}$$

In addition, since Eq. (3.2) also applies for the basis $\mathcal{U}$, we also have

$$b(x, y) = ([x]_{\mathcal{U}})^{\top} [b]_{\mathcal{U}\mathcal{U}} [y]_{\mathcal{U}}.$$

Therefore, $[b]_{\mathcal{U}\mathcal{U}} = ([\text{Id}]_{\mathcal{U}}^{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} [\text{Id}]_{\mathcal{U}}^{\mathcal{E}}$. We proceed similarly to prove the claim for $s \in \text{Sesq}(E)$. □

3.3 Tensor Product

A major motivation to study sesquilinear forms is connected to our desire to define a notion of multiplication between vectors. We begin by discussing the axioms that such a construction should satisfy.

Definition 3.8: Tensor product: axioms

The tensor product operation $\otimes$ maps a pair of vectors $(x, y) \in E \times E$ to an object called the tensor product $x \otimes y$. We can add tensors and scale them. The product satisfies the following properties:

1. **Sesquilinearity.** For all $x, y, z \in E$ and $\lambda, \mu \in \mathbb{K}$

$$\begin{aligned} (\lambda x + \mu y) \otimes z &= \lambda x \otimes z + \mu y \otimes z, \\ x \otimes (\lambda y + \mu z) &= \bar{\lambda} x \otimes y + \bar{\mu} x \otimes z, \end{aligned}$$

2. **Interaction with the zero vector.** For all $x, y \in E$,

$$x \otimes 0 = 0 \otimes 0 = 0 \otimes y.$$

3. **Faithfulness.** For $x, y \in E$, if

$$x \otimes y = 0 \otimes 0,$$

then either $x = 0$ or $y = 0$.

Observe that we have not yet defined in which space the tensor product $x \otimes y$ belongs to. This is because we are going to refine our definition in a moment. Let us consider some familiar binary operations to see whether they satisfy the desired axioms.

Example 3.9

- **Inner product:** It does not generally satisfy the axioms of the tensor product. Consider the case where $E = \mathbb{R}^2$. The inner product for vectors x and $y \in E$ is defined by

$$\langle x, y \rangle : (x, y) \mapsto x^T y.$$

However, it is not faithful, since one can easily find vectors x and y that are not equal to 0, but where $\langle x, y \rangle = 0$. For instance,

$$x = (1, 1)^T, \quad y = (1, -1)^T.$$

Example 3.10: Outer product

We denote the outer product between vectors x and y in $E = \mathbb{R}^n$ by $x \otimes y$. It is defined by

$$\begin{aligned} \otimes : \mathbb{R}^n \times \mathbb{R}^n &\rightarrow \mathbb{R}^{n \times n} \\ (x, y) &\mapsto xy^T = (x_i y_j)_{i,j=1}^n. \end{aligned}$$

and in $E = \mathbb{C}^n$ as

$$\begin{aligned} \otimes : \mathbb{C}^n \times \mathbb{C}^n &\rightarrow \mathbb{C}^{n \times n} \\ (x, y) &\mapsto xy^H = (x_i \bar{y}_j)_{i,j=1}^n. \end{aligned}$$

Let us focus on $E = \mathbb{R}^n$. We verify that the outer product satisfies the tensor product axioms (we would do similarly for the complex case).

1. **Bilinearity.** Since the outer product is a particular case of matrix multiplication (between an $n \times 1$ matrix and a $1 \times n$ matrix) and matrix multiplication is additive, the outer product is also additive. In addition, given $\alpha \in \mathbb{R}$,

$$x(\alpha y)^T = (x_i \alpha y_j)_{i,j=1}^n = (\alpha x_i y_j)_{i,j=1}^n,$$

$$(\alpha x)y^T = (\alpha x_i y_j)_{i,j=1}^n,$$

$$\alpha(xy^T) = (\alpha x_i y_j)_{i,j=1}^n.$$

2. **Interaction with zero.** If either x or y equals 0, then either $x_i = 0$ for all i or $y_j = 0$ for all j , and therefore

$$xy^T = (x_i y_j)_{i,j=1}^n = 0.$$

3. **Faithfulness.** If $xy^T = 0$, then $x_i y_j = 0$ for all i, j . Fix $i = 1$. Either $x_1 = 0$ or $x_1 \neq 0$.

If $x_1 \neq 0$, then $y_j = 0$ for all j , so $y = 0$. Otherwise proceed to $i = 2$, and so on. Eventually, either $y = 0$ or $x_i = 0$ for all i , which implies $x = 0$.

Thus the outer product satisfies all four axioms of the tensor product.

Having defined the tensor product operator as a binary operator following the axioms in Definition 3.8, we now define the tensor product space involving all possible linear combinations of tensor products.

Definition 3.11: Tensor product space

Let E be a $\mathbb{K}$ -vector space. We define its tensor product space as

$$E \otimes E := \left\{ \sum_{i=1}^r \alpha_i x_i \otimes y_i : r \in \mathbb{N}, \alpha_i \in \mathbb{K}, x_i, y_i \in E \text{ for } 1 \leq i \leq r \right\}.$$

Lemma 3.12

$E \otimes E$ is a $\mathbb{K}$ -vector space with neutral element $0_{E \otimes E} = 0_E \otimes 0_E$.

Proof. It holds that $0_{E \otimes E}$ is indeed the neutral element, and that for all $\lambda_1, \lambda_2 \in \mathbb{K}$ and $T_1, T_2 \in E \otimes E$, $\lambda_1 T_1 + \lambda_2 T_2 \in E \otimes E$. $\square$

The concept of bilinear form allows us to build an alternative definition of the tensor product space which is the following.

Definition 3.13

Let E be a vector space. The tensor product space $E \otimes E$ is the algebraic dual of the space of bilinear forms $\text{Bil}(E)$.

What this means is that, given $x, y \in E$, we identify the tensor product $x \otimes y \in E \otimes E$ as a linear functional acting on $\text{Sesq}(E)$,

$$\begin{aligned} x \otimes y : \text{Sesq}(E) &\rightarrow \mathbb{K} \\ b &\mapsto b(x, y). \end{aligned}$$

This can easily be extended to all tensors by linearity as follows:

$$\sum_{i=1}^n \alpha_i x_i \otimes y_i = \sum_{i=1}^n \alpha_i b(x_i, y_i).$$

We next show that this construction satisfies the axioms of Definition 3.8.

1. **Sesquilinearity:** For all $x, y, z \in E$, and all $b \in \text{Bil}(E)$,

$$((x + y) \otimes z)(b) = b(x + y, z) = b(x, z) + b(y, z) = (x \otimes z)(b) + (y \otimes z)(b).$$

For all $\alpha \in \mathbb{K}$, all $x, y \in E$ and all $b \in \text{Bil}(E)$,

$$((\alpha x) \otimes y)(b) = b(\alpha x, y) = \alpha b(x, y) = b(x, \bar{\alpha} y) = (x \otimes (\bar{\alpha} y))(b).$$

2. **Interaction with zero:** Easy to check

3. **Faithfulness:** If $(x \otimes y)(b) = 0$ for every $b \in \text{Sesq}(E)$, then either $x = 0$ or $y = 0$. We show this rigorously by contradiction. Suppose that $(x \otimes y)(b) = 0$ for every $b \in \text{Bil}(E)$, and that x, y are both nonzero. We can build a bilinear form b such that $b(x, y) = \|x\|^2 \|y\|^2$ (this can be done by choosing $[b]_{\mathcal{E}\mathcal{E}} = \bar{x}y$). We have thus build b such that $(x \otimes y)(b) > 0$, which is in contradiction with the assumption that $(x \otimes y)(b) = 0$ for every $b \in \text{Bil}(E)$. We thus conclude that either $x = 0$ or $y = 0$.

We may observe that if E is a finite n -dimensional space with a basis $\mathcal{E}$, then the space of bilinear forms is isomorphic to the space of matrices $\mathbb{K}^{n \times n}$. We can then identify $x \otimes y$ with the outer product matrix $[x]^{\mathcal{E}} ([y]^{\mathcal{E}})^{\top}$ so that

$$(x \otimes y)(b) = [x]^{\mathcal{E}} ([y]^{\mathcal{E}})^{\top} \text{BMat}(b) = [x]^{\mathcal{E}} ([y]^{\mathcal{E}})^{\top} [b]_{\mathcal{E}\mathcal{E}} \in \mathbb{K}.$$

3.4 Exercises

Exercise 3.1. (Bilinear form) Let $\mathbb{E}$ be a $\mathbb{K}$ -vector space, prove the following:

- (i) For all bilinear forms $b, \tilde{b} : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ and for all $\lambda, \mu \in \mathbb{K}$,

$$\begin{aligned} \lambda b + \mu \tilde{b} : \mathbb{E} \times \mathbb{E} &\rightarrow \mathbb{K} \\ (x, y) &\mapsto \lambda b(x, y) + \mu \tilde{b}(x, y) \end{aligned}$$

is a bilinear form. Thus the set $\text{Bil}(\mathbb{E})$ of all bilinear forms on $\mathbb{E}$ is a vector space with these operations of vector sum and scalar multiplication.

- (ii) For linear forms $\phi_1, \dots, \phi_k, \psi_1, \dots, \psi_k \in \mathbb{E}^*$, the function $b := \phi_1 \otimes \psi_1 + \dots + \phi_k \otimes \psi_k$ is a bilinear form. Where $\phi_i \otimes \psi_i : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ is defined by

$$\begin{aligned} \phi_i \otimes \psi_i : \mathbb{E} \times \mathbb{E} &\rightarrow \mathbb{K} \\ (x, y) &\mapsto \phi_i \otimes \psi_i(x, y) = \phi_i(x) \psi_i(y) \end{aligned}$$

Furthermore, for $\mathcal{U}$ basis on $\mathbb{E}$, we have

$$[b]_{\mathcal{U}\mathcal{U}} = \sum_{i=1}^k ([\phi_i]_{\mathcal{U}})^T [\psi_i]_{\mathcal{U}} = ([\phi_1]_{\mathcal{U}})^T [\psi_1]_{\mathcal{U}} + \dots + ([\phi_k]_{\mathcal{U}})^T [\psi_k]_{\mathcal{U}}.$$

Exercise 3.2. (Bilinear form on a subspace)

Let $\mathbb{E}$ be a vector space with $\dim \mathbb{E} = n < \infty$ and a bilinear form $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$. Let $\mathbb{F}$ be a subspace of $\mathbb{E}$ with $\dim \mathbb{F} = k \leq n$. If $\mathcal{U} := \{u_1, \dots, u_k, u_{k+1}, \dots, u_n\}$ is a basis on $\mathbb{E}$ where the first k vectors $\tilde{\mathcal{U}} = \{u_1, \dots, u_k\}$ is a basis of $\mathbb{F}$, show that

$$[b]_{\mathcal{U}\mathcal{U}} = \begin{pmatrix} [b|_{\mathbb{F}}]_{\tilde{\mathcal{U}}\tilde{\mathcal{U}}} & \star \\ \star & \star \end{pmatrix},$$

i.e. show that the upper left $k \times k$ submatrix of $[b]_{\mathcal{U}\mathcal{U}}$ is the matrix representation of $b|_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{K}$ (b restricted to $\mathbb{F}$) with respect to the basis $\tilde{\mathcal{U}}$.

Exercise 3.3. (Bilinear form on polynomial spaces)

Let $\mathbb{E} = \mathbb{R}_2[X]$ be the space of real polynomials with degree less or equal to 2, and let $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ be defined by

$$b(P(X), Q(X)) = P(1)Q'(2) + P(4) \int_{-1}^1 Q(X) dX.$$

Prove that b is a bilinear form and give $[b]_{\mathcal{E}\mathcal{E}}$, where $\mathcal{E} = \{1, X, X^2\}$ is the standard basis on $\mathbb{E}$.

Exercise 3.4. (Inner product defined by matrices) Let $B \in \mathbb{C}^{p \times q}$ be a matrix with $p \geq q$, and let $\omega : \mathbb{C}^q \times \mathbb{C}^q \rightarrow \mathbb{C}$ be the following mapping:

$$\omega(x, y) = (By)^H Bx.$$

Give necessary and sufficient condition on the matrix B such that ω is an inner product.

Exercise 3.5. (Bilinear form from trace)

Let $\mathbb{E} = \mathbb{R}^2$, we define the following function $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ with

$$b\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = \text{Tr} \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}.$$

Is b a bilinear form? If so, give $[b]_{\mathcal{E}\mathcal{E}}$, where $\mathcal{E} = \{e_1, e_2\}$ is the standard basis on $\mathbb{R}^2$.

Exercise 3.6. (Inner product of trigonometric functions)

We define the following vector space:

$$\mathbb{E} = \{f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = \alpha \cos x + b \sin x, \alpha, b \in \mathbb{R}\}.$$

We assume proven that $\mathcal{B} = \{\cos x, \sin x\}$ forms a basis on this space.

(i) On $\mathbb{E}$ we define the following bilinear form:

$$\omega(f, g) = \int_0^\pi f(x)g(x) dx.$$

Give the representation $[\omega]_{\mathcal{B}\mathcal{B}}$. Can we conclude from this matrix representation that ω is an inner product? Is the basis $\mathcal{B}$ an orthogonal basis w.r.t. ω ? Is it orthonormal? Give an orthogonal/orthonormal basis if $\mathcal{B}$ is not.

(ii) Given $y \in \mathbb{R}$, we define the *phase shift operator* $T_y : \mathbb{E} \rightarrow \mathbb{E}$ as in ex. 1.41.

(iii) Show that the mapping $b : \mathbb{E} \rightarrow \mathbb{E}$ where $b(f, g) := \omega(T_y f, T_y g)$ is also an inner product.

(**Indication:** If $T_y : \mathbb{E} \rightarrow \mathbb{E}$ is an automorphism, then we can apply the result from ex. 2.10.)

(iv) From the previous point, prove that ω is *invariant* under the actions of T_y , that is to say:

$$\omega(f, g) = \omega(T_y f, T_y g), \quad \forall f, g \in \mathbb{E}.$$

(**Indication:** instead of evaluating these messy integrals, make use of the matrix representations of $\omega(\cdot, \cdot)$ and $b(\cdot, \cdot) = \omega(T_y \cdot, T_y \cdot)$).

Exercise 3.7. (Bilinear form on the space of hyperbolic functions)

Let $\mathbb{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} : f''(x) = f(x), \forall x \in \mathbb{R}\}$ be the vector space of functions f whose second derivative f'' is equal to itself. Suppose that we have proven that the functions $\mathcal{E} = \{\cosh x, \sinh x\}$ form a basis on $\mathbb{H}$ (recall that $\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$, and we have the *hyperbolic identity* $\cosh^2 x - \sinh^2 x = 1$).

(i) We define a mapping $b : \mathbb{H} \times \mathbb{H} \rightarrow \mathbb{R}$ by

$$b(f, g) = f(0)g(0) - f'(0)g'(0).$$

Verify that B is a symmetric bilinear form.

(ii) Give the matrix representation $[b]_{\mathcal{E}\mathcal{E}}$, based on this matrix, is B an inner product on $\mathbb{H}$? Explain why or why not.

(iii) Let $\mathcal{U} = \{e^x, e^{-x}\}$, is $\mathcal{U}$ a basis on $\mathbb{H}$? If so, compute the matrix $[b]_{\mathcal{U}\mathcal{U}}$.

(iv) For $y \in \mathbb{R}$, let $T_y : \mathbb{H} \rightarrow \mathbb{H}$ be the phase shift operator $T_y(f(x)) = f(x + y)$. Give $[T_y]_{\mathcal{U}}^{\mathcal{U}}$, and prove that b is invariant under the action of T_y . That is, show that for any $f, g \in \mathbb{H}$:

$$b(T_y f, T_y g) = b(f, g).$$

Exercise 3.8. (Bilinear form on polynomial spaces)

Let $\mathbb{E} = \mathbb{R}_2[X]$ be the space of polynomials with degree less or equal to 2, and $\mathcal{E} = \{1, X, X^2\}$ be the standard basis on $\mathbb{E}$.

(i) Let $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ be the bilinear form defined by $b(P, Q) = \int_0^1 P'(X)Q(X) dX$, where $P(X), Q(X) \in \mathbb{E}$. Compute $[b]_{\mathcal{E}\mathcal{E}}$.

(ii) Is b symmetric? If not, give a specific pair of $P, Q \in \mathbb{E}$ such that $b(P, Q) \neq b(Q, P)$.

(iii) Is b *left-degenerate* (i.e. does there exists a non-zero $P \in \mathbb{E}$ such that $b(P, Q) = 0$ for all $Q \in \mathbb{E}$)?

(iv) Is b *right-degenerate* (i.e. does there exists a non-zero $Q \in \mathbb{E}$ such that $b(P, Q) = 0$ for all $P \in \mathbb{E}$)? Justify the existence of such polynomials using an argument of the rank of $[b]_{\mathcal{E}\mathcal{E}}$, and find a specific polynomial $\tilde{Q} \in \mathbb{E}$ that satisfies $b(P, Q) = 0, \forall P \in \mathbb{E}$.

(v) (**Optional**) Check using integration by parts the $\tilde{Q}$ that we have obtained in the previous point satisfies $\int_0^1 P'(X)\tilde{Q}(X) dX = 0$, for all $P(X) \in \mathbb{E}$.

Exercise 3.9. (Bilinear form on rational functions)

Let $\mathbb{E}$ be the vector space of all real rational functions f of the form

$$f(x) = \frac{a + bx}{x^2 - 1}, \quad a, b \in \mathbb{R},$$

equipped with the usual function addition and scalar multiplication.

- (i) Prove that the mapping $\Phi : \mathbb{E} \rightarrow \mathbb{R}^2$ with

$$\Phi\left(\frac{a + bx}{x^2 - 1}\right) = \begin{pmatrix} a \\ b \end{pmatrix}$$

is an isomorphism. Conclude that $\mathbb{E}$ is a two-dimensional $\mathbb{R}$ -vector space and $\left\{\frac{1}{x^2-1}, \frac{x}{x^2-1}\right\}$ is a basis.

- (ii) We define the mapping $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ by

$$b(f, g) = \left[\lim_{x \rightarrow 1} (x - 1) \cdot f(x) \right] \cdot \left[\lim_{x \rightarrow -1} (x + 1) \cdot g(x) \right]$$

Show that b is a bilinear form.

(Indication: Instead of tackling the linearity head-on, can we write $b = \phi \otimes \psi$ for some $\phi, \psi \in \mathbb{E}^*$, as in ex. 3.1?)

- (iii) Prove that functions $u(x) = 1/(x-1)$, $v(x) = 1/(x+1)$ belong to $\mathbb{E}$ and that $\mathcal{U} = \{u, v\}$ form a basis. Give the matrix representation $[b]_{\mathcal{U}\mathcal{U}}$.

- (iv) Find $h \in \mathbb{E}$ such that $b(h, h) \neq 0$.

Exercise 3.10. (Decomposition of bilinear form)

We call a bilinear $\alpha : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ *skew-symmetric* if for all $x, y \in \mathbb{E}$, $\alpha(x, y) = -\alpha(y, x)$.

- (i) Let $\mathcal{S}(\mathbb{E}) \subset \text{Bil}(\mathbb{E})$ be the set of all symmetric bilinear forms, and $\mathcal{A}(\mathbb{E}) \subset \text{Bil}(\mathbb{E})$ be the set of all skew-symmetric bilinear forms. Prove that $\mathcal{S}(\mathbb{E}), \mathcal{A}(\mathbb{E})$ are subspaces of $\text{Bil}(\mathbb{E})$.
- (ii) Show that all bilinear form can be written as the sum of a symmetric bilinear form and a skew-symmetric bilinear form. In other words, for all bilinear form $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$, there exists a symmetric bilinear form $\sigma : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ and a skew-symmetric bilinear form $\alpha : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ such that $b(x, y) = \sigma(x, y) + \alpha(x, y)$, $\forall x, y \in \mathbb{E}$.

(Indication: if $\tilde{b}$ is a symmetric bilinear form, then $\tilde{b}(x, y) = \frac{1}{2} [\tilde{b}(x, y) + \tilde{b}(y, x)]$.)

- (iii) Show that if a bilinear form is simultaneously symmetric and skew-symmetric, then it has to be the zero bilinear form. Conclude that the vector space of bilinear forms $\text{Bil}(\mathbb{E})$ admits the direct-sum decomposition $\text{Bil}(\mathbb{E}) = \mathcal{S}(\mathbb{E}) \oplus \mathcal{A}(\mathbb{E})$, and thus the decomposition that we have obtained in the previous point is unique.

Exercise 3.11. (Self-annihilating bilinear forms) Show that a self-annihilating bilinear form (i.e. a bilinear form $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ where $b(x, x) = 0$, $\forall x \in \mathbb{E}$) is necessarily skew-symmetric.

Exercise 3.12. (Bilinear form on polynomial space)

Let $\mathbb{E} = \mathbb{R}_3[X]$. We define the following mappings:

$$b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R},$$

$$(p, q) \mapsto \int_0^1 p(X)q(X) dX,$$

$$B : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R},$$

$$(p, q) \mapsto \int_0^1 p(X)q'(X) dX.$$

- (i) Show that b, B are bilinear forms.
- (ii) Give $[b]_{\mathcal{E}\mathcal{E}}$ and $[B]_{\mathcal{E}\mathcal{E}}$.

(Indication: You can use the *derivative mapping's* matrix representation $[D]_{\mathcal{E}}^{\mathcal{E}}$ in ex. 1.40.)

- (iii) We now define the bilinear form $\beta(p, q) = p(1)q(1) - p(0)q(0)$, give the matrix representation $[\beta]_{\mathcal{E}\mathcal{E}}$.

(iv) Using the previous points, prove the *integration by part formula*:

$$\int_0^1 p(X)q'(X) dX = p(1)q(1) - p(0)q(0) - \int_0^1 q(X)p'(X) dX, \quad \forall p, q \in \mathbb{E}.$$

Exercise 3.13. (Bilinear form on polynomial space)

Let $\mathbb{E} = \mathbb{R}_1[X]$, we define the following mappings:

$$b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}, \quad \beta : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R},$$

$$(p, q) \mapsto \left(\int_0^1 p(X) dX \right) \left(\int_0^1 q(X) dX \right), \quad (p, q) \mapsto \int_0^1 p(X) \left(\int_0^X q(Y) dY \right) dX.$$

- (i) Show that b, β are bilinear forms.
- (ii) Give $[b]_{\mathcal{E}\mathcal{E}}$ and $[\beta]_{\mathcal{E}\mathcal{E}}$, where $\mathcal{E} = \{1, X\}$ is the standard basis of $\mathbb{E}$.
- (iii) Using the previous point, prove that for all $p, q \in \mathbb{E}$:

$$\left(\int_0^1 p(X) dX \right) \left(\int_0^1 q(X) dX \right) = \int_0^1 p(X) \left(\int_0^X q(Y) dY \right) dX + \int_0^1 q(X) \left(\int_0^X p(Y) dY \right) dX.$$

Exercise 3.14. (Symmetric bilinear form)

On a vector space $\mathbb{E}$ let $\sigma(\cdot, \cdot)$ be a symmetric bilinear form.

- (i) Show that the *polar identity* holds:

$$\sigma(x, y) = \frac{1}{2} [\sigma(x + y, x + y) - \sigma(x, x) - \sigma(y, y)] = \frac{1}{4} [\sigma(x + y, x + y) - \sigma(x - y, x - y)].$$

(**Note:** this is the same identity that we have talked about in Lemma 2.7 regarding inner products.)

- (ii) Using the previous point, show that if $\sigma \neq 0$ (i.e. there exists $x, y \in \mathbb{E}$ such that $\sigma(x, y) \neq 0$), then there must exist some $v \in \mathbb{E}$ such that $\sigma(v, v) \neq 0$.

Exercise 3.15. (Bilinear form on infinite-dimensional space)

Let $\mathbb{E} = \mathcal{F}_{00}(\mathbb{N}, \mathbb{R})$ be the vector space of all real sequences with only a finite amount of non-zero terms. That is, $x := (x_n)_{n=1}^\infty = (x_1, x_2, \dots) \in \mathbb{E}$ if and only if there exists some $N \in \mathbb{N}$ such that $x_n = 0$ for all $n > N$. Let $\mathcal{E} = \{e_1, e_2, \dots\}$ be the standard basis of $\mathbb{E}$, where $e_j = (0, 0, \dots, 0, \overset{(j)}{1}, 0, \dots)$ (**Note:** Do not take our word that $\mathbb{E}$ is a vector space, try proving it).

We define a function $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{R}$ by

$$b(x, y) = \sum_{i=1}^{\infty} x_i y_{i+1}, \quad \text{where } x = (x_n)_{n=1}^\infty, \quad y = (y_n)_{n=1}^\infty.$$

- (i) Explain why the infinite sum in the definition of $b(x, y)$ always converges to a finite real number (and does not diverge to $\pm\infty$) for any $x, y \in \mathbb{E}$. Then prove that b is a bilinear form on $\mathbb{E}$.
- (ii) Is b a symmetric bilinear form? is it skew-symmetric? (**Indication:** Justify your answers by evaluating $B(e_i, e_j)$ for appropriate basis vectors).
- (iii) We know from ex. 3.10 that any bilinear form can be uniquely decomposed into the sum of a symmetric form $\sigma(x, y)$ and a skew-symmetric form $\alpha(x, y)$. Find the explicit formula for $\sigma(x, y)$ and $\alpha(x, y)$ for this specific bilinear form b .

Exercise 3.16. (Bilinear form on Fibonacci sequences)

Let $\mathbb{E} = \text{Fib}(\mathbb{R})$ be the vector space of all real Fibonacci sequences, recall that there exists an isomorphism from $\text{Fib}(\mathbb{R})$ and $\mathbb{R}^2$ (ex. 1.32). On $\mathbb{E}$ we define the *left-shift operator*

$$S : \mathbb{E} \rightarrow \mathbb{E}$$

$$x = (x_0, x_1, x_2, \dots) \mapsto S(x) = (x_1, x_2, x_3, \dots)$$

(i) We define the bilinear form where for each $x, y \in \mathbb{E}$:

$$b(x, y) = x_0 y_1 - x_1 y_0, \quad \text{where} \quad x = (x_n)_{n=0}^{\infty}, \quad y = (y_n)_{n=0}^{\infty}.$$

Prove that

$$b(S(x), S(y)) = -b(x, y).$$

(ii) Using the result from the previous point, for $n \in \mathbb{N}$, express $b(S^n(x), S^n(y))$ in terms of $b(x, y)$ and n .

(iii) Let $v \in \mathbb{E}$ be the *standard Fibonacci sequence*, i.e. $v = (0, 1, 1, 2, 3, 5, 8, 13, \dots)$. Let $v^{(n)} := S^n(v) = (v_n, v_{n+1}, v_{n+2}, \dots)$. Write down the expression of $b(v^{(n-1)}, v^{(n)})$ in terms of v_j and use the result from the previous point to prove the *Cassini's Identity*:

$$v_{n-1} v_{n+1} - v_n^2 = (-1)^n.$$

Exercise 3.17. (Application of orthogonal projection for a minimization problem) Our goal in this exercise is to approximate a cubic polynomial in a given interval using only polynomials of degree less or equal to 2. Specifically, we wish to find the coefficients of the polynomial $Q(x) = ax^2 + bx + c$ such that the value

$$J[Q] = \int_{-1}^1 [x^3 - Q(x)]^2 dx$$

is minimized.

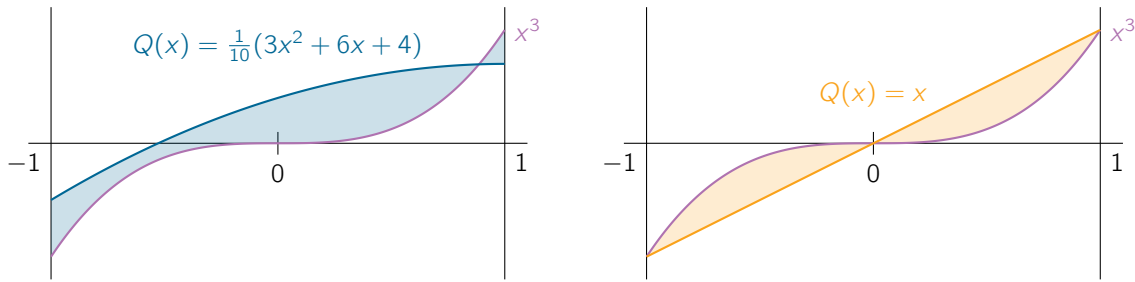


Figure 3.1: The geometric interpretation of minimizing $J[Q]$ is to find the appropriate Q such that the shaded area between Q and $P(x) = x^3$ within the interval $x \in [-1, 1]$ is "as small as possible".

At a glance, this may seem to be an analysis exercise that involves very messy calculations, but we will see that we can use concepts such as inner product and orthogonality to convert it into a simple geometric exercise.

(i) Let $\mathbb{E} = \mathbb{R}_3[X]$ be the real polynomials with degree less or equal to 3. We define the inner product

$$\omega(f, g) = \int_{-1}^1 p(x)q(x) dx.$$

Give the matrix representation $[\omega]_{\mathcal{E}\mathcal{E}}$, where $\mathcal{E} = \{1, x, x^2, x^3\}$ is the standard basis on $\mathbb{E}$.

(ii) Let $R(x) = x^3$, express $J[Q(x)]$ in terms of the norm $\|\cdot\|$ induced by the inner product.

(iii) By definition we know that the polynomial Q belongs to the subspace $\mathbb{F} = \mathbb{R}_2[X]$ of real polynomials of degree less or equal to 2. Let $\mathcal{F} = \{1, X, X^2\}$ be the standard basis of $\mathbb{F}$, find an orthonormal basis $\mathcal{U}$ in $\mathbb{F}$ with respect to the inner product $\omega(\cdot, \cdot)$ by applying Gram-Schmidt orthogonalization on the basis $\mathcal{E}$.

(iv) With the orthogonal basis $\mathcal{U}$, apply the result from ex. 2.14 to obtain the projection of R onto the subspace $\mathbb{F}$. Is this polynomial the one that minimizes $J[Q]$?

Exercise 3.18. (Self adjoint operators) Let $(\mathbb{E}, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space with $\mathbb{K} = \mathbb{C}$, and let $\mathcal{U}$ be an orthonormal basis with respect to $\langle \cdot, \cdot \rangle$.

- (i) Show that $f : \mathbb{E} \rightarrow \mathbb{E}$ is a self adjoint operator if and only if $[f]_{\mathcal{U}}^{\mathcal{U}}$ is a hermitian matrix (Recall that $A \in \mathbb{K}^{n \times n}$ is a hermitian matrix iff $A^H = A$).
- (ii) Show that, for self adjoint operators $f, g : \mathbb{E} \rightarrow \mathbb{E}$, their composition $f \circ g : \mathbb{E} \rightarrow \mathbb{E}$ is a self adjoint operator if and only if f and g commute with each other (i.e. $f \circ g = g \circ f$).

Chapter 4

Determinants

So far, here are the notions/operations that we have defined for vector spaces:

- addition of vectors, and multiplication by scalars (thanks to the axioms of vector spaces)
- product of two and more vectors (thanks to tensor products)
- distances between vectors, and lengths of vectors (thanks to norms)
- angles between vectors and notions of orthogonality (thanks to scalar products)

Somewhat in connection with lengths, one element which is still missing is the one of area, or volume. This idea is actually very connected to the concept that we want to present in this chapter: determinants of square matrices $\mathbb{K}^{n \times n}$. We start by giving their definition, and study their properties. Everything will look very disconnected with the idea of volumes. We will see at the end of the chapter how both ideas are related.

4.1 Definition and first properties

The definition of the determinant of an $n \times n$ matrix involves $\mathcal{S}_n$, the set of permutations of $\{1, \dots, n\}$, which is a set of cardinality $n!$. We recall that, for any permutation $\sigma \in \mathcal{S}_n$, its signature $\varepsilon(\sigma) = 1$ if the permutation can be obtained with an even number of transpositions (exchanges of two entries); otherwise, $\varepsilon(\sigma) = -1$.

Definition 4.1: Leibniz Formula of the determinant

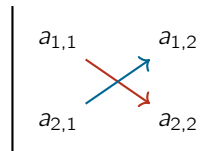
For any $A \in \mathbb{K}^{n \times n}$, its determinant is defined as

$$\det(A) = \begin{vmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,n} \end{vmatrix} := \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)} = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}. \quad (4.1)$$

Here is an observation that helps to understand the Leibniz formula: in each product of the form $a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)}$ we have one coefficient from each row from A . The fact that σ is a permutation (thus a bijection) tells us that each column is represented exactly only once in $\sigma(1), \dots, \sigma(n)$.

Let us express more explicitly how the determinant looks like for matrices for $n = 1, 2, 3$.

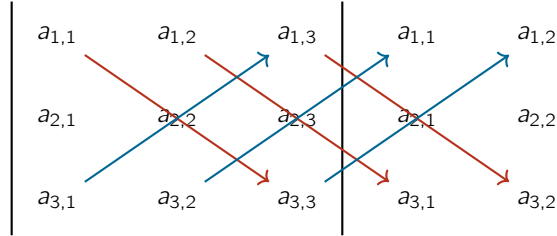
- For $n = 1$, $\det(A) = a_{1,1}$.
- For $n = 2$,

$$\begin{vmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{vmatrix} = a_{1,1}a_{2,2} - a_{2,1}a_{1,2}$$


- For $n = 3$,

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{1,1}a_{2,2}a_{3,3} + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3}$$

and there is the following memory aid to remember it: we multiply with $+$ the red arrows below, and with $-$ the blue arrows:



From the the Leibnitz formula, we can derive a recursive expression for the determinant which is sometimes called the Laplace expansion. The formula involves the following notation: for any matrix $A \in \mathbb{K}^{n \times n}$ and any two indices $1 \leq i, j \leq n$, we denote by $A_{i,j}^*$ the $(n-1) \times (n-1)$ matrix which we obtain when we remove the i -th row and the j -th column of A . For example, for

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

we have

$$A_{1,1}^* = \begin{pmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} \quad \text{and} \quad A_{2,3}^* = \begin{pmatrix} a_{11} & a_{12} & \cancel{a_{13}} \\ \cancel{a_{21}} & \cancel{a_{22}} & \cancel{a_{23}} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{pmatrix}$$

Lemma 4.2: Laplace expansion

The determinant of a matrix $A \in \mathbb{K}^{n \times n}$ is given by

1. If $n = 1$, then $\det(A) = a_{1,1}$.
2. If $n > 1$, then for every $j = 1, \dots, n$,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{i,j} \det(A_{i,j}^*).$$

Proof. We omit it for brevity. □

Example 4.3

We can use the Laplace expansion to compute 3×3 determinants. By being very careful with the sign alternation, we have

$$\begin{vmatrix} 1 & 0 & 9 \\ 0 & 5 & 0 \\ -4 & 0 & -6 \end{vmatrix} = 1 \cdot \begin{vmatrix} 5 & 0 \\ 0 & -6 \end{vmatrix} - 0 \cdot \begin{vmatrix} 0 & 9 \\ 0 & -6 \end{vmatrix} + 4 \cdot \begin{vmatrix} 0 & 9 \\ 5 & 0 \end{vmatrix} = -30 + 0 + 4 \cdot 45 = 150.$$

4.2 Properties

Both the Leibnitz formula and the Laplace expansion of the determinant suggest that the determinant is an object with potentially many “hidden” structures. Here we present the most significant properties.

A first nice property is that the determinant of upper and lower triangular matrices is easy.

Proposition 4.4: Determinant of upper and lower triangular matrices

For any lower or upper triangular matrix $A \in \mathbb{K}^{n \times n}$,

$$\det(A) = \prod_{i=1}^n a_{i,i}.$$

In particular,

$$\det(\text{Id}_n) = 1.$$

Proof. We use the Laplace expansion and proceed by induction. Left as an exercise. □

Lemma 4.5: Determinant of the transpose

$$\det(A^\top) = \det(A).$$

Proof. By the Leibniz formula,

$$\det(A) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}.$$

and

$$\det(A^\top) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{i=1}^n (A^\top)_{i,\sigma(i)} = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{i=1}^n a_{\sigma(i),i}.$$

Now let $\tau = \sigma^{-1}$. Since the map $\sigma \mapsto \sigma^{-1}$ is a bijection on $\mathcal{S}_n$, and since $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$, we rewrite the sum as

$$\det(A^\top) = \sum_{\tau \in \mathcal{S}_n} \varepsilon(\tau) \prod_{j=1}^n a_{j,\tau(j)}.$$

Indeed, if $j = \sigma(i)$, then $i = \sigma^{-1}(j) = \tau(j)$, so

$$\prod_{i=1}^n a_{\sigma(i),i} = \prod_{j=1}^n a_{j,\tau(j)}.$$

Therefore,

$$\det(A^\top) = \sum_{\tau \in \mathcal{S}_n} \varepsilon(\tau) \prod_{j=1}^n a_{j,\tau(j)} = \det(A).$$
□

Another property is the following.

Proposition 4.6

Let $A, B \in \mathbb{K}^{n \times n}$. The mapping

$$\begin{aligned} \mathbb{K} &\rightarrow \mathbb{K} \\ t &\mapsto \det(A + tB) \end{aligned}$$

is a polynomial of degree at most n .

Proof. This follows immediately by applying formula (4.1) to $A + tB$. $\square$

To present further properties, it is useful to think of the determinant by seeing its columns as vectors from $\mathbb{K}^n$. If $u_1, \dots, u_n$ are n vectors of $\mathbb{K}^n$, we write

$$\det(u_1, \dots, u_n) = \det \begin{pmatrix} u_{1,1} & \dots & u_{1,n} \\ \vdots & & \vdots \\ u_{n,1} & \dots & u_{n,n} \end{pmatrix}$$

In this way, we view the determinant as a map from

$$\det : (\mathbb{K}^n) \times \dots \times (\mathbb{K}^n) \rightarrow \mathbb{K}$$

Here are three further properties.

Lemma 4.7

- (i) **Linearity with respect to each column:** Let $j_0 \in \{1, \dots, n\}$, and let $v, w \in \mathbb{K}^n$ and $\lambda, \mu \in \mathbb{K}$. Then

$$\begin{aligned} \det(u_1, \dots, u_{j_0-1}, (\lambda v + \mu w), u_{j_0+1}, \dots, u_n) \\ = \lambda \det(u_1, \dots, u_{j_0-1}, v, u_{j_0+1}, \dots, u_n) + \mu \det(u_1, \dots, u_{j_0-1}, w, u_{j_0+1}, \dots, u_n). \end{aligned} \quad (4.2)$$

- (ii) **If two columns are identical, then the determinant is zero:** If $u_i = u_j$ for two indices $1 \leq i \neq j \leq n$, then

$$\det(u_1, \dots, u_n) = 0.$$

- (iii) **The above two claims also hold for rows instead of columns.**

Proof. (i) Let the matrix $A, B, C \in \mathbb{K}^{n \times n}$ be defined as

$$\begin{aligned} A &= (u_1, \dots, u_{j_0-1}, (\lambda v + \mu w), u_{j_0+1}, \dots, u_n) \\ B &= (u_1, \dots, u_{j_0-1}, v, u_{j_0+1}, \dots, u_n) \\ C &= (u_1, \dots, u_{j_0-1}, w, u_{j_0+1}, \dots, u_n) \end{aligned}$$

The determinant of A is

$$\det(A) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)}.$$

Fix $\sigma \in \mathcal{S}_n$ and consider the product $a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)}$. In this product, exactly one term comes from the column j_0 because there is a unique i_0 such that $\sigma(i_0) = j_0$. Since $a_{i_0,\sigma(i_0)} = a_{i_0,j_0} = \lambda v_{i_0} + \mu w_{i_0}$, it follows that

$$\begin{aligned} a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)} &= \lambda a_{1,\sigma(1)} \dots a_{i_0-1,\sigma(i_0-1)} v_{i_0,j_0} a_{i_0+1,\sigma(i_0+1)} \dots a_{n,\sigma(n)} \\ &\quad + \mu a_{1,\sigma(1)} \dots a_{i_0-1,\sigma(i_0-1)} w_{i_0,j_0} a_{i_0+1,\sigma(i_0+1)} \dots a_{n,\sigma(n)} \\ &= \lambda b_{1,\sigma(1)} \dots b_{i_0-1,\sigma(i_0-1)} b_{i_0,j_0} b_{i_0+1,\sigma(i_0+1)} \dots b_{n,\sigma(n)} \\ &\quad + \mu c_{1,\sigma(1)} \dots c_{i_0-1,\sigma(i_0-1)} c_{i_0,j_0} c_{i_0+1,\sigma(i_0+1)} \dots c_{n,\sigma(n)}. \end{aligned}$$

We now multiply this last expression by $\varepsilon(\sigma)$ and sum over $\sigma \in \mathcal{S}_n$. This leads to Eq. (4.2).

- (ii) Let $A = (u_1, \dots, u_n)$ with $u_i = u_j = v$ for two indices $1 \leq i \neq j \leq n$. Fix $\sigma \in \mathcal{S}_n$, and consider the product $a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{n,\sigma(n)}$. There exist indices $i_0, j_0 \in \{1, \dots, n\}$ such that $\sigma(i_0) = i$ and $\sigma(j_0) = j$. For that fixed σ , there exists $\tilde{\sigma}$ such that $\tilde{\sigma}(i_0) = j$ and $\tilde{\sigma}(j_0) = i$ and $\sigma(k) = \tilde{\sigma}(k)$ for any

other index $k \neq i_0, j_0$. These permutations σ and $\tilde{\sigma}$ differ only by one swap, therefore $\varepsilon(\sigma) = -\varepsilon(\tilde{\sigma})$. Consequently,

$$\begin{aligned} & \varepsilon(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} + \varepsilon(\tilde{\sigma}) a_{1,\tilde{\sigma}(1)} a_{2,\tilde{\sigma}(2)} \cdots a_{n,\tilde{\sigma}(n)} \\ &= \varepsilon(\sigma) (a_{i_0,\sigma(i_0)} a_{j_0,\sigma(j_0)} - a_{i_0,\tilde{\sigma}(i_0)} a_{j_0,\tilde{\sigma}(j_0)}) \prod_{k \neq i_0, j_0} a_{k,\sigma(k)} \\ &= \varepsilon(\sigma) (a_{i_0,i} a_{j_0,j} - a_{i_0,j} a_{j_0,i}) \prod_{k \neq i_0, j_0} a_{k,\sigma(k)} \\ &= 0 \quad (\text{since columns } i \text{ and } j \text{ are identical}). \end{aligned}$$

Summing over all permutations leads to the conclusion that $\det(A) = 0$.

(iii) Since $\det(A) = \det(A^T)$, the claims also apply for rows. □

The properties of Lemma 4.7 have the following direct consequences which are useful to efficiently compute determinants in practice.

Corollary 4.8

- (a) **Adding a multiple of a column or a row to another:** If i and j are two integers in $\{1, \dots, n\}$ with $j \neq i$, and if α is any scalar, then

$$\det(u_1, \dots, u_{j-1}, (u_j + \alpha u_i), u_{j+1}, \dots, u_n) = \det(u_1, \dots, u_n).$$

In other words, an operation of the form

$$C_j \leftarrow C_j + \alpha C_i \quad (i \neq j)$$

does not change the determinant. This also holds for operations per rows: or

$$R_j \leftarrow R_j + \alpha R_i \quad (i \neq j)$$

does not change the determinant.

- (b) **Scaling a single column/row:** It holds that

$$\det(u_1, \dots, u_{j-1}, \alpha u_j, u_{j+1}, \dots, u_n) = \alpha \det(u_1, \dots, u_{j-1}, u_j, u_{j+1}, \dots, u_n)$$

and the same holds for rows.

- (c) **Scaling the whole matrix:**

$$\det(\lambda A) = \lambda^n \det(A).$$

- (d) **Swapping two columns or two rows:** If $n \geq 2$, if $u_1, \dots, u_n$ are vectors of $\mathbb{K}^n$, and if $i < j$ are two distinct integers between 1 and n , then

$$\det(u_1, \dots, u_{i-1}, u_j, u_{i+1}, \dots, u_{j-1}, u_i, u_{j+1}, \dots, u_n) = (-1) \det(u_1, \dots, u_n).$$

So when one swaps two rows of a matrix A , the determinant is replaced by its opposite.

- (e) **The determinant of linearly dependent vectors is zero:** If $u_1, \dots, u_n \in \mathbb{K}^n$ is a linearly dependent set, then $\det(u_1, \dots, u_n) = 0$.

Proof. These are direct consequences of Lemma 4.7. Left as an exercise. □

Example 4.9

From property (e), we immediately conclude that

$$\det \begin{vmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{vmatrix} = 0$$

since the second row is a linear combination of rows 1 and 3 (it is one quarter of their sum).

The operations (a), (b), and (d) are precisely those that occur in the “pivot method” with which you are familiar with. This leads to the following strategy to compute the determinant:

Lemma 4.10: We can compute the determinant by the pivot method

We will give the statement in a narrated way, and in a more mathematical way:

- By a sequence of operations of types (a) to (c) above, one can reduce the computation of the determinant of any matrix $A \in \mathbb{K}^{n \times n}$ to that of an upper triangular matrix.
 - It holds that $\det(A) = (-1)^s \prod_{i=1}^n u_{i,i}$ where U is the upper triangular matrix from the PLU decomposition of A and s is the number of row swaps in the permutation matrix P .
-

Proof.

$$\begin{aligned} \det(A) &= \det(PLU) \\ &= \det(P) \det(L) \det(U) \quad (\text{by Lemma 4.12 stated just below}) \\ &= (-1)^s \prod_{i=1}^n u_{i,i}, \end{aligned}$$

where in the last line we have used that $\det(P) = (-1)^s$, that L is a lower triangular matrix with unity diagonal (hence $\det(L) = 1$), and that U is a triangular matrix. $\square$

Example 4.11

Let us use this method to compute the determinant of

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 3 & 2 & 1 \end{pmatrix}.$$

The previous results allow us to track what happens to the determinant as we “reduce” the matrix:

$$\begin{aligned}
\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 3 & 2 & 1 \end{vmatrix} &= \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -2 \\ 0 & -4 & -8 \end{vmatrix} && (R_2 \leftarrow R_2 - 2R_1 \text{ and } R_3 \leftarrow R_3 - 3R_1) \\
&= (-2) \cdot (-4) \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix} && (\text{the } -2 \text{ comes from } R_2 \text{ and the } -4 \text{ from } R_3) \\
&= 8 \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix} && (\text{here we are simply copying from above}) \\
&= 8 \cdot (-1) \cdot \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} && (\text{row swap}) \\
&= -8 \cdot 1 = -8 && (\text{triangular matrix}) .
\end{aligned}$$

Lemma 4.12: Determinant of the matrix product

Let $A, B \in \mathbb{K}^{n \times n}$. It holds that

$$\det(AB) = \det(A) \det(B)$$

Proof. Let $C = AB = (c_{ij})_{1 \leq i, j \leq n}$, where $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$. The Leibnitz formula for the determinant of C reads

$$\det(C) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n c_{i, \sigma(i)} = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n \left(\sum_{k_i=1}^n a_{ik_i} b_{k_i, \sigma(i)} \right).$$

Expanding the product of sums gives

$$\det(AB) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \sum_{k_1, \dots, k_n=1}^n \prod_{i=1}^n a_{ik_i} b_{k_i, \sigma(i)}.$$

Reordering the sums,

$$\det(AB) = \sum_{k_1, \dots, k_n=1}^n \left(\prod_{i=1}^n a_{ik_i} \right) \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n b_{k_i, \sigma(i)}.$$

We now observe that if two indices $k_r = k_s$ with $r \neq s$, then two rows in the matrix $(b_{k_i, j})$ are equal, so

$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n b_{k_i, \sigma(i)} = 0.$$

Hence only tuples $(k_1, \dots, k_n)$ with distinct entries contribute. Since each $k_i \in \{1, \dots, n\}$, such a tuple must be a permutation $\tau \in S_n$, i.e.

$$k_i = \tau(i).$$

Therefore,

$$\det(AB) = \sum_{\tau \in S_n} \left(\prod_{i=1}^n a_{i, \tau(i)} \right) \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n b_{\tau(i), \sigma(i)}.$$

Now reindex the inner product by $j = \tau(i)$. Then

$$\prod_{i=1}^n b_{\tau(i), \sigma(i)} = \prod_{j=1}^n b_{j, (\sigma \circ \tau^{-1})(j)}.$$

Also,

$$\varepsilon(\sigma) = \varepsilon(\sigma \circ \tau^{-1})\varepsilon(\tau).$$

Thus,

$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n b_{\tau(i), \sigma(i)} = \varepsilon(\tau) \sum_{\pi \in S_n} \varepsilon(\pi) \prod_{j=1}^n b_{j, \pi(j)},$$

where $\pi = \sigma \circ \tau^{-1}$. Hence,

$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n b_{\tau(i), \sigma(i)} = \varepsilon(\tau) \det(B).$$

Substituting back,

$$\det(AB) = \sum_{\tau \in S_n} \left(\prod_{i=1}^n a_{i, \tau(i)} \right) \varepsilon(\tau) \det(B).$$

Factoring out $\det(B)$,

$$\det(AB) = \det(B) \sum_{\tau \in S_n} \varepsilon(\tau) \prod_{i=1}^n a_{i, \tau(i)}.$$

The remaining sum is exactly $\det(A)$. Therefore $\det(AB) = \det(A) \det(B)$. □

Corollary 4.13: Determinant and invertibility

A matrix $A \in \mathbb{K}^{n \times n}$ is invertible if and only if

$$\det(A) \neq 0.$$

In that case,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Proof. • If A is invertible, then

$$A(A^{-1}) = I_n,$$

and therefore

$$\det(A) \det(A^{-1}) = \det(I_n) = 1,$$

so $\det(A)$ cannot be zero, and indeed

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

- If A is not invertible, then there exists a linear dependence relation among its rows, and we have already seen that in this case the determinant of A is zero. □

The above properties allow us to study the determinant of similar matrices in the sense given by the following definition.

Definition 4.14

Two square matrices A and $B \in \mathbb{K}^{n \times n}$ are similar if and only if there exists an invertible matrix $P \in \mathbb{K}^{n \times n}$ such that

$$A = PBP^{-1}.$$

Corollary 4.15: Similar matrices have the same determinant.

If A and B are two similar matrices, then

$$\det(A) = \det(B).$$

Proof. To say that A and B are similar means that there exists an invertible matrix P such that

$$A = PBP^{-1}.$$

Hence

$$\det(A) = \det(P) \det(B) \det(P^{-1}).$$

But we have just seen that

$$\det(P^{-1}) = \frac{1}{\det(P)},$$

and the product

$$\det(P) \det(B) \frac{1}{\det(P)}$$

is just a product of real numbers, so it is of course equal to $\det(B)$. □

Definition 4.16: Determinant of an endomorphism

Let E be an n -dimensional $\mathbb{K}$ -vector space and let $f : E \rightarrow E$ be an endomorphism of E ($f \in \mathcal{L}(E)$). We define the determinant of f as the determinant of its matrix representation $[f]_{\mathcal{E}}^{\mathcal{E}}$ on a basis $\mathcal{E}$ of E . The determinant is independent to the choice of the basis,

$$\det(f) := \det([f]_{\mathcal{E}}^{\mathcal{E}}), \quad \forall \mathcal{E} \text{ basis of } E.$$

The fact that the determinant is independent of the choice of basis comes from the fact that if $\mathcal{F}$ and $\mathcal{E}$ are two basis of an n -dimensional space E then, by Proposition 1.42,

$$[f]_{\mathcal{F}}^{\mathcal{F}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [f]_{\mathcal{E}}^{\mathcal{E}} [\text{Id}]_{\mathcal{F}}^{\mathcal{E}},$$

which means that $[f]_{\mathcal{F}}^{\mathcal{F}}$ and $[f]_{\mathcal{E}}^{\mathcal{E}}$ are similar. By Corollary 4.15, they have the same determinant.

4.3 The overarching structure: alternating n -linear forms

The overarching concept for determinants is that they are alternating linear forms

Definition 4.17: Alternating n -linear form

Let E be a vector space over $\mathbb{K}$, let n be a positive integer, and let

$$\Phi : E^n \rightarrow \mathbb{K}, \quad (u_1, \dots, u_n) \mapsto \Phi(u_1, \dots, u_n)$$

be a map which, starting from n vectors in E , produces a scalar.

- (a) We say that Φ is a *n -linear form* if it is linear in each of its variables: whenever one fixes an $i \in \{1, \dots, n\}$ and vectors

$$u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_n \in E,$$

the map

$$E \rightarrow \mathbb{K}, \quad v \mapsto \Phi(u_1, \dots, u_{i-1}, v, u_{i+1}, \dots, u_n)$$

is linear.

- (b) We say that Φ is *alternating* if, whenever two of the vectors among $(u_1, \dots, u_n)$ are identical, then

$$\Phi(u_1, \dots, u_n) = 0.$$

Corollary 4.18

The map

$$(u_1, \dots, u_n) \mapsto \det(u_1, \dots, u_n)$$

is an alternating n -linear form on the space $E = \mathbb{K}^n$.

It is easy to check that if Φ_1 and Φ_2 are two alternating n -linear forms on a $\mathbb{K}$ -vector space E , and if λ_1, λ_2 are two scalars in $\mathbb{K}$, then the map

$$\lambda_1 \Phi_1 + \lambda_2 \Phi_2$$

is an alternating n -linear form. Thus the set of alternating n -linear forms on E also contains the zero map. It is therefore a vector subspace of $\mathcal{F}(E^n, \mathbb{K})$, and it is traditionally denoted by $\Lambda^n(E)$.

The following theorem may appear abstract, but the most concrete properties of the determinant that we have proven earlier follow from it. We will not detail its proof.

Theorem 4.19: All alternating n -linear forms on an n -dimensional space are proportional.

Let E be a finite-dimensional vector space, and let n denote the dimension of E . Then the space $\Lambda^n(E)$ has dimension 1. Consequently, if Φ is an alternating n -linear form on $\mathbb{K}^n$, then there exists a scalar $\lambda \in \mathbb{K}$ such that

$$\Phi = \lambda \det$$

As an example of application, we next prove Lemma 4.12 about $\det(AB) = \det(A)\det(B)$ using the more axiomatic properties of multilinear forms.

Alternative proof of Lemma 4.12. Let $A, B \in \mathbb{K}^{n \times n}$, and write the columns of B as

$$B = (b_1 \mid b_2 \mid \dots \mid b_n),$$

where $b_1, \dots, b_n \in \mathbb{K}^n$. Then the columns of AB are $Ab_1, \dots, Ab_n$, so

$$AB = (Ab_1 \mid Ab_2 \mid \dots \mid Ab_n).$$

Define a map

$$\omega : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

by

$$\omega(v_1, \dots, v_n) = \det(Av_1, \dots, Av_n).$$

Since the map $v \mapsto Av$ is linear and the determinant is n -linear in the columns, the map ω is n -linear. Moreover, if $v_i = v_j$ for some $i \neq j$, then

$$Av_i = Av_j,$$

so the matrix $(Av_1, \dots, Av_n)$ has two equal columns. Hence,

$$\omega(v_1, \dots, v_n) = 0.$$

Therefore, ω is an alternating n -linear form on $\mathbb{K}^n$. By the Theorem 4.19 for alternating n -linear forms on $E = \mathbb{K}^n$, there exists a scalar $c \in \mathbb{K}$ such that

$$\omega(v_1, \dots, v_n) = \lambda \det(v_1, \dots, v_n)$$

for all $v_1, \dots, v_n \in \mathbb{K}^n$. To determine λ , evaluate at the standard basis vectors $e_1, \dots, e_n$:

$$\lambda = \lambda \det(e_1, \dots, e_n) = \omega(e_1, \dots, e_n).$$

Since

$$\det(e_1, \dots, e_n) = 1,$$

we obtain

$$\lambda = \omega(e_1, \dots, e_n) = \det(Ae_1, \dots, Ae_n).$$

But $Ae_1, \dots, Ae_n$ are exactly the columns of A . Therefore,

$$\lambda = \det(A).$$

Hence, for all $v_1, \dots, v_n \in \mathbb{K}^n$,

$$\det(Av_1, \dots, Av_n) = \det(A) \det(v_1, \dots, v_n).$$

Now apply this to the columns $b_1, \dots, b_n$ of B :

$$\det(AB) = \det(Ab_1, \dots, Ab_n) = \det(A) \det(b_1, \dots, b_n) = \det(A) \det(B).$$

□

4.4 Determinant and volume

If u and v are two vectors in $\mathbb{K}^2$, the parallelogram spanned by u and v is the set

$$\mathcal{P}(u, v) := \{\alpha u + \beta v ; \alpha \in [0, 1], \beta \in [0, 1]\}.$$

More generally, in dimension $n \in \mathbb{N}$, the parallelotope spanned by $u_1, \dots, u_n \in \mathbb{K}^n$ is

$$\mathcal{P}(u_1, \dots, u_n) := \left\{ \sum_{i=1}^n \alpha_i u_i : \alpha_i \in [0, 1], \forall 1 \leq i \leq n \right\}.$$

The determinant of $(u_1, \dots, u_n)$ is linked to the volume of $\mathcal{P}(u_1, \dots, u_n)$ in the Euclidean space $(\mathbb{K}, \langle \cdot, \cdot \rangle_2)$. To understand the connection, let us work in $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$ and fix two vectors u and v . Using coordinates in the canonical basis $\mathcal{E}$, we write

$$[u]^\mathcal{E} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{and} \quad [v]^\mathcal{E} = \begin{pmatrix} c \\ d \end{pmatrix},$$

and picture this parallelogram in Fig. 4.1. Recall that its area is given by the formula “base $\times$ height”: with the notation of the figure, the area of the parallelogram spanned by u and v is

$$\text{Vol}(\mathcal{P}(u, v)) = \|u\|_2 \|v\|_2 \sin(\theta).$$

Taking squares,

$$\begin{aligned} (\text{Vol}(\mathcal{P}(u, v)))^2 &= \|u\|_2^2 \|v\|_2^2 \sin^2(\theta) \\ &= \|u\|_2^2 \|v\|_2^2 (1 - \cos^2(\theta)) \\ &= \|u\|_2^2 \|v\|_2^2 \left(1 - \frac{\langle u, v \rangle_2^2}{\|u\|_2^2 \|v\|_2^2} \right) \quad \text{by Eq. (2.19)} \\ &= \|u\|_2^2 \|v\|_2^2 - \langle u, v \rangle_2^2 \\ &= (a^2 + b^2)(c^2 + d^2) - (ac + bd)^2 \\ &= (ad - bc)^2 \\ &= \left(\det([u]^\mathcal{E}, [v]^\mathcal{E}) \right)^2. \end{aligned}$$

Therefore $\text{Vol}(\mathcal{P}(u, v)) = |\det([u]^\mathcal{E}, [v]^\mathcal{E})|$. This development can be generalized to higher dimensions n . It leads to the following definition of volume on any general n -dimensional inner-product vector space E as follows.

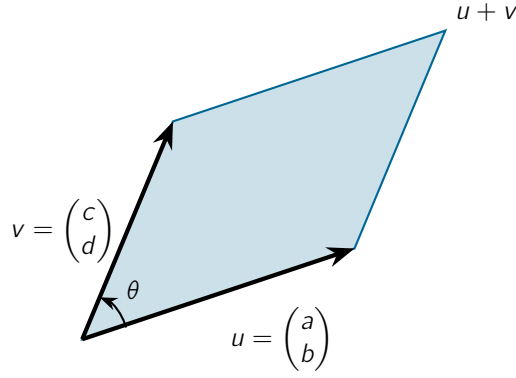


Figure 4.1: Parallelogram $\mathcal{P}(u, v)$ spanned by $u = (a, b)^\top$ and $v = (c, d)^\top$. Its volume is $|\det(u, v)|$.

Definition 4.20: Volume of a parallelotope

Let E be an n -dimensional inner-product vector space and let $\mathcal{E}$ be an orthonormal basis. The volume of the parallelotope associated to a set of n vectors $u_1, \dots, u_n \in E$ is defined as

$$\text{Vol}(\mathcal{P}(u_1, \dots, u_n)) := |\det([u_1]^\mathcal{E}, \dots, [u_n]^\mathcal{E})|. \quad (4.3)$$

For any non-orthonormal basis $\mathcal{F}$, it holds that

$$\text{Vol}(\mathcal{P}(u_1, \dots, u_n)) = \left| \det([\text{Id}]_\mathcal{F}^\mathcal{E}) \right| \left| \det([u_1]^\mathcal{F}, \dots, [u_n]^\mathcal{F}) \right|. \quad (4.4)$$

Proof. To prove Eq. (4.4), it suffices to recall that, by Eq. (1.6), $[u_i]^\mathcal{E} = [\text{Id}]_\mathcal{F}^\mathcal{E} [u_i]^\mathcal{F}$. Thus

$$\begin{aligned} \text{Vol}(\mathcal{P}(u_1, \dots, u_n)) &= |\det([u_1]^\mathcal{E}, \dots, [u_n]^\mathcal{E})| \\ &= \left| \det([\text{Id}]_\mathcal{F}^\mathcal{E} [u_1]^\mathcal{F}, \dots, [\text{Id}]_\mathcal{F}^\mathcal{E} [u_n]^\mathcal{F}) \right| \\ &= \left| \det([\text{Id}]_\mathcal{F}^\mathcal{E} ([u_1]^\mathcal{F} \mid \dots \mid [u_n]^\mathcal{F})) \right| \\ &= \left| \det([\text{Id}]_\mathcal{F}^\mathcal{E}) \right| \left| \det([u_1]^\mathcal{F}, \dots, [u_n]^\mathcal{F}) \right|. \end{aligned}$$

□

This definition nicely aligns with some intuitive facts, for example, if we double the length of one of the vectors, then the volume is doubled as well. In general, for any $\lambda \in \mathbb{K}$,

$$\text{Vol}(\mathcal{P}(u_1, \dots, \lambda u_j, \dots, u_n)) = |\lambda| \text{Vol}(\mathcal{P}(u_1, \dots, u_j, \dots, u_n)).$$

The above result is a good complement to the fact that the volume of the parallelotope is zero if and only if there exist u_i and u_j that are linearly dependent.

Also, we may note that the interpretation of the determinant as the actual Euclidean volume crucially relies on working with an orthonormal basis for the vector representation (see Eq. (4.3)). Working with a non-orthonormal basis distorts lengths and angles, and one needs to account for this fact as Eq. (4.4) reflects.

Objects that are not parallelotopes have a volume as well and it is also related to the determinant, although in a more complicated way. Measure theory and the change-of-variables formula in multiple integrals will explain this later in your studies.

An interesting question is to understand how the volume of a parallelotope changes under the action of an endomorphism. The following result brings light to this question.

Lemma 4.21

Let E be an n -dimensional inner-product space, let $f : E \rightarrow E$ be an endomorphism, and let $u_1, \dots, u_n$ be a set of vectors in E . It holds that for any orthonormal basis $\mathcal{E}$ of E ,

$$\text{Vol}(\mathcal{P}(f(u_1), \dots, f(u_n))) = \left| \det([f]_{\mathcal{E}}^{\mathcal{E}}) \right| \text{Vol}(\mathcal{P}(u_1, \dots, u_n))$$

Proof. Let $\mathcal{E}$ be an orthonormal basis of E . We have that

$$\begin{aligned} \text{Vol}(\mathcal{P}(f(u_1), \dots, f(u_n))) &= \left| \det([f(u_1)]^{\mathcal{E}}, \dots, [f(u_n)]^{\mathcal{E}}) \right| \\ &= \left| \det([f]_{\mathcal{E}}^{\mathcal{E}} ([u_1]^{\mathcal{E}}, \dots, [u_n]^{\mathcal{E}})) \right| \\ &= \left| \det([f]_{\mathcal{E}}^{\mathcal{E}}) \right| \left| \det([u_1]^{\mathcal{E}}, \dots, [u_n]^{\mathcal{E}}) \right| \\ &= \left| \det([f]_{\mathcal{E}}^{\mathcal{E}}) \right| \text{Vol}(\mathcal{P}(u_1, \dots, u_n)). \end{aligned}$$

□

We finish this section with a result that bounds the volume of a parallelotope by the one of the rectangle generated by edges whose lengths are equal to the lengths of each of the vectors.

Theorem 4.22: Hadamard's inequality

Let $A = (a_1 | \dots | a_n) \in \mathbb{K}^{n \times n}$, then

$$|\det(A)| \leq \prod_{i=1}^n \|a_i\|_2.$$

and equality holds if and only if the columns are pairwise orthogonal, that is,

$$\langle a_i, a_j \rangle_2 = 0, \quad \text{for } 1 \leq i \neq j \leq n.$$

The result also holds by rows instead of columns.

Proof. The proof requires eigenvalue theory which we present in the next chapter. Let $A = (a_1 | \dots | a_n) \in \mathbb{K}^{n \times n}$, and define $B \in \mathbb{K}^{n \times n}$ as the matrix A with normalized columns,

$$B := \left(\frac{a_1}{\|a_1\|_2} \mid \dots \mid \frac{a_n}{\|a_n\|_2} \right).$$

We next consider the matrix $P = B^H B = (\langle b_i, b_j \rangle_2)_{1 \leq i, j \leq n}$. The matrix P is Hermitian and positive definite, therefore diagonalizable and with real, nonnegative eigenvalues. Let $\lambda_1, \dots, \lambda_n \in \mathbb{R}_+$ be the eigenvalues. Using the inequality of arithmetic-geometric means,

$$\det(P) = \prod_{i=1}^n \lambda_i \leq \left(\frac{1}{n} \sum_{i=1}^n \lambda_i \right)^n = \left(\frac{1}{n} \text{Tr}(P) \right)^n = \left(\frac{1}{n} n \right)^n = 1.$$

Finally, since

$$\det(P) = \det(B)^2 = \frac{1}{\prod_{i=1}^n \|a_i\|_2^2} \det(A)^2$$

we infer that $|\det(A)| \leq \prod_{i=1}^n \|a_i\|_2$ which is the desired inequality.

We finish the proof by showing that

$$|\det(A)| = \prod_{i=1}^n \|a_i\|_2 \iff \langle a_i, a_j \rangle_2 = 0, \quad \text{for } 1 \leq i \neq j \leq n.$$

The direction $\Leftarrow$ follows very directly: If $\langle a_i, a_j \rangle_2 = 0$ for all $i \neq j$, then $P = \text{Id}$ and equality readily follows. To show the direction $\Rightarrow$, we observe that in this case it holds that the eigenvalues of P are such that

$$\prod_{i=1}^n \lambda_i = \left(\frac{1}{n} \sum_{i=1}^n \lambda_i \right)^n = 1$$

This implies that $\lambda_1 = \dots = \lambda_n = 1$. Consequently, $P = \text{Id}$ and since $P = B^H B$, we infer that $\langle b_i, b_j \rangle_2 = 0$ for $i \neq j$. Therefore $\langle a_i, a_j \rangle_2 = 0$ for $i \neq j$. $\square$

4.5 Cramer's rule to solve linear systems

Determinants bring a new formula for solving linear systems.

Proposition 4.23: (Cramer's rule)

Let $A = (a_1 | \dots | a_n) \in \mathbb{K}^{n \times n}$ be an invertible matrix with columns $a_i \in \mathbb{K}^n$, and let $b \in \mathbb{K}^n$. The solution to the linear system $Ax = b$ is the vector $x = (x_i)_{i=1}^n$ where

$$x_i = \frac{\det(a_1 | \dots | a_{i-1} | b | a_{i+1} | \dots | a_n)}{\det A}, \quad 1 \leq i \leq n. \quad (4.5)$$

Proof. Since the determinant is an alternating multilinear form, it holds that

$$\det \left(a_1 | \dots | a_{i-1} | \lambda_i a_i + \sum_{j \neq i} \lambda_j a_j | a_{i+1} | \dots | a_n \right) = \lambda_i \det A,$$

for all $\lambda_1, \dots, \lambda_n$ in $\mathbb{K}$. Since $Ax = b$ is equivalent to writing $b = \sum_{i=1}^n x_i a_i$, we can take $\lambda_i = x_i$ in the previous formula and obtain

$$\det(a_1 | \dots | a_{i-1} | b | a_{i+1} | \dots | a_n) = x_i \det A,$$

which proves the formula. $\square$

Although theoretically very elegant, Cramer's rule (4.5) is unfortunately highly inefficient for practical implementation, since it requires an enormous number of operations. Let c_n denote the cost of computing the determinant of an $n \times n$ matrix by expansion along rows (or columns). One has

$$c_n = n(1 + c_{n-1}) \geq nc_{n-1},$$

and therefore

$$c_n \geq n!.$$

Since one must compute $n + 1$ determinants to solve the linear system $Ax = b$, the total number of operations is therefore greater than $(n + 1)!$. For $n = 50$, we thus obtain

$$c_{50} \geq 50! \approx 3 \times 10^{64}.$$

If a computer performs computations at 100 Gigaflips (that is, 10^{11} operations per second), then the computation using Cramer's method would take on the order of 10^{46} years!

For comparison, solving the linear system with Gaussian elimination (*PLU* decomposition) has complexity of the order $n^3/3$, which means that for $n = 50$ the computation would be done in about 4.10^{-7} seconds.

4.6 Exercises

Exercise 4.1. (Determinants and Gaussian eliminations)

Calculate the determinants of the following matrices using Gaussian eliminations:

$$a = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 2 & h \\ -1 & h & 2 \end{pmatrix}, \quad c = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 3 & 3 \\ 0 & 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 & 5 \end{pmatrix}, \quad d = \begin{pmatrix} -1 & -1 & -1 \\ 4 & 5 & -3 \\ -1 & -6 & 3 \end{pmatrix}$$

Exercise 4.2. (Properties of determinant)

Solve the following determinants:

$$\begin{pmatrix} 1 & 4 & 8 & 0 & 0 & 0 \\ 4 & 3 & 2 & 0 & 0 & 0 \\ 1 & 2 & 4 & 0 & 0 & 0 \\ 5 & 8 & 2 & 1 & 0 & 0 \\ 5 & 8 & 2 & 1 & 4 & 0 \\ 5 & 8 & 2 & 1 & 4 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \\ 5 & 6 & 7 \end{pmatrix}$$

Exercise 4.3. (Properties of determinant)

Consider the following matrices

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, \quad B = \begin{pmatrix} a & 5g & d \\ b & 5h & e \\ 2c & 10i & 2f \end{pmatrix}.$$

Determine $\det A$ if $\det B = 10$.

Exercise 4.4. (Properties of determinant)

Solve the following equations for $\lambda \in \mathbb{R}$

$$\det \begin{pmatrix} 2 & \lambda & 1 \\ \lambda & 2 & 1 \\ 3 & 4 & \lambda \end{pmatrix} = 0, \quad \det \begin{pmatrix} 4-\lambda & 1 & 0 \\ 2 & 6-\lambda & 1 \\ 0 & 1 & 4-\lambda \end{pmatrix} = 0, \quad \det \begin{pmatrix} \lambda & 1 & \lambda+1 \\ 2 & \lambda & 3 \\ \lambda+1 & 4 & \lambda \end{pmatrix} = -2\lambda^2 + 11.$$

Exercise 4.5. (Properties of determinants)

Calculate the determinant of the following matrix

$$M_n = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ 1+n & 2+n & 3+n & \dots & n+n \\ 1+2n & 2+2n & 3+2n & \dots & n+2n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1+(n-1)n & 2+(n-1)n & 3+(n-1)n & \dots & n+(n-1)n \end{pmatrix},$$

where $n \geq 3$. What happens when $n < 3$?

Exercise 4.6. (Rank and determinant of a matrix)

Let $A \in \mathbb{K}^{n \times n}$, prove that $\text{rank } A$ is equal to the maximum of the natural numbers $m \leq n$ such that there exists an invertible minor $A_m \in \mathbb{K}^{m \times m}$ within A .

Exercise 4.7. (Vandermonde Matrix and Vandermonde Polynomial)

Let $n, m \in \mathbb{N}$ and $x = (x_0, x_1, \dots, x_m) \in \mathbb{K}^{m+1}$, we define the *Vandermonde matrix* as the $\mathbb{K}^{(m+1) \times (n+1)}$ matrix

$$V(x) = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{pmatrix},$$

where x_j^k denotes the k -th power of the scalar x_j .

In the following parts we let $n = m$.

- (i) Prove that $\det V(x) = 0$ if and only if $x_i = x_j$ for some $i \neq j$.
- (ii) Using the previous result, show that $\det V(x)$ is divisible by $x_i - x_j$ for all $0 \leq i < j \leq n$. (**Indication:** the *Factor Theorem* states that if $f(x, y)$ and $g(y)$ are two polynomials, then $x - g(y)$ is a factor of $f(x, y)$ if and only if $f(g(y), y)$ is the zero polynomial.)

- (iii) Using the previous result and by comparing the degree of $\prod_{0 \leq i < j \leq m} (x_j - x_i)$ and $\det V(x)$, arrive at the result

$$\det V(x) = C \cdot \prod_{0 \leq i < j \leq m} (x_j - x_i),$$

where $C \in \mathbb{K}$ is some scalar.

- (iv) Conclude that $C = 1$, thus

$$\det V(x) = \prod_{0 \leq i < j \leq m} (x_j - x_i).$$

Exercise 4.8. (Vandermonde Polynomial for integer sequence)

Let $n \in \mathbb{N}$ and $x = (x_1, \dots, x_n) \in \mathbb{N}^n$. We define the matrix

$$M(x) = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \binom{x_1}{1} & \binom{x_2}{1} & \dots & \binom{x_n}{1} \\ \binom{x_1}{2} & \binom{x_2}{2} & \dots & \binom{x_n}{2} \\ \vdots & \vdots & \ddots & \vdots \\ \binom{x_1}{n-1} & \binom{x_2}{n-1} & \dots & \binom{x_n}{n-1} \end{pmatrix}, \quad \text{where} \quad \binom{m}{k} = \frac{m!}{k!(m-k)!}.$$

- (i) Show that $\det M(x) \in \mathbb{Z}$.
(ii) Using properties of the determinant, prove that

$$\det M(x) = \left[\prod_{i=1}^n \frac{1}{(i-1)!} \right] \cdot \det V(x),$$

where $V(x)$ denotes the Vandermonde matrix as defined in ex. 4.7.

- (iii) Conclude that

$$\prod_{1 \leq i < j \leq n} \frac{x_j - x_i}{j - i}$$

is an integer for all $x_1, \dots, x_n \in \mathbb{N}$.

Exercise 4.9. Product and Determinant of a matrices defined by blocks

Let $A, B \in \mathbb{K}^{n \times n}$ be two matrices and let $p \leq n$. The two matrices are defined by blocks as

$$A = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix} \quad B = \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix}.$$

with $A_{1,1} \in \mathbb{K}^{p \times p}$, $A_{1,2} \in \mathbb{K}^{p \times (n-p)}$, $A_{2,1} \in \mathbb{K}^{(n-p) \times p}$ and $A_{2,2} \in \mathbb{K}^{(n-p) \times (n-p)}$.

- (i) Prove that

$$AB = \begin{pmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{pmatrix}.$$

- (ii) Show that, if $A_{2,1} = 0$, then $\det A = \det(A_{1,1}) \det(A_{2,2})$.
(iii) Show that, if $\det A_{1,1} \neq 0$, then $\det A = \det(A_{1,1}) \det(A_{2,2} - A_{2,1}A_{1,1}^{-1}A_{1,2})$. (**Indication:** The matrix

$$M = \begin{pmatrix} \text{Id}_p & 0 \\ -A_{2,1}A_{1,1}^{-1} & \text{Id}_{n-p} \end{pmatrix}$$

may be of help.)

Exercise 4.10. (Sylvester's Criterion, Part I) Let $\mathbb{E}$ be a $\mathbb{K}$ -vector space equipped with basis $\mathcal{E}$, and let $b : \mathbb{E} \times \mathbb{E} \rightarrow \mathbb{K}$ be a symmetric bilinear form. Given $i \leq n$ and for $B = [b]_{\mathcal{E}\mathcal{E}}$, let B_i be the i -th leading principal minor of the matrix B as shown below:

$$B = \begin{pmatrix} B_{11} & B_{12} & \dots & B_{1n} \\ B_{21} & B_{22} & \dots & B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \dots & B_{nn} \end{pmatrix}, \quad B_i = \begin{pmatrix} B_{11} & B_{12} & \dots & B_{1i} \\ B_{21} & B_{22} & \dots & B_{2i} \\ \vdots & \vdots & \ddots & \vdots \\ B_{i1} & B_{i2} & \dots & B_{ii} \end{pmatrix}.$$

- (i) Prove that if b is positive definite symmetric then $\det B > 0$. (**Indication:** If $\mathcal{E}$ is an orthogonal basis with respect to b then the result is trivial, how can we extend the result to non orthogonal bases?)
- (ii) Is the reverse implication of the previous statement true?
- (iii) Prove one of the implication of the *Sylvester Criterion*: if b is positive definite, then $\det B_i > 0$ for all $1 \leq i \leq n$. (**Indication:** An inner product restricted to a vector subspace is an inner product on the subspace.)

Exercise 4.11. (Sylvester's Criterion, Part II)

We return to the settings of ex. 4.10, and we wish to prove the reverse implication of the Sylvester's Criterion: if $\det B_i > 0$ for all $i \leq n$, then the associated symmetric bilinear form b is positive definite.

We proceed in an inductive manner:

- (i) (*Base case*) Check that the implication holds true for $n = 1$.
- (ii) (*Inductive hypothesis*) Suppose now that the implication holds true for $n - 1$. Let b be a symmetric bilinear form on an n -dimensional vector space $\mathbb{E}$, with $\det B_i > 0$ for all $i \leq n$. Writing

$$B_n = \begin{pmatrix} B_{n-1} & v \\ v^\top & c \end{pmatrix}, \quad x = \begin{pmatrix} y \\ z \end{pmatrix} \quad \text{with } v, y \in \mathbb{K}^{n-1} \text{ and } c, z \in \mathbb{K},$$

express $x^\top B_n x$ in terms of B_{n-1} , v , c , y and z .

- (iii) With the help of the change of variable(s) $w = y + zB_{n-1}^{-1}v$, arrive at the formula

$$x^\top B_n x = w^\top B_{n-1} w + z^2 \sigma, \tag{4.6}$$

determine the exact algebraic expression for $\sigma \in \mathbb{K}$.

- (iv) Verify that

$$\begin{pmatrix} \text{Id}_{n-1} & 0 \\ -v^\top B_{n-1}^{-1} & 1 \end{pmatrix} B_n \begin{pmatrix} \text{Id}_{n-1} & -B_{n-1}^{-1}v \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} B_{n-1} & 0 \\ 0 & \sigma \end{pmatrix}.$$

With this formula, reason that σ has to be strictly positive.

- (v) Going back to Eq. (4.6), argue why the right hand side has to be strictly positive for any $x \neq 0$ (**Indication:** consider the cases where $z = 0$ and $z \neq 0$). With this, conclude that the implication is true for n .

Exercise 4.12. (Alternative formulation of the Laplace expansion)

From ex. 4.12, deduce the following alternative expression for the determinant of a matrix $A \in \mathbb{K}^{n \times n}$:

$$\det A = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(A_{ij}^*), \quad \text{where } i \in \{1, \dots, n\}.$$

Exercise 4.13. (Adjugate matrix)

Let $A \in \mathbb{K}^{n \times n}$ with $n \in \mathbb{N}$ with elements a_{ij} , $1 \leq i, j \leq n$. For each pair of indices (i, j) , we define the cofactor C_{ij} associated to the element a_{ij} as

$$C_{ij} = (-1)^{i+j} \det(A_{ij}^*),$$

where $A_{ij}^* \in \mathbb{K}^{(n-1) \times (n-1)}$ is the submatrix obtained by deleting the i -th row and j -column of A .

We define the adjugate matrix $\text{adj}(A)$ as the matrix whose (i, j) -th element is C_{ji} ($\leftarrow$ the subscript here is not a mistake).

- (i) Let $P = A \cdot \text{adj}(A)$. Write out the expression for the i -th diagonal element P_{ii} and prove that $P_{ii} = \det A$, $\forall i \in \{1, \dots, n\}$.
- (ii) Write out the expression for P_{ik} with $i \neq k$, and prove that $P_{ik} = 0 \ \forall i \neq k$. (**Indication:** the expression for the P_{ik} should look like a Laplace expansion formula for the determinant of a matrix that is not invertible.)
- (iii) Conclude that when A is invertible, we have the following formula for the inverse matrix A^{-1} :

$$A^{-1} = \frac{1}{\det A} \text{adj}(A).$$

What is $\det[\text{adj}(A)]$?

Exercise 4.14. (Multilinearity of the determinant)

Let $w = (w_1, \dots, w_n)^T \in \mathbb{K}^n$, $\lambda \in \mathbb{K}$, and let $s = \sum_{i=1}^n w_i$, prove that

$$\Delta := \det(w + \lambda e_1, w + \lambda e_2, \dots, w + \lambda e_n) = \lambda^n + \lambda^{n-1}s.$$

(**Indication:** title of this exercise.)

Chapter 5

Eigenvalues

So far, we have been insisting quite a lot on the fact that the action of linear mappings in finite dimension is independent of its basis representation. This may give the impression that the representation does not play a role at all in any front. In this chapter, we will see that representation matters when we wish to express the action of linear mappings in the simplest possible way. This has practical implications because, if we have a simple representation, then we save in numerical complexity. Even more importantly, it turns out that the question of “simple representations” has also very deep connections with crucial physical concepts, or the efficient solution to numerous applications. The main idea that we will develop in this course about such representations is the one of eigenvalues.

5.1 Eigenvalues, eigenvectors and eigenspaces of an endomorphism

5.1.1 Eigenvalues and eigenvectors

Eigenvalues and eigenvectors are concepts that are attached to endomorphisms on vector spaces which can be finite, or infinite dimensional.

Definition 5.1: Eigenvalue and eigenvector of an endomorphism

Let E be a $\mathbb{K}$ -vector space and $f \in \mathcal{L}(E)$ be an endomorphism. We say that $(\lambda, x) \in \mathbb{K} \times E$ with $x \neq 0$ is an eigenpair of f if and only if

$$f(x) = \lambda x.$$

The scalar λ is called an eigenvalue and $x \neq 0$ is an eigenvector.

In finite dimension, we can deal with eigenvalues and eigenvectors through matrices as the following proposition shows.

Proposition 5.2

Let E be a finite dimensional space, and $f \in \mathcal{L}(E)$. Then $(\lambda, x) \in \mathbb{K} \times E$ with $x \neq 0$ is an eigenpair of f if and only if

$$[f]_{\mathcal{E}}^{\mathcal{E}} [x]^{\mathcal{E}} = \lambda [x]^{\mathcal{E}}, \quad \text{for any basis } \mathcal{E} \text{ of } E. \quad (5.1)$$

Proof. $\Rightarrow$ The result is simply due to the fact that the matrix representation of $f(x) = \lambda x$ is $[f(x)]^{\mathcal{E}} = [\lambda x]^{\mathcal{E}}$ and therefore $[f]_{\mathcal{E}}^{\mathcal{E}} [x]^{\mathcal{E}} = \lambda [x]^{\mathcal{E}}$.

$\Leftarrow$ We have that $[f(x)]_{\mathcal{E}}^{\mathcal{E}} = [f]_{\mathcal{E}}^{\mathcal{E}} [x]^{\mathcal{E}} = \lambda [x]^{\mathcal{E}} = [\lambda x]^{\mathcal{E}}$. Therefore, since the coordinates are in bijection with the underlying elements (see Proposition 1.17), we deduce that $f(x) = \lambda x$. $\square$

It is important to realize that Eq. (5.1) holds independently of the basis in which we represent the endomorphism f . In that formula, the eigenvalues stay independent of the matrix representation, and only the coordinate representation of the eigenvectors changes. This is in perfect alignment with our original Definition 6.8 of eigenvalues: these are scalars connected to the mapping f , but not to its representation.

It is instructive to confirm that equality (5.1) does not change for two different basis representations via the following calculation. Suppose that we have $\mathcal{E}$ and $\mathcal{F}$ two bases of E , and that we have proven that Eq. (5.1) holds for $\mathcal{E}$. We now want to check that the same holds for a representation in $\mathcal{F}$ by doing the appropriate basis transformations. For this, using the change of basis formula, we have that

$$\begin{aligned} [f]_{\mathcal{F}}^{\mathcal{F}} [x]_{\mathcal{F}}^{\mathcal{F}} &= [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [f]_{\mathcal{E}}^{\mathcal{E}} [\text{Id}]_{\mathcal{F}}^{\mathcal{E}} [x]_{\mathcal{F}}^{\mathcal{F}} \\ &= [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [f]_{\mathcal{E}}^{\mathcal{E}} [x]_{\mathcal{E}}^{\mathcal{E}} \quad (\text{since } [\text{Id}]_{\mathcal{F}}^{\mathcal{E}} [x]_{\mathcal{F}}^{\mathcal{F}} = [x]_{\mathcal{E}}^{\mathcal{E}}) \\ &= \lambda [\text{Id}]_{\mathcal{E}}^{\mathcal{F}} [x]_{\mathcal{E}}^{\mathcal{E}} \quad (\text{since } [f]_{\mathcal{E}}^{\mathcal{E}} [x]_{\mathcal{E}}^{\mathcal{E}} = \lambda [x]_{\mathcal{E}}^{\mathcal{E}}) \\ &= \lambda [x]_{\mathcal{F}}^{\mathcal{F}}. \end{aligned}$$

This confirms that Eq. (5.1) holds indeed for any basis of E .

As soon as E is finite dimensional, discussions about eigenvalues and eigenvectors are often based on their matrix representation. This is the reason why we very often directly speak about eigenvalues and eigenvectors of matrices. While, strictly speaking, this is an abuse of terminology, it is justified by the fact that the space of endomorphisms $\mathcal{L}(E)$ is in bijection with the space $\mathbb{K}^{n \times n}$ of square matrices (we proved this in Lemma 1.35). Indeed, that lemma states that as soon as we fix a basis representation $\mathcal{E}$ of E , the mapping $\text{Mat} : \mathcal{L}(E) \rightarrow \mathbb{K}^{n \times n}$ such that $\text{Mat}(f) = [f]_{\mathcal{E}}^{\mathcal{E}}$ for all $f \in \mathcal{L}(E)$ is an isomorphism. The following definition establishes the rigorous connection between matrices and eigenpairs.

Definition 5.3: Eigenvalue and eigenvector of a matrix

Given a matrix $A \in \mathbb{K}^{n \times n}$, we say that $(\lambda, x) \in \mathbb{K} \times \mathbb{K}^n$ with $x \neq 0$ is an eigenpair of A if and only if

$$Ax = \lambda x.$$

If we fix a basis $\mathcal{E}$ of E , this is equivalent to saying that $(\lambda, \text{Vec}_{\mathcal{E}}^{-1}(x)) \in \mathbb{K} \times E$ is an eigenpair of the associated endomorphism $f_A = \text{Mat}_{\mathcal{E}}^{-1}(A)$ on a vector space E .

From this definition, we deduce that similar matrices have the same eigenvalues, and their eigenvectors are related by a matrix multiplication.

Corollary 5.4

Let A and $B \in \mathbb{K}^{n \times n}$ be two similar matrices such that $B = PAP^{-1}$ for an invertible matrix $P \in \mathbb{K}^{n \times n}$. Then, if $(\lambda, x) \in \mathbb{K} \times (\mathbb{K}^n \setminus \{0\})$ is an eigenpair of A , then (λ, Px) is an eigenpair of B .

Proof. $Ax = \lambda x \iff PAx = \lambda Px \iff PAP^{-1}y = \lambda y$ for $y = Px$. □

Definition 5.5

The set of eigenvalues of an endomorphism $f \in \mathcal{L}(E)$ is called the spectrum and it is defined as

$$\sigma(f) := \{\lambda \in \mathbb{K} \mid \det(f - \lambda \text{Id}_E) = 0\}.$$

Similarly, the spectrum of a matrix $A \in \mathbb{K}^{n \times n}$ is

$$\sigma(A) := \{\lambda \in \mathbb{K} \mid \det(A - \lambda \text{Id}_{\mathbb{K}^n}) = 0\}.$$

Example 5.6

1. Let $E = \mathbb{R}^2$ and $\mathbb{K} = \mathbb{R}$, and consider the matrix

$$A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

We have

$$f_A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, \quad \text{and} \quad f_A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

thus 4 and 0 are eigenvalues of f_A and $(1, 1)^\top$ and $(1, -1)^\top$ are the associated eigenvectors of $f_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Alternatively, we can say that these are the eigenpairs of the matrix A .

2. We again work in $E = \mathbb{R}^2$ and $\mathbb{K} = \mathbb{R}$. We observe that the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has no real eigenvalue. However, if we consider $E = \mathbb{C}^2$ and $\mathbb{K} = \mathbb{C}$, the same matrix has two eigenvalues $\lambda = i$ and $\lambda = -i$ with eigenvectors $(1, -i)^\top$ and $(1, i)^\top$ respectively.

3. Let us now place ourselves in $E = C^\infty(\mathbb{R}, \mathbb{R})$ and $\mathbb{K} = \mathbb{R}$. For every ω in $\mathbb{R}$, let $s_\omega \in E$ denote the function

$$s_\omega(x) := \sin(\omega x), \quad \forall x \in \mathbb{R}.$$

Consider the endomorphism $\Phi \in \mathcal{L}(E)$ defined as

$$\Phi(f) = f'', \quad \forall f \in E.$$

We observe that

$$\Phi(s_\omega) = -\omega^2 s_\omega,$$

thus, s_ω is an eigenvector of Φ for the eigenvalue $-\omega^2$.

5.1.2 Eigenproblems from real-world applications

The PageRank algorithm: A very famous applications of eigenvalues in data science is the *PageRank* algorithm, originally developed by Google to rank webpages according to their importance. The starting point is to model the web as a directed graph:

- each node represents a webpage,
- an arrow from page i to page j indicates that page i contains a hyperlink to page j .

Suppose that the web contains n pages. We define the adjacency matrix $A \in \mathbb{R}^{n \times n}$ by

$$A_{ij} = \begin{cases} 1, & \text{if page } i \text{ links to page } j, \\ 0, & \text{otherwise.} \end{cases}$$

We define the PageRank index r_i of page i as

$$r_i = \sum_{j \in C_i} \frac{r_j}{l_j}$$

where C_i is the set of pages that out-link to page i , and l_j is the amount of pages that page j is out-linking to. Gathering the PageRanks in a vector $r \in \mathbb{R}^n$, we observe that

$$r = P^\top r$$

where $P \in \mathbb{R}^{n \times n}$ is defined as

$$P_{ij} = \begin{cases} \frac{1}{l_j}, & \text{if page } i \text{ links to page } j, \\ 0, & \text{otherwise.} \end{cases}$$

The matrix P has the following interpretation: we imagine a user navigating the web, and suppose that it is currently on page i . If the user surfs by clicking uniformly at random on the available hyperlinks, then $p_{i,j}$ can be interpreted as the probability that the user goes from page i to page j . We observe that each row of P sums to one

$$\sum_{j=1}^n P_{ij} = 1, \quad 1 \leq i \leq n.$$

Equivalently, this means that

$$P \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix},$$

so $\lambda = 1$ is an eigenvalue of P associated to the eigenvector $v = (1, \dots, 1)^\top$. A matrix with such a property is called a *stochastic matrix*.

In practice, the web contains problematic pages (for example, pages with no outgoing links). To avoid this issue, we introduce a modified transition matrix (called the Google matrix)

$$G := \alpha P + \frac{(1-\alpha)}{n} \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix},$$

where $0 < \alpha < 1$ is called a damping factor. It seems it is typically chosen as $\alpha = 0.85$. We observe that G is also a stochastic matrix. This modification means that:

- with probability α , the user follows a hyperlink,
- with probability $1 - \alpha$, the user jumps to a random page.

We now update our definition of PageRank as the vector solving $r = G^\top r$. Under suitable assumptions (for example, if the web graph is sufficiently connected), the Perron–Frobenius theorem (which we will not have time to prove in this course) guarantees that:

- $\lambda = 1$ is the largest eigenvalue of $G^\top$ in modulus,
- it admits a positive eigenvector, which will be our PageRank vector r .
- this eigenvector is unique up to multiplication by a scalar.

This makes the search for the ranking r a well-defined task.

For very large matrices, computing the full eigendecomposition of $P^\top$ is impossible. Instead, one uses the so-called power-method, which is the iterative scheme

$$x^{(k+1)} = G^\top x^{(k)},$$

which provably converges to the dominant eigenvector r . If time permits, we will discuss the power-method in the lecture.

Mechanical systems: Any mechanical system has the ability to oscillate. Analogous phenomena also exist in many other systems, such as electrical, optical, or chemical systems. Focusing on the mechanical case, it is well known that if the system is excited at specific frequencies, it will enter resonance (i.e., it will vibrate with large amplitudes). Depending on the situation, this may have beneficial consequences, but it can also have catastrophic effects. A classical example of it is the collapse of the Tacoma Bridge under the action of a moderate wind blowing at one of the bridge's resonance frequencies:

<https://www.youtube.com/watch?v=j-zczJXSxw&spfreload=10>

For these reasons, understanding and computing the vibration frequencies of a system (whether mechanical, electrical, or otherwise) is crucial.

A simple model describing these oscillatory phenomena is given by the following problem, known as a Sturm–Liouville problem. We seek $\lambda \in \mathbb{C}$ and a function $u(x)$ satisfying the partial differential equation

$$\begin{cases} -u''(x) - \lambda c(x)u(x) = 0, & x \in]0, 1[, \\ u(0) = u(1) = 0, \end{cases} \quad (5.2)$$

with $c(x) > 0$, $\forall x \in [0, 1]$. In the case where the function c is constant, we are dealing with a similar eigenvalue problem as in Example 5.6 (see point 3). Here, we consider the space

$$E = \{f \in \mathcal{C}^\infty([0, 1], \mathbb{R}) : f(0) = f(1) = 0\}.$$

We observe that this is a subspace of $\mathcal{F}([0, 1], \mathbb{R})$. Defining the endomorphism

$$\Phi(f) = -\frac{1}{c}f'', \quad \forall f \in E,$$

problem (5.2) can be expressed as finding $(\lambda, u) \in \mathbb{R} \times (E \setminus \{0\})$ such that $\Phi(u) = \lambda u$. The solutions to this problem are explicit:

$$\begin{cases} \lambda = (k\pi/c)^2, \\ u(x) = \sin(k\pi x), \quad k = 1, 2, \dots \end{cases}$$

When c is no longer constant, explicit solutions can no longer be obtained. One therefore resorts to discretization: we seek values u_i approximating the exact solution $u(x_i)$ for $i = 0, \dots, n$, with points x_i uniformly distributed on $[0, 1]$, i.e.,

$$x_i = i/n, \quad i = 0, \dots, n.$$

Approximating the second derivative with the finite difference formula

$$u''(x_i) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}, \quad \text{for } 0 < i < n,$$

and with $h = 1/n$, we arrive at a linear system of size $n - 1$

$$Av - \lambda Cv = 0,$$

with

$$A = n^2 \begin{pmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & -1 & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \dots & 0 & -1 & 2 \end{pmatrix}, \quad C = \text{diag}(c(x_i))_{i=1}^{n-1}, \quad v = \begin{pmatrix} u_1 \\ \vdots \\ u_{n-1} \end{pmatrix}$$

Assuming that the entries of $c(x_i) \neq 0$, we can invert C and write the task as finding the eigenvalues (λ, v) of the matrix $B = C^{-1}A$, which satisfy $Bv = \lambda v$.

More on oscillations:

- In electronics, an RLC circuit is an electrical circuit containing these three basic components:
 1. R (Resistor): dissipates energy, typically as heat, and introduces damping into the system.
 2. L (Inductor): stores energy in a magnetic field and resists changes in current.
 3. C (Capacitor): stores energy in an electric field and resists changes in voltage.

The interplay between the inductor and the capacitor creates oscillatory behavior, since energy can be exchanged back and forth between the magnetic field of the inductor and the electric field of the capacitor. The oscillations behave like a harmonic oscillator, which is mathematically modeled with equations similar to Eq. (5.2). The resistor dissipates part of this energy, and damps the oscillations. RLC electrical circuits can enter resonance when excited at certain frequencies. This property makes them useful for frequency filtering in many devices (for example, analog radio).

- Analogously to RLC circuits, it is possible to construct optical cavities that allow only a specific wavelength to pass through incident light. This is achieved through the creation of interference between waves. For some wavelengths, this interference is constructive, producing outgoing light. For others, it is destructive, and that portion of the light is blocked. This is essential, for example, in the construction of lasers.
- The general underlying idea in all these examples is that waves of any nature can be filtered through interference phenomena. Devices that perform this function are called interferometers.

5.1.3 Eigenspaces

The following result gives several equivalent characterizations of eigenvalues and eigenvectors. It also introduces the concept of eigenspaces.

Lemma 5.7: Characterizations of eigenvalues and vectors

At the level of endomorphisms, the following are equivalent

$$\begin{aligned}
 & (\lambda, x) \in \mathbb{K} \times (E \setminus \{0_E\}) \text{ is an eigenpair of } f \in \mathcal{L}(E) \\
 \iff & (f - \lambda \text{Id}_E) \text{ is not injective} \\
 \iff & \ker(f - \lambda \text{Id}_E) \neq \{0_E\} \\
 \iff & \det(f - \lambda \text{Id}_E) = 0
 \end{aligned} \tag{5.3}$$

The subspace

$$V_\lambda^f := \ker(f - \lambda \text{Id}_E) \subset E$$

is called the eigensubspace of f associated to the eigenvalue λ .

The same holds at the level of matrices:

$$\begin{aligned}
 & (\lambda, x) \in \mathbb{K} \times (\mathbb{K}^n \setminus \{0_{\mathbb{K}^n}\}) \text{ is an eigenpair of } A \in \mathbb{K}^{n \times n} \\
 \iff & (A - \lambda \text{Id}_{\mathbb{K}^n}) \text{ is not injective} \\
 \iff & \ker(A - \lambda \text{Id}_{\mathbb{K}^n}) \neq \{0_{\mathbb{K}^n}\} \\
 \iff & \det(A - \lambda \text{Id}_{\mathbb{K}^n}) = 0
 \end{aligned}$$

The subspace

$$V_\lambda^A := \ker(A - \lambda \text{Id}_{\mathbb{K}^n}) \subset \mathbb{K}^n$$

is called the eigensubspace of A associated to the eigenvalue λ .

As a consequence of the above characterizations, we see that if 0 is an eigenvalue of f , then f is not injective, and the associated eigensubspace is $V_0^f = \ker(f)$.

Example 5.8

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$. We see that $f_A((1, -1, 0)^\top) = (1, -1, 0)^\top$ therefore 1 is an eigenvalue of A .

- The associated subspace is $V_1 = \ker(A - \text{Id}) = \ker(B)$ with $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$
 - Matrix B has rank 1, thus the dimension of its kernel is 2. By observing the relationships between the columns of B , we infer that $V_1 = \ker(B) = \text{span}\{(1, -1, 0)^\top, (0, 1, -1)^\top\}$.
-

5.1.4 The characteristic polynomial

Lemma 5.9

Let E be an n -dimensional $\mathbb{K}$ -vector space, $\mathcal{E}$ be a basis of E and $f \in \mathcal{L}(E)$. The function

$$p_f : \mathbb{K} \rightarrow \mathbb{K} \\ t \mapsto p_f(t) := \det(f - t \text{Id}_E) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t \text{Id}_{\mathbb{K}^n})$$

is a polynomial of degree n , called the characteristic polynomial of f . It has the following properties:

1. p_f is independent of the basis $\mathcal{E}$,
2. The coefficients relative to t^0 , t^{n-1} and t^n are $\det(f)$, $(-1)^{n-1} \text{Tr}(f)$ and $(-1)^n$.
3. It holds that

$$p_f(\lambda) = 0 \iff \lambda \text{ is an eigenvalue of } f.$$

One can define in the same way the characteristic polynomial of a matrix $A \in \mathbb{K}^{n \times n}$, and it will have the same properties.

Before proving the lemma, it is instructive to manipulate the characteristic polynomial to get familiar with it. Observe that for a given matrix $A = (a_1 | \dots | a_n) \in \mathbb{K}^{n \times n}$,

$$p_A(t) = \det(A - t \text{Id}) = \det(a_1 - te_1 | \dots | a_n - te_n) = \det \begin{pmatrix} a_{1,1} - t & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} - t & \dots & a_{2,n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n,1} & \dots & \dots & a_{n,n} - t \end{pmatrix}$$

Developping $p_A(t) = \det(a_1 - te_1 | \dots | a_n - te_n)$ by using linearity in each of the columns. Since each column is a sum of two vectors, the fully developed expression of $p_A(t)$ is composed of 2^n summands, two of them being

$$\det(a_1 | \dots | a_n) = \det(A), \quad \text{and} \quad \det(-te_1 | \dots | -te_n) = (-t)^n. \quad (5.4)$$

The other terms are of the form $\det(\dots | -te_i | \dots | a_j | \dots)$ where the $-te_i$ appear in k columns and the $n - k$ other columns are of the form a_j . For example, in the case $n = 3$

$$p_A(t) = \det(a_1 | a_2 | a_3) + \det(-te_1 | a_2 | a_3) + \det(a_1 | -te_2 | a_3) + \det(a_1 | a_2 | -te_3) \\ + \det(-te_1 | -te_2 | a_3) + \det(-te_1 | a_2 | -te_3) + \det(a_1 | -te_2 | -te_3) \\ + \det(-te_1 | -te_2 | -te_3). \quad (5.5)$$

Proof. 1. The mapping p_f is a polynomial by direct application of Proposition 4.6. In addition, by Definition 4.16, $\det(f - t \text{Id}_E) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t \text{Id}_{\mathbb{K}^n})$ and the fact that this holds for any basis $\mathcal{E}$ was proven below Definition 4.16 (the argument was that for any two bases $\mathcal{E}$ and $\mathcal{F}$, $[f]_{\mathcal{E}}^{\mathcal{E}}$ and $[f]_{\mathcal{F}}^{\mathcal{F}}$ are similar matrices, and the determinant of such matrices has the same value.)

2. By Eq. (5.4), we readily see that the coefficient in front of t^n is $(-1)^n$, and that $\det(A)$ is the

coefficient associated to t^0 . To derive the coefficient in front of t^{n-1} , we develop

$$\begin{aligned}
& \sum_{i=1}^n \det(-te_1 | \dots | -te_{i-1} | a_i | -te_{i+1} | \dots | -te_n) \\
&= (-t)^{n-1} \sum_{i=1}^n \det(e_1 | \dots | e_{i-1} | a_i | e_{i+1} | \dots | e_n) \\
&= (-t)^{n-1} \sum_{i=1}^n \det(e_1 | \dots | e_{i-1} | \sum_{k=1}^n a_{k,i} e_k | e_{i+1} | \dots | e_n) \\
&= (-t)^{n-1} \sum_{i=1}^n \sum_{k=1}^n a_{k,i} \det(e_1 | \dots | e_{i-1} | e_k | e_{i+1} | \dots | e_n) \\
&= (-t)^{n-1} \sum_{i=1}^n a_{i,i} \det(e_1 | \dots | e_{i-1} | e_i | e_{i+1} | \dots | e_n) \quad (\text{two identical columns make the det zero}) \\
&= \text{Tr}(A)(-t)^{n-1}
\end{aligned}$$

3. The fact that eigenvalues are the roots of the characteristic polynomial holds directly by the definition (or by characterization Eq. (5.3)). □

Corollary 5.10

For $n = 2$, $p_f(x) = x^2 - \text{Tr}(f)x + \det(f)$.

Example 5.11

1. The characteristic polynomial of $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ is $p_A(t) = (t-4)(t-1)^2$ so 4 and 1 are eigenvalues with multiplicities 1 and 2.
2. Let $E = \mathbb{R}^{2 \times 2}$ be the space of 2×2 matrices with real coefficients (meaning that $\mathbb{K} = \mathbb{R}$). Fix $U = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and define the endomorphism

$$\Phi(A) = UA - AU, \quad \forall A \in \mathbb{R}^{2 \times 2}.$$

The characteristic polynomial is $p_\Phi(t) = t^2(t-2)(t+2)$ so 0, 2, -2 are eigenvalues with multiplicities 2, 1, 1.

5.2 Diagonalization

5.2.1 Definition and main result

For an endomorphism $f \in \mathcal{L}(E)$ in an n -dimensional space E , we want to find a basis in which its matrix representation $[f]_{\mathcal{E}} = ([f(e_i)]^{\mathcal{E}} | \dots | [f(e_n)]^{\mathcal{E}})$ is as simple as possible. Of course, “simple” is a vague word. It means that we want that the matrix has as many zeros as possible. This is why we search for an $\mathcal{E}$ such that $[f]_{\mathcal{E}} = ([f(e_i)]^{\mathcal{E}} | \dots | [f(e_n)]^{\mathcal{E}})$ is a diagonal matrix. In the same spirit, if our focus is a matrix $A \in \mathbb{K}^{n \times n}$, we would like to find a change of basis which provides a diagonal matrix.

It turns out that, depending on f (resp. A) finding such a basis is not always going to be possible. We need certain conditions on the endomorphism (resp. the matrix) which we are going to discuss in this section. Let us first start by defining very clearly what we understand by diagonalization.

Definition 5.12

1. Let E be an n -dimensional $\mathbb{K}$ -vector space, and $f \in \mathcal{L}(E)$. We say that f is diagonalizable if and only if there exists a basis $\mathcal{E}$ of E made of eigenvectors of f .
2. We say that a matrix $A \in \mathbb{K}^{n \times n}$ is diagonalizable if and only if there exists a basis $\mathcal{E}$ of $\mathbb{K}^n$ made of eigenvectors of A .

Lemma 5.13

Let $\mathcal{B}$ be any basis of E :

1. f is diagonalizable if and only if $[f]_{\mathcal{B}}^{\mathcal{B}}$ is similar to a diagonal matrix.
2. A is diagonalizable if and only if it is similar to a diagonal matrix, namely, there exists an invertible matrix P and a diagonal matrix $D \in \mathbb{K}^{n \times n}$ such that

$$A = PDP^{-1}$$

Proof. We only prove point 1 of the lemma. Point 2 is obtained by leveraging the bijection between the space of endomorphisms $\mathcal{L}(E)$ and the space of matrices $\mathbb{C}^{n \times n}$ as we proved in Eq. (1.3).

$\Rightarrow$ Suppose $f \in \mathcal{L}(E)$ is diagonalizable. Then there exists a basis $\mathcal{E}$ of E made of eigenvectors of f . On that basis, $[f]_{\mathcal{E}}^{\mathcal{E}}$ is diagonal. By the change of basis formula, $[f]_{\mathcal{B}}^{\mathcal{B}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{B}} [f]_{\mathcal{E}}^{\mathcal{E}} [\text{Id}]_{\mathcal{B}}^{\mathcal{E}}$, which means that $[f]_{\mathcal{E}}^{\mathcal{E}}$ and $[f]_{\mathcal{B}}^{\mathcal{B}}$ are similar.

$\Leftarrow$ Suppose that $[f]_{\mathcal{B}}^{\mathcal{B}}$ is similar to a diagonal matrix. Then, there exists an invertible matrix P and a diagonal matrix $D \in \mathbb{K}^{n \times n}$ such that $[f]_{\mathcal{B}}^{\mathcal{B}} = PDP^{-1}$. We define the set of vectors $\{e_1, \dots, e_n\}$ of E where e_j is the vector whose coordinate representation in $\mathcal{E}$ is the j -th column of P . In other words, $e_j = \sum_{i=1}^n p_{i,j} b_i$ for all $1 \leq j \leq n$, and $[\text{Id}]_{\mathcal{E}}^{\mathcal{B}} = P$. The set of vectors $\mathcal{E}$ is linearly independent since P is invertible. They are therefore a basis of E . In addition, we have that $[f]_{\mathcal{B}}^{\mathcal{B}} = [\text{Id}]_{\mathcal{E}}^{\mathcal{B}} D [\text{Id}]_{\mathcal{B}}^{\mathcal{E}}$. Therefore $D = [\text{Id}]_{\mathcal{B}}^{\mathcal{E}} [f]_{\mathcal{B}}^{\mathcal{B}} [\text{Id}]_{\mathcal{E}}^{\mathcal{B}} = [f]_{\mathcal{E}}^{\mathcal{E}}$. From this equality, it follows that $f(e_j) = d_{j,j} e_j$ so the e_j are eigenvectors of f with eigenvalues $\lambda_j = d_{j,j}$. We have thus proven that there exists a basis $\mathcal{E}$ of E made of eigenvectors of f , so f is diagonalizable. $\square$

Our ability to diagonalize an endomorphism deeply relies on the interplay of the so-called algebraic and geometric multiplicities of the associated eigenvalues.

Definition 5.14: Algebraic and Geometric Multiplicity

Let λ be an eigenvalue of $f \in \mathcal{L}(E)$:

1. The algebraic multiplicity of λ , denoted m_{λ} , is its multiplicity as a root of the characteristic polynomial p_f . It is the largest value m such that $(X - \lambda)^m$ divides p_f .
2. The geometric multiplicity of λ is defined as the dimension of its eigenspace, $d_{\lambda} := \dim(V_{\lambda}^f)$.

It is important to realize that both multiplicities can differ. For example, the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad p_A(t) = (t - 1)^2, \quad \lambda = 1, \quad V_1^A = \text{span}\{(1, 0)^T\}, \quad d_1 = 1, \quad m_1 = 2.$$

It will turn out that for an endomorphism (resp. a matrix) to be diagonalizable, we need that both multiplicities agree for all eigenvalues. To come to this result, we need to prove some intermediate statements which are valuable in their own right. We start by defining the concept of a stable subspace.

Definition 5.15

Let E be a vector space, F be a subspace of E , and $f \in \mathcal{L}(E)$. We say that F is stable under f if and only if $f(F) \subset F$.

The most straightforward examples of stable subspaces of $f \in \mathcal{L}(E)$ are $F = \{0\}$, E , $\text{im}(f)$ and $\ker(f)$. One of particular interest to us in the following is $V_\lambda^f = \ker(f - \lambda \text{Id})$ when λ is an eigenvalue of f . This is a stable subspace since for every $x \in V_\lambda^f$, $f(x) = \lambda x \in V_\lambda^f$.

The following result holds for any stable subspace.

Theorem 5.16

If F is a stable subspace of $f \in \mathcal{L}(E)$, then the characteristic polynomial $p_{f|_F}$ of the restriction $f|_F : F \rightarrow F$ divides the characteristic polynomial p_f of f .

Proof. We take a basis $\mathcal{E} = \{e_1, \dots, e_m, e_{m+1}, \dots, e_n\}$ of E , with $\mathcal{F} = \{e_1, \dots, e_m\}$ being a basis of the subspace F . Since F is stable under f , for $1 \leq j \leq m$, $f(e_j)$ must be a linear combination of the vectors $\{e_1, \dots, e_m\}$. Thus

$$[f]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} A & B \\ 0_{(n-m) \times m} & C \end{pmatrix}, \quad A \in \mathbb{K}^{m \times m}, \quad B \in \mathbb{K}^{m \times (n-m)}, \quad C \in \mathbb{K}^{(n-m) \times (n-m)}.$$

Computing by blocks the characteristic polynomial $p_f(t) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t \text{Id}_n)$ (see ex. 4.9), it follows that

$$p_f(t) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t \text{Id}_n) = \det(A - t \text{Id}_m) \det(C - t \text{Id}_{(n-m)}).$$

However $A = [f|_F]_{\mathcal{F}}^{\mathcal{F}}$ since it is the matrix of the restriction of f to F . Therefore $\det(A - t \text{Id}_m) = p_{f|_F}(t)$, which implies that

$$p_f(t) = p_{f|_F}(t) \det(C - t \text{Id}_{(n-m)}).$$

This shows that $p_{f|_F}$ divides p_f . □

Theorem 5.16 allows us to prove the following general result connecting algebraic and geometric multiplicities.

Theorem 5.17

For every eigenvalue $\lambda \in \mathbb{K}$ of an endomorphism $f \in \mathcal{L}(E)$, $d_\lambda \leq m_\lambda$.

Proof. We apply Theorem 5.16 to the stable subspace V_λ^f whose dimension is, by definition, the geometric multiplicity d_λ . Denoting $f_\lambda : V_\lambda^f \rightarrow V_\lambda^f$ the restriction of f to V_λ^f , we observe that its characteristic polynomial is $p_{f_\lambda}(x) = (x - \lambda)^{d_\lambda}$. By Theorem 5.16, there exists $r \in \mathbb{P}[\mathbb{K}]$ such that $p_f(x) = p_{f_\lambda}(x)r(x)$. Now, the definition of the algebraic multiplicity of λ is that it is the largest value m_λ such that $(x - \lambda)^{m_\lambda}$ divides p_f . Therefore $d_\lambda \leq m_\lambda$. □

One of the essential results to prove our main theorem for diagonalization is the following.

Theorem 5.18

Let $\{\lambda_1, \dots, \lambda_k\}$ be a subset of distinct eigenvalues of $f \in \mathcal{L}(E)$, namely $\lambda_i \neq \lambda_j, 1 \leq i \neq j \leq k$. Then $V_{\lambda_1}^f + \dots + V_{\lambda_k}^f$ is in direct sum. As a consequence, if $x_1, \dots, x_k$ are eigenvectors associated to $\lambda_1, \dots, \lambda_k$ respectively, then they are linearly independent.

Proof. We proceed by induction. The base case $k = 1$ is trivially true.

For $k > 1$, we assume that $V_{\lambda_1} + \dots + V_{\lambda_{k-1}}$ is in direct sum. Our goal is to show that this implies that $V_{\lambda_1} + \dots + V_{\lambda_k}$ is also in direct sum. For this, we will use point 2 of Theorem 1.24, and prove that if $0 = x_1 + \dots + x_k$ with $x_i \in V_{\lambda_i}^f$ for $1 \leq i \leq k$, then $x_i = 0$. So let $x_i \in V_{\lambda_i}^f$ be such that $0 = x_1 + \dots + x_k$. Then $x_k = -x_1 - \dots - x_{k-1}$. Applying f to this equality, and using that $f(x_k) = \lambda_k x_k$, we deduce that $\lambda_k x_k = -\lambda_k x_1 - \dots - \lambda_k x_{k-1}$. Since we also have that $f(x_i) = \lambda_i x_i$ for $1 \leq i < k$, we also have that $\lambda_k x_k = -\lambda_k x_1 - \dots - \lambda_k x_{k-1} = -\lambda_1 x_1 - \dots - \lambda_{k-1} x_{k-1}$. Therefore

$$(\lambda_1 - \lambda_k)x_1 + \dots + (\lambda_{k-1} - \lambda_k)x_{k-1} = 0.$$

By our induction hypothesis, $(\lambda_1 - \lambda_k)x_1, \dots, (\lambda_{k-1} - \lambda_k)x_{k-1}$ are in direct sum. Therefore $(\lambda_i - \lambda_k)x_i = 0$ for $1 \leq i \leq k-1$. Since the eigenvalues are all distinct, it follows that $x_1 = \dots = x_{k-1} = 0$, and from one of the previous equalities this implies that $x_k = 0$ as well. This shows that $V_{\lambda_1} + \dots + V_{\lambda_k}$ is in direct sum and we write $V_{\lambda_1} \oplus \dots \oplus V_{\lambda_{k-1}}$.

From our derivation, we directly obtain that if $x_1, \dots, x_k$ are eigenvectors associated to $\lambda_1, \dots, \lambda_k$ respectively, then they are linearly independent. Indeed, suppose that $c_1, \dots, c_k \in \mathbb{K}$ are such that $c_1 x_1 + \dots + c_k x_k = 0$. Since the eigensubspaces are in direct sum, then $c_i x_i = 0$ for all $1 \leq i \leq k$, and since the $x_i \neq 0$ (because they are eigenvectors, and by definition they are nonzero), this implies that $c_i = 0$, thus proving linear independence. $\square$

From Theorem 5.18, we can already deduce a first simple diagonalization theorem.

Lemma 5.19

Let $f \in \mathcal{L}(E)$ be an endomorphism of an n -dimensional space. If f has n distinct eigenvalues $\lambda_1 \neq \lambda_2 \neq \dots \neq \lambda_n$, then f is diagonalizable.

Proof. This is the topic of ex. 5.14. $\square$

In Lemma 5.19, we observe that we are in a case in which the algebraic and geometric multiplicities of the eigenvalues are equal and their value is 1, $d_{\lambda_i} = m_{\lambda_i} = 1$ for $1 \leq i \leq n$. The theorem below addresses the general case in which these dimensions can be larger, and it is the key result about the diagonalization of endomorphisms in finite-dimensional spaces.

Theorem 5.20: Diagonalization of an endomorphism

Let $f \in \mathcal{L}(E)$ be an endomorphism in an n -dimensional $\mathbb{K}$ -vector space E with characteristic polynomial $p_f \in \mathbb{P}_n[\mathbb{K}]$, and with $\sigma(f) = \{\lambda_1, \dots, \lambda_k\}$ being the set of its eigenvalues. The following statements are equivalent:

1. f is diagonalizable, that is, there exists a basis $\mathcal{E}$ of eigenvectors of f in which $[f]_{\mathcal{E}}^{\mathcal{E}}$ is diagonal.
2. The characteristic polynomial can be written as

$$p_f(t) = (-1)^n (t - \lambda_1)^{m_{\lambda_1}} \dots (t - \lambda_k)^{m_{\lambda_k}},$$

and the geometric multiplicity m_{λ_i} is equal to the algebraic multiplicity $d_i = \dim(V_{\lambda_i}^f)$.

3. E can be expressed as a direct sum of eigenspaces, $E = V_{\lambda_1}^f \oplus \dots \oplus V_{\lambda_k}^f$.

Proof. We will prove that $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$.

$1 \Rightarrow 2$: We assume that there exists a basis $\mathcal{E}$ such that $[f]_{\mathcal{E}}^{\mathcal{E}}$ is diagonal. Up to re-indexing the basis elements, we cluster the identical eigenvectors in such a way that

$$[f]_{\mathcal{E}}^{\mathcal{E}} = \text{diag}(\underbrace{\lambda_1, \dots, \lambda_1}_{m_{\lambda_1} \text{ times}}, \dots, \underbrace{\lambda_k, \dots, \lambda_k}_{m_{\lambda_k} \text{ times}}).$$

We then directly obtain the factorized formula of the characteristic polynomial

$$p_f(t) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t\text{Id}) = (-1)^n \prod_{i=1}^k (t - \lambda_i)^{m_{\lambda_i}},$$

We next show that $d_{\lambda_i} = m_{\lambda_i}$ for $1 \leq i \leq k$. Since we know that p_f is a polynomial of degree n , it holds that $n = m_{\lambda_1} + \dots + m_{\lambda_k}$. Furthermore, we observe that the rank of the matrix $[f]_{\mathcal{E}}^{\mathcal{E}} - \lambda_i \text{Id}$ is $n - m_{\lambda_i}$ since this matrix has exactly m_{λ_i} zeros columns. Therefore,

$$d_{\lambda_i} = \dim(\ker(f - \lambda_i \text{Id})) = \dim(E) - \text{rank}(f - \lambda_i \text{Id}) = n - (n - m_{\lambda_i}) = m_{\lambda_i}.$$

$2 \Rightarrow 3$: We assume that point 2 holds. The factorization of p_f gives us the roots $\lambda_1, \dots, \lambda_k$ which, by Lemma 5.9, are eigenvalues of f . Since these roots have distinct values, Lemma 5.19 gives us the direct sum of eigenspaces $V_{\lambda_1}^f \oplus \dots \oplus V_{\lambda_k}^f$. In addition,

$$\dim(V_{\lambda_1}^f \oplus \dots \oplus V_{\lambda_k}^f) = \dim(V_{\lambda_1}^f) + \dots + \dim(V_{\lambda_k}^f) = d_{\lambda_1} + \dots + d_{\lambda_k} = m_{\lambda_1} + \dots + m_{\lambda_k} = n = \dim(E).$$

Therefore $E = V_{\lambda_1}^f \oplus \dots \oplus V_{\lambda_k}^f$.

$3 \Rightarrow 1$: For every eigenspace V_{λ_i} , we can extract a basis of d_{λ_i} linearly independent eigenvectors of f associated to the eigenvalue λ_i . We call $\mathcal{E}_i = \{e_{i,1}, \dots, e_{i,d_{\lambda_i}}\}$ the set of these eigenvectors. We then consider the union of all the eigenvectors, obtaining $\mathcal{E} = \bigcup_{i=1}^k \mathcal{E}_i$. This set has exactly n vectors since $m_{\lambda_i} = d_{\lambda_i}$ for $1 \leq i \leq k$. Therefore, it forms a basis of E , and $[f]_{\mathcal{E}}^{\mathcal{E}}$ is diagonal. $\square$

5.2.2 How to study diagonalizability in practice?

For a given matrix $A \in \mathbb{K}^{n \times n}$, the following steps allow us to study if A is diagonalizable:

1. Compute the characteristic polynomial of A , find its roots and write it in factorized form. If $\mathbb{K} = \mathbb{R}$, the roots have to be real, and if $\mathbb{K} = \mathbb{C}$ the roots are complex.
2. For every eigenvalue $\lambda \in \sigma(A)$, describe the associated eigenspace $V_{\lambda}^A = \ker(A - \lambda \text{Id})$. The best way to express the eigenspace is as the span of basis vectors $\{u_1, \dots, u_{d_{\lambda}}\}$. The dimension of V_{λ}^A is d_{λ} .
3. Check if the dimension d_{λ} of each V_{λ}^A is equal to the power m_{λ} that appears in the term $(t - \lambda)^{m_{\lambda}}$ of the characteristic polynomial. If $d_{\lambda} = m_{\lambda}$ for all eigenvalues $\lambda \in \sigma(A)$, then A is diagonalizable.
4. If A is diagonalizable, gather all eigenbases together to form in a basis $\mathcal{U} = \{u_1, \dots, u_n\}$ of $\mathbb{K}^n$.

$$A = P \text{diag}(\lambda_1, \dots, \lambda_n) P^{-1}, \quad \text{with} \quad P = (u_1 | \dots | u_n).$$

For a given endomorphism $f \in \mathcal{L}(E)$ of an n -dimensional space E , fix a basis $\mathcal{E} = \{e_1, \dots, e_n\}$ of E . To study its diagonalizability through the one of its matrix representation $[f]_{\mathcal{E}}^{\mathcal{E}}$. If the matrix is diagonalizable, then:

1. the eigenvalues of $[f]_{\mathcal{E}}^{\mathcal{E}}$ are equal to the ones of f (see Definition 5.3).
2. from the eigenvectors of $[f]_{\mathcal{E}}^{\mathcal{E}}$ gathered in the matrix $P = (u_1 | \dots | u_n)$, we deduce the eigenvalues v_j of f by using the fact that $[v_j]_{\mathcal{E}}^{\mathcal{E}} = u_j$. Thus $v_j = \sum_{i=1}^n p_{i,j} e_i$ for all $1 \leq j \leq n$.

5.3 The Cayley-Hamilton Theorem

We start this section by a flash-back to Section 1.2.3 about polynomial spaces $\mathbb{P}[X]$. In that section, we saw that polynomials can be defined over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, but we can also define polynomials over the matrices from $\mathbb{K}^{n \times n}$ (see Eq. (1.1)) and endomorphisms from $\mathcal{L}(E)$ with E being a vector space (see Eq. (1.2)).

For every $f \in \mathcal{L}(E)$, we have defined its characteristic polynomial in $\mathbb{K}$,

$$p_f : \mathbb{K} \rightarrow \mathbb{K}$$

$$t \mapsto p_f(t) = \det(f - t\text{Id}_E) = \det([f]_{\mathcal{E}}^{\mathcal{E}} - t\text{Id}_{\mathbb{K}^{n \times n}}) = \sum_{i=0}^n c_i t^i,$$

where the coefficients $c_i \in \mathbb{K}$ are deduced by developping the determinant in the spirit of the computations revolving around Eq. (5.4) and Eq. (5.5). Let us now consider the polynomial with the same coefficients, but we define it for endomorphisms. We overload the notation p_f , and write

$$p_f : \mathcal{L}(E) \rightarrow \mathbb{K}$$

$$T \mapsto p_f(T) := \sum_{i=0}^n c_i T^i, \quad \forall T \in \mathcal{L}(E)$$

and also for matrices,

$$p_f : \mathbb{K}^{n \times n} \rightarrow \mathbb{K}$$

$$A \mapsto p_f(A) := \sum_{i=0}^n c_i A^i, \quad \forall A \in \mathbb{K}^{n \times n}.$$

The Cayley-Hamilton theorem states that $f \in \mathcal{L}(E)$ (resp. $[f]_{\mathcal{E}}^{\mathcal{E}}$) is a zero of the characteristic polynomial when defined for endomorphisms (resp. for matrices).

Theorem 5.21: (Cayley-Hamilton)

Let $f \in \mathcal{L}(E)$. Then

$$p_f(f) = 0_E, \quad \text{and} \quad p_f([f]_{\mathcal{E}}^{\mathcal{E}}) = 0_{\mathbb{K}^{n \times n}} \quad \text{for any basis } \mathcal{E} \text{ of } E.$$

An elegant proof of this theorem relies on the concept of companion matrix, and cyclic subspaces which we define next.

Lemma 5.22

The companion matrix to a monic polynomial

$$p(X) = c_0 + c_1 X + \cdots + c_{n-1} X^{n-1} + X^n$$

is the square $n \times n$ matrix

$$C = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 & -c_0 \\ 1 & 0 & 0 & \cdots & 0 & 0 & -c_1 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -c_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & -c_{n-2} \\ 0 & 0 & 0 & \cdots & 0 & 1 & -c_{n-1} \end{pmatrix}.$$

This matrix is such that $p_C(X) = (-1)^n p(x)$.

Proof. Left as an exercise. □

The companion matrix guarantees that for every polynomial $p \in \mathbb{P}_n[\mathbb{K}]$, there always exists a matrix (the companion matrix) whose characteristic polynomial is exactly p . Since matrices are connected to endomorphisms, we now ask what type of endomorphism is connected to the companion matrix. The next lemma answers this question through the concept of cyclic subspaces.

Lemma 5.23: (Cyclic subspaces)

Let E be an n dimensional space, and let:

1. $f \in \mathcal{L}(E)$, and $v \in E$.
2. $1 \leq k \leq n$ be the smallest integer such that $\{w, f(v), f^2(v), \dots, f^{k-1}(v)\}$ is a linearly indepen-

dent set.

3. $c_0, \dots, c_{k-1} \in \mathbb{K}$ be such that

$$f^k(v) = -c_0 f^0(v) - c_1 f^1(v) - c_2 f^2(v) - \dots - c_{k-1} f^{k-1}(v). \quad (5.6)$$

The space

$$Z(v, f) := \text{span}\{f^0(v), f^1(v), f^2(v), \dots, f^{k-1}(v)\}$$

is called the *cyclic subspace generated by v and f* , and

$$m_v(X) := c_0 + c_1 X + \dots + c_{k-1} X^{k-1} + X^k$$

is a polynomial of degree k called the f -annihilator of v . The following properties hold:

1. $\dim(Z(v, f)) = k$ and $\mathcal{B} = \{f^0(v), f^1(v), f^2(v), \dots, f^{k-1}(v)\}$ is a basis of $Z(v, f)$,
2. $Z(v, f)$ is a stable subspace of f ,
3. $m_v(f)(v) = 0_E$.
4. Let $f_v : Z(v, f) \rightarrow Z(v, f)$ be the restriction of f onto $Z(v, f)$. Its matrix representation in $\mathcal{B}$ is the companion matrix of m_v , namely

$$[f_v]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 & -c_0 \\ 1 & 0 & 0 & \cdots & 0 & 0 & -c_1 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -c_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & -c_{k-2} \\ 0 & 0 & 0 & \cdots & 0 & 1 & -c_{k-1} \end{pmatrix} \quad \text{and} \quad p_{f_v}(X) = (-1)^k m_v(X).$$

Proof. 1. The vectors $\{w, f(v), f^2(v), \dots, f^{k-1}(v)\}$ are linearly independent since k is the smallest integer for which we start having linear dependence. Therefore $\dim(Z(v, f)) = k$ and $\mathcal{B}$ is a basis of $Z(v, f)$.

2. For every $z = \sum_{i=0}^{k-1} \alpha_i f^i(v) \in Z(v, f)$, we have that $f(z) = \sum_{i=1}^k \alpha_i f^i(v)$. Since $f^k(v) = \sum_{i=0}^{k-1} -c_i f^i(v)$, we have that $f(z) \in Z(v, f)$. This proves that $Z(v, f)$ is stable under f .
3. We have that $m_v(f) := c_0 \text{Id}_E + c_1 f + \dots + c_{k-1} f^{k-1} + f^k$ and therefore $m_v(f)(v) := c_0 v + c_1 f(v) + \dots + c_{k-1} f^{k-1}(v) + f^k(v) = 0_E$ thanks to Eq. (5.6).
4. By definition of the matrix representation of f_v in $\mathcal{B}$,

$$[f_v]_{\mathcal{B}}^{\mathcal{B}} = \left([f_v(v)]^{\mathcal{B}} \mid \dots \mid [f_v(f^{k-1}(v))]^{\mathcal{B}} \right) = \left([f(v)]^{\mathcal{B}} \mid \dots \mid [f^k(v)]^{\mathcal{B}} \right) = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 & -c_0 \\ 1 & 0 & 0 & \cdots & 0 & 0 & -c_1 \\ 0 & 1 & 0 & \cdots & 0 & 0 & -c_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & -c_{k-2} \\ 0 & 0 & 0 & \cdots & 0 & 1 & -c_{k-1} \end{pmatrix}.$$

□

Summarizing Lemma 5.23: companion matrices are the matrix representation of linear maps acting on cyclic subspaces.

Cyclic subspaces play a crucial role in state of the art solvers of matrix equations (Krylov subspaces, and algorithms like Arnoldi, Lanczos, and GMRES). These methods arise naturally in data science whenever one needs to handle large matrices without explicitly factorizing them. Their core strength is that they only require matrix–vector products, making them ideal for large-scale machine learning.

Cyclic spaces also provide a simple proof of the Cayley–Hamilton theorem as we see next.

Proof. (of the Cayley-Hamilton Theorem) Let p_f be the characteristic polynomial of $f \in \mathcal{L}(E)$, and let $v \in E$ be nonzero. We must show that $p_f(f)(v) = 0_E$. Since $Z(v, f)$ is a stable subspace under f , we have seen in Theorem 5.16 that

$$p_f(X) = p_v(X) q(X)$$

for some polynomial $q(X)$. By Lemma 5.23, $p_v(X) = (-1)^k m_v(X)$ (see point 4) and $m_v(f)(v) = 0_E$ (see point 3). Therefore, $p_f(f)(v) = 0$, which concludes the proof. $\square$

The Cayley Hamilton theorem is useful for:

1. Computing matrix inverses
2. Reducing high powers of matrices to low powers
3. Proving structural results about linear operators

Example 5.24

As an example, let us derive the formula of the inverse of a 2×2 matrix by using the theorem. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

be an invertible matrix. The Cayley-Hamilton theorem gives

$$A^2 - \text{Tr}(A)A + \det(A)\text{Id} = 0.$$

Therefore

$$A(A - \text{Tr}(A)\text{Id}) = \det(A)\text{Id} \iff A^{-1} = \frac{\text{Tr}(A)\text{Id} - A}{\det(A)} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

5.4 Matrix Functions

In this section, we address the following general question: given a function $f : \mathbb{C} \rightarrow \mathbb{C}$, is there a canonical way to extend it to square matrices and endomorphisms defined on finite dimensional spaces? If f is a polynomial of a certain degree m , $f(z) = \sum_{k=0}^m a_k z^k$ with $a_k \in \mathbb{C}$ for all $0 \leq k \leq m$, we have already seen that it is possible to define the canonical extension

$$f : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{n \times n}, \quad f(A) = \sum_{k=0}^m a_k A^k.$$

More generally, if $f(z)$ can be expressed by a convergent power series of the type $f(z) = \sum_{k=0}^{\infty} a_k z^k$, a natural step is to define

$$f : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}^{n \times n}, \quad f(A) = \sum_{k=0}^{\infty} a_k A^k, \tag{5.7}$$

and for endomorphisms

$$f : \mathcal{L}(E) \rightarrow \mathcal{L}(E), \quad f(T) = \sum_{k=0}^{\infty} a_k T^k.$$

In the following, we concentrate in the matrix case (Eq. (5.7)), since we can also study endomorphisms through their matrix representation. For Eq. (5.7) to make sense we need to discuss convergence issues. Indeed, to define a series, we need to specify in which norm we want it to converge and for which $A \in \mathbb{C}^{n \times n}$ this holds. When the power series is defined for complex numbers, convergence is defined with respect to

the module, which is the most straightforward notion of norm in $\mathbb{C}$. In the space of matrices $\mathbb{C}^{n \times n}$, we can choose any norm that we want since it is a finite dimensional space, and we said in Theorem 1.57 that all norms are equivalent in this case. For our purposes, the Frobenius norm arising from the inner product (2.4) will be practical. Working with this norm means that a sequence (A_k) of matrices converges if and only if each entry of the matrix converges to a value in $\mathbb{C}$. We are first going to show that formula (5.7) is well defined for diagonal matrices.

Proposition 5.25

Let $D = \text{diag}(d_1, \dots, d_n) \in \mathbb{C}^{n \times n}$ be a diagonal matrix. If the power series $f(z) = \sum_{k=0}^{\infty} a_k z^k$ converges for $|z| \leq \rho$ with $\rho > 0$ and if $\{d_1, \dots, d_n\} \subset \{z \in \mathbb{C} : |z| < \rho\}$, then the matrix series converges to

$$f(D) := \lim_{m \rightarrow \infty} \sum_{k=0}^m a_k D^k = \sum_{k=0}^{\infty} a_k D^k = \text{diag}(f(d_1), \dots, f(d_n)).$$

Proof. For every $m \in \mathbb{N}$, we define the partial sum $f_m(z) = \sum_{k=0}^m a_k z^k$ and $f_m(D) = \sum_{k=0}^m a_k D^k$. Using that $D^k = D \dots D = \text{diag}(d_1^k, \dots, d_n^k)$, and putting the summation over k inside each coefficient of the matrix $f_m(D)$, we obtain $f_m(D) = \text{diag}(f_m(d_1), \dots, f_m(d_n))$. Since, by assumption, for all $1 \leq k \leq n$ we have that $|d_k| \leq \rho$, then $f_m(d_k) \rightarrow f(d_k)$ as $m \rightarrow \infty$. Therefore

$$\lim_{m \rightarrow \infty} f_m(D) = \text{diag}(f(d_1), \dots, f(d_n)),$$

and we define $f(D) := \text{diag}(f(d_1), \dots, f(d_n))$. □

We are now going to extend Proposition 5.25 to any matrix $A \in \mathbb{C}^{n \times n}$.

Theorem 5.26

Let $A \in \mathbb{C}^{n \times n}$. If the power series $f(z) = \sum_{i=0}^{\infty} a_i z^i$ converges for $|z| \leq \rho$ with $\rho > 0$ and if $\sigma(A) \subset \{z \in \mathbb{C} : |z| < \rho\}$, then the matrix series converges absolutely to $f(A) = \sum_{k=0}^{\infty} a_k A^k$. If A is diagonalizable with $A = PDP^{-1}$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ the diagonal matrix with the eigenvalues, then

$$f(A) = Pf(D)P^{-1} = P \text{diag}(f(\lambda_1), \dots, f(\lambda_n))P^{-1}.$$

Proof. The proof of this result for a general matrix $A \in \mathbb{C}^{n \times n}$ requires the so-called Jordan decomposition of the matrix, which is a notion which we will not have time to discuss in the lecture. We therefore give the proof in the case when A is diagonalizable. In that case, there exists an invertible matrix $P \in \mathbb{C}^{n \times n}$ and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that $A = PDP^{-1}$.

Let $f_m(A) = \sum_{k=0}^m a_k A^k$. Since for every $k \in \mathbb{N}$,

$$A^k = A \cdot A \dots A = (PDP^{-1})(PDP^{-1}) \dots (PDP^{-1}) = PD^k P^{-1},$$

we have

$$f_m(A) = \sum_{k=0}^m a_k PD^k P^{-1} = P \text{diag}(f_m(\lambda_1), \dots, f_m(\lambda_n))P^{-1}.$$

Since all $|\lambda_k| \leq \rho$ for all $1 \leq k \leq n$, then $f_m(\lambda_k) \rightarrow f(\lambda_k)$ as $m \rightarrow \infty$. Thus

$$\lim_{m \rightarrow \infty} f_m(A) = P \text{diag}(f(\lambda_1), \dots, f(\lambda_n))P^{-1} = f(A). \quad \square$$

Theorem 5.26 allows us to give a matrix definition of many known functions $f : \mathbb{C} \rightarrow \mathbb{C}$ that are defined as power series. Here are some examples.

Function $f(z)$	Power Series	If $A = PDP^{-1}$, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$
e^z	$\sum_{k=0}^{\infty} \frac{z^k}{k!}$	$e^A = P \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) P^{-1}$
$\sin(z)$	$\sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$	$\sin(A) = P \text{diag}(\sin(\lambda_1), \dots, \sin(\lambda_n)) P^{-1}$
$\cos(z)$	$\sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$	$\cos(A) = P \text{diag}(\cos(\lambda_1), \dots, \cos(\lambda_n)) P^{-1}$
$\sinh(z)$	$\sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}$	$\sinh(A) = P \text{diag}(\sinh(\lambda_1), \dots, \sinh(\lambda_n)) P^{-1}$
$\cosh(z)$	$\sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!}$	$\cosh(A) = P \text{diag}(\cosh(\lambda_1), \dots, \cosh(\lambda_n)) P^{-1}$
$\text{erf}(z)$	$\frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{k!(2k+1)}$	$\text{erf}(A) = P \text{diag}(\text{erf}(\lambda_1), \dots, \text{erf}(\lambda_n)) P^{-1}$
$J_m(z)$	$\sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+m+1)} \left(\frac{z}{2}\right)^{2k+m}$	$J_m(A) = P \text{diag}(J_m(\lambda_1), \dots, J_m(\lambda_n)) P^{-1}$
General entire function	$f(z) = \sum_{k=0}^{\infty} a_k z^k$	$f(A) = P \text{diag}(f(\lambda_1), \dots, f(\lambda_n)) P^{-1}$

Table 5.1: Examples of power functions with an infinite convergence radius ρ , and their extension to diagonalizable matrices.

Function	If $A = PDP^{-1}$, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$
$\log(A)$	$\log(A) = P \text{diag}(\log(\lambda_1), \dots, \log(\lambda_n)) P^{-1}$
$A^{1/2}$	$A^{1/2} = P \text{diag}(\lambda_1^{1/2}, \dots, \lambda_n^{1/2}) P^{-1}$
A^α	$A^\alpha = P \text{diag}(\lambda_1^\alpha, \dots, \lambda_n^\alpha) P^{-1}$

Table 5.2: The expressions from this table are well-defined whenever the scalar function can be expressed as an infinite power sum (=is holomorphic) on a neighborhood of the spectrum of A .

5.5 Triangularization over $\mathbb{C}$

In Section 5.2 we saw that not every endomorphism or matrix is diagonalizable. This is not surprising, since diagonalization requires the existence of a basis in which the matrix representation is diagonal, and this is a highly restrictive condition. In this section, we show that if we relax this requirement and merely ask for a basis in which the matrix representation is tridiagonal, then such a basis always exists, provided that the underlying field is $\mathbb{K} = \mathbb{C}$.

Definition 5.27

Let E be an n -dimensional $\mathbb{K}$ -vector space over a field $\mathbb{K}$, and let $f \in \mathcal{L}(E)$.

1. We say that f is *triangularizable* if there exists a basis $\mathcal{E}$ of E in which the matrix $[f]_{\mathcal{E}}^{\mathcal{E}}$ is upper triangular.
2. We say that a matrix $A \in \mathbb{K}^{n \times n}$ is *triangularizable* if there exists an invertible matrix $P \in \mathbb{K}^{n \times n}$ and an upper triangular matrix $T \in \mathbb{K}^{n \times n}$ such that

$$A = PTP^{-1}.$$

Theorem 5.28: Triangularization over $\mathbb{C}$

Let E be a finite-dimensional $\mathbb{C}$ -vector space (the assumption that the base field is $\mathbb{K} = \mathbb{C}$ is essential). Then:

1. every endomorphism of E is triangularizable,
 2. every matrix $A \in \mathbb{C}^{n \times n}$ is triangularizable.
-

Proof. We proceed by induction on $n = \dim(E)$. For each $n \geq 1$, let $\mathcal{S}(n)$ denote the statement:

“For every complex vector space V of dimension n and every endomorphism f of V , there exists a basis of V in which the matrix of φ is upper triangular.”

- If $n = 1$, the statement is obvious.
- Assume that $\mathcal{S}(n)$ holds, and let E be a complex vector space of dimension $n + 1$. Let f be an endomorphism of E . Since the characteristic polynomial p_f has complex coefficients, it has at least one root. Therefore f has at least one eigenvalue λ . Let u be a corresponding eigenvector and choose a basis $\mathcal{E} = \{u, u_2, \dots, u_{n+1}\}$ of E whose first vector is u . Since $f(u) = \lambda u$, the matrix of f in this basis has the form

$$\begin{pmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & M & \\ 0 & & & \end{pmatrix},$$

where $M \in \mathbb{C}^{n \times n}$. Let

$$V = \text{span}\{u_2, \dots, u_{n+1}\}.$$

Then

$$E = \text{span}\{u\} \oplus V.$$

Let $\pi : E \rightarrow V$ denote the projection onto V along $\text{span}\{u\}$. The map

$$\varphi = \pi \circ f : V \rightarrow V$$

defines an endomorphism of V and $[\varphi]_V^V = M$. By the induction hypothesis $\mathcal{S}(n)$, there exists a basis $(v_2, \dots, v_{n+1})$ of V in which the matrix N of φ is upper triangular. Then

$$\mathcal{E}' = (u, v_2, \dots, v_{n+1})$$

is a basis of E , and the matrix of f in this basis has the form

$$\begin{pmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & N & \\ 0 & & & \end{pmatrix},$$

which is upper triangular. Hence $\mathcal{S}(n + 1)$ holds, completing the induction. □

Theorem 5.28 brings very practical consequences, namely that the determinant and the trace of an endomorphism/matrix are, respectively, the product and the sum of its eigenvalues.

Corollary 5.29: Determinant/Trace as the product/sum of the eigenvalues

Let E be a finite-dimensional complex vector space, and let $f \in \mathcal{L}(E)$. Suppose that the characteristic polynomial of f factors over $\mathbb{C}$ as

$$p_f(X) = (X - \lambda_1)^{m_1} \cdots (X - \lambda_k)^{m_k}.$$

Then

$$\text{Tr}(f) = m_1\lambda_1 + \cdots + m_k\lambda_k,$$

and

$$\det(f) = \lambda_1^{m_1} \cdots \lambda_k^{m_k}.$$

Proof. By the triangularization theorem, there exists a basis $\mathcal{E}$ of E in which the matrix representation $[f]_{\mathcal{E}}^{\mathcal{E}}$ of f is upper triangular. Since

$$p_f(X) = (X - \lambda_1)^{m_1} \cdots (X - \lambda_k)^{m_k},$$

the diagonal of $[f]_{\mathcal{E}}^{\mathcal{E}}$ contains λ_1 exactly m_1 times, λ_2 exactly m_2 times, and so on. Because

$$\text{Tr}(f) = \text{Tr}([f]_{\mathcal{E}}^{\mathcal{E}}), \quad \det(f) = \det([f]_{\mathcal{E}}^{\mathcal{E}}),$$

and because the trace of a triangular matrix is the sum of its diagonal entries while its determinant is the product of its diagonal entries, we immediately obtain the stated formulas. $\square$

Remark 5.30. There is an alternative proof of Corollary 5.29 which does not involve tridiagonalization:

Proof. Consider the characteristic polynomial of A , $p_A(t) = \det(A - t\text{Id})$. Since A is a complex matrix, the Fundamental Theorem of Algebra implies that p_A splits completely over $\mathbb{C}$. Since the roots of the characteristic polynomial are precisely the eigenvalues of A , counted with algebraic multiplicity, we have

$$p_A(t) = (-1)^n \prod_{i=1}^n (t - \lambda_i). \quad (5.8)$$

Evaluating at $t = 0$, we obtain $p_A(0) = \prod_{i=1}^n \lambda_i$. On the one hand, by point 2 of Lemma 5.9, we know that $p_A(0) = \det(A)$. Therefore $\det(A) = \prod_{i=1}^n \lambda_i$.

To derive the result for the trace, developing (5.8), we observe that the coefficient of t^{n-1} is $(-1)^{n-1} \sum_{i=1}^n \lambda_i$. On the other hand, by point 2 of Lemma 5.9, we know that the coefficient in front of power t^{n-1} is $(-1)^{n-1} \text{Tr}(A)$. This yields the desired equality for the trace. $\square$

Remark 5.31 (Practical use). When computing and factoring the characteristic polynomial of a matrix or endomorphism, this result provides an easy consistency check. For example, suppose that a 3×3 matrix A has characteristic polynomial

$$p_A(X) = (X - 1)^2(X + 3).$$

Then

$$\text{Tr}(A) = 1 + 1 + (-3) = -1.$$

If the actual trace of A is not equal to -1 , then there must be an error either in the computation of the characteristic polynomial or in its factorization.

Remark 5.32 (The case of real matrices). Let A be a real matrix. Its characteristic polynomial can always be factored over $\mathbb{C}$ as

$$p_A(X) = (X - \lambda_1)^{m_1} \cdots (X - \lambda_k)^{m_k},$$

where the λ_i are the complex roots of p_A . Viewing A as the matrix of a complex-linear endomorphism of $\mathbb{C}^n$, the previous corollary implies

$$\text{Tr}(A) = m_1\lambda_1 + \cdots + m_k\lambda_k,$$

and

$$\det(A) = \lambda_1^{m_1} \cdots \lambda_k^{m_k}.$$

Thus the formulas remain valid provided that all complex roots of the characteristic polynomial are included with their multiplicities, not only the real eigenvalues.

Example 5.33

Consider the matrix

$$A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 1 & -1 & 4 \end{pmatrix}.$$

Since each row sums to 4, we have

$$A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix},$$

so 4 is an eigenvalue of A . Let

$$p_A(X) = (X - 4)(X - \alpha)(X - \beta).$$

Using the previous corollary,

$$4 + \alpha + \beta = \text{Tr}(A) = 8,$$

and

$$4\alpha\beta = \det(A) = 16.$$

Therefore

$$\alpha + \beta = 4, \quad \alpha\beta = 4.$$

The scalars α and β are thus roots of

$$X^2 - 4X + 4 = (X - 2)^2,$$

which implies

$$\alpha = \beta = 2.$$

Hence

$$p_A(X) = (X - 4)(X - 2)^2,$$

and the spectrum of A is

$$\sigma(A) = \{2, 4\}.$$

5.6 Exercises

Exercise 5.1. Let

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 2 & 4 & 2 \\ -1 & 0 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 & -8 \\ 0 & 2 & 0 \\ 4 & 0 & 12 \end{pmatrix}.$$

- (i) Compute the characteristic polynomial of A and B , and find their eigenvalues.
- (ii) Find the eigensubspaces associated to each eigenvalue.
- (iii) Are the matrices diagonalizable?
- (iv) Compute A^n and B^n for $n \in \mathbb{N}$.

Exercise 5.2. (Spectrum of a particular complex matrix I)

For $n \in \mathbb{N}$, $\mathbb{K} = \mathbb{C}$, determine the eigenvalues and eigenspaces of $A \in \mathbb{C}^{2 \times 2}$:

$$A = \begin{pmatrix} i & i & 0 & 0 & \dots & 0 & 0 \\ 0 & i & i & 0 & \dots & 0 & 0 \\ 0 & 0 & i & i & \dots & 0 & 0 \\ 0 & 0 & 0 & i & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & i & i \\ 0 & 0 & 0 & 0 & \dots & 0 & i \end{pmatrix}, \quad \text{where } i = \sqrt{-1}.$$

For which value(s) of n is A diagonalizable?

Exercise 5.3. (Shift of spectrum)

Let $A \in \mathbb{K}^{n \times n}$ and $\alpha \in \mathbb{K}$, show that $\alpha \cdot \text{Id}_n + A$ is diagonalizable if and only if A is diagonalizable. What are the eigenvalues and associated eigenspaces of $\alpha \cdot \text{Id}_n + A$?

Exercise 5.4. (Diagonalization of a matrix)

Let

$$A = \begin{pmatrix} \alpha & 1 & -1 \\ 1 & \alpha & 1 \\ 1 & -1 & \alpha \end{pmatrix},$$

where $\alpha \in \mathbb{K}$. Give the eigenvalues of A . Is A diagonalizable over $\mathbb{K} = \mathbb{R}$? What about $\mathbb{K} = \mathbb{C}$?

Exercise 5.5. Let A be the $n \times n$ matrix defined by

$$\begin{pmatrix} \alpha & 1 & 1 & 1 & \dots & 1 \\ 1 & \alpha & 1 & 1 & \dots & 1 \\ 1 & 1 & \alpha & 1 & \dots & 1 \\ 1 & 1 & 1 & \alpha & \dots & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \dots & \alpha \end{pmatrix}$$

Is A diagonalizable over $\mathbb{R}$? If so, obtain the diagonalized form. (**Indication:** ex. 4.14 or ex. 5.3 may be of help.)

Exercise 5.6. Let $\mathbb{E}$ be a vector space, and let $L, M \in \mathcal{L}(E)$ with L being an isomorphism. Prove that $L \circ M$ and $M \circ L$ have the same characteristic polynomial.

Exercise 5.7. (Diagonalizability) Let $A \in \mathbb{K}^{n \times n}$ be an upper triangular matrix with the all equal terms of the diagonal and some non-zero coefficient above the diagonal. Prove that A is not diagonalizable.

Exercise 5.8. (Spectrum of a particular complex matrix II)

Let $p, q \in \mathbb{R}$ with $q \neq 0$. We consider the matrix

$$A = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}.$$

Determine the eigenvalues and eigenvectors of A . Is A diagonalizable over $\mathbb{R}$? and $\mathbb{C}$?

Exercise 5.9. Let $A = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}$. We define the endomorphism

$$\begin{aligned} \Phi : \mathbb{R}^{2 \times 2} &\rightarrow \mathbb{R}^{2 \times 2} \\ M &\mapsto AM \end{aligned}$$

- (i) Recall what is the canonical basis $\mathcal{E}$ of the real vector space $\mathbb{R}^{2 \times 2}$.
- (ii) Give the matrix representation of Φ in this basis, $[\Phi]_{\mathcal{E}}$.
- (iii) Find the eigenvalues of Φ and associated eigensubspaces.
- (iv) Is Φ diagonalizable?

Exercise 5.10. Let $\mathbb{E}$ be the space of all 2×2 upper-triangular matrices, and

$$L : \mathbb{E} \rightarrow \mathbb{E}, \quad L \begin{pmatrix} p & q \\ 0 & r \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p & q \\ 0 & r \end{pmatrix}$$

Obtain the eigenvalue(s) and associated eigenspace(s) of L . Is L diagonalizable?

Exercise 5.11. Let $f \in \mathcal{L}(\mathbb{E})$ be an endomorphism of an n -dimensional space $\mathbb{E}$. Prove that f has at most n distinct eigenvalues.

Exercise 5.12. (Characteristic polynomial)

Prove that for any $n \in \mathbb{N}$, the constant term of the characteristic polynomial of a matrix $A \in \mathbb{K}^{n \times n}$ is always the determinant of A .

Exercise 5.13. (Characteristic polynomial of two dimensional matrices) Let $A \in \mathbb{K}^{2 \times 2}$, prove that

$$p_A(\lambda) = \lambda^2 - \text{Tr}(A) \cdot \lambda + \det(A).$$

Exercise 5.14. Prove Lemma 5.19.

Exercise 5.15. (Diagonalizability of $\text{SO}(2)$)

Let $A \in \text{SO}(2) := \{A \in \mathbb{R}^{2 \times 2} \mid A^\top = A^{-1} \text{ and } \det A = 1\}$. Prove that if A is diagonalizable (over $\mathbb{K} = \mathbb{R}$), then either $A = \text{Id}$ or $A = -\text{Id}$.

Exercise 5.16. (Diagonalizability of projection and symmetric operators)

Let $P, S : \mathbb{E} \rightarrow \mathbb{E}$ be (respectively) a projection and a symmetry operator (not necessarily orthogonal) on a $\mathbb{K}$ -vector space $\mathbb{E}$.

- (i) Prove that the eigenvalues of P can only be 1 or 0.
- (ii) Prove that the eigenvalues of S can only be ± 1 .
- (iii) Prove that P and S are always diagonalizable.

Exercise 5.17. (Closed expression for the Fibonacci sequences)

We are well aware of the recursive definition of a Fibonacci sequence (see ex. 1.32). In the same exercise, we have also solved for the closed form of expression for the n -th term of the standard Fibonacci sequence $(0, 1, 1, 2, 3, 5, 8, 13, \dots)$. We shall now try and obtain the closed expression for any Fibonacci sequence.

- (i) Let $x = (x_i)_{i=0}^\infty$ be a real Fibonacci sequence. Determine the matrix $A \in \mathbb{R}^{2 \times 2}$ such that

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = A \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix}, \quad \forall n \in \mathbb{N}.$$

- (ii) Obtain an eigenbasis for A and therefore diagonalize A .
- (iii) With the information above, obtain the closed form of the n -th entry of the sequence x_n :

$$x_n = \frac{1}{\sqrt{5}} [(x_1 - \psi x_0)\phi^n - (x_1 - \phi x_0)\psi^n], \quad \text{where } \phi = \frac{1 + \sqrt{5}}{2} \text{ and } \psi = \frac{1 - \sqrt{5}}{2}.$$

Exercise 5.18. (Inverse of a matrix) Let $A \in \mathbb{K}^{n \times n}$ be an invertible matrix. Show that there exist $c_0, \dots, c_{n-1}$ such that

$$A^{-1} = c_0 \text{Id} + \dots + c_{n-1} A^{n-1}.$$

Exercise 5.19. (Eigenvalues on an infinite dimensional space) Let $\mathbb{E} = \mathcal{F}(\mathbb{N}_0, \mathbb{C}) = \{(x_n)_{n=0}^\infty : x_n \in \mathbb{C}, \forall n \in \mathbb{N}\}$ be the space of all complex sequences. On $\mathbb{E}$ we define the *left* and *right shift operators*:

$$L(x_0, x_1, x_2, \dots) = (x_1, x_2, x_3, \dots), \quad R(x_0, x_1, x_2, \dots) = (0, x_0, x_1, x_2, \dots).$$

- (i) Prove that $\sigma(L) = \mathbb{C}$ (i.e. any $\lambda \in \mathbb{C}$ is an eigenvalue of L). Determine for each $\lambda \in \mathbb{C}$ its associated eigenspace.
- (ii) Prove that $\sigma(R) = \emptyset$.

Exercise 5.20. (Simultaneous diagonalization and commutativity)

Let $A, B \in \mathbb{K}^{n \times n}$ be two diagonalizable matrices. We say that A and B are simultaneously diagonalizable if there exists $P \in \mathbb{K}^{n \times n}$ with $\det P \neq 0$ such that both $P^{-1}AP$ and $P^{-1}BP$ are diagonal matrices.

Prove that A, B are simultaneously diagonalizable if and only if $AB = BA$. (**Indication:** to see that commutativity implies simultaneous diagonalizability, Theorem 5.20 may be useful.)

Exercise 5.21. (Eigenvalue of hermitian matrices)

- (i) Let $M \in \mathbb{C}^{n \times n}$ be a hermitian matrix (i.e. $M_{ij} = \bar{M}_{ji}$). Prove that the eigenvalues of M are real numbers.

- (ii) Let $Q \in \mathbb{C}^{n \times n}$ be a skew-hermitian matrix (i.e. $Q_{ij} = -\bar{Q}_{ji}$). Prove that the eigenvalues of Q are purely imaginary numbers.

Exercise 5.22. (Diagonalization on polynomial space I)

Let $\mathbb{E} = \mathbb{R}_4[X]$, and let $L : \mathbb{E} \rightarrow \mathbb{E}$ be defined as $L(P) := P' + P''$. Determine whether L is diagonalizable.

Exercise 5.23. (Fundamental matrix and ODE)

Let $A \in \mathbb{C}^{n \times n}$, consider the following linear system of ordinary differential equations (ODE):

$$\dot{x}(t) = Ax(t), \quad \text{where } x(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} \quad \text{and} \quad \dot{x}(t) \equiv \frac{d}{dt}x(t) = \begin{pmatrix} \dot{x}_1(t) \\ \vdots \\ \dot{x}_n(t) \end{pmatrix}. \quad (5.9)$$

The goal is to find differentiable functions $x_1, \dots, x_n : \mathbb{R} \rightarrow \mathbb{C}$ such that $x(t) = (x_1, \dots, x_n)^\top$ satisfy the equation above. For this we define the *fundamental matrix associated to A* to be:

$$\Phi_A(t) := \exp(tA) = \sum_{k=0}^{\infty} \frac{(tA)^k}{k!}. \quad (5.10)$$

- (i) Prove that $\frac{d}{dt}\Phi_A(t) = A \cdot \Phi_A(t)$. What is $\Phi_A(0)$? (**Indication:** Assume proven that the infinite sum (5.10) is absolutely convergent for all $t \in \mathbb{R}$).
- (ii) Conclude that for any $x_0 \in \mathbb{C}^n$, the vector function $x(t) := \Phi_A(t)x_0$ is a solution to the system of ODEs in (5.9). What is $x(0)$ in this case?
- (iii) Solve the equation subject to the initial condition:

$$\begin{cases} \dot{x}(t) = y(t) \\ \dot{y}(t) = x(t) \end{cases} \quad \text{with} \quad x(0) = 1, \quad y(0) = 0.$$

- (iv) Solve the equation subject to the initial condition:

$$\begin{cases} \dot{x}(t) = -y(t) \\ \dot{y}(t) = x(t) \end{cases} \quad \text{with} \quad x(0) = 2, \quad y(0) = 5.$$

Exercise 5.24. (System of ODEs)

Solve the following system of ODEs:

$$\dot{x}(t) = \begin{pmatrix} 2 & 3 \\ 2 & -3 \end{pmatrix} x(t), \quad \text{with} \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

Exercise 5.25. (Commutativity and matrix exponential) Let $A, B \in \mathbb{C}^{n \times n}$.

- (i) Prove that if $AB = BA$ then the *Binomial theorem* holds:

$$(A + B)^n = \sum_{k=0}^n A^k B^{n-k}.$$

- (ii) Prove that if $AB = BA$ then $\exp(A + B) = \exp(A)\exp(B)$

(**Indication:** *Cauchy Product:* $(\sum_{i=0}^{\infty} a_i) (\sum_{j=0}^{\infty} b_j) = \sum_{k=0}^{\infty} c_k$ with $c_k = \sum_{h=0}^k a_{k-h} b_h$).

- (iii) Is the reverse implication true (i.e. for matrices that satisfy $\exp(A)\exp(B) = \exp(A + B)$ we always have $AB = BA$)?

Exercise 5.26. (Fundamental matrix for a non-diagonalizable matrix)

Let

$$A = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ 0 & 0 & \lambda & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{pmatrix} \in \mathbb{C}^{n \times n}.$$

Calculate $\Phi_A(t)$. (**Indication:** ex. 5.26 and ex. 1.28 are useful here).

Exercise 5.27. (Fundamental matrix for a particular Jordan form)

Let

$$B = \begin{pmatrix} \lambda & 1 & 0 & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 \\ 0 & 0 & \lambda & 0 & 0 \\ 0 & 0 & 0 & \mu & 1 \\ 0 & 0 & 0 & 0 & \mu \end{pmatrix}.$$

Calculate $\Phi_B(t)$.

Exercise 5.28. (Diagonalization on polynomial space I)

Let $\mathbb{E} = \mathbb{C}_1[X] = \{p : \mathbb{C} \rightarrow \mathbb{C} : p(X) = \alpha X + \beta, \alpha, \beta \in \mathbb{C}\}$, and let $L : \mathbb{E} \rightarrow \mathbb{E}$ be the mapping

$$L(\alpha X + \beta) := (\alpha + i\beta)X + i\beta.$$

Determine the eigenvalues and associated eigenspaces, and whether L is diagonalizable.

Exercise 5.29. (Nilpotent matrix and characteristic polynomial)

Let $P(X)$ be a polynomial and $A \in \mathbb{K}^{n \times n}$, we define $B := P(A) \in \mathbb{K}^{n \times n}$. Prove for eigenvector v of A with eigenvalues λ , then v is also an eigenvector of B with eigenvalue $P(\lambda)$. Apply this result to prove:

- (i) If $A \neq 0$ is nilpotent, then A is not diagonalizable.
- (ii) If A is diagonalizable, then $p_A(A) = 0^{n \times n}$, i.e. substituting the matrix in its characteristic polynomial, the resulting matrix is the zero matrix. (**Note:** Here we have assumed A to be diagonalizable, but the result can be generalized to all square matrices, which is the Cayley-Hamilton Theorem.)

Exercise 5.30. (Differentiation of a particular function)

Calculate the 2026-th derivative of the function $3e^t \cos t + 4e^t \sin t$. I.e., find

$$\frac{d^{2026}}{dt^{2026}} f(t), \quad \text{where} \quad f(t) = 3e^t \cos t + 4e^t \sin t.$$

Exercise 5.31. (Chebyshev polynomials of the first and second kinds)

Let $x \in \mathbb{R}$ be a fixed parameter. Consider the matrix

$$A(x) = \begin{pmatrix} 2x & -1 \\ 1 & 0 \end{pmatrix}.$$

Some well known families of polynomials can be generated by recurrence relations that share this underlying matrix structure. In this exercise, we will explore two such families.

- (i) Obtain an eigenbasis for $A(x)$ and therefore diagonalize $A(x)$ in terms of its eigenvalues $\lambda_-(x)$ and $\lambda_+(x)$.
- (ii) Consider a general sequence of functions $P_n(x)$ satisfying the recurrence relation $P_{n+1}(x) = 2xP_n(x) - P_{n-1}(x)$ for $n \geq 1$. Express the vector $(P_n(x), P_{n-1}(x))^T$ in terms of $A(x)$, n , and the initial conditions $P_1(x)$ and $P_0(x)$.
- (iii) Let $T_n(x)$ denote the Chebyshev polynomials of the *first kind*, defined by the initial conditions $T_0(x) = 1$ and $T_1(x) = x$. Using the diagonalization from (i), obtain the closed algebraic form of $T_n(x)$.

- (iv) By restricting $x \in [-1, 1]$ and making the substitution $x = \cos \theta$, show that $T_n(\cos \theta) = \cos(n\theta)$ (**Indication:** de Moivre's formula.)
- (v) Let $U_n(x)$ denote the Chebyshev polynomials of the *second kind*, defined by the initial conditions $U_0(x) = 1$ and $U_1(x) = 2x$. Using the diagonalization from (i), obtain the closed algebraic form of $U_n(x)$.
- (vi) Using the same substitution $x = \cos \theta$, show that $U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}$.

Exercise 5.32. (Chebyshev polynomials of the first and second kinds)

Let $x \in \mathbb{R}$ be a fixed parameter. Consider the matrix

$$A(x) = \begin{pmatrix} 2x & -1 \\ 1 & 0 \end{pmatrix}.$$

Many families of orthogonal polynomials can be generated by recurrence relations that share this underlying matrix structure. We will explore two such families.

- (i) Obtain an eigenbasis for $A(x)$ and therefore diagonalize $A(x)$ in terms of its eigenvalues $\lambda_-(x)$ and $\lambda_+(x)$.
- (ii) Consider a general sequence of functions $P_n(x)$ satisfying the recurrence relation $P_{n+1}(x) = 2xP_n(x) - P_{n-1}(x)$ for $n \geq 1$. Express the vector $(P_n(x), P_{n-1}(x))^T$ in terms of $A(x)$, n , and the initial conditions $P_1(x)$ and $P_0(x)$.
- (iii) Let $T_n(x)$ denote the *Chebyshev polynomials of the first kind*, defined by the initial conditions $T_0(x) = 1$ and $T_1(x) = x$. Using the diagonalization from (i), obtain the closed algebraic form of $T_n(x)$.
- (iv) By restricting $x \in [-1, 1]$ and making the substitution $x = \cos \theta$, show that $T_n(\cos \theta) = \cos(n\theta)$ (**Indication:** de Moivre's formula.)
- (v) Let $U_n(x)$ denote the *Chebyshev polynomials of the second kind*, defined by the initial conditions $U_0(x) = 1$ and $U_1(x) = 2x$. Using the diagonalization from (i), obtain the closed algebraic form of $U_n(x)$.
- (vi) Using the same substitution $x = \cos \theta$, show that $U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}$.

Chapter 6

Spectral Theorems in Inner Product Spaces

This chapter is a continuation of Chapter 5. Here we discuss further results on diagonalization and triangularization where we add an extra structural component to the $\mathbb{K}$ -vector space E : the presence of an inner product.

6.1 Schur's lemma and diagonalization of normal endomorphisms

The first result of this chapter is about triangularization. We saw in Theorem 5.28 that we can triangularize every endomorphism $f \in \mathcal{L}(E)$ of a complex vector space E (resp. every matrix from $\mathbb{C}^{n \times n}$). Somewhat unsurprisingly, if E is an inner product space, there exists an orthonormal basis $\mathcal{E}$ in which $[f]_{\mathcal{E}}^{\mathcal{E}}$ is upper triangular. This is called Schur's lemma¹.

Lemma 6.1: Schur's Lemma

Let $(E, \langle \cdot, \cdot \rangle)$ be a complex inner product space ($\mathbb{K} = \mathbb{C}$). Then:

1. every endomorphism $f \in \mathcal{L}(E)$ is triangularizable in an orthonormal basis,
2. every matrix $A \in \mathbb{C}^{n \times n}$ is unitary triangularizable, that is, there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ and a triangular matrix $T \in \mathbb{C}^{n \times n}$ such that

$$A = UTU^*.$$

Proof. The proof is very similar to the proof of Theorem 5.28. Like before, we proceed by induction on $n = \dim(E)$. For each $n \geq 1$, let $\mathcal{S}(n)$ denote the statement:

"For every complex vector space V of dimension n and every endomorphism f of V , there exists an orthonormal basis of V in which the matrix of f is upper triangular."

- If $n = 1$, the statement is obvious.
- Assume that $\mathcal{S}(n)$ holds, and let E be a complex vector space of dimension $n + 1$. Let f be an endomorphism of E . Since the characteristic polynomial p_f has complex coefficients, it has at least one root. Therefore f has at least one eigenvalue λ . Let u be a corresponding eigenvector which we choose to have norm 1 ($\|u\|_E = 1$). We next extend u to an orthonormal basis $\mathcal{E} = \{u, u_2, \dots, u_{n+1}\}$ of E whose first vector is u . Since $f(u) = \lambda u$, the matrix of f in this basis has the form

$$[f]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & M & \\ 0 & & & \end{pmatrix},$$

¹There are several Schur's lemmas in linear algebra, so one always has to specify which one is referred to in discussions!

where $M \in \mathbb{C}^{n \times n}$. Let

$$V = \text{span}\{u_2, \dots, u_{n+1}\}.$$

Then

$$E = \text{span}\{u\} \oplus V.$$

Let $\pi : E \rightarrow V$ denote the orthogonal projection onto V . The map $\varphi = \pi \circ f$ defines an endomorphism of V . By the induction hypothesis $\mathcal{S}(n)$, there exists an orthonormal basis $(v_2, \dots, v_{n+1})$ of V in which the matrix N of φ is upper triangular. Then

$$\mathcal{E}' = (u, v_2, \dots, v_{n+1})$$

is an orthonormal basis of E , and the matrix of f in this basis has the form

$$\begin{pmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & N & \\ 0 & & & \end{pmatrix},$$

which is upper triangular. Hence $\mathcal{S}(n+1)$ holds, completing the induction.

□

6.2 Diagonalization of normal endomorphisms

In this section, we show in Theorem 6.6 that when $f \in \mathcal{L}(E)$ is a normal operator (i.e. it commutes with its adjoint $f \circ f^* = f^* \circ f$), then the upper triangular matrix from the Schur's lemma is actually a diagonal matrix carrying the eigenvalues of f . This implies that normal endomorphisms of complex vector spaces can be diagonalized in $\mathbb{C}$ in an orthonormal basis of eigenvectors. This result is very important because many relevant linear operators are normal. From the table which we gave in Section 2.5.3, we see that the following operators are normal:

1. Unitary operators: $f^* \circ f = f \circ f^* = \text{Id}$
2. Self-Adjoint operators: $f = f^*$
3. Skew-adjoint operators: $f = -f^*$
4. Orthogonal projections: $f^2 = f$ and $f = f^*$

The proof of Theorem 6.6 uses certain properties of normal operators which we give next.

Lemma 6.2

Let f be normal. Then for $v, w \in V$ and $\lambda, \lambda_1, \lambda_2 \in \mathbb{C}$,

1. $f(v) = 0 \iff f^*(v) = 0$;
 2. $f - \lambda \text{Id}_E$ is normal for any $\lambda \in \mathbb{C}$;
 3. $f(v) = \lambda v \iff f^*(v) = \bar{\lambda} v$;
 4. $f(v) = \lambda_1 v, f(w) = \lambda_2 w, \lambda_1 \neq \lambda_2 \implies \langle v, w \rangle = 0$.
-

Proof. Let $v, w \in V$ and $\lambda, \lambda_1, \lambda_2 \in \mathbb{C}$.

1. $\|f(v)\|^2 = \langle f(v), f(v) \rangle = \langle v, f^* \circ f(v) \rangle = \langle v, f \circ f^*(v) \rangle = \langle f^*(v), f^*(v) \rangle = \|f^*(v)\|^2$ and hence

$$f(v) = 0 \iff f^*(v) = 0.$$

2. We have that $(f - \lambda \text{Id})^* = f^* - \bar{\lambda} \text{Id}$ commutes with $f - \lambda \text{Id}$.
3. We have that $(f - \lambda \text{Id})$ is a normal operator by point 2. By applying point 1 to it, it follows that $(f - \lambda \text{Id})(v) = 0 \iff (f - \lambda \text{Id})^*(v) = 0 \iff (f^* - \bar{\lambda} \text{Id})(v) = 0$.
4. On the one hand

$$\langle f(v), w \rangle = \langle \lambda_1 v, w \rangle = \lambda_1 \langle v, w \rangle.$$

On the other hand, by the adjoint property,

$$\langle f(v), w \rangle = \langle v, f^*(w) \rangle \underset{(\text{by 3})}{=} \langle v, \bar{\lambda}_2 w \rangle = \bar{\lambda}_2 \langle v, w \rangle$$

Hence $(\lambda_2 - \lambda_1) \langle v, w \rangle = 0$, which implies that $\langle v, w \rangle = 0$ if $\lambda_1 \neq \lambda_2$.

□

We are now ready to prove the main theorem of this section.

Theorem 6.3: Spectral theorem for normal endomorphisms

Let $(E, \langle \cdot, \cdot \rangle)$ be a complex n -dimensional inner product space (it is crucial that $\mathbb{K} = \mathbb{C}$). Then:

- every normal endomorphism $f \in \mathcal{L}(E)$ is diagonalizable in an orthonormal basis of eigenvectors.
- every normal matrix $A \in \mathbb{C}^{n \times n}$ is unitarily diagonalizable, that is, there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ containing the eigenvalues of A such that

$$A = UDU^*.$$

Proof. We proceed very similarly to the proof of Lemma 6.1. The fundamental theorem of algebra ensures that f has at least one eigenvalue $\lambda \in \mathbb{C}$, and we let u be the corresponding eigenvector which we choose to have norm 1. We then extend u to an orthonormal basis of E which we call $\mathcal{E} = \{u, u_2, \dots, u_n\}$. By point 3 of Lemma 6.2, the subspace $U = \text{span}\{u\}$ is stable under f and also under f^* . Similarly, $U^\perp = \text{span}\{u_2, \dots, u_n\}$ is also stable under both f and f^* . Therefore, the matrix representation of f in this basis is of the form

$$[f]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & M & \\ 0 & & & \end{pmatrix}.$$

Compared to the proof of Lemma 6.1, observe that the first row is full of zeros except from the diagonal element. This is thanks to the stability property of U and $U^\perp$ that the normal operator presents. We once again proceed by induction on the dimension of E to get the result.

□

The following is a reformulation of the above theorem.

Theorem 6.4: Spectral theorem for normal endomorphisms

Let $f : E \rightarrow E$ be a normal operator on a complex n -dimensional inner product space. Then there exist eigenvalues $\lambda_1, \dots, \lambda_k \in \mathbb{C}$ and orthogonal projections $P_i : E \rightarrow V_{\lambda_i}$ for $1 \leq i \leq k$ such that:

1. $f = \lambda_1 P_1 + \cdots + \lambda_k P_k$
 2. $P_1 + \cdots + P_k = \text{Id}_E$
 3. $P_i \circ P_j = 0_E$ for $1 \leq i \neq j \leq k$.
-

As a direct consequence of the spectral theorem, most operators which we have defined in the table from Section 2.5.3 are diagonalizable. We record this in the following lemma. The lemma also records the fact that the eigenvalues of these operators have special properties as well.

Lemma 6.5

Let E be a complex n -dimensional vector space. Then the following operators are diagonalizable and their eigenvalues have the following properties:

- unitary: $|\lambda| = 1$
- self-adjoint: $\lambda \in \mathbb{R}$
- skew-adjoint: $\lambda \in i\mathbb{R}$
- orthogonal projection: $\lambda \in \{0, 1\}$

Proof. The fact that the operators are diagonalizable follows directly from Theorem 6.6. To prove the properties of the eigenvalues, we proceed as follows:

- Let $f \in \mathcal{L}(E)$ be a unitary operator, and let (λ, v) be an eigenpair. Then

$$f^* \circ f(v) = f^*(\lambda v) = \lambda f^*(v) = \lambda \bar{\lambda} v = |\lambda|^2 v,$$

and since f is unitary $f^* \circ f(v) = v$. Therefore $(1 - |\lambda|^2)v = 0$, and since $v \neq 0$ because it is an eigenvector, then $|\lambda|^2 = 1$.

- Let now f be a self-adjoint operator, and let (λ, v) be an eigenpair. Then

$$\bar{\lambda} v = f^*(v) = f(v) = \lambda v,$$

thus $\lambda = \bar{\lambda}$, which implies that $\lambda \in \mathbb{R}$.

- We proceed similarly for the skew-adjoint case and the one of the orthogonal projection.

□

The previous lemma shows that the eigenvalues of self-adjoint operators are always real despite the fact that we work on a complex vector space. Eigenvectors have no particular property, though, and will in general be complex. The reality of the eigenvalues leads to wonder if self-adjoint operators on real vector spaces are diagonalizable, and hence admit a basis of real eigenvectors. The next theorem shows that this is indeed the case.

Theorem 6.6: Spectral theorem for real, self-adjoint endomorphisms

Let $(E, \langle \cdot, \cdot \rangle)$ be a real n -dimensional inner product space. Then:

- every self-adjoint endomorphism $f \in \mathcal{L}(E)$ is diagonalizable in an orthonormal basis of real eigenvectors.
- every symmetric matrix $A \in \mathbb{R}^{n \times n}$ is orthogonally diagonalizable, that is, there exists an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^{n \times n}$ containing the real eigenvalues of A such that

$$A = QDQ^T.$$

Proof. We provide a proof which emphasizes that the real spectral theorem is a direct consequence of the complex spectral theorem 6.6. For this, we let $f \in \mathcal{L}(E)$ be a self-adjoint operator on a real inner product space $(E, \langle \cdot, \cdot \rangle_E)$. We are going to lift the real operator into the complex plane by a technique called complexification. For this we let

$$E_{\mathbb{C}} := \{u + iv : u, v \in E\},$$

equipped with the inner product

$$\langle u_1 + iv_1, u_2 + iv_2 \rangle_{\mathbb{C}} := \langle u_1, u_2 \rangle_E + \langle v_1, v_2 \rangle_E + i(\langle v_1, u_2 \rangle_E - \langle u_1, v_2 \rangle_E).$$

We next define the complex-linear extension of f ,

$$f_{\mathbb{C}} : E_{\mathbb{C}} \rightarrow E_{\mathbb{C}}, \quad f_{\mathbb{C}}(u + iv) = f(u) + if(v).$$

We claim that $f_{\mathbb{C}}$ is self-adjoint. Indeed, for $u_j, v_j \in E$,

$$\begin{aligned} \langle f_{\mathbb{C}}(u_1 + iv_1), u_2 + iv_2 \rangle_{\mathbb{C}} &= \langle f(u)_1, u_2 \rangle_E + \langle f(v)_1, v_2 \rangle_E \\ &\quad + i(\langle f(v)_1, u_2 \rangle_E - \langle f(u)_1, v_2 \rangle_E). \end{aligned}$$

Since f is self-adjoint,

$$\langle f(x), y \rangle_E = \langle x, f(y) \rangle_E \quad \forall x, y \in E,$$

and therefore

$$\langle f_{\mathbb{C}}(x), y \rangle_{\mathbb{C}} = \langle x, f_{\mathbb{C}}(y) \rangle_{\mathbb{C}} \quad \forall x, y \in E_{\mathbb{C}}.$$

Hence $f_{\mathbb{C}}^* = f_{\mathbb{C}}$ so $f_{\mathbb{C}}$ is self-adjoint.

Since self-adjoint operators are normal, the complex spectral theorem 6.6 applies. Therefore there exists an orthonormal basis $\mathcal{Z} = \{z_1, \dots, z_n\}$ of $E_{\mathbb{C}}$ consisting of eigenvectors of $f_{\mathbb{C}}$ such that

$$f_{\mathbb{C}} z_j = \lambda_j z_j, \quad 1 \leq j \leq n.$$

Since $f_{\mathbb{C}}$ is self-adjoint, all eigenvalues are real by Lemma 6.5, so $\lambda_j \in \mathbb{R}$.

Let

$$V_{\lambda} := \ker(f - \lambda \text{Id}) \subseteq E$$

denote the real eigenspace of f associated with λ . We claim that

$$(V_{\lambda})_{\mathbb{C}} = \ker(f_{\mathbb{C}} - \lambda \text{Id}).$$

Indeed, if $u, v \in V_{\lambda}$, then

$$f_{\mathbb{C}}(u + iv) = f(u) + if(v) = \lambda u + i\lambda v = \lambda(u + iv),$$

showing that

$$(V_{\lambda})_{\mathbb{C}} \subseteq \ker(f_{\mathbb{C}} - \lambda \text{Id}).$$

Conversely, if

$$f_{\mathbb{C}}(u + iv) = \lambda(u + iv),$$

then

$$f(u) + if(v) = \lambda u + i\lambda v.$$

Comparing real and imaginary parts gives

$$f(u) = \lambda u, \quad f(v) = \lambda v,$$

so that $u, v \in V_{\lambda}$. Hence

$$\ker(f_{\mathbb{C}} - \lambda \text{Id}) \subseteq (V_{\lambda})_{\mathbb{C}},$$

proving the claim.

Since the eigenspaces of $f_{\mathbb{C}}$ span $E_{\mathbb{C}}$, we obtain

$$E_{\mathbb{C}} = \bigoplus_{\lambda} (V_{\lambda})_{\mathbb{C}}.$$

Taking real dimensions, it follows that

$$E = \bigoplus_{\lambda} V_{\lambda}.$$

Finally, eigenspaces corresponding to distinct eigenvalues are orthogonal. Indeed, if

$$f(u) = \lambda u, \quad f(v) = \mu v, \quad \lambda \neq \mu,$$

then

$$\lambda \langle u, v \rangle_E = \langle f(u), v \rangle_E = \langle u, f(v) \rangle_E = \mu \langle u, v \rangle_E,$$

and therefore

$$\langle u, v \rangle_E = 0.$$

Choosing an orthonormal basis of each eigenspace V_λ and taking their union yields an orthonormal basis of V consisting entirely of eigenvectors of f . $\square$

We finish this section by discussing the case of positive operators. On complex vector spaces, such operators are self-adjoint, thus normal and diagonalizable and their study falls into the category of this section. On real vector spaces, these operators need not be normal nor diagonalizable, making them out of the scope of our section on normal operators. In both cases (complex or real vector spaces), positive operators have the property that their eigenvalues are real and positive. We record these facts in the following theorem.

Theorem 6.7: Positive operators have positive eigenvalues

Let $f \in \mathcal{L}(E)$ be a positive endomorphism over an n -dimensional $\mathbb{K}$ -vector space E . That is,

$$\langle f(x), x \rangle \geq 0, \quad \forall x \in E.$$

Then:

- If $\mathbb{K} = \mathbb{C}$, f is self-adjoint (thus normal), and its eigenvalues are real nonnegative.
 - If $\mathbb{K} = \mathbb{R}$, f is not necessarily normal, nor diagonalizable. If it has eigenvalues, they are real nonnegative.
-

Proof. We distinguish the cases $\mathbb{K} = \mathbb{C}$ and $\mathbb{K} = \mathbb{R}$.

Assume first that $\mathbb{K} = \mathbb{C}$. Since

$$\langle f(x), x \rangle \geq 0 \quad \forall x \in E,$$

the quantity $\langle f(x), x \rangle$ is real for every $x \in E$. Hence the mapping

$$q(x) := \langle f(x), x \rangle$$

is real-valued. Proceeding in the same way that we did for the polarization identity Eq. (2.7), we deduce that for all $x, y \in E$,

$$\langle f(x), y \rangle = \frac{1}{4} \left(q(x+y) - q(x-y) - iq(x+iy) + iq(x-iy) \right).$$

Since q is real-valued, this identity implies

$$\langle f(x), y \rangle = \langle x, f(y) \rangle.$$

Thus f is self-adjoint. In particular, f is normal and thus diagonalizable thanks to Theorem 6.6.

Let now λ be an eigenvalue of f , and let $v \neq 0$ be an eigenvector associated with λ . Since f is self-adjoint, $\lambda \in \mathbb{R}$. Moreover,

$$0 \leq \langle f(v), v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle = \lambda \|v\|^2.$$

Since $\|v\|^2 > 0$, it follows that $\lambda \geq 0$. Therefore all eigenvalues of f are real and nonnegative.

Assume now that $\mathbb{K} = \mathbb{R}$. If $\lambda \in \mathbb{R}$ is an eigenvalue of f with eigenvector $v \neq 0$, then the same argument gives

$$0 \leq \langle f(v), v \rangle = \langle \lambda v, v \rangle = \lambda \|v\|^2.$$

Since $\|v\|^2 > 0$, we again obtain $\lambda \geq 0$. Thus every real eigenvalue of f is nonnegative.

However, over a real vector space, the condition

$$\langle f(x), x \rangle \geq 0 \quad \forall x \in E$$

does not imply that f is self-adjoint. For instance, on $\mathbb{R}^2$ with the Euclidean inner product, consider the mapping $f_A(x) = Ax$ with

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Then, for every $x = (x_1, x_2) \in \mathbb{R}^2$,

$$\langle Ax, x \rangle = x^\top A^\top x = x_1^2 + x_2^2 = \|x\|^2 \geq 0.$$

Thus A is positive in the above sense. But

$$A^\top = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \neq A,$$

so A is not self-adjoint, and therefore not normal. Its characteristic polynomial is

$$p_A(\lambda) = (\lambda - 1)^2 + 1,$$

so its complex eigenvalues are $1 \pm i$. In particular, A is not diagonalizable over $\mathbb{R}$.

This shows that, in the real case, positivity alone does not imply normality or diagonalizability. $\square$

6.3 Singular Value Decomposition

One of the main limitations of the eigenvalue decomposition is that it only applies to endomorphisms, that is, to linear maps where the input and output domains are the same. The notion of Singular Value Decomposition (SVD) extends this type of decomposition to general linear maps $f : E \rightarrow F$ where $(E, \langle \cdot, \cdot \rangle_E)$ and $(F, \langle \cdot, \cdot \rangle_F)$ are two different inner product spaces. The spaces have dimensions $\dim(E) = m$ and $\dim(F) = n$ respectively, and they can be defined over a field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. It is one of the most important tools in linear algebra, and it is used in countless applications in data science.

For endomorphisms $f : E \rightarrow E$, the goal was to find a basis $\mathcal{E}$ where the matrix representation $[f]_{\mathcal{E}}^{\mathcal{E}} \in \mathbb{R}^{m \times m}$ was diagonal. The main goal of the SVD is the same. However, since we have two different spaces, we now want to find a pair of bases $\mathcal{E}$ and $\mathcal{F}$ of E and F respectively such that $[f]_{\mathcal{F}}^{\mathcal{E}} \in \mathbb{K}^{m \times n}$ is diagonal. Since $[f]_{\mathcal{F}}^{\mathcal{E}} \in \mathbb{K}^{m \times n}$ is a rectangular matrix, what we mean by diagonal is that its diagonal elements may be nonzero. The following theorem guarantees that we can always find such a pair of bases for any linear map $f \in \mathcal{L}(E, F)$, both when $\mathbb{K} = \mathbb{R}$ and $\mathbb{C}$. In addition, the bases are orthonormal.

Definition 6.8: Singular value and singular vectors

Let $(E, \langle \cdot, \cdot \rangle_E)$ and $(F, \langle \cdot, \cdot \rangle_F)$ be two inner product spaces of dimension $\dim(E) = m$ and $\dim(F) = n$ and let $f \in \mathcal{L}(E, F)$. We say that $\sigma \geq 0$ is a singular value associated to the singular vectors $u \in E$ and $v \in F$ (with $u, v \neq 0$) if and only if

$$f(u) = \sigma v.$$

Theorem 6.9: Singular Value Decomposition

Let $(E, \langle \cdot, \cdot \rangle_E)$ and $(F, \langle \cdot, \cdot \rangle_F)$ be two inner product spaces of dimension $\dim(E) = m$ and $\dim(F) = n$ which are defined over a field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then:

- for every linear map $f \in \mathcal{L}(E, F)$, there exist r singular values $\sigma_1 \geq \dots \geq \sigma_r \geq 0$ with $r \leq n$, and

orthonormal bases $\mathcal{E}$ and $\mathcal{F}$ of the spaces E and F respectively such that

$$[f]_{\mathcal{E}}^{\mathcal{F}} = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \\ & & & & \ddots \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

- for every matrix $A \in \mathbb{K}^{m \times n}$ there exists r singular values $\sigma_1 \geq \dots \geq \sigma_r \geq 0$, and unitary matrices $U \in \mathbb{K}^{m \times m}$ and $V \in \mathbb{K}^{n \times n}$ such that

$$A = U\Sigma V^*,$$

where

$$\Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \\ & & & & \ddots \end{pmatrix} \in \mathbb{R}^{m \times n}$$

Proof. The mapping $f^* \circ f : E \rightarrow E$ is positive and self-adjoint (make sure you know how to prove it!). By the spectral theorems that we have given in Section 6.2, such an operator is diagonalizable in an orthonormal basis $\mathcal{E} = \{v_1, \dots, v_n\}$ of E and it has real, nonnegative eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. Suppose that the first r eigenvalues are strictly positive and the rest are zero. Define

$$\sigma_i := \lambda_i^{1/2}, \quad 1 \leq i \leq n$$

and

$$u_i := \frac{1}{\sigma_i} f(v_i) \in F, \quad 1 \leq i \leq r.$$

Then

$$\langle u_i, u_j \rangle_F = \frac{1}{\sigma_i \sigma_j} \langle f(v_i), f(v_j) \rangle_F = \frac{1}{\sigma_i \sigma_j} \langle v_i, f^* \circ f(v_j) \rangle_E = \frac{\lambda_j}{\sigma_i \sigma_j} \langle v_i, v_j \rangle_E = \delta_{i,j}, \quad 1 \leq i, j \leq r.$$

We now extend the vectors $\{u_1, \dots, u_r\}$ to an orthonormal basis of F which we call $\mathcal{F} = \{u_1, \dots, u_n\}$. By construction,

$$[f]_{\mathcal{E}}^{\mathcal{F}} = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \\ & & & & \ddots \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

This proves the theorem. □

The SVD is fundamental in data compression techniques. To understand why, let $f \in \mathcal{L}(E, F)$ and let $\mathcal{E} = \{v_1, \dots, v_n\}$ and $\mathcal{F} = \{u_1, \dots, u_m\}$ be the orthonormal bases of E and F provided by the SVD decomposition. For every $x = \sum_{i=1}^n \langle x, v_i \rangle_E v_i \in E$, we can express $f(x)$ as

$$\begin{aligned} f(x) &= \sum_{i=1}^n \langle x, v_i \rangle_E f(v_i) \\ &= \sum_{i=1}^r \langle x, v_i \rangle_E \sigma_i u_i \quad (\text{since } f(v_i) = \sigma_i u_i \text{ for } 1 \leq i \leq r \text{ and } 0 \text{ else}). \end{aligned} \tag{6.1}$$

This writing shows that $r = \text{rank}(f)$ since every $f(x)$ can be expressed as a linear combination of the set of linearly independent vectors $\{u_1, \dots, u_r\}$.

In many relevant applications, n and m are large and r is usually expected to be large as well. We can thus consider that expressing the action of f can be costly because the sum in Eq. (6.1) is large. The goal of data compression techniques is to find another map $\tilde{f} \in \mathcal{L}(E, F)$ which approximates f and which is less computationally expensive. If we assume that the singular values are given in decreasing order such that $\sigma_1 \geq \dots \geq \sigma_r > 0$, and if we assume that they present a fast decay, then one would hope that keeping only the first $k \leq r$ elements of the expansion in Eq. (6.1) would still provide a good approximation of $f(x)$. We thus consider the mapping $\tilde{f}_k \in \mathcal{L}(E, F)$ such that

$$\tilde{f}_k(x) := \sum_{i=1}^k \langle x, v_i \rangle_E \sigma_i u_i.$$

This mapping has rank $k \leq r$, and if $k \ll r$ then we can consider that evaluating $\tilde{f}_k(x)$ comes at a low computational cost. The main question that arises is the one of accuracy: how accurate is $\tilde{f}_k(x)$ as an approximation of $f(x)$ and what are the trade-offs between the loss of accuracy and the computational savings? The following theorem, called Eckart-Young-Mirsky, shows that the truncated mapping $\tilde{f}_k$ is the best approximation of f among all linear mappings from E to F that have rank $\leq k$. This means that the SVD provides a pathway to optimally reduce the complexity of endomorphisms.

Theorem 6.10: Eckart-Young-Mirsky

Let $f \in \mathcal{L}(E, F)$ be a linear map with SVD decomposition given by singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$ ($r \leq \min(n, m)$) and orthonormal bases $\mathcal{E}$ and $\mathcal{F}$ of E and F respectively, such that

$$f(x) = \sum_{i=1}^r \langle x, v_i \rangle_E \sigma_i u_i.$$

For $1 \leq k \leq r$, let

$$\tilde{f}_k(x) := \sum_{i=1}^k \langle x, v_i \rangle_E \sigma_i u_i.$$

Then $\tilde{f}_k$ is the best rank k approximation of f in the sense that:

- In operator norm:

$$\|f - \tilde{f}_k\|_{\text{op}} = \min_{\substack{g \in \mathcal{L}(E, F) \\ \text{rank}(g) \leq k}} \|f - g\|_2 = \sigma_{k+1} \quad \left(\text{and remember that } \|f - \tilde{f}_k\|_2 := \max_{x \in E} \frac{\|f(x) - \tilde{f}_k(x)\|_F}{\|x\|_E} \right).$$

- In the Fröbenius (Hilbert-Schmidt) norm:

$$\|f - \tilde{f}_k\|_F = \min_{\substack{g \in \mathcal{L}(E, F) \\ \text{rank}(g) \leq k}} \|f - g\|_F = \left(\sum_{i=k+1}^r \sigma_i^2 \right)^{1/2} \quad \left(\text{and remember that } \|f - \tilde{f}_k\|_F := \text{Tr}(f^* \circ f) \right).$$

Singular Value Decomposition: $A = U\Sigma V^*$

A linear operator $A : \mathbb{K}^n \rightarrow \mathbb{K}^m$ decomposes into rotations and a stretch.

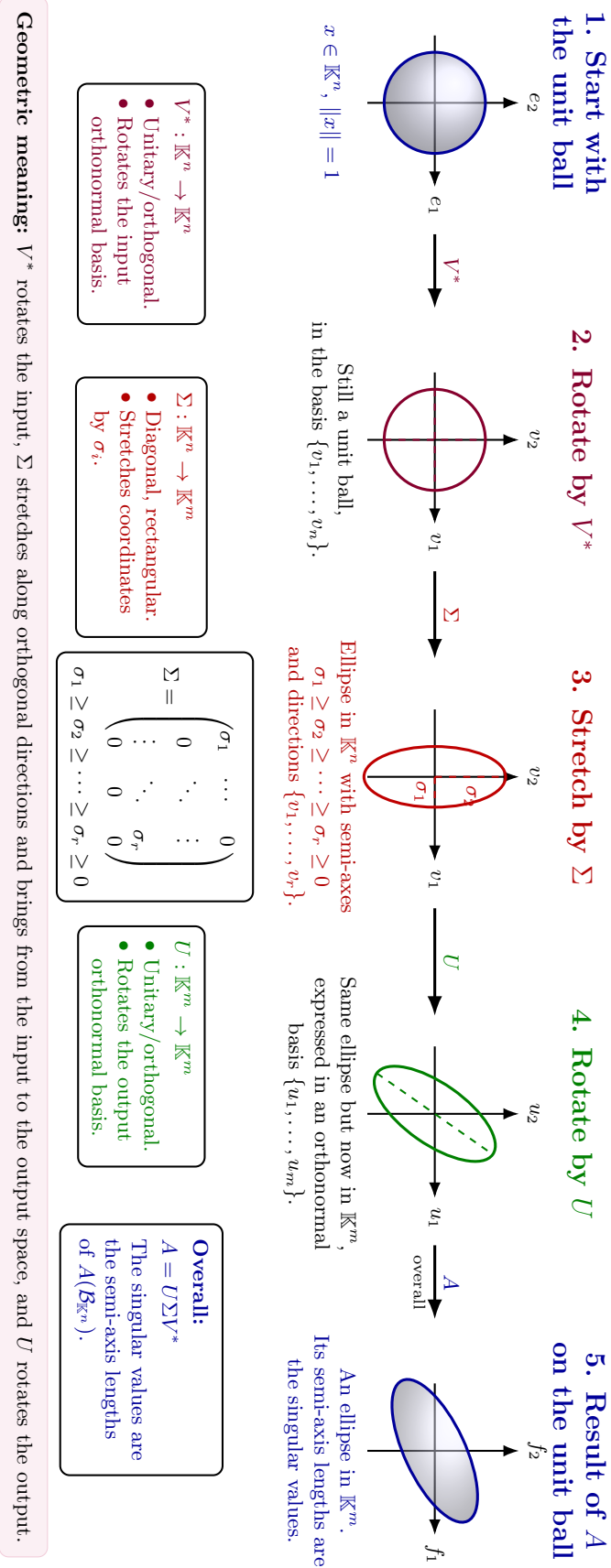


Figure 6.1: A summary of the Singular Value Decomposition.

Grading of Pro-Seminar

Grading Scheme and its motivations

Motivations: Our grading scheme is our best attempt to reflect on the following:

- First and foremost, we want that the PS is a place of exchange of ideas where we openly talk about the content of the lecture, the structure of the proofs of the exercises, and we all help each other out to become better mathematicians. We would like that each person presenting solutions at the black-board gets constructive feedback from the peers and the teacher. It is not our intention to penalize very heavily wrong answers if you show that you are making real efforts to understand.
- At present, the solution to most of the problems can be obtained via AI chatbots. There is no way for us to control how you make use of it. We believe that simply asking the chatbots for the solutions is not going to help you develop enough as a mathematician. In the mid/long run, your ability to critically solve problems will be very diminished. Therefore, even if we will value positively that you present correct answers at the black-board, we will value even more that you show that you understand the content that you are presenting. We will evaluate this by your explanations, and also through the constructive questions that you will receive from the teacher and your peers. The questions will be related to the exercise, but also to the main content of the course.

Grading Scheme: Suppose the grade goes from 0 points to 100 points (best grade). Then the points are allocated as follows:

Activity	Grade Baseline	Conditions	Based on	Evaluation criteria
Attendance	40	if ≤ 2 absences	roll call in class	Grade is 40 points.
	30	if ≥ 3 absences	roll call in class	Let $n+3$ be the number of absences with $n = 0, \dots, 12$. Grade is $30(1 - n/12)$.
Homework	30	if $\geq 70\%$ of ticked exercises	2 black-board presentations	correctness and structure of solution, oral explanation, insightful comments, reaction to questions from audience
	15	if $< 70\%$ of ticked exercises	2 black-board presentations	
Oral Participation	30			constructive questions, feedback and comments to students presenting

Remarks:

- In 2026, we will have 15 PS sessions.
- If you are facing personal issues that prevent you from attending, please let us know. We will make exceptions if the case is justified.

Acknowledged flaws in this grading system:

- Attendance: We feel that we may be a bit too generous with your absences, but this is the scheme that we will go for this year.

- Homework: The difficulty of the exercises may vary so some students will be evaluated on harder exercises. However, since we are focusing very much on how you explain and structure your comments, rather than only on the correctness, presenting harder exercises may actually turn out to be to your advantage. What happens if there is not enough time for everyone to present twice? The teachers will do their best to prevent this from happening but we will be generous if in the end a few people did not manage to present twice.
- Oral Participation: Shy people may not engage as much as more outgoing people. The teachers will do their best to make sure that everyone has an equal chance to speak. Also, what is considered constructive may be subjective. Again, the teachers will do their best in applying the same criteria to everyone.

We welcome any constructive feedback to improve our grading system in subsequent years.

Examples of computations for a grade

- Anna missed 5 sessions, ticked 100% of exercises, and did a great job in presenting 2 homework exercises. She regularly participated in giving constructive feedback. Her grade is then:

- Attendance (baseline is 30 points): $30 \text{ points} \times (1 - 2/12) = 25$ (here $n = 2$)
- Homework (baseline is 30 points): 30 points
- Oral Participation (baseline is 30 points): 30 points

Final grade: $25 + 30 + 30 = 84$ points

- John missed 2 sessions, ticked 50% of exercises, and did not do a great job in presenting 2 homework exercises (unstructured proofs, poor oral presentation, could not answer very well to questions). He sometimes asked questions and gave constructive feedback to other students:

- Attendance: 40 points
- Homework (baseline is 15 points): 5 points
- Oral Participation: 20 points

Final grade: 65 points

Back Matter

List of Acronyms

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